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(54) **RECOMBINANT YEAST CELL**

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(57) **ABSTRACT**

A recombinant yeast cell functionally expressing: —a
nucleic acid sequence encoding a native protein having
transketolase activity (EC 2.2.1.1); and —a nucleic acid
sequence encoding a heterologous protein having transke-
tolase activity (EC 2.2.1.1).

Specification includes a Sequence Listing.

RECOMBINANT YEAST CELL

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a U.S. National Stage Application under 35 U.S.C. § 371 of International Application No. PCT/EP2022/068917, filed Jul. 7, 2022, which claims priority to U.S. Application No. 63/220,896, filed Jul. 12, 2021, and European Application No. 21185148.0, filed Jul. 12, 2021, all of which are hereby incorporated by reference in their entireties.

FIELD OF THE INVENTION

[0002] The invention relates to a recombinant yeast cell and to a process for the production of ethanol wherein said recombinant yeast cell is used.

INCORPORATION BY REFERENCE OF THE SEQUENCE LISTING

[0003] The present application is being filed with a Sequence Listing in electronic format. The Sequence Listing is provided as a file entitled IFF34056-US-PCT_SEQ_Listing.xml, created on Jul. 31, 2024, which is 188,359 bytes in size. The information in the electronic format of the Sequence Listing is incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

[0004] Microbial fermentation processes are applied to industrial production of a broad and rapidly expanding range of chemical compounds from renewable carbohydrate feedstocks. Especially in anaerobic fermentation processes, redox balancing of the cofactor couple NADH/NAD⁺ can cause important constraints on product yields. This challenge is exemplified by the formation of glycerol as major by-product in the industrial production of—for instance—fuel ethanol by *Saccharomyces cerevisiae*, a direct consequence of the need to re-oxidize NADH formed in biosynthetic reactions.

[0005] Ethanol production by *Saccharomyces cerevisiae* is currently, by volume, the single largest fermentation process in industrial biotechnology. Various approaches have been proposed to improve the fermentative properties of organisms used in industrial biotechnology by genetic modification. A major challenge relating to the stoichiometry of yeast-based production of ethanol, is that substantial amounts of NADH-dependent side-products such as glycerol are generally formed as a by-product, especially under anaerobic and oxygen-limited conditions or under conditions where respiration is otherwise constrained or absent. It has been estimated that, in typical industrial ethanol processes, up to about 4 wt. % of the sugar feedstock is converted into glycerol (Nissen et al., “Anaerobic and aerobic batch cultivations of *Saccharomyces cerevisiae* mutants impaired in glycerol synthesis”, (2000), Yeast, vol. 16, pages 463-474). Under conditions that are ideal for anaerobic growth, the conversion into glycerol may even be higher, up to about 10%.

[0006] Glycerol production under anaerobic conditions is primarily linked to redox metabolism. During anaerobic growth of *S. cerevisiae*, sugar dissimilation occurs via alcoholic fermentation. In this process, the NADH formed in the glycolytic glyceraldehyde-3-phosphate dehydrogenase reaction is reoxidized by converting acetaldehyde, formed

by decarboxylation of pyruvate to ethanol via NAD⁺-dependent alcohol dehydrogenase. The fixed stoichiometry of this redox-neutral dissimilatory pathway causes problems when a net reduction of NAD⁺ to NADH occurs elsewhere in metabolism. Under anaerobic conditions, NADH reoxidation in *S. cerevisiae* is strictly dependent on reduction of sugar to glycerol. Glycerol formation is initiated by reduction of the glycolytic intermediate dihydroxyacetone phosphate (DHAP) to glycerol 3-phosphate (glycerol-3P), a reaction catalyzed by NAD⁺-dependent glycerol 3-phosphate dehydrogenase. Subsequently, the glycerol 3-phosphate formed in this reaction is hydrolysed by glycerol-3-phosphatase to yield glycerol and inorganic phosphate. Consequently, glycerol is a major by-product during anaerobic production of ethanol by *S. cerevisiae*, which is undesired as it reduces overall conversion of sugar to ethanol. Further, the presence of glycerol in effluents of ethanol production plants may impose costs for waste-water treatment.

[0007] In literature, several different approaches have been reported that could help to reduce the amount of the byproduct glycerol and divert carbon to ethanol resulting in an increased yield of ethanol per gram of fermented carbohydrate.

[0008] In WO2011/010923, WO2014/081803, WO2014/129898, WO2015/107496, WO2015/148272, WO2018/172328 and WO2018/228836, several alternative reduction pathways to reoxidize the NADH produced, also referred to as alternative “redox sinks”, have been suggested and implemented into (recombinant) yeast cells.

[0009] For example, in WO 2011/010923, the NADH-related side-product (glycerol) formation in a process for the production of ethanol from a carbohydrate containing feedstock is addressed by providing a recombinant yeast cell comprising one or more recombinant nucleic acid sequences encoding an NAD⁺ dependent acetylating acetaldehyde dehydrogenase (EC 1.2.1.10) activity. The cell may for example lack enzymatic activity needed for the NADH dependent glycerol synthesis or the cell may have a reduced enzymatic activity with respect to the NADH dependent glycerol synthesis compared to its corresponding wild-type yeast cell.

[0010] WO2014/129898 describes a recombinant cell functionally expressing heterologous nucleic acid sequences encoding for ribulose-1,5-phosphate carboxylase/oxygenase (EC 4.1.1.39; herein abbreviated as “Rubisco”), and optionally molecular chaperones for Rubisco, and phosphoribulokinase (EC 2.7.1.19; herein abbreviated as “PRK”). In addition, reference is made to the use of carbon dioxide as an electron acceptor in a recombinant chemotrophic micro-organism.

[0011] Although the described processes and yeast cells are advantageous, there is a continuing need for improvement. In an industrial environment the reduction in glycerol production by the above recombinant yeast cells can potentially affect their osmotolerance and their stress response to the external environment. Especially under challenging process conditions, for example when applying a fermentation medium having a high dry solids content and/or a high fermentation temperature, this may lead to a decline of the cell population and/or cell activity at the end of the fermentation period. It would be an advancement in the art to provide a process, and yeast cells for use therein, wherein the yeast cells have an improved robustness under high dry

solids/high dry matter conditions and/or high temperatures. In addition, it would be an advancement in the art to provide yeast cells that have a reduced accumulation of glucose and/or total sugar content within the yeast cell. That is, it would be an advancement in the art to achieve a continued performance of the yeast cell and/or a low concentration of remaining glucose at the end of the fermentation, even where a high concentration of glucose is present at the start and/or throughout the fermentation.

SUMMARY OF THE INVENTION

[0012] The inventors have now surprising found that the processes and yeast cells of the prior art can be improved by using a specific combination of transketolase proteins.

[0013] Accordingly the invention provides a recombinant yeast cell functionally expressing:

[0014] a nucleic acid sequence encoding a native protein having transketolase activity; and

[0015] a nucleic acid sequence encoding a heterologous protein having transketolase activity.

In addition, the invention provides a process for the production of ethanol, comprising converting a carbon source, such as a carbohydrate or another organic carbon source, using the above recombinant yeast cell, thereby suitably forming ethanol.

[0016] Advantageously, use of the above recombinant yeast cell and/or the above process results in an improved robustness. Such is especially advantageous when a medium having a high dry solids content is applied and/or if a high fermentation temperature is applied.

[0017] A process for the production of ethanol from a carbon source, such as a carbohydrate, can advantageously be carried out in the presence of a saccharolytic enzyme,

such as glucoamylase, to convert polysaccharides and/or oligosaccharides into glucose. When the process is carried out in a medium with a high dry matter content, for example after starting the process with a high concentration of corn mash, the concentration of glucose in the medium can become very high. Without wishing to be bound by any kind of theory, it is believed that a high concentration of glucose can cause osmotic stress for the yeast cell, causing the yeast cell to stop performing and even die.

[0018] Without wishing to be bound by any kind of theory it is believed that, compared to a yeast cell not comprising the claimed combination of transketolase proteins, the above recombinant yeast cell allows for reduced accumulation of glucose and/or other sugars within the yeast cell, thereby suitably allowing for an improved robustness.

[0019] The advantages are illustrated by the examples. In the examples fermentation is carried out at a high dry matter content of 36% w/w. As illustrated by the examples the recombinant yeast cell according to the invention, and the process according to the invention, allow for a continued performance of the yeast cell and/or continued conversion of the glucose. Even in a medium comprising a concentration of glucose as high as 36% w/w and/or temperatures as high as 3200, the recombinant yeast cell is still alive and converting carbohydrates into ethanol after 66 hours. As a result a low concentration of remaining glucose can be obtained at the end of the fermentation, even where a high concentration of glucose is present at the start and/or throughout the fermentation.

BRIEF DESCRIPTION OF THE SEQUENCE LISTING

[0020] This application contains a Sequence Listing in computer readable form, which is incorporated herein by reference. An overview is provided by Table 1 below.

TABLE 1

Overview of sequence listings:			
SEQ ID No:	Enzyme Name	Origin	Type
SEQ ID No: 1	Transketolase	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID No: 2	Transketolase	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID No: 3	Transketolase (DAS1)	<i>Komagataella phaffii</i> CBS 7435	protein
SEQ ID No: 4	Transketolase (DAS2)	<i>Komagataella phaffii</i> CBS 7435	protein
SEQ ID NO: 5	Transketolase	<i>Candida tenuis</i>	protein
SEQ ID NO: 6	Transketolase	<i>Debaryomyces</i>	protein
SEQ ID NO: 7	Transketolase	<i>Kuraishia capsulata</i> CBS 1993	protein
SEQ ID NO: 8	Transketolase	<i>Candida arabinofementans</i> NRRL YB-2248	protein
SEQ ID NO: 9	Transketolase	<i>Metschnikowia pulcherrima</i>	protein
SEQ ID NO: 10	Transketolase	<i>Meyerozyma</i>	protein
SEQ ID NO: 11	Transketolase	<i>Diutina rugosa</i>	protein
SEQ ID NO: 12	Transketolase	<i>Pachysolen tannophilus</i> NRRL Y-2460	protein
SEQ ID NO: 13	Transketolase	<i>Rasamsonia emersonii</i> CBS 393.64	protein
SEQ ID NO: 14	Transketolase	<i>Kluyveromyces marxianus</i> (strain DMKU3-1042/BCC 29191/NBRC 104275)	protein
SEQ ID NO: 15	Transketolase	<i>Ogataea parapolymorpha</i> DL-1	protein
SEQ ID NO: 16	Transketolase	Artificial sequence; based on Transketolase protein from <i>Komagataella phaffii</i>	protein
SEQ ID NO: 17	Transketolase	<i>Komagataella phaffii</i> (<i>Pichia pastoris</i>)	protein

TABLE 1-continued

Overview of sequence listings:			
SEQ ID No:	Enzyme Name	Origin	Type
SEQ ID NO: 18	Transketolase	<i>Komagataella phaffii</i> (<i>Pichia pastoris</i>)	DNA
SEQ ID NO: 19	Transketolase	<i>Kluyveromyces lactis</i>	protein
SEQ ID NO: 20	Transketolase	<i>Kluyveromyces lactis</i>	DNA
SEQ ID NO: 21	Promoter Motif	Artificial sequence	DNA
SEQ ID NO: 22	Promoter Motif	Artificial sequence	DNA
SEQ ID NO: 23	Promoter ANB1	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID NO: 24	Promoter DAN1	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID NO: 25	Ribulose-1,5-biphosphate carboxylase oxygenase (RuBisCO) as previously described in WO2014/129898	<i>Thiobacillus denitrificans</i>	protein
SEQ ID NO: 26	Ribulose-1,5-biphosphate carboxylase oxygenase (RuBisCO) as previously described in WO2014/129898	Artificial sequence; based on cbbM gene from <i>Thiobacillus</i> <i>denitrificans</i> - codon optimized	DNA
SEQ ID NO: 27	Phosphoribulokinase (PRK) as previously described in WO2014/129898	<i>Spinacea oleracea</i>	protein
SEQ ID NO: 28	Phosphoribulokinase (PRK) as previously described in WO2014/129898	Artificial sequence; based on prk gene from <i>Spinacea</i> <i>oleracea</i> , codon optimized	DNA
SEQ ID NO: 29	(long) chaperone groEL as previously described in WO2014/129898	Artificial sequence; Translated protein sequence based on GroEL of <i>Escherichia coli</i>	protein
SEQ ID NO: 30	(long) chaperone groEL as previously described in WO2014/129898	Artificial sequence; based on GroEL gene from <i>Escherichia</i> <i>coli</i> , codon optimized	DNA
SEQ ID NO: 31	(short) chaperone groES as previously described in WO2014/129898	Artificial sequence; Translated protein sequence based on GroES of <i>Escherichia coli</i>	protein
SEQ ID NO: 32	(short) chaperone groES as previously described in WO2014/129898	Artificial sequence; based on GroES gene from <i>Escherichia</i> <i>coli</i> , codon optimized	DNA
SEQ ID NO: 33	Phosphoketolase	<i>Bifidobacterium animalis</i>	protein
SEQ ID NO: 34	Phosphoketolase	<i>Bifidobacterium adolescentis</i>	protein
SEQ ID NO: 35	Phosphoketolase	<i>Bifidobacterium lactis</i>	protein
SEQ ID NO: 36	Phosphoketolase	<i>Leuconostoc mesenteroides</i>	protein
SEQ ID NO: 37	Phosphotransacetylase	<i>Bacillus subtilis</i>	protein
SEQ ID NO: 38	Phosphotransacetylase	<i>Lactobacillus plantarum</i>	protein
SEQ ID NO: 39	Phosphotransacetylase	<i>Bifidobacterium adolescentis</i>	protein
SEQ ID NO: 40	Phosphotransacetylase	<i>Methosarcina thermophila</i>	protein
SEQ ID NO: 41	Acetate kinase	<i>Bifidobacterium adolescentis</i>	protein
SEQ ID NO: 42	Acetate kinase	<i>Aspergillus nidulans</i>	protein
SEQ ID NO: 43	Bifunctional NAD ⁺ dependent acetylating acetaldehyde/alcohol dehydrogenase (adhE)	<i>Escherichia coli</i>	protein
SEQ ID NO: 44	Ethanolamine utilizing protein (eutE)	<i>Escherichia coli</i>	protein
SEQ ID NO: 45	acetaldehyde dehydrogenase (acdH)	<i>Lactobacillus plantarum</i>	protein
SEQ ID NO: 46	acetaldehyde dehydrogenase (acdH)	<i>Listeria innocua</i>	protein
SEQ ID NO: 47	acetaldehyde/alcohol dehydrogenase (adhE)	<i>Staphylococcus aureus</i>	protein
SEQ ID NO: 48	acetylating acetaldehyde dehydrogenase (mhpF)	<i>Escherichia coli</i>	protein
SEQ ID NO: 49	Acetyl-coenzyme A synthetase 1 (ACS1)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 50	Acetyl-coenzyme A synthetase 2 (ACS2)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 51	Alcohol dehydrogenase 1 (ADH1)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 52	Alcohol dehydrogenase 2 (ADH2)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 53	Alcohol dehydrogenase 3 (ADH3)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 54	Alcohol dehydrogenase 4 (ADH4)	<i>Saccharomyces cerevisiae</i>	protein

TABLE 1-continued

Overview of sequence listings:			
SEQ ID No:	Enzyme Name	Origin	Type
SEQ ID NO: 55	Alcohol dehydrogenase 5 (ADH5)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 56	Glucoamylase (GA)	<i>Punctularia strigosozonata</i>	protein
SEQ ID NO: 57	Glucoamylase (GA)	<i>Punctularia strigosozonata</i>	protein
SEQ ID NO: 58	Glucoamylase (GA)	Artificial sequence, based on <i>Punctularia strigosozonata</i>	protein
SEQ ID NO: 59	Glycerol dehydrogenase (Kpne_gldA)	<i>Klebsiella pneumoniae</i>	protein
SEQ ID NO: 60	Glycerol dehydrogenase (Eaer_gldA)	<i>Enterococcus aerogenes</i>	protein
SEQ ID NO: 61	Glycerol dehydrogenase (Ya_gldA)	<i>Yersinia aldovae</i>	protein
SEQ ID NO: 62	Glycerol dehydrogenase (Ec_gldA)	<i>Escherichia coli</i>	protein
SEQ ID NO: 63	Glycerol dehydrogenase (Ec_gldA)	Artificial sequence; based on gldA gene from <i>Escherichia coli</i> - codon optimized	DNA
SEQ ID NO: 64	Dihydroxyacetone kinase (DAK1)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 65	Dihydroxyacetone kinase (DAK2)	<i>Saccharomyces cerevisiae</i>	protein
SEQ ID NO: 66	Dihydroxyacetone kinase (Kpne_dhaK)	<i>Klebsiella pneumoniae</i>	protein
SEQ ID NO: 67	Dihydroxyacetone kinase (Ylip_DAK1)	<i>Yarrowia lipolytica</i>	protein
SEQ ID NO: 68	Dihydroxyacetone kinase (Spom_DAK1)	<i>Schizosaccharomyces pombe</i>	protein
SEQ ID NO: 69	Dihydroxyacetone kinase (DAK1)	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID NO: 70	Dihydroxyacetone kinase (DAK2)	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID NO: 71	Glycerol transporter Aquaporin 9 (Drer_T3)	<i>Danio rerio</i>	protein
SEQ ID NO: 72	Glycerol transporter ZYRO0E01210p (Zrou_T5)	<i>Zygosaccharomyces rouxii</i>	protein
SEQ ID NO: 73	“fragmentA”	Artificial Sequence	DNA
SEQ ID NO: 74	“fragmentB”	Artificial Sequence	DNA
SEQ ID NO: 75	“fragmentC”	Artificial Sequence	DNA
SEQ ID NO: 76	“fragmentD”	Artificial Sequence	DNA
SEQ ID NO: 77	“fragmentE”	Artificial Sequence	DNA
SEQ ID NO: 78	repair DNA fragment	Artificial Sequence	DNA
SEQ ID NO: 79	Terminator EBO1 (Sc_EBO1t)	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID NO: 80	Promoter ENO2 (Sc_ENO2p)	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID NO: 81	<i>S. cerevisiae</i> YOX1 terminator (Sc_YOX1t)	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID NO: 82	INT14.02 protospacer	Artificial Sequence	DNA
SEQ ID NO: 83	Sc_INT14.02_FLANK5	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID NO: 84	Sc_INT14.02_FLANK3	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID NO: 85	INT95 protospacer	Artificial Sequence	DNA
SEQ ID NO: 86	Sc_INT95B_FLANK5	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID NO: 87	Sc_INT95B_FLANK3	<i>Saccharomyces cerevisiae</i>	DNA
SEQ ID NO: 88	GPD2 protospacer	Artificial Sequence	DNA
SEQ ID No: 89	NR1 nitrate reductase [Uniprot accession No: Q9XGW5]	<i>Zea mays</i>	Protein (amino acid sequence)
SEQ ID No: 90	NIA Nitrate reductase [Uniprot accession No: P23312]	<i>Spinacia oleracea</i>	Protein
SEQ ID No: 91	nirD Nitrite reductase (small subunit)	<i>Escherichia coli</i>	Protein
SEQ ID No: 92	nirB Nitrite reductase (large subunit)	<i>Escherichia coli</i>	Protein
SEQ ID No: 93	niiA Nitrite reductase	<i>Emericella nidulans</i> (<i>Aspergillus nidulans</i>)	Protein
SEQ ID No: 94	YNT1 Nitrate/Nitrite transporter	<i>Pichia angusta</i> (<i>Hansenula polymorpha</i>)	Protein

[0021] In the context of this patent application, each of the above protein/amino acid sequences is preferably encoded by a DNA/nucleic acid sequence that is codon-pair optimized for expression in a yeast, more preferably for expression in a *Saccharomyces cerevisiae* yeast.

DETAILED DESCRIPTION OF THE INVENTION

Definitions

[0022] Unless defined otherwise or clearly indicated by context, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art.

[0023] Throughout the present specification and the accompanying claims, the words “comprise” and “include” and variations such as “comprises”, “comprising”, “includes” and “including” are to be interpreted inclusively. That is, these words are intended to convey the possible inclusion of other elements or integers not specifically recited, where the context allows.

[0024] The articles “a” and “an” are used herein to refer to one or to more than one (i.e. to one or at least one) of the grammatical object of the article. By way of example, “an element” may mean one element or more than one element. When referring to a noun (e.g. a compound, an additive, etc.) in the singular, the plural is meant to be included. Thus, when referring to a specific moiety, e.g. “gene”, this means “at least one” of that gene, e.g. “at least one gene”, unless specified otherwise.

[0025] When referring to a compound of which several isomers exist (e.g. a D and an L enantiomer), the compound in principle includes all enantiomers, diastereomers and cis/trans isomers of that compound that may be used in the particular aspect of the invention; in particular when referring to such as compound, it includes the natural isomer(s).

[0026] Unless explicitly indicated otherwise, the various embodiments of the invention described herein can be cross-combined.

[0027] The term “carbon source” refers to a source of carbon, preferably a compound or molecule comprising carbon. Preferably the carbon source is a carbohydrate. A carbohydrate is understood herein to be an organic compound made of carbon, oxygen and hydrogen. Suitably the carbon source may be selected from the group consisting of mono-, di- and/or polysaccharides, acids and acid salts. More preferably the carbon source is a compound selected from the group consisting of glucose, arabinose, xylose, galactose, mannose, rhamnose, fructose, glycerol, and acetic acid or a salt thereof.

[0028] The terms “dry matter” and “dry solids”, abbreviated respectively as “DM” and “DS”, are used interchangeably herein and refer to material remaining after removal of water. Dry matter content can be determined by any method known to the person skilled in the art therefore.

[0029] The term “ferment”, and variations thereof such as “fermenting”, “fermentation” and/or “fermentative”, is used herein in a classical sense, i.e. to indicate that a process is or has been carried out under anaerobic conditions. An anaerobic fermentation is herein defined to be a fermentation carried out under anaerobic conditions. Anaerobic conditions are herein defined as conditions without any oxygen or in which essentially no oxygen is consumed by the yeast cell. Conditions in which essentially no oxygen is consumed

suitably corresponds to an oxygen consumption of less than 5 mmol/l·h⁻¹, in particular to an oxygen consumption of less than 2.5 mmol/l·h⁻¹, or less than 1 mmol/l·h⁻¹. More preferably 0 mmol/L/h is consumed (i.e. oxygen consumption is not detectable). This suitably corresponds to a dissolved oxygen concentration in a culture broth of less than 5% of air saturation, more suitably to a dissolved oxygen concentration of less than 1% of air saturation, or less than 0.2% of air saturation.

[0030] The term “fermentation process” refers to a process for the preparation or production of a fermentation product.

[0031] The term “cell” refers to a eukaryotic or prokaryotic organism, preferably occurring as a single cell. In the present invention the cell is a recombinant yeast cell. That is, the recombinant cell is selected from the group of genera consisting of yeast.

[0032] The terms “yeast” and “yeast cell” are used herein interchangeably and refer to a phylogenetically diverse group of single-celled fungi, most of which are in the division of Ascomycota and Basidiomycota. The budding yeasts (“true yeasts”) are classified in the order Saccharomycetales. The yeast cell according to the invention is preferably a yeast cell derived from the genus of *Saccharomyces*. More preferably the yeast cell is a yeast cell of the species *Saccharomyces cerevisiae*.

[0033] The term “recombinant”, for example referring to a “recombinant yeast”, a “recombinant cell”, “recombinant micro-organism” and/or “recombinant strain” as used herein, refers to a yeast, cell, micro-organism or strain, respectively, containing nucleic acid which is the result of one or more genetic modifications. Simply put the yeast, cell, micro-organism or strain contains a different combination of nucleic acid from (either of) its parent(s). To construe a recombinant yeast, cell, micro-organism or strain, recombinant DNA technique(s) and/or another mutagenic technique(s) can be used. For example a recombinant yeast and/or a recombinant yeast cell may comprise nucleic acid not present in the corresponding wild-type yeast and/or cell, which nucleic acid has been introduced into that yeast and/or yeast cell using recombinant DNA techniques (i.e. a transgenic yeast and/or cell), or which nucleic acid not present in said wild-type yeast and/or cell is the result of one or more mutations—for example using recombinant DNA techniques or another mutagenesis technique such as UV-irradiation—in a nucleic acid sequence present in said wild-type yeast and/or yeast cell (such as a gene encoding a wild-type polypeptide) or wherein the nucleic acid sequence of a gene has been modified to target the polypeptide product (encoding it) towards another cellular compartment. Further, the term “recombinant” may suitably relate to a yeast, cell, micro-organism or strain from which nucleic acid sequences have been removed, for example using recombinant DNA techniques.

[0034] By a recombinant yeast comprising or having a certain activity is herein understood that the recombinant yeast may comprise one or more nucleic acid sequences encoding for a protein having such activity. Hence allowing the recombinant yeast to functionally express such a protein or enzyme.

[0035] The term “functionally expressing” means that there is a functioning transcription of the relevant nucleic acid sequence, allowing the nucleic acid sequence to actually be transcribed, for example resulting in the synthesis of a protein.

[0036] The term “transgenic” as used herein, for example referring to a “transgenic yeast” and/or a “transgenic cell”, refers to a yeast and/or cell, respectively, containing nucleic acid not naturally occurring in that yeast and/or cell and which has been introduced into that yeast and/or cell using for example recombinant DNA techniques, such as a recombinant yeast and/or cell.

[0037] The term “mutated” as used herein regarding proteins or polypeptides means that, as compared to the wild-type or naturally occurring protein or polypeptide sequence, at least one amino acid has been replaced with a different amino acid, inserted into, or deleted from the amino acid sequence. The replacement, insertion or deletion of the amino acid can for example be achieved via mutagenesis of nucleic acids encoding these amino acids. Mutagenesis is a well-known method in the art, and includes, for example, site-directed mutagenesis by means of PCR or via oligonucleotide-mediated mutagenesis as described in Sambrook et al., *Molecular Cloning-A Laboratory Manual*, 2nd ed., Vol. 1-3 (1989), published by Cold Spring Harbor Publishing).

[0038] The term “mutated” as used herein regarding genes means that, as compared to the wild-type or naturally occurring nucleic acid sequence, at least one nucleotide in the nucleic acid sequence of a gene or a regulatory sequence thereof, has been replaced with a different nucleotide, inserted into, or deleted from the nucleic acid sequence. The replacement, insertion or deletion of the amino acid can for example be achieved via mutagenesis, resulting for example in the transcription of a protein sequence with a qualitatively or quantitatively altered function or the knock-out of that gene. In the context of this invention an “altered gene” has the same meaning as a mutated gene.

[0039] The term “gen” or “gene”, as used herein, refers to a nucleic acid sequence that can be transcribed into mRNAs that are then translated into protein. A gene encoding for a certain protein refers to the one or more nucleic acid sequence(s) encoding for such a protein.

[0040] The term “nucleic acid” or “nucleotide” as used herein, refers to a monomer unit in a deoxyribonucleotide or ribonucleotide polymer, i.e. a polynucleotide, in either single or double-stranded form, and unless otherwise limited, encompasses known analogues having the essential nature of natural nucleotides in that they hybridize to single-stranded nucleic acids in a manner similar to naturally occurring nucleotides (e. g., peptide nucleic acids). For example, a certain enzyme that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to the reference nucleotide sequence encoding the enzyme. A polynucleotide can be full-length or a subsequence of a native or heterologous structural or regulatory gene. Unless otherwise indicated, the term includes reference to the specified sequence as well as the complementary sequence thereof. Thus, DNAs or RNAs with backbones modified for stability or for other reasons are “polynucleotides” as that term is intended herein. Moreover, DNAs or RNAs comprising unusual bases, such as inosine, or modified bases, such as tritylated bases, to name just two examples, are polynucleotides as the term is used herein. It will be appreciated that a great variety of modifications have been made to DNA and RNA that serve many useful purposes known to those of skill in the art. The term polynucleotide as it is employed herein embraces such chemically, enzymatically or metabolically modified

forms of polynucleotides, as well as the chemical forms of DNA and RNA characteristic of viruses and cells, including among other things, simple and complex cells.

[0041] The terms “nucleotide sequence” and “nucleic acid sequence” are used interchangeably herein. An example of a nucleic acid sequence is a DNA sequence.

[0042] The terms “polypeptide”, “peptide” and “protein” are used interchangeably herein to refer to a polymer of amino acid residues, for example illustrated by an amino acid sequence. The terms apply to amino acid polymers in which one or more amino acid residue is an artificial chemical analogue of a corresponding naturally occurring amino acid, as well as to naturally occurring amino acid polymers. The essential nature of such analogues of naturally occurring amino acids is that, when incorporated into a protein, that protein is specifically reactive to antibodies elicited to the same protein but consisting entirely of naturally occurring amino acids. The terms “polypeptide”, “peptide” and “protein” are also inclusive of modifications including, but not limited to, glycosylation, lipid attachment, sulphation, gamma-carboxylation of glutamic acid residues, hydroxylation and ADP-ribosylation.

[0043] The term “enzyme” refers herein to a protein having a catalytic function. Where a protein catalyzes a certain biological reaction, the terms “protein” and “enzyme” may be used interchangeably herein. When an enzyme is mentioned with reference to an enzyme class (EC), the enzyme class is a class wherein the enzyme is classified or may be classified, on the basis of the Enzyme Nomenclature provided by the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology (NC-IUBMB), which nomenclature may be found at <http://www.chem.qmul.ac.uk/iubmb/enzyme/>. Other suitable enzymes that have not (yet) been classified in a specified class but may be classified as such, are meant to be included.

[0044] If referred herein to a protein or a nucleic acid sequence, such as a gene, by reference to an accession number, this number in particular is used to refer to a protein or nucleic acid sequence (gene) having a sequence as can be found via www.ncbi.nlm.nih.gov/, (as available on 1 Oct. 2020) unless specified otherwise.

[0045] Every nucleic acid sequence herein that encodes a polypeptide also includes any conservatively modified variants thereof. This includes that, by reference to the genetic code, it describes every possible silent variation of the nucleic acid. The term “conservatively modified variants” applies to both amino acid and nucleic acid sequences. With respect to particular nucleic acid sequences, conservatively modified variants refers to those nucleic acids which encode identical or conservatively modified variants of the amino acid sequences due to the degeneracy of the genetic code. The term “degeneracy of the genetic code” refers to the fact that a large number of functionally identical nucleic acids encode any given protein. For instance, the codons GCA, GCC, GCG and GCU all encode the amino acid alanine. Thus, at every position where an alanine is specified by a codon, the codon can be altered to any of the corresponding codons described without altering the encoded polypeptide. Such nucleic acid variations are “silent variations” and represent one species of conservatively modified variation.

[0046] The term “functional homologue” (or in short “homologue”) of a polypeptide and/or amino acid sequence having a specific sequence (e.g. “SEQ ID NO: X”), as used

herein, refers to a polypeptide and/or amino acid sequence comprising said specific sequence with the proviso that one or more amino acids are mutated, substituted, deleted, added, and/or inserted, and which polypeptide has (qualitatively) the same enzymatic functionality for substrate conversion.

[0047] The term “functional homologue” (or in short “homologue”) of a polynucleotide and/or nucleic acid sequence having a specific sequence (e.g. “SEQ ID NO: X”), as used herein, refers to a polynucleotide and/or nucleic acid sequence comprising said specific sequence with the proviso that one or more nucleic acids are mutated, substituted, deleted, added, and/or inserted, and which polynucleotide encodes for a polypeptide sequence that has (qualitatively) the same enzymatic functionality for substrate conversion. With respect to nucleic acid sequences, the term functional homologue is meant to include nucleic acid sequences which differ from another nucleic acid sequence due to the degeneracy of the genetic code and encode the same polypeptide sequence.

[0048] Sequence identity is herein defined as a relationship between two or more amino acid (polypeptide or protein) sequences or two or more nucleic acid (polynucleotide) sequences, as determined by comparing the sequences. Usually, sequence identities or similarities are compared over the whole length of the sequences compared. In the art, “identity” also means the degree of sequence relatedness between amino acid or nucleic acid sequences, as the case may be, as determined by the match between strings of such sequences.

[0049] Amino acid or nucleotide sequences are said to be homologous when exhibiting a certain level of similarity. Two sequences being homologous indicate a common evolutionary origin. Whether two homologous sequences are closely related or more distantly related is indicated by “percent identity” or “percent similarity”, which is high or low respectively. Although disputed, to indicate “percent identity” or “percent similarity”, “level of homology” or “percent homology” are frequently used interchangeably. A comparison of sequences and determination of percent identity between two sequences can be accomplished using a mathematical algorithm. The skilled person will be aware of the fact that several different computer programs are available to align two sequences and determine the homology between two sequences (Kruskal et al., “*An overview of sequence comparison: Time warps, string edits, and macromolecules*”, (1983), Society for Industrial and Applied Mathematics (SIAM), Vol 25, No. 2, pages 201-237 and D. and the handbook edited by Sankoff and J. B. Kruskal, (ed.), “*Time warps, string edits and macromolecules: the theory and practice of sequence comparison*”, (1983), pp. 1-44, published by Addison-Wesley Publishing Company, Massachusetts USA).

[0050] The percent identity between two amino acid sequences can be determined using the Needleman and Wunsch algorithm for the alignment of two sequences. (Needleman et al “*A General Method Applicable to the Search for Similarities in the Amino Acid Sequence of Two Proteins*” (1970) J. Mol. Biol. Vol. 48, pages 443-453). The algorithm aligns amino acid sequences as well as nucleotide sequences. The Needleman-Wunsch algorithm has been implemented in the computer program NEEDLE. For the purpose of this invention the NEEDLE program from the EMBOSS package is used (version 2.8.0 or higher, see Rice

et al, “*EMBOSS: The European Molecular Biology Open Software Suite*” (2000), Trends in Genetics vol. 16, (6) pages 276-277, <http://emboss.bioinformatics.nl/>). For protein sequences, EBLOSUM62 is used for the substitution matrix. For nucleotide sequences, EDNAFULL is used. Other matrices can be specified. The optional parameters used for alignment of amino acid sequences are a gap-open penalty of 10 and a gap extension penalty of 0.5. The skilled person will appreciate that all these different parameters will yield slightly different results but that the overall percentage identity of two sequences is not significantly altered when using different algorithms.

[0051] The homology or identity is the percentage of identical matches between the two full sequences over the total aligned region including any gaps or extensions. The homology or identity between the two aligned sequences is calculated as follows: Number of corresponding positions in the alignment showing an identical amino acid in both sequences divided by the total length of the alignment including the gaps. The identity defined as herein can be obtained from NEEDLE and is labelled in the output of the program as “IDENTITY”.

[0052] The homology or identity between the two aligned sequences is calculated as follows: Number of corresponding positions in the alignment showing an identical amino acid in both sequences divided by the total length of the alignment after subtraction of the total number of gaps in the alignment. The identity defined as herein can be obtained from NEEDLE by using the NOBRIEF option and is labelled in the output of the program as “longest-identity”.

[0053] A variant of a nucleotide or amino acid sequence disclosed herein may also be defined as a nucleotide or amino acid sequence having one or more substitutions, insertions and/or deletions as compared to the nucleotide or amino acid sequence specifically disclosed herein (e.g. in the sequence listing).

[0054] Optionally, in determining the degree of amino acid similarity, the skilled person may also take into account so-called “conservative” amino acid substitutions, as will be clear to the skilled person. Conservative amino acid substitutions refer to the interchangeability of residues having similar side chains. For example, a group of amino acids having aliphatic side chains is glycine, alanine, valine, leucine, and isoleucine; a group of amino acids having aliphatic-hydroxyl side chains is serine and threonine; a group of amino acids having amide-containing side chains is asparagine and glutamine; a group of amino acids having aromatic side chains is phenylalanine, tyrosine, and tryptophan; a group of amino acids having basic side chains is lysine, arginine, and histidine; and a group of amino acids having sulphur-containing side chains is cysteine and methionine. In an embodiment, conservative amino acid substitution groups are: valine-leucine-isoleucine, phenyl-alanine-tyrosine, lysine-arginine, alanine-valine, and asparagine-glutamine. Substitutional variants of the amino acid sequence disclosed herein are those in which at least one residue in the disclosed sequences has been removed and a different residue inserted in its place. Preferably, the amino acid change is conservative. In an embodiment, conservative substitutions for each of the naturally occurring amino acids are as follows: Ala to Ser; Arg to Lys; Asn to Gln or His; Asp to Glu; Cys to Ser or Ala; Gln to Asn; Glu to Asp; Gly to Pro; His to Asn or Gln; Ile to Leu or Val; Leu to Ile or Val; Lys

to Arg; Gln or Glu; Met to Leu or Ile; Phe to Met, Leu or Tyr; Ser to Thr; Thr to Ser; Trp to Tyr; Tyr to Trp or Phe; and, Val to Ile or Leu.

[0055] Nucleotide sequences of the invention may also be defined by their capability to hybridise with parts of specific nucleotide sequences disclosed herein, respectively, under moderate, or preferably under stringent hybridisation conditions. Stringent hybridisation conditions are herein defined as conditions that allow a nucleic acid sequence of at least about 25, preferably about 50 nucleotides, 75 or 100 and most preferably of about 200 or more nucleotides, to hybridise at a temperature of about 65° C. in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength, and washing at 65° C. in a solution comprising about 0.1 M salt, or less, preferably 0.2×SSC or any other solution having a comparable ionic strength. Preferably, the hybridisation is performed overnight, i.e. at least for 10 hours and preferably washing is performed for at least one hour with at least two changes of the washing solution. These conditions will usually allow the specific hybridisation of sequences having about 90% or more sequence identity. Moderate conditions are herein defined as conditions that allow a nucleic acid sequences of at least 50 nucleotides, preferably of about 200 or more nucleotides, to hybridise at a temperature of about 45° C. in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength, and washing at room temperature in a solution comprising about 1 M salt, preferably 6×SSC or any other solution having a comparable ionic strength. Preferably, the hybridisation is performed overnight, i.e. at least for 10 hours, and preferably washing is performed for at least one hour with at least two changes of the washing solution. These conditions will usually allow the specific hybridisation of sequences having up to 50% sequence identity. The person skilled in the art will be able to modify these hybridisation conditions in order to specifically identify sequences varying in identity between 50% and 90%.

[0056] “Expression” refers to the transcription of a gene into structural RNA (rRNA, tRNA) or messenger RNA (mRNA) with subsequent translation into a protein.

[0057] “Overexpression” refers to expression of a gene, respectively a nucleic acid sequence, by a recombinant cell in excess to its expression in a corresponding wild-type cell. Such overexpression can for example be arranged for by: increasing the frequency of transcription of one or more nucleic acid sequences, for example by operational linking of the nucleic acid sequence to a promoter functional within the recombinant cell; and/or by increasing the number of copies of a certain nucleic acid sequence.

[0058] The terms “upregulate”, “upregulated” and “upregulation” refer to a process by which a cell increases the quantity of a cellular component, such as RNA or protein. Such an upregulation may be in response to or caused by a genetic modification.

[0059] By the term “pathway” or “metabolic pathway” is herein understood a series of chemical reactions in a cell that build and breakdown molecules.

[0060] Nucleic acid sequences (i.e. polynucleotides) or proteins (i.e. polypeptides) may be native or heterologous to the genome of the host cell.

[0061] “Native”, “homologous” or “endogenous” with respect to a host cell, means that the nucleic acid sequence does naturally occur in the genome of the host cell or that the

protein is naturally produced by that cell. The terms “native”, “homologous” and “endogenous” are used interchangeably herein.

[0062] As used herein, “heterologous” may refer to a nucleic acid sequence or a protein. For example, “heterologous”, with respect to the host cell, may refer to a polynucleotide that does not naturally occur in that way in the genome of the host cell or that a polypeptide or protein is not naturally produced in that manner by that cell. A heterologous nucleic acid sequence is a nucleic acid that originates from a foreign species, or, if from the same species, is substantially modified from its native form in composition and/or genomic locus by deliberate human intervention. For example, a promoter operably linked to a native structural gene is from a species different from that from which the structural gene is derived, or, if from the same species, one or both are substantially modified from their original form. A heterologous protein may originate from a foreign species or, if from the same species, is substantially modified from its original form by deliberate human intervention. That is, heterologous protein expression involves expression of a protein that is not naturally expressed in that way in the host cell. The term “heterologous expression” refers to the expression of heterologous nucleic acids in a host cell. The expression of heterologous proteins in eukaryotic host cell systems such as yeast are well known to those of skill in the art. A polynucleotide comprising a nucleic acid sequence of a gene encoding a certain protein or enzyme with a specific activity can be expressed in such a eukaryotic system. In some embodiments, transformed/transfected cells may be employed as expression systems for the expression of the enzymes. Expression of heterologous proteins in yeast is well known. Sherman, F., et al., *Methods in Yeast Genetics*, (1986), published by Cold Spring Harbor Laboratory, is a well-recognized work describing the various methods available to express proteins in yeast. Two widely utilized yeasts are *Saccharomyces cerevisiae* and *Pichia pastoris*. Vectors, strains, and protocols for expression in *Saccharomyces* and *Pichia* are known in the art and available from commercial suppliers (e.g., Invitrogen). Suitable vectors usually have expression control sequences, such as promoters, including 3-phosphoglycerate kinase or alcohol oxidase, and an origin of replication, termination sequences and the like as desired.

[0063] As used herein “promoter” is a DNA sequence that directs the transcription of a (structural) gene or other (part of) nucleic acid sequence. Suitably, a promoter is located in the 5'-region of a gene, proximal to the transcriptional start site of a (structural) gene. Promoter sequences may be constitutive, inducible or repressible. In an embodiment there is no (external) inducer needed.

[0064] The term “vector” as used herein, includes reference to an autosomal expression vector and to an integration vector used for integration into the chromosome.

[0065] The term “expression vector” refers to a DNA molecule, linear or circular, that comprises a segment encoding a polypeptide of interest under the control of (i.e. operably linked to) additional nucleic acid segments that provide for its transcription. Such additional segments may include promoter and terminator sequences, and may optionally include one or more origins of replication, one or more selectable markers, an enhancer, a polyadenylation signal, and the like. Expression vectors are generally derived from plasmid or viral DNA, or may contain elements of both. In particular an expression vector comprises a nucleic acid

sequence that comprises in the 5' to 3' direction and operably linked: (a) a yeast-recognized transcription and translation initiation region, (b) a coding sequence for a polypeptide of interest, and (c) a yeast-recognized transcription and translation termination region.

[0066] “Plasmid” refers to autonomously replicating extrachromosomal DNA which is not integrated into a microorganism’s genome and is usually circular in nature.

[0067] An “integration vector” refers to a DNA molecule, linear or circular, that can be incorporated in a microorganism’s genome and provides for stable inheritance of a gene encoding a polypeptide of interest. The integration vector generally comprises one or more segments comprising a gene sequence encoding a polypeptide of interest under the control of (i.e. operably linked to) additional nucleic acid segments that provide for its transcription. Such additional segments may include promoter and terminator sequences, and one or more segments that drive the incorporation of the gene of interest into the genome of the target cell, usually by the process of homologous recombination. Typically, the integration vector will be one which can be transferred into the target cell, but which has a replicon which is nonfunctional in that organism. Integration of the segment comprising the gene of interest may be selected if an appropriate marker is included within that segment.

[0068] By “host cell” is herein understood a cell, such as a yeast cell, that is to be transformed with one or more nucleic acid sequences encoding for one or more heterologous proteins, to construe a transformed cell, also referred to as a recombinant cell. For example, the transformed cell may contain a vector and may support the replication and/or expression of the vector.

[0069] “Transformation” and “transforming”, as used herein, refers to the insertion of an exogenous polynucleotide into a host cell, irrespective of the method used for the insertion, for example, direct uptake, transduction, f-mating or electroporation. The exogenous polynucleotide may be maintained as a non-integrated vector, for example, a plasmid, or alternatively, may be integrated into the host cell genome. “Transformation” and “transforming”, as used herein, refers to the insertion of an exogenous polynucleotide (i.e. an exogenous nucleic acid sequence) into a host cell, irrespective of the method used for the insertion, for example, direct uptake, transduction, f-mating or electroporation. The exogenous polynucleotide may be maintained as a non-integrated vector, for example, a plasmid, or alternatively, may be integrated into the host cell genome.

[0070] By “constitutive expression” and “constitutively expressing” is herein understood that there is a continuous transcription of a nucleic acid sequence. That is, the nucleic acid sequence is transcribed in an ongoing manner. Constitutively expressed genes are always “on”.

[0071] By “anaerobic constitutive expression” is herein understood that nucleic acid sequence is constitutively expressed in an organism under anaerobic conditions. That is, under anaerobic conditions the nucleic acid sequence is transcribed in an ongoing manner, i.e. under such anaerobic conditions the genes are always “on”.

[0072] By “disruption” is herein understood any disruption of activity, including, but not limited to, deletion, mutation and reduction of the affinity of the disrupted gene and expression of RNA complementary to such disrupted gene. It includes all nucleic acid modifications such as nucleotide deletions or substitutions, gene knock-outs, and

other actions which affect the translation or transcription of the corresponding polypeptide and/or which affect the enzymatic (specific) activity, its substrate specificity, and/or or stability. It also includes modifications that may be targeted on the coding sequence or on the promoter of the gene. A gene disruptant is a cell that has one or more disruptions of the respective gene. Native to yeast herein is understood as that the gene is present in the yeast cell before the disruption.

[0073] The term “encoding” has the same meaning as “coding for”. Thus, by way of example, “one or more genes encoding a protein having activity X” has the same meaning as “one or more genes coding for a protein having activity X”.

[0074] As far as genes or nucleic acid sequences encoding a protein or an enzyme are concerned, the phrase “a nucleic acid sequence encoding a X”, respectively “one or more nucleic acid sequences encoding a X”, wherein X denotes a certain protein or (enzymatic) activity, has the same meaning as “a nucleic acid sequence encoding a protein having X activity”, respectively “one or more nucleic acid sequences encoding a protein having X activity”. Thus, by way of example, “one or more nucleic acid sequences encoding a transketolase” has the same meaning as “one or more nucleic acid sequences encoding a protein having transketolase activity”. As indicated above, the article “a” refers to “one or more”.

[0075] By a “redox sink” is herein understood a metabolic pathway that, overall, consumes or oxidizes NADH into NAD⁺ and/or prevents or reduces the consumption or reduction of NAD⁺ into NADH. A non-native metabolic pathway is a metabolic pathway that does not occur in the corresponding wild-type cell. Hence, a non-native metabolic pathway forming a redox sink is preferably a non-native metabolic pathway that, as compared to a corresponding wild-type yeast cell, increases NADH consumption and/or decreases NAD⁺ consumption. By increasing NADH consumption and/or decreasing NAD⁺ consumption advantageously an (additional) non-native redox sink can be created within the cell.

[0076] The abbreviation “NADH” refers to reduced, hydrogenated form of nicotinamide adenine dinucleotide. The abbreviation “NAD⁺” refers to the oxidized form of nicotinamide adenine dinucleotide. Nicotinamide adenine dinucleotide may act as a so-called cofactor, assisting in biochemical reactions and/or transformations in a cell.

[0077] “NADH dependent” or “NAD⁺ dependent” is herein equivalent to NADH specific and “NADH dependency” or “NAD⁺ dependency” is herein equivalent to NADH specificity.

[0078] By a “NADH dependent” or “NAD⁺ dependent” enzyme is herein understood an enzyme that is exclusively dependent on NADH/NAD⁺ as a co-factor or that is predominantly dependent on NADH/NAD⁺ as a cofactor, i.e. as contrasted to other types of co-factor. By an “exclusive NADH/NAD⁺ dependent” enzyme is herein understood an enzyme that has an absolute requirement for NADH/NAD⁺ over NADPH/NADP⁺. That is, it is only active when NADH/NAD⁺ is applied as cofactor. By a “predominantly NADH/NAD⁺-dependent” enzyme is herein understood an enzyme that has a higher specificity and/or a higher catalytic efficiency for NADH/NAD⁺ as a cofactor than for NADPH/NADP⁺ as a cofactor.

[0079] The enzyme's specificity characteristics can be described by the formula:

$$1 < K_m \text{ NADP}^+ / K_m \text{ NAD}^+ < \infty (\text{infinity})$$

wherein K_m is the so-called Michaelis constant.

[0080] For a predominantly NADH-dependent enzyme, preferably $K_m \text{ NADP}^+ / K_m \text{ NAD}^+$ is between 1 and 1000, between 1 and 500, between 1 and 200, between 1 and 100, between 1 and 50, between 1 and 10, between 5 and 100, between 5 and 50, between 5 and 20 or between 5 and 10.

[0081] The K_m 's for the enzymes herein can be determined as enzyme specific, for NAD^+ and NADP^+ respectively, using known analysis techniques, calculations and protocols. These are described for instance in Lodish et al., Molecular Cell Biology 6th Edition, Ed. Freeman, pages 80 and 81, e.g. Figure 3-22. For an predominantly NADH-dependent enzyme, preferably the ratio of the catalytic efficiency for $\text{NADPH}/\text{NADP}^+$ as a cofactor (k_{cat}/K_m)^{NADP⁺} to NADH/NAD^+ as cofactor (k_{cat}/K_m)^{NAD⁺}, i.e. the catalytic efficiency ratio (k_{cat}/K_m)^{NADP⁺}:(k_{cat}/K_m)^{NAD⁺}, is more than 1:1, more preferably equal to or more than 2:1, still more preferably equal to or more than 5:1, even more preferably equal to or more than 10:1, yet even more preferably equal to or more than 20:1, even still more preferably equal to or more than 100:1, and most preferably equal to or more than 1000:1. There is no upper limit, but for practical reasons the predominantly NADH-dependent enzyme may have a catalytic efficiency ratio (k_{cat}/K_m)^{NADP⁺}:(k_{cat}/K_m)^{NAD⁺} of equal to or less than 1.000.000.000:1 (i.e. 1.10^9 :1).

The Yeast Cell

[0082] The recombinant yeast cell is preferably a yeast cell, or derived from a yeast cell, from the genus of Saccharomycetaceae or the genus of Schizosaccharomycetaceae. That is, preferably the host cell from which the recombinant yeast cell is derived is a yeast cell from the genus of Saccharomycetaceae or the genus of Schizosaccharomycetaceae.

[0083] Examples of suitable yeast cells include *Saccharomyces*, such as *Saccharomyces cerevisiae*, *Saccharomyces eubayanus*, *Saccharomyces jurei*, *Saccharomyces pastorianus*, *Saccharomyces beticus*, *Saccharomyces fermentati*, *Saccharomyces paradoxus*, *Saccharomyces uvarum* and *Saccharomyces bayanus*.

[0084] Examples of suitable yeast cells further include *Schizosaccharomyces*, such as *Schizosaccharomyces pombe*, *Schizosaccharomyces japonicus*, *Schizosaccharomyces octosporus* and *Schizosaccharomyces cryophilus*.

[0085] Other exemplary yeasts include *Torulaspora* such as *Torulaspora delbrueckii*; *Kluyveromyces* such as *Kluyveromyces marxianus*; *Pichia* such as *Pichia stipitis*, *Pichia pastoris* or *Pichia angusta*; *Zygosaccharomyces* such as *Zygosaccharomyces bailii*; *Brettanomyces* such as *Brettanomyces intermedius*; *Brettanomyces bruxellensis*, *Brettanomyces anomalus*, *Brettanomyces custersianus*, *Brettanomyces naardenensis*, *Brettanomyces nanus*, *Dekkera bruxellensis* and *Dekkera anomala*; *Metschnikowia*, *Issatchenkia*, such as *Issatchenkia orientalis*, *Kloeckera* such as *Kloeckera apiculata*; and *Aureobasidium* such as *Aureobasidium pullulans*.

[0086] The yeast cell is preferably a yeast cell of the genus *Schizosaccharomyces*, herein also referred to as a *Schizosaccharomyces* yeast cell, or a yeast cell of the genus *Saccharomyces*, herein also referred to as a *Saccharomyces*

yeast cell. More preferably the yeast cell is a yeast cell derived from a yeast cell of the species *Saccharomyces cerevisiae*, herein also referred to as a *Saccharomyces cerevisiae* yeast cell. That is, preferably the host cell from which the recombinant yeast cell is derived is a yeast cell from the species *Saccharomyces cerevisiae*.

[0087] Preferably the yeast cell is an industrial yeast cell. The living environments of yeast cells in industrial processes are significantly different from that in the laboratory. Industrial yeast cells must be able to perform well under multiple environmental conditions which may vary during the process. Such variations include changes in nutrient sources, pH, ethanol concentration, temperature, oxygen concentration, etc., which together have potential impact on the cellular growth and ethanol production of the yeast cell. An industrial yeast cell can be understood to refer to a yeast cell that, when compared to a laboratory counterpart, has a more robust performance. That is, when compared to a laboratory counterpart, the industrial yeast cell shows less variation in performance when one or more environmental conditions selected from the group of nutrient sources, pH, ethanol concentration, temperature, oxygen concentration, are varied during fermentation. Preferably, the yeast cell is constructed on the basis of an industrial yeast cell as a host, wherein the construction is conducted as described herein-after. Examples of industrial yeast cells are Ethanol Red® (Fermentis) Fermiol® (DSM) and Thermosacc® (Lallemand).

[0088] The recombinant yeast cell described herein may be derived from any host cell capable of producing a fermentation product. Preferably the host cell is a yeast cell, more preferably an industrial yeast cell as described herein above. Preferably the yeast cell described herein is derived from a host cell having the ability to produce ethanol.

[0089] The yeast cell described herein may be derived from the host cell through any technique known by one skilled in the art to be suitable therefore. Such techniques may include any one or more of mutagenesis, recombinant DNA technology (including, but not limited to, CRISPR-CAS techniques), selective and/or adaptive evolution, mating, cell fusion, and/or cytoduction between yeast strains. Suitably the one or more desired genes are incorporated in the yeast cell by a combination of one or more of the above techniques.

[0090] The recombinant yeast cells according to the invention are preferably inhibitor tolerant, i.e. they can withstand common inhibitors at the level that they typically have with common pretreatment and hydrolysis conditions, so that the recombinant yeast cells can find broad application, i.e. it has high applicability for different feedstock, different pretreatment methods and different hydrolysis conditions. In an embodiment the recombinant yeast cell is inhibitor tolerant. Inhibitor tolerance is resistance to inhibiting compounds. The presence and level of inhibitory compounds in lignocellulose may vary widely with variation of feedstock, pretreatment method hydrolysis process. Examples of categories of inhibitors are carboxylic acids, furans and/or phenolic compounds. Examples of carboxylic acids are lactic acid, acetic acid or formic acid. Examples of furans are furfural and hydroxy-methylfurfural. Examples of phenolic compounds are vanillin, syringic acid, ferulic acid and coumaric acid. The typical amounts of inhibitors are for carboxylic acids: several grams per liter, up to 20 grams per liter or more, depending on the feedstock, the pretreatment

and the hydrolysis conditions. For furans: several hundreds of milligrams per liter up to several grams per liter, depending on the feedstock, the pretreatment and the hydrolysis conditions. For phenolics: several tens of milligrams per liter, up to a gram per liter, depending on the feedstock, the pretreatment and the hydrolysis conditions.

[0091] In an embodiment, the recombinant yeast cell is a cell that is naturally capable of alcoholic fermentation, preferably, anaerobic alcoholic fermentation. A recombinant yeast cell preferably has a high tolerance to ethanol, a high tolerance to low pH (i.e. capable of growth at a pH lower than about 5, about 4, about 3, or about 2.5) and towards organic and/or a high tolerance to elevated temperatures.

Native Transketolase

[0092] The recombinant yeast cell is suitably functionally expressing:

[0093] a nucleic acid sequence encoding a native protein having transketolase activity; and

[0094] a nucleic acid sequence encoding a heterologous protein having transketolase activity.

[0095] A protein having transketolase activity is herein also referred to as “transketolase protein”, “transketolase enzyme” or simply “transketolase”. The “transketolase” is herein abbreviated as “TKL”.

[0096] In analogy, a native protein having transketolase activity is herein also referred to as “native transketolase protein”, “native transketolase enzyme” or simply “native transketolase”.

[0097] Transketolase is an enzyme that is active within the pentose phosphate pathway of a yeast cell. The genes encoding for this pentose phosphate pathway are herein also referred to as the “PPP” genes. Preferably references in this specification to the pentose phosphate pathway are to be understood as references to the non-oxidative part of the pentose phosphate pathway. The enzymes active within the pentose phosphate pathway include the enzymes ribulose-5-phosphate isomerase (RKI), ribulose-5-phosphate epimerase (RPE), transketolase (TKL) and transaldolase (TAL).

[0098] The enzyme “transketolase” (belonging to enzyme class EC 2.2.1.1) is herein defined as an enzyme that catalyses the reaction: D-ribose 5-phosphate+D-xylulose 5-phosphate $\leftrightarrow$ sedoheptulose 7-phosphate+D-glyceraldehyde 3-phosphate; and vice versa.

[0099] The enzyme is also known as glycolaldehydetransferase or sedoheptulose-7-phosphate:D-glyceraldehyde-3-phosphate glycolaldehydetransferase. A certain transketolase can be further defined by its amino acid sequence. Likewise a transketolase can be further defined by a nucleotide sequence encoding the transketolase. As explained in detail above under definitions, a certain transketolase that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the transketolase.

[0100] The nucleic acid sequences encoding the native protein having transketolase activity can in itself be homologous nucleic acid sequences, heterologous nucleic acid sequences or a mixture of homologous and heterologous nucleic acid sequences, provided that the protein that is encoded by such nucleic acid sequences is a transketolase protein that is native, i.e. endogenous, to the host cell. *Saccharomyces cerevisiae* is a preferred host cell. Hence,

preferably the recombinant yeast cell is a recombinant *Saccharomyces cerevisiae* yeast cell, functionally expressing a nucleic acid sequence encoding a protein having transketolase activity that is native to the *Saccharomyces cerevisiae* yeast cell.

[0101] More preferably the nucleic acid sequence encoding the native protein having transketolase activity is native to the host cell.

[0102] The recombinant yeast cell is therefore preferably functionally expressing:

[0103] a native nucleic acid sequence encoding a native protein having transketolase activity; and

[0104] a heterologous nucleic acid sequence encoding a heterologous protein having transketolase activity.

[0105] Wild-type yeasts may comprise one or two native transketolase genes. In addition to a first transketolase gene “TKL1”, some yeasts, such as for example *Saccharomyces cerevisiae*, comprises the paralog “TKL2”, a second transketolase gene.

[0106] Suitably the recombinant yeast cells according to the invention may comprise a native TKL1 gene and/or a native TKL2 gene.

[0107] That is, suitably the recombinant yeast cell may comprise:

[0108] a nucleic acid sequence encoding for native TKL1 (e.g. a gene “TKL1”); or

[0109] a nucleic acid sequence encoding for native TKL2 (e.g. a gene “TKL2”); or

[0110] both a nucleic acid sequence encoding for native TKL1 (e.g. a gene “TKL1”) and a nucleic acid sequence encoding for native TKL2 (e.g. a gene “TKL2”).

[0111] Preferably the recombinant yeast cell comprises a native nucleotide sequence encoding for native transketolase TKL1. That is, preferably the recombinant yeast cell comprises a native TKL1 gene.

[0112] The recombinant yeast cell may comprise one or more copies, suitably in the range from equal to or more than 1 to equal to or less than 30 copies, preferably in the range equal to or more than 1 to equal to or less than 20 copies, of a native gene encoding a transketolase. More preferably the recombinant yeast cell comprises one, two, three, four, five, six, seven, eight, nine, ten, eleven or twelve copies of a native gene encoding a transketolase.

[0113] The recombinant yeast cell can be a recombinant yeast cell, wherein the native nucleic acid sequence encoding for the native protein having transketolase activity is under control of a TKL promoter, as described in detail herein below.

[0114] As indicated above, host cells from the species *Saccharomyces cerevisiae* are preferred. The amino acid sequence of native transketolase 1 of *Saccharomyces cerevisiae* is illustrated by SEQ ID NO: 1. The native nucleic acid sequence encoding transketolase 1 in *Saccharomyces cerevisiae* is illustrated by SEQ ID NO: 2.

[0115] Preferably the native protein having transketolase activity comprises or consists of: the amino acid sequence of SEQ ID NO: 1; or a functional homologue of SEQ ID NO: 1, suitably native to the host cell, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 1.

[0116] Preferably the, preferably native, nucleic acid sequence encoding the native transketolase protein therefore comprises or consists of:

[0117] the nucleic acid sequence of SEQ ID NO: 2; or

[0118] a functional homologue of SEQ ID NO: 2, encoding for a native transketolase, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 2.

[0119] More preferably the recombinant yeast cell is a recombinant *Saccharomyces cerevisiae* yeast cell comprising, respectively functionally expressing, a native protein having transketolase activity, which protein comprises or consists of the amino acid sequence of SEQ ID NO: 1. Most preferably the recombinant yeast cell is a recombinant *Saccharomyces cerevisiae* yeast cell comprising, respectively functionally expressing, a nucleic acid sequence comprising or consisting of SEQ ID NO: 2.

Heterologous Transketolase

[0120] As indicated above, the recombinant yeast cell is suitably functionally expressing:

[0121] a nucleic acid sequence encoding a native protein having transketolase activity; and

[0122] a nucleic acid sequence encoding a heterologous protein having transketolase activity.

[0123] A heterologous protein having transketolase activity is herein also referred to as “heterologous transketolase protein”, “heterologous transketolase enzyme” or simply “heterologous transketolase”.

[0124] The recombinant yeast cell is suitably functionally expressing a heterologous nucleic acid sequence encoding the heterologous protein having transketolase activity.

[0125] Such heterologous nucleic acid sequence encoding for the heterologous protein having transketolase activity, respectively the heterologous transketolase, is suitably present in addition to the, preferably native, nucleic acid sequence encoding for the native protein having transketolase activity, respectively the native transketolase.

[0126] Preferably the recombinant yeast cell comprises a heterologous nucleic acid sequence encoding for a heterologous transketolase in addition to a native nucleic acid sequence encoding for a native transketolase.

[0127] Such heterologous nucleic acid sequence encoding for the transketolase is preferably under control of a TKL promoter as detailed herein below.

[0128] The recombinant yeast cell can comprise, respectively functionally express, one or more heterologous nucleic acid sequence(s) encoding for one or more heterologous transketolase protein(s). That is the recombinant yeast cell can comprise one or more heterologous transketolase(s).

[0129] Preferably the one or more heterologous transketolase(s) comprise(s) or consist(s) of:

[0130] the amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19; or

[0131] a functional homologue of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO:

7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19; or

[0132] a functional homologue of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19, comprising an amino acid sequence having one or more mutations, substitutions, insertions and/or deletions when compared with the amino sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19, where more preferably the amino acid sequence of any such functional homologue has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions as compared to the amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19.

[0133] Preferably the recombinant yeast cell comprises, respectively functionally expresses:

[0134] one or more heterologous nucleic acid sequences encoding for one or more amino acid sequence(s) chosen from the group consisting of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19; and/or

[0135] functional homologues thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with any of those; and/or

[0136] functional homologues thereof comprising a nucleic acid sequence having one or more mutations, substitutions, insertions and/or deletions when compared therewith.

[0137] More preferably the nucleic acid sequence of any such functional homologues has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no

more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions as compared to such nucleic acid sequences.

[0138] More preferably the heterologous transketolase is derived from a *Komagataella phaffii*, a yeast species also referred to as "*Pichia pastoris*". For example, preferably the heterologous transketolase comprises or consists of an amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 16 or SEQ ID NO: 17; or a functional homologues thereof comprising an amino acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 16 or SEQ ID NO: 17; or a functional homologue thereof comprising an amino acid sequence having one or more mutations, substitutions, insertions and/or deletions when compared with the amino sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 16 or SEQ ID NO: 17, where more preferably the amino acid sequence of any such functional homologue has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions as compared to the amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 16 or SEQ ID NO: 17.

[0139] In order to allow for a good expression of the heterologous transketolase in the host cell, it can be advantageous to use a heterologous transketolase that may have an amino acid sequence having equal to or more than 30%, equal to or more than 35%, equal to or more than 40%, equal to or more than 45%, equal to or more than 50%, equal to or more than 55%, equal to or more than 60%, equal to or more than 65%, equal to or more than 70%, equal to or more than 75%, equal to or more than 80%, equal to or more than 85%, equal to or more than 90% equal to or more than 95%, equal to or more than 98% or equal to or more than 99% sequence identity with the amino acid sequence of the native transketolase of the host cell.

[0140] However, it may also be preferred for the heterologous transketolase to be a heterologous transketolase that is not regulated by native (i.e. endogenous) regulators of the host cell. That is, preferably the heterologous transketolase is a transketolase enzyme of which the activity cannot be increased or decreased by molecules that are natively produced by the host cell. In order to avoid native regulators, it can be advantageous to use a heterologous transketolase in the host cell that may have an amino acid sequence having equal to or less than 99%, equal to or less than 98%, equal to or less than 95%, equal to or less than 90%, equal to or less than 85%, equal to or less than 80%, equal to or less than 75%, equal to or less than 70%, or equal to or less than 65% sequence identity with the amino acid sequence of the native transketolase of the host cell.

[0141] Therefore, more preferably the heterologous transketolase comprises or consists of an amino acid sequence having a percentage identity with the amino acid sequence of the native transketolase of the host cell in the range of equal to or more than 30% to equal to or less than 80%, more preferably in the range of equal to or more than 30%, or equal to or more than 35%, to equal to or less than 75%, and

most preferably in the range of equal to or more than 35% to equal to or less than 70% or even equal to or less than 65%.

[0142] More preferably any heterologous nucleic acid sequence encoding for the heterologous transketolase comprises or consists of a nucleic acid sequence having a percentage identity with the nucleic acid sequence encoding for the native transketolase of the host cell in the range of equal to or more than 30% to equal to or less than 80%, more preferably in the range of equal to or more than 30%, or equal to or more than 35%, to equal to or less than 75%, and most preferably in the range of equal to or more than 35% to equal to or less than 70% or even equal to or less than 65%.

[0143] Host cells from the species *Saccharomyces cerevisiae* are preferred. As indicated above, the amino acid sequence of native transketolase 1 of *Saccharomyces cerevisiae* is illustrated by SEQ ID NO: 1, the native nucleic acid sequence encoding transketolase 1 in *Saccharomyces cerevisiae* is illustrated by SEQ ID NO: 2.

[0144] The recombinant yeast cell is therefore preferably a recombinant *Saccharomyces cerevisiae* yeast cell, functionally expressing:

[0145] a nucleic acid sequence encoding a native transketolase, wherein the native transketolase preferably comprises or consists of an amino acid sequence of SEQ ID NO: 1; and

[0146] a nucleic acid sequence encoding a heterologous transketolase, wherein the heterologous transketolase preferably comprises or consists of an amino acid sequence having in the range of equal to or more than 30% to equal to or less than 80%, more preferably in the range of equal to or more than 30%, or equal to or more than 35%, to equal to or less than 75%, and most preferably in the range of equal to or more than 35% to equal to or less than 70% or even equal to or less than 65%, sequence identity with the amino acid sequence of SEQ ID NO: 1.

[0147] Similarly, the recombinant yeast cell is preferably a recombinant *Saccharomyces cerevisiae* yeast cell, functionally expressing:

[0148] a native nucleic acid sequence encoding a native transketolase, wherein the native nucleic acid preferably comprises or consists of the nucleic acid sequence of SEQ ID NO: 2; and

[0149] a heterologous nucleic acid sequence encoding a heterologous transketolase, wherein the heterologous nucleic acid sequence preferably comprises or consists of a nucleic acid sequence having in the range of equal to or more than 30% to equal to or less than 80%, more preferably in the range of equal to or more than 35%, or equal to or more than 35%, to equal to or less than 75%, and most preferably in the range of equal to or more than 35% to equal to or less than 70% or even equal to or less than 65%, sequence identity with the nucleic acid sequence of SEQ ID NO: 2.

[0150] Most preferably the recombinant yeast cell comprises a, suitably heterologous, nucleic acid sequence encoding the heterologous protein having transketolase activity, wherein the nucleic acid sequence comprises or consists of:

[0151] an nucleic acid sequence of SEQ ID NO: 18 or SEQ ID NO: 20; or

[0152] a functional homologue of SEQ ID NO: 18 or SEQ ID NO: 20, having at least 40%, at least 45%, at

least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 18 or SEQ ID NO: 20; or

[0153] a functional homologue of SEQ ID NO: 18 or SEQ ID NO: 20, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 20 or SEQ ID NO: 22, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 18 or SEQ ID NO: 20.

[0154] The recombinant yeast cell may comprise one, two, or more copies of a heterologous nucleic acid sequence (e.g. a heterologous gene) encoding for the heterologous transketolase and/or one, two, or more copies of a native nucleic acid sequence (e.g. a native gene) encoding for the native transketolase.

[0155] Most preferably the recombinant yeast cell may comprise one, two, three, four, five, six, seven, eight, nine, ten, eleven or twelve copies of a heterologous nucleic acid sequence (e.g. a heterologous gene) encoding for the heterologous transketolase and/or one, two, three, four, five, six, seven, eight, nine, ten, eleven or twelve copies of a native nucleic acid sequence (e.g. a native gene) encoding for the native transketolase.

[0156] Preferably the recombinant yeast cell is a recombinant yeast cell comprising, respectively functionally expressing:

[0157] one, two or more copies of a heterologous nucleic acid sequence comprises or consists of a nucleic acid sequence having in the range of equal to or more than 30% to equal to or less than 80%, more preferably in the range of equal to or more than 35% to equal to or less than 75%, and most preferably in the range of equal to or more than 35% to equal to or less than 70% or even equal to or less than 65%, sequence identity with the nucleic acid sequence encoding for the native transketolase; and/or

[0158] one, two or more copies of a nucleic acid sequence encoding for any of the above mentioned heterologous transketolases; and/or

[0159] one, two or more copies of a nucleic acid sequence of SEQ ID NO: 18 and/or SEQ ID NO: 20;

[0160] and/or

[0161] one, two or more copies of a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 18 and/or SEQ ID NO: 20; and/or

[0162] one, two or more copies of a nucleic acid sequence having one or more mutations, substitutions, insertions and/or deletions as compared to the nucleic acid sequence of respectively SEQ ID NO: 18 and/or SEQ ID NO: 20, wherein more preferably this nucleic acid sequence has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100,

no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions as compared to the nucleic acid sequence of respectively SEQ ID NO: 18 and/or SEQ ID NO: 20.

[0163] Most preferably the recombinant yeast cell is a recombinant *Saccharomyces cerevisiae* yeast cell comprising, respectively functionally expressing:

[0164] i) one, two or more copies of the nucleic acid sequence of SEQ ID NO: 10; and

[0165] ii) one, two or more copies of a nucleic acid sequence of SEQ ID NO: 18 and/or SEQ ID NO: 20; and/or one, two or more copies of a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 18 and/or SEQ ID NO: 20; and/or one, two or more copies of a nucleic acid sequence having one or more mutations, substitutions, insertions and/or deletions as compared to the nucleic acid sequence of respectively SEQ ID NO: 18 and/or SEQ ID NO: 20, wherein more preferably this nucleic acid sequence has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions as compared to the nucleic acid sequence of respectively SEQ ID NO: 18 and/or SEQ ID NO: 20.

Optional Overexpression of One or More Other Enzymes of the PPP Pathway

[0166] The recombinant yeast cell may further optionally comprise one or more genetic modifications in the other PPP-genes, i.e. RKI, RPE and TAL, that increase the flux of the pentose phosphate pathway. Advantageously, such genetic modification(s) may lead to a further increased flux through the non-oxidative part of the pentose phosphate pathway.

[0167] The recombinant yeast cell may thus optionally comprise one or more additional genetic modifications to overexpress one or more other enzymes of the (non-oxidative part of) the pentose phosphate pathway. For example, the recombinant yeast cell may comprise one or more nucleic acid sequences to overexpress one or more of the enzymes selected from the group consisting of ribulose-5-phosphate isomerase, ribulose-5-phosphate epimerase and transaldolase.

[0168] The enzyme "ribulose 5-phosphate epimerase" (EC 5.1.3.1) is herein defined as an enzyme that catalyses the epimerisation of D-xylulose 5-phosphate into D-ribulose 5-phosphate and vice versa. The enzyme is also known as phosphoribulose epimerase; erythrose-4-phosphate isomerase; phosphoketopentose 3-epimerase; xylulose phosphate 3-epimerase; phosphoketopentose epimerase; ribulose 5-phosphate 3-epimerase; D-ribulose phosphate-3-epimerase; D-ribulose 5-phosphate epimerase; D-ribulose-5-P 3-epimerase; D-xylulose-5-phosphate 3-epimerase; pentose-5-phosphate 3-epimerase; or D-ribulose-5-phosphate 3-epimerase. A ribulose 5-phosphate epimerase may be further defined by its amino acid sequence. Likewise a

ribose 5-phosphate epimerase may be defined by a nucleotide sequence encoding the enzyme as well as by a nucleotide sequence hybridising to a reference nucleotide sequence encoding a ribulose 5-phosphate epimerase. The nucleotide sequence encoding for ribulose 5-phosphate epimerase is herein designated as RPE or RPE1.

[0169] The enzyme “ribulose 5-phosphate isomerase” (EC 5.3.1.6) is herein defined as an enzyme that catalyses direct isomerisation of D-ribose 5-phosphate into D-ribulose 5-phosphate and vice versa. The enzyme is also known as phosphopentosisomerase; phosphoriboisomerase; ribose phosphate isomerase; 5-phosphoribose isomerase; D-ribose 5-phosphate isomerase; D-ribose-5-phosphate ketol-isomerase; or D-ribose-5-phosphate aldose-ketose-isomerase. A ribulose 5-phosphate isomerase may be further defined by its amino acid sequence. Likewise a ribulose 5-phosphate isomerase may be defined by a nucleotide sequence encoding the enzyme as well as by a nucleotide sequence hybridising to a reference nucleotide sequence encoding a ribulose 5-phosphate isomerase. The nucleotide sequence encoding for ribulose 5-phosphate isomerase is herein designated RKI or RKI1.

[0170] The enzyme “transaldolase” (EC 2.2.1.2) is herein defined as an enzyme that catalyses the reaction: sedoheptulose 7-phosphate+D-glyceraldehyde 3-phosphate $\leftrightarrow$ D-erythrose 4-phosphate+D-fructose 6-phosphate and vice versa. The enzyme is also known as dihydroxyacetone transferase; dihydroxyacetone synthase; formaldehyde transketolase; or sedoheptulose-7-phosphate:D-glyceraldehyde-3-phosphate glyceronetransferase. A transaldolase may be further defined by its amino acid sequence. Likewise a transaldolase may be defined by a nucleotide sequence encoding the enzyme as well as by a nucleotide sequence hybridising to a reference nucleotide sequence encoding a transaldolase. The nucleotide sequence encoding for transketolase from is herein designated TAL or TAL1.

TKL Promoter

[0171] The recombinant yeast cell is preferably a recombinant yeast cell, wherein the nucleic acid sequence encoding for the native protein having transketolase activity and/or the nucleic acid sequence encoding for the heterologous protein having transketolase activity is/are under control of a promoter (the “TKL promoter”), which TKL promoter has an anaerobic/aerobic expression ratio for the transketolase of 2 or more. Herewith is suitably meant that the expression of the heterologous and/or native transketolase (“TKL”) is at least a factor 2 higher under anaerobic conditions than under aerobic conditions. The above can alternatively be phrased as the recombinant yeast cell functionally expressing a nucleic acid sequence encoding the native transketolase, respectively the heterologous transketolase, wherein this native transketolase, respectively this heterologous transketolase, is under control of a promoter (the “TKL promoter”) which has a TKL expression ratio $\frac{\text{anaerobic}}{\text{aerobic}}$ of 2 or more.

[0172] The TKL promoter can suitably be operably linked to the nucleic acid sequence encoding the protein having transketolase activity. Preferably, the TKL promoter is located in the 5-region of a TKL gene, more preferably it is located proximal to the transcriptional start site of a TKL gene.

[0173] Preferably the TKL promoter is ROX1 repressed. ROX1 is herein Heme-dependent repressor of hypoxic gene

(s); that mediates aerobic transcriptional repression of hypoxia induced genes such as COX5b and CYC7; the repressor function is regulated through decreased promoter occupancy in response to oxidative stress; and contains an HMG domain that is responsible for DNA bending activity; involved in the hyperosmotic stress resistance. ROX1 is regulated by oxygen.

[0174] Without wishing to be limited by any kind of theory it is believed that the regulation of ROX1 may function as follows: According to Kwast et al., “*Genomic Analysis of Anaerobically induced genes in Saccharomyces cerevisiae: Functional roles of ROX1 and other factors in mediating the anoxic response*”, (2002), Journal of bacteriology vol 184, no 1 pages 250-265, herein incorporated by reference.; “Although Rox1 functions in an O₂-independent manner, its expression is oxygen (heme) dependent, activated by the heme-dependent transcription factor Hap1 [19]. Thus, as oxygen levels fall to those that limit heme biosynthesis [20], ROX1 is no longer transcribed [21], its protein levels fall [22], and the genes it regulates are derepressed”. Further details and suitable motifs are provided by Keng, T. (1992), “*HAP1 and ROX1 form a regulatory pathway in the repression of HEM13 transcription in Saccharomyces cerevisiae*”, Mol. Cell. Biol. 12: pages 2616-2623, and Ter Kinde and de Steensma, “*A microarray-assisted screen for potential Hap1 and Rox1 target genes in Saccharomyces cerevisiae*”, (2002), Yeast 19: pages 825-840, incorporated herein by reference.

[0175] Preferably, the TKL promoter comprises a ROX1 binding motif. The TKL promoter may suitably comprise one or more ROX1 binding motif(s).

[0176] More preferably the TKL promoter can comprise in its nucleic acid sequence one or more copies of the motif NNNATTGTTNNN. Herein “N” represents a nucleic acid chosen from the group consisting of Adenine (A), Guanine (G), Cytosine (C) and Thymine (T). Such motif is illustrated by SEQ ID NO: 21.

[0177] More preferably, the TKL promoter comprises or consists of a nucleic acid sequence that is identical to the nucleic acid sequence of a, preferably native, promoter of a gene selected from the list consisting of: FET4, ANB1, YHR048W, DAN1, AAC3, TIR2, DIP5, HEM13, YNR014W, YAR028W, FUN 57, COX5B, OYE2, SUR2, FRDS1, PIS1, LAC1, YGR035C, YAL028W, EUG1, HEM14, ISU2, ERG26, YMR252C and SML1, more preferably FET4, ANB1, YHR048W, DAN1, AAC3, TIR2, DIP5 and HEM13, or a functional homologue thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity therewith. The reference to a native promoter is herein to the promoter that is native to the host cell.

[0178] Preferably the recombinant yeast cell is a recombinant *Saccharomyces cerevisiae* yeast cell and preferably the TKL promoter is a native promoter of a *Saccharomyces cerevisiae* gene selected from the list consisting of: FET4, ANB1, YHR048W, DAN1, AAC3, TIR2, DIP5, HEM13, YNR014W, YAR028W, FUN 57, COX5B, OYE2, SUR2, FRDS1, PIS1, LAC1, YGR035C, YAL028W, EUG1, HEM14, ISU2, ERG26, YMR252C and SML1.

[0179] In addition or in the alternative, the TKL promoter preferably comprises in its nucleic acid sequence one or more copies of the motifs: TCGTTYAG and/or AAAAAT-

TGTTGA. Herein “Y” represents C or T. The AAAAAT-TGTTGA motif is illustrated by SEQ ID NO: 22.

[0180] The TKL promoter can also comprise or consist of a nucleic acid sequence that is identical to the nucleic acid sequence of a, preferably native, promoter of a DAN, TIR or PAU gene. For example, the TKL promoter can suitably comprise or consist of a nucleic acid sequence of a, preferably native, promoter of a gene selected from the list consisting of: TIR2, DAN1, TIR4, TIR3, PAU7, PAU5, YLL064C, YGR294W, DAN3, YIL176C, YGL261C, YOL161C, PAU1, PAU6, DAN2, YDR542W, YIR041W, YKL224C, PAU3, YLL025W, YOR394W, YHL046C, YMR325W, YAL068C, YPL282C, PAU2, and PAU4 or a functional homologue thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity therewith. The reference to a native promoter is herein to the promoter that is native to the host cell.

[0181] Preferably the recombinant yeast cell is a recombinant *Saccharomyces cerevisiae* yeast cell and preferably the TKL promoter is a native promoter of a *Saccharomyces cerevisiae* gene selected from the list consisting of: TIR2, DAN1, TIR4, TIR3, PAU7, PAU5, YLL064C, YGR294W, DAN3, YIL176C, YGL261C, YOL161C, PAU1, PAU6, DAN2, YDR542W, YIR041W, YKL224C, PAU3, YLL025W, YOR394W, YHL046C, YMR325W, YAL068C, YPL282C, PAU2, and PAU4.

[0182] More preferably, the TKL promoter can comprise or consist of a sequence that is identical to the nucleic acid sequence of a, preferably native, promoter of a gene selected from the list consisting of: TIR2, DAN1, TIR4, TIR3, PAU7, PAU5, YLL064C, YGR294W, DAN3, YIL176C, YGL261C, YOL161C, PAU1, PAU6, DAN2, YDR542W, YIR041 W, YKL224C, PAU3, and YLL025W or a functional homologue thereof comprising a nucleic acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity therewith.

[0183] The nucleic acid sequence of the *S. cerevisiae* ANB1 promoter is illustrated in SEQ ID NO: 23. The nucleic acid sequence of the *S. cerevisiae* DAN1 promoter is illustrated in SEQ ID NO: 24.

[0184] Preferred TKL promoters can thus comprise or consist of:

[0185] a nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24; or

[0186] a functional homologue of the nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24; or

[0187] a functional homologue of the nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24, having one or more mutations, substitutions, insertions and/or deletions as compared to the nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24, wherein more preferably the nucleic acid sequence has no more than 300, no more than 250, no more than 200, no more than

150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions as compared to the nucleic acid sequence of SEQ ID NO: 23 or SEQ ID NO: 24.

[0188] The TKL promoter can also be a synthetic oligonucleotide. That is, the TKL promoter may be a product of artificial oligonucleotide synthesis. Artificial oligonucleotide synthesis is a method in synthetic biology that is used to create artificial oligonucleotides, such as genes, in the laboratory. Commercial gene synthesis services are now available from numerous companies worldwide, some of which have built their business model around this task. Current gene synthesis approaches are most often based on a combination of organic chemistry and molecular biological techniques and entire genes may be synthesized “de novo”, without the need for precursor template DNA.

[0189] The TKL promoter has a TKL expression ratio $\frac{\text{anaerobic}}{\text{aerobic}}$ of 2 or more, preferably of 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 20 or more or 50 or more. By a TKL expression ratio $\frac{\text{anaerobic}}{\text{aerobic}}$ of 2 or more is suitably meant that the expression of the enzyme transketolase (“TKL”) is, under further identical expression conditions, at least a factor 2 higher under anaerobic conditions than under aerobic conditions.

[0190] There is no upper limit, and the TKL promoter can be a TKL promoter that allows the promoted transketolase gene to be expressed only at anaerobic conditions and not at aerobic conditions.

[0191] For practical reasons a TKL expression ratio $\frac{\text{anaerobic}}{\text{bicaerobic}}$ in the range from equal to or more than 2 to equal to or less than $10 \exp 10$ (i.e. 10^{10}) or to or less than $10 \exp 4$ (i.e. 104) can be considered.

[0192] As indicated above, “Expression” herein refers to the transcription of a gene into structural RNA (rRNA, tRNA) or messenger RNA (mRNA) with subsequent translation into a protein.

[0193] The TKL expression ratio can for example be determined by measuring the amount of Transketolase (TKL) protein of cells grown under aerobic and anaerobic conditions. The amount of TKL protein can be determined by proteomics or any other method known to quantify protein amounts.

[0194] It is also possible to determine the level or transketolase (TKL) expression ratio by measuring the transketolase (TKL) activity of cells grown under aerobic and anaerobic conditions, e.g. in a cell-free extract.

[0195] In addition or in the alternative to the above, the level or TKL expression ratio can be determined by measuring the transcription level (e.g. as amount of mRNA) of the TKL gene of cells grown under aerobic and anaerobic conditions. The skilled person knows how to determine translation levels using methods commonly known in the art, e.g. Q-PCR, real-time PCR, northern blot, RNA-seq.

[0196] The TKL promoter advantageously enables higher expression of transketolase during anaerobic conditions than under aerobic conditions. In the process according to the invention, the recombinant yeast cell preferably expresses transketolase, where the amount of transketolase expressed under anaerobic conditions is a multiplication factor higher than the amount of transketolase expressed under aerobic conditions and wherein this multiplication factor is prefer-

ably 2 or more, more preferably 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 20 or more or 50 or more.

Increased Flux

[0197] Preferably the genetic modification(s) made in respect of the PPP-genes, i.e. with respect to TKL1 and optionally RKL, RPE and TAL, cause an increased flux of the non-oxidative part of the pentose phosphate pathway is herein understood to mean a modification that increases the flux by at least a factor of about 1.1, about 1.2, about 1.5, about 2, about 5, about 10 or about 20 as compared to the flux in a strain which is genetically identical except for the genetic modification causing the increased flux. The flux of the non-oxidative part of the pentose phosphate pathway may be measured by growing the modified host on xylose as sole carbon source, determining the specific xylose consumption rate and subtracting the specific xylitol production rate from the specific xylose consumption rate, if any xylitol is produced. However, the flux of the non-oxidative part of the pentose phosphate pathway is proportional with the growth rate on xylose as sole carbon source, preferably with the anaerobic growth rate on xylose as sole carbon source. There is a linear relation between the growth rate on xylose as sole carbon source (μ_{max}) and the flux of the non-oxidative part of the pentose phosphate pathway. The specific xylose consumption rate (Q_s) is equal to the growth rate (μ) divided by the yield of biomass on sugar (Y_{xs}) because the yield of biomass on sugar is constant (under a given set of conditions: anaerobic, growth medium, pH, genetic background of the strain, etc.; i.e. $Q_s = \mu/Y_{xs}$). Therefore the increased flux of the non-oxidative part of the pentose phosphate pathway may be deduced from the increase in maximum growth rate under these conditions unless transport (uptake is limiting).

[0198] One or more genetic modifications that increase the flux of the pentose phosphate pathway may be introduced in the host cell in various ways. These including e.g. achieving higher steady state activity levels of xylulose kinase and/or one or more of the enzymes of the non-oxidative part pentose phosphate pathway and/or a reduced steady state level of unspecific aldose reductase activity. These changes in steady state activity levels may be effected by selection of mutants (spontaneous or induced by chemicals or radiation) and/or by recombinant DNA technology e.g. by overexpression or inactivation, respectively, of genes encoding the enzymes or factors regulating these genes.

[0199] In a preferred host cell, the genetic modification comprises overexpression of at least one enzyme of the (non-oxidative part) pentose phosphate pathway. Preferably the enzyme is selected from the group consisting of the enzymes encoding for ribulose-5-phosphate isomerase, ribulose-5-phosphate epimerase, transketolase and transaldolase. Various combinations of enzymes of the (non-oxidative part) pentose phosphate pathway may be overexpressed. E.g. the enzymes that are overexpressed may be at least the enzymes ribulose-5-phosphate isomerase and ribulose-5-phosphate epimerase; or at least the enzymes ribulose-5-phosphate isomerase and transketolase; or at least the enzymes ribulose-5-phosphate isomerase and transaldolase; or at least the enzymes ribulose-5-phosphate epimerase and transaldolase; or at least the enzymes transketolase and transaldolase; or at least the enzymes ribulose-

5-phosphate epimerase, transketolase and transaldolase; or at least the enzymes ribulose-5-phosphate isomerase, transketolase and transaldolase; or at least the enzymes ribulose-5-phosphate isomerase, ribulose-5-phosphate epimerase, and transaldolase; or at least the enzymes ribulose-5-phosphate isomerase, ribulose-5-phosphate epimerase, and transketolase. In one embodiment of the invention each of the enzymes ribulose-5-phosphate isomerase, ribulose-5-phosphate epimerase, transketolase and transaldolase are overexpressed in the host cell. More preferred is a host cell in which the genetic modification comprises at least overexpression of both the enzymes transketolase and transaldolase as such a host cell is already capable of anaerobic growth on xylose. In fact, under some conditions host cells overexpressing only the transketolase and the transaldolase already have the same anaerobic growth rate on xylose as do host cells that overexpress all four of the enzymes, i.e. the ribulose-5-phosphate isomerase, ribulose-5-phosphate epimerase, transketolase and transaldolase. Moreover, host cells overexpressing both of the enzymes ribulose-5-phosphate isomerase and ribulose-5-phosphate epimerase are preferred over host cells overexpressing only the isomerase or only the epimerase as overexpression of only one of these enzymes may produce metabolic imbalances.

Redox Sink

[0200] Preferably the recombinant yeast cell can further comprise one or more genetic modifications to functionally express a protein that functions in a metabolic pathway forming a non-native redox sink.

[0201] For example, these one or more genetic modifications can be one or more genetic modifications for the functional expression of one or more, optionally heterologous, nucleic acid sequences encoding for one or more NAD⁺/NADH dependent proteins that function in a metabolic pathway to convert NADH to NAD⁺. Several examples of such metabolic pathways exist, as illustrated further below.

[0202] For example, the “one or more genetic modifications to functionally express a protein that functions in a metabolic pathway forming a non-native redox sink” can be chosen from the group consisting of:

[0203] a) one or more genetic modifications comprising or consisting of:

[0204] a, preferably heterologous, nucleic acid sequence encoding a protein comprising phosphoketolase activity (EC 4.1.2.9 or EC 4.1.2.22, PKL); and/or

[0205] a, preferably heterologous, nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8); and/or

[0206] a, preferably heterologous, nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

[0207] and/or

[0208] b) one or more genetic modifications comprising or consisting of:

[0209] a, preferably heterologous, nucleic acid sequence encoding for a protein having ribulose-1, 5-biphosphate carboxylase oxygenase (Rubisco) activity; and/or

[0210] a, preferably heterologous, nucleic acid sequences encoding for a protein having phosphoribulokinase (PRK) activity; and

[0211] optionally, a, preferably heterologous, nucleic acid sequence encoding for one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity, and/or

[0212] c) one or more genetic modifications comprising or consisting of:

[0213] a, preferably heterologous, nucleic acid sequence encoding a protein comprising NADH dependent acetylating acetaldehyde dehydrogenase activity.

[0214] For example, WO2014/081803 describes a recombinant microorganism expressing a heterologous phosphoketolase, phosphotransacetylase or acetate kinase and bifunctional acetaldehyde-alcohol dehydrogenase, incorporated herein by reference; and WO2015/148272 describes a recombinant *S. cerevisiae* strain expressing a heterologous phosphoketolase, phosphotransacetylase and acetylating acetaldehyde dehydrogenase, incorporated herein by reference. Further WO2018172328A1 describes a recombinant cell that may comprise one or more (heterologous) genes coding for an enzyme having phosphoketolase activity. The phosphoketolase (PKL) routes described in WO2014/081803, WO2015/148272 and WO2018172328A1, all incorporated herein by reference, provide preferred metabolic pathways to convert NADH to NAD⁺ and the NADH dependent phosphoketolase described therein is a preferred NADH dependent protein for application in the current invention.

Rubisco

[0215] As indicated above, the recombinant yeast cell may advantageously functionally express one or more, preferably heterologous, nucleic acid sequences encoding for ribulose-1,5-phosphate carboxylase/oxygenase (EC4.1.1.39; Rubisco), and optionally one or more molecular chaperones for Rubisco.

[0216] More preferably the recombinant yeast cell functionally expresses:

[0217] a heterologous nucleic acid sequence encoding a protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity; and/or

[0218] a heterologous nucleic acid sequence encoding a protein having phosphoribulokinase (PRK) activity; and/or

[0219] optionally one or more heterologous nucleic acid sequence encoding one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.

[0220] The protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity is herein also referred to as “ribulose-1,5-biphosphate carboxylase oxygenase”, “ribulose-1,5-biphosphate carboxylase oxygenase protein”, “ribulose-1,5-biphosphate carboxylase oxygenase enzyme”, “Rubisco enzyme”, “Rubisco protein” or simply “Rubisco”. A ribulose-1,5-biphosphate carboxylase oxygenase may be further defined by its amino acid sequence. Likewise a ribulose-1,5-biphosphate carboxylase oxygenase may be further defined by a nucleotide sequence encoding the ribulose-1,5-biphosphate carboxylase oxygenase. As explained in detail above under definitions, a certain ribulose-1,5-biphosphate carboxylase oxygenase that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridis-

ing to such nucleotide sequence encoding the ribulose-1,5-biphosphate carboxylase oxygenase. Preferences for the Rubisco protein and the nucleic sequences encoding for such are as described in WO2014/129898, incorporated herein by reference.

[0221] The Rubisco protein may suitably be selected from the group of eukaryotic and prokaryotic Rubisco proteins. The Rubisco protein is preferably from a non-phototrophic organism. For example, the Rubisco protein may be from a chemolithoautotrophic microorganism. Good results have been achieved with a bacterial Rubisco protein. Preferably, the Rubisco protein originates from a *Thiobacillus*, in particular, *Thiobacillus denitrificans*, which is chemolithoautotrophic.

[0222] The Rubisco protein may be a single-subunit Rubisco protein or a Rubisco protein having more than one subunit. Preferably the Rubisco protein is a single-subunit Rubisco protein. Good results have been obtained with a Rubisco protein that is a so-called form-II Rubisco protein.

[0223] Especially good results were achieved with a Rubisco protein encoded by a *cbbM* gene, also referred to as *CbbM*.

[0224] A preferred Rubisco protein is the Rubisco protein encoded by the *cbbM* gene from *Thiobacillus denitrificans*. SEQ ID NO: 25 shows the amino acid sequence of a suitable Rubisco protein, encoded by the *cbbM* gene from *Thiobacillus denitrificans*. SEQ ID NO: 26 illustrates the nucleic acid sequence of the *cbbM* gene from *Thiobacillus denitrificans*, codon optimized for *S. cerevisiae*.

[0225] Preferably the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity thus comprises or consists of:

[0226] an amino acid sequence of SEQ ID NO: 25; or

[0227] a functional homologue of SEQ ID NO: 25, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 25; or

[0228] a functional homologue of SEQ ID NO: 25, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 25, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 25.

[0229] Preferable the nucleic acid sequence encoding the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity comprises or consists of:

[0230] a nucleic acid sequence of SEQ ID NO: 26; or

[0231] a functional homologue of SEQ ID NO: 26, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 26; or

[0232] a functional homologue of SEQ ID NO: 26, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 26, more preferably a func-

tional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 26.

[0233] Examples of other suitable Rubisco polypeptides and their origin are given in Table 1 of WO2014/129898, incorporated herein by reference, and in Table 2 below, with reference to the sequence identity with the amino acid sequence of SEQ ID NO:25.

[0234] The nucleic acid sequence (e.g. the gene) encoding for the ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) protein may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2014/129898 and by the article of Guadalupe-Medina et al., “Carbon dioxide fixation by Calvin-Cycle enzymes improves ethanol yield in yeast”, published in Biotechnol. Biofuels, 2013, vol. 6, p. 125, both herein incorporated by reference.

TABLE 2

Natural Rubisco polypeptides suitable for expression		
Source	Accession no.	MAX ID (%)
<i>Thiobacillus denitrificans</i>	AAA99178.2	100
<i>Sideroxydans lithotrophicus</i> ES-1	YP_003522651.1	94
<i>Thiothrix nivea</i> DSM 5205	ZP_10101642.1	91
<i>Halothiobacillus neapolitanus</i> c2	YP_003262978.1	90
<i>Acidithiobacillus ferrooxidans</i> ATCC 53993	YP_002220242.1	88
<i>Rhodospirillum rubrum</i> T118	YP_522655.1	86
<i>Thiorhodococcus dresvii</i> AZ1	ZP_08824342.1	85
uncultured prokaryote	AGE14067.1	82

[0235] As indicated above, the Rubisco protein is suitably functionally expressed in the recombinant yeast cell, at least during use in a fermentation process.

[0236] The nucleic acid sequence encoding for the Rubisco protein can be present in one, two or more copies with the recombinant yeast cell. Without wishing to be bound by any kind of theory it is believed that the robustness of the recombinant yeast cell is best served when the nucleic acid sequence (e.g. the gene) encoding for the Rubisco protein is present in the recombinant yeast cell in less than 12 copies, more preferably less than 8 copies. Preferably the recombinant yeast cell therefore comprises in the range from equal to or more than 1 copy, more preferably equal to or more than 2 copies, to equal to or less than 7 copies, more preferably equal to or less than 6 copies of a nucleic acid sequence (e.g. a gene) encoding for a Rubisco protein. The recombinant yeast cell may for example comprise one, two, three, four, five, six or seven copies of a nucleic acid sequence encoding for ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco).

[0237] To increase the likelihood that the Rubisco protein is expressed at sufficient levels and in active form in the transformed (recombinant) host cells of the invention, the nucleic acid sequence encoding the Rubisco protein and other proteins as described herein (see below), are preferably adapted to optimise their codon usage to that of the host cell in question. The adaptiveness of a nucleic acid sequence

encoding an enzyme to the codon usage of a host cell may be expressed as codon adaptation index (CAI). The codon adaptation index is herein defined as a measurement of the relative adaptiveness of the codon usage of a gene towards the codon usage of highly expressed genes in a particular host cell or organism. The relative adaptiveness (w) of each codon is the ratio of the usage of each codon, to that of the most abundant codon for the same amino acid. The CAI index is defined as the geometric mean of these relative adaptiveness values. Non-synonymous codons and termination codons (dependent on genetic code) are excluded. CAI values range from 0 to 1, with higher values indicating a higher proportion of the most abundant codons (see Sharp and Li, “The codon adaptation index—a measure of directional synonymous codon usage bias, and its potential applications”, (1987), published in Nucleic Acids Research vol. 15, pages 1281-1295; also see: Jansen et al., “Revisiting the codon adaptation index from a whole-genome perspective: analyzing the relationship between gene expression and codon occurrence in yeast using a variety of models”, (2003), Nucleic Acids Res. Vol. 31(8), pages 2242-51). An adapted nucleic acid sequence preferably has a CAI of at least 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8 or 0.9. Preferably, the sequences have been codon optimized for expression in the fungal host cell in question, such as for example *Saccharomyces cerevisiae* cells.

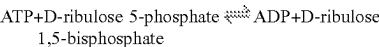
[0238] Preferably the functionally expressed Rubisco protein has an activity, defined by the rate of ribulose-1,5-bisphosphate-dependent ¹⁴C-bicarbonate incorporation by cell extracts of at least 1 nmol·min⁻¹·(mg protein)⁻¹, in particular an activity of at least 2 nmol·min⁻¹·(mg protein)⁻¹, more in particular an activity of at least 4 nmol·min⁻¹·(mg protein)⁻¹. The upper limit for the activity is not critical. In practice, the activity may be about 200 nmol·min⁻¹·(mg protein)⁻¹ or less, in particular 25 nmol·min⁻¹·(mg protein)⁻¹, more in particular 15 nmol·min⁻¹·(mg protein)⁻¹ or less, e.g. about 10 nmol·min⁻¹·(mg protein)⁻¹ or less. The conditions for an assay for determining this Rubisco activity are as found in the Examples (e.g. Example 4) of WO2014/129898, incorporated herein by reference.

Phosphoribulokinase

[0239] Preferably recombinant yeast cell is also functionally expressing a heterologous nucleic acid sequence encoding a protein having phosphoribulokinase (PRK) activity (EC2.7.1.19; PRK).

[0240] The protein having phosphoribulokinase (PRK) activity is herein also referred to as “phosphoribulokinase protein”, “phosphoribulokinase enzyme”, “phosphoribulokinase”, “PRK enzyme”, “PRK protein” or simply “PRK”. Preferences for the PRK protein and the nucleic sequences encoding for such are as described in WO2014/129898, incorporated herein by reference.

[0241] A functionally expressed phosphoribulokinase (PRK, (EC 2.7.1.19)) according to the invention is capable of catalyzing the chemical reaction



Thus, the two substrates of this enzyme are ATP and D-ribulose 5-phosphate; its two products are ADP and D-ribulose 1,5-bisphosphate.

[0242] The PRK protein belongs to the family of transferases, specifically those transferring phosphorus-contain-

ing groups (phosphotransferases) with an alcohol group as acceptor. The systematic name of this enzyme class is ATP:D-ribulose-5-phosphate 1-phosphotransferase. Other names in common use include phosphopentokinase, ribulose-5-phosphate kinase, phosphopentokinase, phosphoribulokinase (phosphorylating), 5-phosphoribulose kinase, ribulose phosphate kinase, PKK, PRuK, and PRK. The PRK enzyme participates in carbon fixation. A phosphoribulokinase (PRK) protein may be further defined by its amino acid sequence. Likewise a phosphoribulokinase (PRK) protein may be further defined by a nucleotide sequence encoding the phosphoribulokinase (PRK). As explained in detail above under definitions, a certain phosphoribulokinase (PRK) that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the phosphoribulokinase (PRK).

[0243] The PRK can be from a prokaryote or a eukaryote. Good results have been achieved with a PRK originating from a eukaryote. Preferably the PRK protein originates from a plant selected from Caryophyllales, in particular from Amaranthaceae, more in particular from *Spinacia*.

[0244] A preferred PRK protein is the PRK protein from *Spinacia*. SEQ ID NO: 27 shows the amino acid sequence of such PRK protein from *Spinacia*. SEQ ID NO: 28 illustrates the nucleic acid sequence of the *prk* gene from *Spinacia oleracea*—codon optimized for *S. cerevisiae*.

[0245] Preferably the protein having phosphoribulokinase (PRK) activity thus comprises or consists of:

[0246] an amino acid sequence of SEQ ID NO: 27; or

[0247] a functional homologue of SEQ ID NO: 27, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 27; or

[0248] a functional homologue of SEQ ID NO: 27, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 27, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 27.

[0249] Preferable the nucleic acid sequence encoding the protein having phosphoribulokinase (PRK) activity comprises or consists of:

[0250] a nucleic acid sequence of SEQ ID NO: 28; or

[0251] a functional homologue of SEQ ID NO: 28, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 28; or

[0252] a functional homologue of SEQ ID NO: 28, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 28, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more

than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 28.

[0253] The nucleic acid sequence (e.g. the gene) encoding for the protein having phosphoribulokinase (PRK) activity may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2014/129898, herein incorporated by reference.

[0254] Examples of suitable PRK polypeptides and their origin are given in Table 2 of WO2014/129898, incorporated herein by reference, and in Table 3 below, with reference to the sequence identity with the amino acid sequence of SEQ ID NO:27.

TABLE 3

Natural PRK polypeptides suitable for expression with identity to PRK from <i>Spinacia</i>		
Source	Accession no.	MAX ID (%)
<i>Spinacia oleracea</i>	P09559.1	100
<i>Medicago truncatula</i>	XP_003612664.1	88
<i>Arabidopsis thaliana</i>	NP_174486.1	87
<i>Vitis vinifera</i>	XP_002263724.1	86
<i>Closterium peracerosum</i>	BAL03266.1	82
<i>Zea mays</i>	NP_001148258.1	78

[0255] The nucleic acid sequences encoding for the PRK protein may be under the control of a promoter (the “PRK promoter”) that enables higher expression under anaerobic conditions than under aerobic conditions. Examples of such promoters are described in WO2017/216136A1 and WO2018/228836, both herein incorporated by reference. More preferably such promoter has a PRK expression ratio *anaerobic/aerobic* of 2 or more, 3 or more, 4 or more, 5 or more, 6 or more, 7 or more, 8 or more, 9 or more, 10 or more, 20 or more or 50 or more. Further preferences are as described in WO2018/228836, incorporated herein by reference.

Rubisco Chaperones

[0256] Optionally, the recombinant yeast cell further comprises one or more, preferably heterologous, nucleic acid sequences encoding for one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.

[0257] Suitably such molecular chaperones are also referred herein as “chaperone protein”, “chaperonin” or simply “chaperone”. Preferences for the chaperones and the nucleic sequences encoding for such are as described in WO2014/129898, incorporated herein by reference.

[0258] Preferably the recombinant yeast cell comprises one or more heterologous nucleic acid sequences encoding for one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.

[0259] Chaperonins are proteins that provide favorable conditions for the correct folding of other proteins, thus preventing aggregation. Newly made proteins usually must fold from a linear chain of amino acids into a three-dimensional form. Chaperonins belong to a large class of molecules that assist protein folding, called molecular chaperones. The energy to fold proteins is supplied by adenosine

triphosphate (ATP). A review article about chaperones that is useful herein is written by Yébenes et al., “*Chaperonins: two rings for folding*” (2011), Trends in Biochemical Sciences, Vol. 36, No. 8, pages 424-432, incorporated herein by reference.

[0260] The chaperone or chaperones may be prokaryotic chaperones or eukaryotic chaperones. In addition, the chaperones may be homologous or heterologous. For example, the recombinant yeast cell may comprises one or more nucleic acid sequence encoding one or more homologous or heterologous, prokaryotic or eukaryotic, molecular chaperones, which—when expressed—are capable of functionally interacting with an enzyme in the recombinant yeast cell, in particular with at least one of Rubisco and PRK.

[0261] Suitably the chaperone or chaperones are derived from a bacterium, more preferably from *Escherichia*, in particular *E. coli*. Preferred chaperones are GroEL and GroES from *E. coli*. Other preferred chaperones are chaperones from *Saccharomyces*, in particular *Saccharomyces cerevisiae* Hsp10 and Hsp60.

[0262] If the chaperones are naturally expressed in an organelle such as a mitochondrion (examples are Hsp60 and Hsp10 of *Saccharomyces cerevisiae*) relocation to the cytosol can be achieved e.g. by modifying the native signal sequence of the chaperonins. In eukaryotes the proteins Hsp60 and Hsp10 are structurally and functionally nearly

identical to GroEL and GroES, respectively. Thus, it is contemplated that Hsp60 and Hsp10 from any recombinant yeast cell may serve as a chaperone for the Rubisco. This is described for example by Zeilstra-Ryalls et al., “*The universally conserved GroE (Hsp60) chaperonins*”, (1991), Annu Rev Microbiol. vol. 45, pages 301-325; and Horwich et al., “*Two Families of Chaperonin: Physiology and Mechanism*” (2007), Annu. Rev. Cell. Dev. Biol. Vol. 23, pages 115-145, both herewith incorporated by reference.

[0263] Good results have been achieved with a recombinant yeast cell comprising both the heterologous chaperones GroEL and GroES.

[0264] As an alternative to GroES a functional homologue of GroES may be present, in particular a functional homologue comprising an amino acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of GroES, respectively the amino sequence of SEQ ID NO: 31.

[0265] SEQ ID NO:31 provides a preferred translated protein sequence, based on GroES of *Escherichia coli*. SEQ ID NO: 32 provides a synthetic nucleic acid sequence, based on GroES from *Escherichia coli*, codon optimized for expression in *Saccharomyces cerevisiae*.

[0266] Examples of suitable natural chaperones polypeptide homologous to GroES are given in Table 4.

TABLE 4

Natural chaperones homologous to GroES polypeptides suitable for expression

>gi 115388105 ref XP_001211558.1 :2-101 10 kDa heat shock protein, mitochondrial [<i>Aspergillus terreus</i> NIH2624]
>gi 116196854 ref XP_001224239.1 :1-102 conserved hypothetical protein [<i>Chaetomium globosum</i> CBS 148.51]
>gi 119175741 ref XP_001240050.1 :3-102 hypothetical protein CIMG_09671 [<i>Coccidioides immitis</i> RS]
>gi 119471607 ref XP_001258195.1 :12-111 chaperonin, putative [<i>Neosartorya fischeri</i> NRRL181]
>gi 121699818 ref XP_001268174.1 :8-106 chaperonin, putative [<i>Aspergillus clavatus</i> NRRL 1]
>gi 126274604 ref XP_001387607.1 :2-102 predicted protein [<i>Scheffersomyces stipitis</i> CBS 6054]
>gi 146417701 ref XP_001484818.1 :5-106 conserved hypothetical protein [<i>Meyerozyma guilliermondii</i> ATCC 6260]
>gi 154303611 ref XP_001552212.1 :1-102 10 kDa heat shock protein, mitochondrial [<i>Botryotinia fuckeliana</i> B05.10]
>gi 156049571 ref XP_001590752.1 :1-102 hypothetical protein SS1G_08492 [<i>Sclerotinia sclerotiorum</i> 1980]
>gi 156840987 ref XP_001643870.1 :1-103 hypothetical protein KpoL_495p10 [<i>Vanderwaltozyma polyspora</i> DSM 70924]
>gi 169608295 ref XP_001797567.1 :1-101 hypothetical protein SNOG_07218 [<i>Phaeosphaeria nodorum</i> SN15]
>gi 171688384 ref XP_001909132.1 :1-102 hypothetical protein [<i>Podospira anserina</i> S mat+]
>gi 189189366 ref XP_001931022.1 :71-168 10 kDa chaperonin [<i>Pyrenophora tritici-repentis</i> Pt-1C-BFP]
>gi 19075598 ref NP_588098.1 :1-102 mitochondrial heat shock protein Hsp10 (predicted) [<i>Schizosaccharomyces pombe</i> 972h-]
>gi 212530240 ref XP_002145277.1 :3-100 chaperonin, putative [<i>Talaromyces marneffe</i> ATCC 18224]
>gi 212530242 ref XP_002145278.1 :3-95 chaperonin, putative [<i>Talaromyces marneffe</i> ATCC 18224]
>gi 213404320 ref XP_002172932.1 :1-102 mitochondrial heat shock protein Hsp10 [<i>Schizosaccharomyces japonicus</i> yFS275]
>gi 225557301 gb EEH05587.1 :381-478 pre-mRNA polyadenylation factor fip1 [<i>Ajellomyces capsulatus</i> G186AR]
>gi 225684092 gb EEH22376.1 :3-100 heat shock protein [<i>Paracoccidioides brasiliensis</i> Pb03]
>gi 238490530 ref XP_002376502.1 :2-104 chaperonin, putative [<i>Aspergillus flavus</i> NRRL3357]
>gi 238878220 gb EEQ41858.1 :1-106 10 kDa heat shock protein, mitochondrial [<i>Candida albicans</i> WO-1]
>gi 240280207 gb EER4371.1 :426-523 pre-mRNA polyadenylation factor fip1 [<i>Ajellomyces capsulatus</i> H143]

TABLE 4-continued

Natural chaperones homologous to GroES polypeptides suitable for expression
>gi 241950445 ref XP_002417945.1 :1-103 10 kDa chaperonin, putative; 10 kDa heat shock protein mitochondrial (hsp10), putative [<i>Candida dubliniensis</i> CD36]
>gi 242819222 ref XP_002487273.1 :90-182 chaperonin, putative [<i>Talaromyces stipitatus</i> ATC]
>gi 254566327 ref XP_002490274.1 :1-102 Putative protein of unknown function [<i>Komagataella pastoris</i> GS115]
>gi 254577241 ref XP_002494607.1 :1-103 ZYRO0A05434p [<i>Zygosaccharomyces rouxii</i>]
>gi 255717999 ref XP_002555280.1 :1-103 KLTH0G05588p [<i>Lachancea thermotolerans</i>]
>gi 255956581 ref XP_002569043.1 :2-101 Pc21g20560 [<i>Penicillium chrysogenum</i> Wisconsin 54-1255]
>gi 258572664 ref XP_002545094.1 :16-108 chaperonin GroS [<i>Uncinocarpus reesii</i> 1704]
>gi 261190594 ref XP_002621706.1 :3-100 chaperonin [<i>Ajellomyces dermatitidis</i> SLH14081]
>gi 295664909 ref XP_002793006.1 :3-100 10 kDa heat shock protein, mitochondrial [<i>Paracoccidioides</i> sp. 'lutzi' Pb01]
>gi 296412657 ref XP_002836039.1 :76-177 hypothetical protein [<i>Tuber melanosporum</i> Mel28]
>gi 302307854 ref NP_984626.2 :2-102 AEL235Wp [<i>Ashbya gossypii</i> ATCC 10895]
>gi 302894117 ref XP_003045939.1 :1-102 predicted protein [<i>Nectria haematococca</i> mpVI 77-13-4]
>gi 303318351 ref XP_003069175.1 :3-100 10 kDa heat shock protein, mitochondrial, putative [<i>Coccidioides posadasii</i> C735 delta SOWgp]
>gi 310795300 gb EFQ30761.1 :1-102 chaperonin 10 kDa subunit [<i>Glomerella graminicola</i> M1.001]
>gi 315053085 ref XP_003175916.1 :12-109 chaperonin GroS [<i>Arthroderma gypseum</i> CBS 118893]
>gi 317032114 ref XP_001394060.2 :334-433 heat shock protein [<i>Aspergillus niger</i> CBS 513.88]
>gi 317032116 ref XP_001394059.2 :2-101 heat shock protein [<i>Aspergillus niger</i> CBS 513.88]
>gi 320583288 gb EFW97503.1 :6-106 chaperonin, putative heat shock protein, putative [<i>Ogataea parapolyomorpha</i> DL-1]
>gi 320591507 gb EFX03946.1 :1-102 heat shock protein [<i>Grosmannia clavigera</i> kw1407]
>gi 322700925 gb EFY92677.1 :1-102 chaperonin [<i>Metarhizium acridum</i> CQMa 102]
>gi 325096696 gb EGC50006.1 :409-506 pre-mRNA polyadenylation factor fip1 [<i>Ajellomyces capsulatus</i> H88]
>gi 326471604 gb EGD95613.1 :14-111 chaperonin 10 Kd subunit [<i>Trichophyton tonsurans</i> CBS112818]
>gi 327293056 ref XP_003231225.1 :3-100 chaperonin [<i>Trichophyton rubrum</i> CBS 118892]
>gi 330942654 ref XP_003306155.1 :37-136 hypothetical protein PTT_19211 [<i>Pyrenophora teres</i> f. <i>teres</i> 0-1]
>gi 336268042 ref XP_003348786.1 :47-147 hypothetical protein SMAC_01809 [<i>Sordaria macrospora</i> khell]
>gi 340519582 gb EGR49820.1 :1-109 predicted protein [<i>Trichoderma reesei</i> QM6a]
>gi 340960105 gb EGS21286.1 :3-103 putative mitochondrial 10 kDa heat shock protein [<i>Chaetomium thermophilum</i> var. <i>thermophilum</i> DSM 1495]
>gi 342883802 gb EGU84224.1 :1-102 hypothetical protein FOXB_05181 [<i>Fusarium oxysporum</i> Fo5176]
>gi 344302342 gb EGW32647.1 :2-102 hypothetical protein SPADRAFT_61712 [<i>Spathaspora passalidarum</i> NRRL Y-27907]
>gi 345570750 gb EGX53571.1 :1-102 hypothetical protein AOL_s00006g437 [<i>Arthrobotrys oligospora</i> ATCC 24927]
>gi 346321154 gb EGX90754.1 :1-102 chaperonin [<i>Cordyceps militaris</i> CM01]
>gi 346970393 gb EGY13845.1 :1-102 heat shock protein [<i>Verticillium dahliae</i> VdLs.17]
>gi 354548296 emb CCE45032.1 :1-106 hypothetical protein CPAR2_700360 [<i>Candida parapsilosis</i>]
>gi 358385052 gb EHK22649.1 :1-102 hypothetical protein TRIVIDRAFT_230640 [<i>Trichoderma virens</i> Gv 29-8]
>gi 358393422 gb EHK42823.1 :1-101 hypothetical protein TRIATDRAFT_258186 [<i>Trichoderma atroviride</i> IMI 206040]
>gi 361126733 gb EHK98722.1 :1-97 putative 10 kDa heat shock protein, mitochondrial [<i>Glare lozoyensis</i> 74030]
>gi 363753862 ref XP_003647147.1 :2-102 hypothetical protein Ecym_5593 [<i>Eremothecium cymbalariae</i> DBVPG#7215]
>gi 365758401 gb EHN00244.1 :1-106 Hsp10p [<i>Saccharomyces cerevisiae</i> x <i>Saccharomyces kudriavzevii</i> VIN7]
>gi 365987664 ref XP_003670663.1 :1-103 hypothetical protein NDAI_0F01010 [<i>Naumovozyma dairenensis</i> CBS 421]
>gi 366995125 ref XP_003677326.1 :1-103 hypothetical protein NCAS_0G00860 [<i>Naumovozyma castellii</i> CBS 4309]
>gi 366999797 ref XP_003684634.1 :1-103 hypothetical protein TPHA_0C00430 [<i>Tetrapistispora phaffii</i> CBS 4417]
>gi 367009030 ref XP_003679016.1 :1-103 hypothetical protein TDEL_0A04730 [<i>Torulaspora delbrueckii</i>]
>gi 367023138 ref XP_003660854.1 :1-104 hypothetical protein MYCTH_59302 [<i>Myceliophthora thermophila</i> ATCC 42464]
>gi 367046344 ref XP_003653552.1 :1-102 hypothetical protein THITE_2116070 [<i>Thielavia terrestris</i> NRRL8126]

TABLE 4-continued

Natural chaperones homologous to GroES polypeptides suitable for expression
>gi 378726440 gb EHY52899.1 :9-109 chaperonin GroES [<i>Exophiala dermatitidis</i> NIH/UT8656]
>gi 380493977 emb CCF33483.1 :1-102 chaperonin 10 kDa subunit [<i>Colletotrichum higginsianum</i>]
>gi 385305728 gb EIF49680.1 :1-102 10 kDa heat shock mitochondrial [Dekkera bruxellensis AWR11499]
>gi 389628546 ref XP_003711926.1 :1-102 hsp10-like protein [Magnaporthe oryzae 70-15]
>gi 396462608 ref XP_003835915.1 :1-101 similar to 10 kDa heat shock protein [Leptosphaeria maculans JN3]
>gi 398392541 ref XP_003849730.1 :1-102 hypothetical protein MYCGRDRAFT_105721 [Zymoseptoria tritici IPO323]
>gi 400597723 gb EJP65453.1 :24-124 chaperonin 10 kDa subunit [Beauveria bassiana ARSEF 2860]
>gi 401623646 gb EJS41738.1 :1-106 hsp10p [Saccharomyces arboricola H-6]
>gi 401842164 gb EJT44422.1 :1-92 HSP10-like protein [Saccharomyces kudriavzevii IFO 1802]
>gi 402084027 gb EJT79045.1 :1-102 hsp10-like protein [Gaeumannomyces graminis var. tritici]
>gi 403215209 emb CCK69709.1 :1-104 hypothetical protein KNAG_OC06130 [Kazachstania naganishii CBS 8797]
>gi 406604629 emb CCH43969.1 :4-100 hypothetical protein BN7_3524 [Wickerhamomyces ciferrii]
>gi 406867021 gb EKD20060.1 :56-156 hypothetical protein MBM_02012 [Marssonina brunnea f. sp. 'multigermtubi' MB_m1]
>gi 407926227 gb EKG19196.1 :74-174 GroES-like protein [Macrophomina phaseolina MS6]
>gi 408398157 gb EKJ77291.1 :11-111 hypothetical protein FPSE_02566 [Fusarium pseudograminearum CS3096]
>gi 410082063 ref XP_003958610.1 :1-103 hypothetical protein KAFR_0H00660 [Kazachstania africana CBS2517]
>gi 425777664 gb EKV15823.1 :58-157 Chaperonin, putative [Penicillium digitatum Pd1]
>gi 440639680 gb ELR09599.1 :1-102 chaperonin GroES [Geomyces destructans 20631-21]
>gi 444323906 ref XP_004182593.1 :1-105 hypothetical protein TBLA_0J00760 [Tetrapisporablatta CBS 6284]
>gi 448083208 ref XP_004195335.1 :2-101 Piso0_005888 [Milleromyces farinosa CBS 7064]
>gi 448087837 ref XP_004196425.1 :2-102 Piso0_005888 [Milleromyces farinosa CBS 7064]
>gi 448534948 ref XP_003870866.1 :1-106 Hsp10 protein [Candida orthopsilosis Co 90-125]
>gi 449295977 gb EMC91998.1 :1-102 hypothetical protein BAUCODRAFT_39148 [Baudoinia compn]
>gi 46123659 ref XP_386383.1 :3-103 hypothetical protein FG06207.1 [Gibberella zeae PH-1]
>gi 50289455 ref XP_447159.1 :1-103 hypothetical protein [Candida glabrata CBS 138]
>gi 50308731 ref XP_454370.1 :1-103 hypothetical protein [Kluyveromyces lactis NRRL Y-1140]
>gi 50411066 ref XP_457014.1 :1-106 DEHA2B01122p [Debaryomyces hansenii CBS767]
>gi 50545998 ref XP_500536.1 :1-102 YALI0B05610p [Yarrowia lipolytica]
>gi 51013895 gb AAT93241.1 :1-106 YOR020C [Saccharomyces cerevisiae]
>gi 6324594 ref NP_014663.1 :1-106 Hsp10p [Saccharomyces cerevisiae S288c]
>gi 67523953 ref XP_660036.1 :2-101 hypothetical protein AN2432.2 [Aspergillus nidulans FGSC A4]
>gi 70992219 ref XP_750958.1 :12-106 chaperonin [Aspergillus fumigatus Af293]
>gi 85079266 ref XP_956315.1 :1-104 hypothetical protein NCU04334 [Neurospora crassa OR74A]

[0267] Table 4: Natural chaperones homologous to GroES polypeptides suitable for expression

[0268] As an alternative to GroEL a functional homologue of GroEL may be present, in particular a functional homologue comprising an amino acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99%

sequence identity with the amino acid sequence of GroEL, respectively the amino sequence of SEQ ID NO: 29.

[0269] SEQ ID NO:29 provides a preferred translated protein sequence, based on GroEL of *Escherichia coli*. SEQ ID NO: 30 provides a synthetic nucleic acid sequence, based on GroEL from *Escherichia coli*, codon optimized for expression in *Saccharomyces cerevisiae*.

[0270] Suitable natural chaperones polypeptides homologous to GroEL are given in Table 5.

TABLE 5

Natural chaperones homologous to GroEL polypeptides suitable for expression
>gi 115443330 ref XP_001218472.1 heat shock protein 60, mitochondrial precursor [Aspergillus terreus NIH2624]
>gi 114188341 gb EAU30041.1 heat shock protein 60, mitochondrial precursor [Aspergillus terreus NIH2624]
>gi 119480793 ref XP_001260425.1 antigenic mitochondrial protein HSP60, putative [Neosartorya fischeri NRRL 181] >gi 119408579 gb EAW18528.1 antigenic mitochondrial protein HSP60, putative [Neosartorya fischeri NRRL 181]

TABLE 5-continued

Natural chaperones homologous to GroEL polypeptides suitable for expression
>gi126138730 reflXP_001385888.1 hypothetical protein PICST_90190 [<i>Scheffersomyces stipitis</i> CBS 6054] >gi126093166 gb ABN67859.1 mitochondrial groEL-type heat shock protein [<i>Scheffersomyces stipitis</i> CBS 6054]
>gi145246630 reflXP_001395564.1 heat shock protein 60 [<i>Aspergillus niger</i> CBS 513.88]
>gi134080285 emb CAK46207.1 unnamed protein product [<i>Aspergillus niger</i>]
>gi350636909 gb EHA25267.1 hypothetical protein ASPNIDRAFT_54001 [<i>Aspergillus niger</i> ATCC 1015]
>gi146413148 reflXP_001482545.1 heat shock protein 60, mitochondrial precursor [<i>Meyerozyma guilliermondii</i> ATCC 6260]
>gi154277022 reflXP_001539356.1 heat shock protein 60, mitochondrial precursor [<i>Ajellomyces capsulatus</i> NAm1] >gi150414429 gb EDN09794.1 heat shock protein 60, mitochondrial precursor [<i>Ajellomyces capsulatus</i> NAm1]
>gi154303540 reflXP_001552177.1 heat shock protein 60 [<i>Botryotinia fuckeliana</i> B05.10]
>gi347840915 emb CCD55487.1 similar to heat shock protein 60 [<i>Botryotinia fuckeliana</i>]
>gi156063938 reflXP_001597891.1 heat shock protein 60, mitochondrial precursor [<i>Sclerotinia sclerotiorum</i> 1980] >gi154697421 gb EDN97159.1 heat shock protein 60, mitochondrial precursor [<i>Sclerotinia sclerotiorum</i> 1980 UF-70]
>gi156844469 reflXP_001645297.1 hypothetical protein Kpol_1037p35 [<i>Vanderwaltozyma polyspora</i> DSM 70294] >gi156115957 gb EDO17439.1 hypothetical protein Kpol_1037p35 [<i>Vanderwaltozyma polyspora</i> DSM 70294]
>gi16416029 emb CAB91379.2 probable heat-shock protein hsp60 [<i>Neurospora crassa</i>]
>gi350289516 gb EGZ70741.1 putative heat-shock protein hsp60 [<i>Neurospora tetrasperma</i> FGSC 2509]
>gi169626377 reflXP_001806589.1 hypothetical protein SNOG_16475 [<i>Phaeosphaeria nodorum</i> SN15] >gi111055053 gb EAT76173.1 hypothetical protein SNOG_16475 [<i>Phaeosphaeria nodorum</i> SN15]
>gi169783766 reflXP_001826345.1 heat shock protein 60 [<i>Aspergillus oryzae</i> RIB40]
>gi238493601 reflXP_002378037.1 antigenic mitochondrial protein HSP60, putative [<i>Aspergillus flavus</i> NRRL3357] >gi83775089 dbj BAE65212.1 unnamed protein product [<i>Aspergillus oryzae</i> RIB40] >gi220696531 gb EED52873.1 antigenic mitochondrial protein HSP60, putative [<i>Aspergillus flavus</i> NRRL3357] >gi391869413 gb EIT78611.1 chaperonin, Cpn60/Hsp60p [<i>Aspergillus oryzae</i> 3.042]
>gi189190432 reflXP_001931555.1 heat shock protein 60, mitochondrial precursor [<i>Pyrenophora tritici-repentis</i> Pt-1C-BFP] >gi187973161 gb EDU40660.1 heat shock protein 60, mitochondrial precursor [<i>Pyrenophora tritici-repentis</i> Pt-1C-BFP]
>gi190348913 gb EDK41467.2 heat shock protein 60, mitochondrial precursor [<i>Meyerozyma guilliermondii</i> ATCC 6260]
>gi225554633 gb EEH02929.1 hsp60-like protein [<i>Ajellomyces capsulatus</i> G186AR]
>gi238880068 gb EEQ43706.1 heat shock protein 60, mitochondrial precursor [<i>Candida albicans</i> WO-1]
>gi239613490 gb EEQ90477.1 chaperonin GroL [<i>Ajellomyces dermatitidis</i> ER-3]
>gi240276977 gb EER40487.1 hsp60-like protein [<i>Ajellomyces capsulatus</i> H143]
>gi241958890 reflXP_002422164.1 heat shock protein 60, mitochondrial precursor, putative [<i>Candida dubliniensis</i> CD36] >gi223645509 emb CAX40168.1 heat shock protein 60, mitochondrial precursor, putative [<i>Candida dubliniensis</i> CD36]
>gi254572906 reflXP_002493562.1 Tetradecameric mitochondrial chaperonin [<i>Komagataella pastoris</i> GS115] >gi238033361 emb CAY71383.1 Tetradecameric mitochondrial chaperonin [<i>Komagataella pastoris</i> GS115]
>gi254579947 reflXP_002495959.1 ZYRO0C07106p [<i>Zygosaccharomyces rouxii</i>]
>gi238938850 emb CAR27026.1 ZYRO0C07106p [<i>Zygosaccharomyces rouxii</i>]
>gi255712781 reflXP_002552673.1 KLTH0C10428p [<i>Lachancea thermotolerans</i>]
>gi238934052 emb CAR22235.1 KLTH0C10428p [<i>Lachancea thermotolerans</i> CBS 6340]
>gi255721795 reflXP_002545832.1 heat shock protein 60, mitochondrial precursor [<i>Candida tropicalis</i> MYA-3404] >gi240136321 gb EER35874.1 heat shock protein 60, mitochondrial precursor [<i>Candida tropicalis</i> MYA-3404]
>gi255941288 reflXP_002561413.1 Pc16g11070 [<i>Penicillium chrysogenum</i> Wisconsin 54-1255] >gi211586036 emb CAP93777.1 Pc16g11070 [<i>Penicillium chrysogenum</i> Wisconsin 54-1255]
>gi259148241 emb CAY81488.1 Hsp60p [<i>Saccharomyces cerevisiae</i> EC1118]
>gi260950325 reflXP_002619459.1 heat shock protein 60, mitochondrial precursor [<i>Clavisporea lusitanae</i> ATCC 42720] >gi238847031 gb EEQ36495.1 heat shock protein 60, mitochondrial precursor [<i>Clavisporea lusitanae</i> ATCC 42720]
>gi261194577 reflXP_002623693.1 chaperonin GroL [<i>Ajellomyces dermatitidis</i> SLH14081]
>gi239588231 gb EEQ70874.1 chaperonin GroL [<i>Ajellomyces dermatitidis</i> SLH14081]
>gi327355067 gb EGE83924.1 chaperonin GroL [<i>Ajellomyces dermatitidis</i> ATCC 18188]
>gi296422271 reflXP_002840685.1 hypothetical protein [<i>Tuber melanosporum</i> Mel28]
>gi295636906 emb CAZ84876.1 unnamed protein product [<i>Tuber melanosporum</i>]
>gi296809035 reflXP_002844856.1 heat shock protein 60 [<i>Arthroderma otae</i> CBS 113480]
>gi238844339 gb EEQ34001.1 heat shock protein 60 [<i>Arthroderma otae</i> CBS 113480]
>gi302308696 reflNP_985702.2 AFR155Wp [<i>Ashbya gossypii</i> ATCC 10895]
>gi299790751 gb AAS53526.2 AFR155Wp [<i>Ashbya gossypii</i> ATCC 10895]
>gi374108933 gb AEY97839.1 EAFR155Wp [<i>Ashbya gossypii</i> FDAG1]
>gi302412525 reflXP_003004095.1 heat shock protein [<i>Verticillium albo-atrum</i> VaMs. 102]
>gi261356671 gb EEY19099.1 heat shock protein [<i>Verticillium albo-atrum</i> VaMs. 102]
>gi302505585 reflXP_003014499.1 hypothetical protein ARB_07061 [<i>Arthroderma</i>

TABLE 5-continued

Natural chaperones homologous to GroEL polypeptides suitable for expression
<i>benhamiae</i> CBS 112371] >gi 291178320 gb EFE34110.1 hypothetical protein ARB_07061 [<i>Arthroderma benhamiae</i> CBS 112371] >gi 302656385 ref XP_003019946.1 hypothetical protein TRV_05992 [<i>Trichophyton verrucosum</i> HKI 0517] >gi 291183723 gb EFE39322.1 hypothetical protein TRV_05992 [<i>Trichophyton verrucosum</i> HKI 0517] >gi 302915513 ref XP_003051567.1 predicted protein [<i>Nectria haematococca</i> mpVI 77-13-4] >gi 256732506 gb EEU45854.1 predicted protein [<i>Nectria haematococca</i> mpVI 77-13-4] >gi 310794550 gb EFQ30011.1 chaperonin GroL [<i>Glomerella graminicola</i> M1.001] >gi 315048491 ref XP_003173620.1 chaperonin GroL [<i>Arthroderma gypseum</i> CBS 118893] >gi 311341587 gb EFR00790.1 chaperonin GroL [<i>Arthroderma gypseum</i> CBS 118893] >gi 320580028 gb EFW94251.1 Tetradameric mitochondrial chaperonin [<i>Ogataea parapolyomorpha</i> DL-1] >gi 320586014 gb EFW98693.1 heat shock protein mitochondrial precursor [<i>Grosmanina clavigera</i> kw1407] >gi 322692465 gb EFY84374.1 Heat shock protein 60 precursor (Antigen HIS-62) [<i>Metarhizium acridum</i> CQMa 102] >gi 322705285 gb EFY96872.1 Heat shock protein 60 (Antigen HIS-62) [<i>Metarhizium anisopliae</i> ARSEF 23] >gi 323303806 gb EGA57589.1 Hsp60p [<i>Saccharomyces cerevisiae</i> FostersB] >gi 323307999 gb EGA61254.1 Hsp60p [<i>Saccharomyces cerevisiae</i> FostersO] >gi 323332364 gb EGA73773.1 Hsp60p [<i>Saccharomyces cerevisiae</i> AWRI796] >gi 326468648 gb EGD92657.1 heat shock protein 60 [<i>Trichophyton tonsurans</i> CBS 112818] >gi 326479866 gb EGE03876.1 chaperonin GroL [<i>Trichophyton equinum</i> CBS 127.97] >gi 330915493 ref XP_003297052.1 hypothetical protein PTT_07333 [<i>Pyrenophora teres</i> f. <i>teres</i> 0-1] >gi 311330479 gb EFQ94847.1 hypothetical protein PTT_07333 [<i>Pyrenophora teres</i> f. <i>teres</i> 0-1] >gi 336271815 ref XP_003350665.1 hypothetical protein SMAC_02337 [<i>Sordaria macrospora</i> k-hell] >gi 380094827 emb CCC07329.1 unnamed protein product [<i>Sordaria macrospora</i> k-hell] >gi 336468236 gb EGO56399.1 hypothetical protein NEUTE1DRAFT_122948 [<i>Neurospora tetrasperma</i> FGSC 2508] >gi 340522598 gb EGR52831.1 hsp60 mitochondrial precursor-like protein [<i>Trichoderma reesei</i> QM6a] >gi 341038907 gb EGS23899.1 mitochondrial heat shock protein 60-like protein [<i>Chaetomium thermophilum</i> var. <i>thermophilum</i> DSM 1495] >gi 342886297 gb EGU86166.1 hypothetical protein FOXB_03302 [<i>Fusarium oxysporum</i> Fo5176] >gi 344230084 gb EGV61969.1 chaperonin GroL [<i>Candida tenuis</i> ATCC 10573] >gi 344303739 gb EGW33988.1 hypothetical protein SPAPADRAFT_59397 [<i>Spathaspora passalidarum</i> NRRL Y-27907] >gi 345560428 gb EGX43553.1 hypothetical protein AOL_s00215g289 [<i>Arthrobotrys oligospora</i> ATCC 24927] >gi 346323592 gb EGX93190.1 heat shock protein 60 (Antigen HIS-62) [<i>Cordyceps militaris</i> CM01] >gi 346975286 gb EGY18738.1 heat shock protein [<i>Verticillium dahliae</i> VdLs. 17] >gi 354545932 emb CCE42661.1 hypothetical protein CPAR2_203040 [<i>Candida parapsilosis</i>] >gi 358369894 dbj GAA86507.1 heat shock protein 60, mitochondrial precursor [<i>Aspergillus kawachii</i> IFO 4308] >gi 358386867 gb EHK24462.1 hypothetical protein TRIVIDRAFT_79041 [<i>Trichoderma virens</i> Gv29-8] >gi 358399658 gb EHK48995.1 hypothetical protein TRIATDRAFT_297734 [<i>Trichoderma atroviride</i> IMI 206040] >gi 363750488 ref XP_003645461.1 hypothetical protein Ecym_3140 [<i>Eremothecium cymbalariae</i> DBVPG#7215] >gi 356889095 gb AET38644.1 Hypothetical protein Ecym_3140 [<i>Eremothecium cymbalariae</i> DBVPG#7215] >gi 365759369 gb EHN01160.1 Hsp60p [<i>Saccharomyces cerevisiae</i> × <i>Saccharomyces kudriavzevii</i> VIN7] >gi 365764091 gb EHN05616.1 Hsp60p [<i>Saccharomyces cerevisiae</i> × <i>Saccharomyces kudriavzevii</i> VIN7] >gi 365985626 ref XP_003669645.1 hypothetical protein NDAI_0D00880 [<i>Naumovozyma dairenensis</i> CBS 421] >gi 343768414 emb CCD24402.1 hypothetical protein NDAI_0D00880 [<i>Naumovozyma dairenensis</i> CBS 421] >gi 366995970 ref XP_003677748.1 hypothetical protein NCAS_0H00890 [<i>Naumovozyma castellii</i> CBS 4309] >gi 342303618 emb CCC71399.1 hypothetical protein NCAS_0H00890 [<i>Naumovozyma castellii</i> CBS 4309] >gi 367005154 ref XP_003687309.1 hypothetical protein TPHA_0J00520 [<i>Tetrapisispora phaffii</i> CBS 4417] >gi 357525613 emb CCE64875.1 hypothetical protein TPHA_0J00520 [<i>Tetrapisispora phaffii</i> CBS 4417] >gi 367017005 ref XP_003683001.1 hypothetical protein TDEL_0G04230 [<i>Torulaspora delbrueckii</i>] >gi 359750664 emb CCE93790.1 hypothetical protein TDEL_0G04230 [<i>Torulaspora delbrueckii</i>]

TABLE 5-continued

Natural chaperones homologous to GroEL polypeptides suitable for expression
>gi 367035486 ref XP_003667025.1 hypothetical protein MYCTH_2097570 [<i>Myceliophthora thermophila</i> ATCC 42464]
>gi 347014298 gb AEO61780.1 hypothetical protein MYCTH_2097570 [<i>Myceliophthora thermophila</i> ATCC 42464]
>gi 367055018 ref XP_003657887.1 hypothetical protein THITE_127923 [<i>Thielavia terrestris</i> NRRL 8126]
>gi 347005153 gb AEO71551.1 hypothetical protein THITE_127923 [<i>Thielavia terrestris</i> NRRL 8126]
>gi 378728414 gb EHY54873.1 heat shock protein 60 [<i>Exophiala dermatitidis</i> NIH/UT8656]
>gi 380494593 emb CCF33032.1 heat shock protein 60 [<i>Colletotrichum higginsianum</i>]
>gi 385305893 gb EIF49836.1 heat shock protein 60 [<i>Dekkera bruxellensis</i> AWRI1499]
>gi 389638386 ref XP_003716826.1 heat shock protein 60 [<i>Magnaporthe oryzae</i> 70-15]
>gi 351642645 gb EHA50507.1 heat shock protein 60 [<i>Magnaporthe oryzae</i> 70-15]
>gi 440474658 gb ELQ43388.1 heat shock protein 60 [<i>Magnaporthe oryzae</i> Y34]
>gi 440480475 gb ELQ61135.1 heat shock protein 60 [<i>Magnaporthe oryzae</i> P131]
>gi 393243142 gb EJD50658.1 chaperonin GroL [<i>Auricularia delicata</i> TFB-10046 SS5]
>gi 396494741 ref XP_003844378.1 similar to heat shock protein 60 [<i>Leptosphaeria maculans</i> JN3]
>gi 312220958 emb CBY00899.1 similar to heat shock protein 60 [<i>Leptosphaeria maculans</i> JN3]
>gi 398393428 ref XP_003850173.1 chaperone ATPase HSP60 [<i>Zymoseptoria tritici</i> IPO323]
>gi 339470051 gb EGP85149.1 hypothetical protein MYCGRDRAFT_75170 [<i>Zymoseptoria tritici</i> IPO323]
>gi 401624479 gb EJS42535.1 hsp60p [<i>Saccharomyces arboricola</i> H-6]
>gi 401842294 gb EJT44530.1 HSP60-like protein [<i>Saccharomyces kudriavzevii</i> IFO 1802]
>gi 402076594 gb EJT72017.1 heat shock protein 60 [<i>Gaeumannomyces graminis</i> var. <i>tritici</i> R3-111a-1]
>gi 403213867 emb CCK68369.1 hypothetical protein KNAG_0A07160 [<i>Kazachstania naganishii</i> CBS 8797]
>gi 406606041 emb CCH42514.1 Heat shock protein 60, mitochondrial [<i>Wickerhamomyces ciferrii</i>]
>gi 406863285 gb EKD16333.1 heat shock protein 60 [<i>Marssonina brunnea</i> f. sp. 'multigermutubi' MB_m1]
>gi 407922985 gb EKG16075.1 Chaperonin Cpn60 [<i>Macrophomina phaseolina</i> MS6]
>gi 408399723 gb EKJ78816.1 hypothetical protein FPSE_00959 [<i>Fusarium pseudograminearum</i> CS3096]
>gi 410083028 ref XP_003959092.1 hypothetical protein KAFR_0101760 [<i>Kazachstania africana</i> CBS 2517]
>gi 372465682 emb CCF59957.1 hypothetical protein KAFR_0101760 [<i>Kazachstania africana</i> CBS 2517]
>gi 444315528 ref XP_004178421.1 hypothetical protein TBLA_0B00580 [<i>Tetrapisispora blattae</i> CBS 6284]
>gi 387511461 emb CCH58902.1 hypothetical protein TBLA_0B00580 [<i>Tetrapisispora blattae</i> CBS 6284]
>gi 448090588 ref XP_004197110.1 Piso0_004347 [<i>Milleriozyma farinosa</i> CBS 7064]
>gi 448095015 ref XP_004198141.1 Piso0_004347 [<i>Milleriozyma farinosa</i> CBS 7064]
>gi 359378532 emb CCE84791.1 Piso0_004347 [<i>Milleriozyma farinosa</i> CBS 7064]
>gi 359379563 emb CCE83760.1 Piso0_004347 [<i>Milleriozyma farinosa</i> CBS 7064]
>gi 448526196 ref XP_003869293.1 Hsp60 heat shock protein [<i>Candida orthopsilosis</i> Co 90-125]
>gi 380353646 emb CCG23157.1 Hsp60 heat shock protein [<i>Candida orthopsilosis</i>]
>gi 46123737 ref XP_386422.1 HS60_AJECA Heat shock protein 60, mitochondrial precursor (Antigen HIS-62) [<i>Gibberella zeae</i> PH-1]
>gi 50292099 ref XP_448482.1 hypothetical protein [<i>Candida glabrata</i> CBS 138]
>gi 49527794 emb CAG61443.1 unnamed protein product [<i>Candida glabrata</i>]
>gi 503109751 ref XP_455510.1 hypothetical protein [<i>Kluyveromyces lactis</i> NRRL Y-1140]
>gi 49644646 emb CAG98218.1 KLLA0F09449p [<i>Kluyveromyces lactis</i>]
>gi 50422027 ref XP_459575.1 DEHA2E05808p [<i>Debaryomyces hansenii</i> CBS767]
>gi 49655243 emb CAG87802.1 DEHA2E05808p [<i>Debaryomyces hansenii</i> CBS767]
>gi 50555023 ref XP_504920.1 YALI0F02805p [<i>Yarrowia lipolytica</i>]
>gi 49650790 emb CAG77725.1 YALI0F02805p [<i>Yarrowia lipolytica</i> CLIB122]
>gi 6323288 ref NP_013360.1 Hsp60p [<i>Saccharomyces cerevisiae</i> S288c]
>gi 123579 sp P19882.1 HSP60_YEAST RecName: Full = Heat shock protein 60, mitochondrial; AltName: Full = Cpn60; AltName: Full = P66; AltName: Full = Stimulator factor I 66 kDa component; Flags: Precursor >gi 171720 gb AAA34690.1 heat shock protein 60 (HSP60) [<i>Saccharomyces cerevisiae</i>] >gi 577181 gb AAB67380.1 Hsp60p: Heat shock protein 60 [<i>Saccharomyces cerevisiae</i>] >gi 151941093 gb EDN59473.1 chaperonin [<i>Saccharomyces cerevisiae</i> YJM789] >gi 190405319 gb EDV08586.1 chaperonin [<i>Saccharomyces cerevisiae</i> RM11-1a] >gi 207342889 gb EDZ70518.1 YLR259Cp-like protein [<i>Saccharomyces cerevisiae</i> AWRI1631] >gi 256271752 gb EEU06789.1 Hsp60p [<i>Saccharomyces cerevisiae</i> JAY291]
>gi 285813676 tpg DAA09572.1 TPA: chaperone ATPase HSP60 [<i>Saccharomyces cerevisiae</i> S288c] >gi 323353818 gb EGA85673.1 Hsp60p [<i>Saccharomyces cerevisiae</i> VL3]
>gi 349579966 dbj GAA25127.1 K7_Hsp60p [<i>Saccharomyces cerevisiae</i> Kyokai no. 7]
>gi 392297765 gb EIW08864.1 Hsp60p [<i>Saccharomyces cerevisiae</i> CEN.PK113-7D]
>gi 226279 prf 1504305A mitochondrial assembly factor
>gi 68485963 ref XP_713100.1 heat shock protein 60 [<i>Candida albicans</i> SC5314]
>gi 68486010 ref XP_713077.1 heat shock protein 60 [<i>Candida albicans</i> SC5314]
>gi 6016258 sp 074261.1 HSP60_CANAL RecName: Full = Heat shock protein 60, mitochondrial; AltName: Full = 60 kDa chaperonin; AltName: Full = Protein Cpn60; Flags:

TABLE 5-continued

Natural chaperones homologous to GroEL polypeptides suitable for expression	
Precursor	>gi 3552009 gb AAC34885.1 heat shock protein 60 [<i>Candida albicans</i>]
	>gi 46434552 gb EAK93958.1 heat shock protein 60 [<i>Candida albicans</i> SC5314]
	>gi 46434577 gb EAK93982.1 heat shock protein 60 [<i>Candida albicans</i> SC5314]
	>gi 71001164 reflXP_755263.1 antigenic mitochondrial protein HSP60 [<i>Aspergillus fumigatus</i> Af293]
	>gi 66852901 gb EAL93225.1 antigenic mitochondrial protein HSP60, putative [<i>Aspergillus fumigatus</i> Af293]
	>gi 159129345 gb EDP54459.1 antigenic mitochondrial protein HSP60, putative [<i>Aspergillus fumigatus</i> A1163]
	>gi 90970323 gb ABE02805.1 heat shock protein 60 [<i>Rhizophagus intraradices</i>]

[0271] The recombinant yeast cell preferably comprises, respectively functionally expresses, a GroES chaperone and a GroEL chaperone. Preferably a 10 kDa chaperone (“GroES”) from Table 4 is combined with a matching 60 kDa chaperone (“GroEL”) from Table 5 of the same organism genus or species for expression in the recombinant yeast cell.

[0272] For instance: >gi|189189366|reflXP_001931022.1:71-168 10 kDa chaperonin [*Pyrenophora tritici-repentis*] expressed together with matching >gi|189190432|reflXP_001931555.1| heat shock protein 60, mitochondrial precursor [*Pyrenophora tritici-repentis* Pt-1C-BFP]. All other combinations from Table 4 and 5 similarly made with same organism source are also available to the skilled person for expression. Furthermore, one may combine a chaperone from Table 4 from one organism with a chaperone from Table 5 from another organism, or one may combine GroES with a chaperone from Table 5, or one may combine GroEL with a chaperone from Table 4.

[0273] Preferably the molecular chaperone(s) thus comprise or consist of:

[0274] an amino acid sequence of SEQ ID NO: 29 and/or SEQ ID NO: 31; or

[0275] one or more functional homologue(s) of SEQ ID NO: 29 and/or SEQ ID NO: 31, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of respectively SEQ ID NO: 29 and/or SEQ ID NO: 31; or

[0276] one or more functional homologue(s) of SEQ ID NO: 29 and/or SEQ ID NO: 31, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of respectively SEQ ID NO: 29 and/or SEQ ID NO: 31, more preferably one or more functional homologue(s) that has/have no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of respectively SEQ ID NO: 29 and/or SEQ ID NO: 31.

[0277] Preferable the nucleic acid sequence(s) encoding the molecular chaperones comprise or consist of:

[0278] a nucleic acid sequence of SEQ ID NO: 30 and/or SEQ ID NO: 32; or

[0279] one or more functional homologue(s) of SEQ ID NO: 30 and/or SEQ ID NO: 32, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at

least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of respectively SEQ ID NO: 30 and/or SEQ ID NO: 32; or

[0280] one or more functional homologue(s) of SEQ ID NO: 30 and/or SEQ ID NO: 32, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of respectively SEQ ID NO: 30 and/or SEQ ID NO: 32, more preferably one or more functional homologue(s) of that has/have no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of respectively SEQ ID NO: 30 and/or SEQ ID NO: 32.

[0281] The nucleic acid sequence(s) encoding for the molecular chaperones may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2014/129898, herein incorporated by reference.

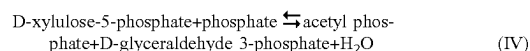
Phosphoketolase

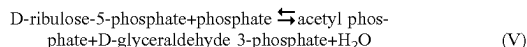
[0282] As indicated above, the recombinant yeast cell can advantageously comprise a, preferably heterologous, nucleic acid sequence encoding a protein comprising phosphoketolase (PKL) activity (EC 4.1.2.9 or EC 4.1.2.22) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

[0283] The recombinant cell may comprise one or more heterologous genes coding for a protein having phosphoketolase activity. Such a protein having phosphoketolase activity is herein also referred to as “phosphoketolase protein”, “phosphoketolase enzyme” or simply as “phosphoketolase”. Phosphoketolase is further herein abbreviated as “PKL” or “XFP”.

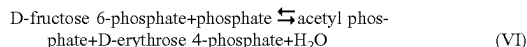
[0284] As used herein, a phosphoketolase catalyzes at least the conversion of D-xylulose 5-phosphate to D-glyceraldehyde 3-phosphate and acetyl phosphate. The phosphoketolase is involved in at least one of the following reactions:

EC 4.1.2.9:





EC 4.1.2.22:



[0285] A suitable enzymatic assay to measure phosphoketolase activity is described e.g. in Sonderegger et al., “*Metabolic Engineering of a Phosphoketolase Pathway for Pentose Catabolism in Saccharomyces cerevisiae*”, (2004), Applied & Environmental Microbiology, vol. 70(5), pages 2892-2897, incorporated herein by reference.

[0286] Preferably the protein having phosphoketolase (PKL) activity comprises or consists of:

[0287] an amino acid sequence of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36; or

[0288] a functional homologue of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36; or

[0289] a functional homologue of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35 or SEQ ID NO: 36.

[0290] Suitable nucleic acid sequences coding for an phosphoketolase protein may in be found in an organism selected from the group of *Aspergillus niger*, *Neurospora crassa*, *L. casei*, *L. plantarum*, *L. plantarum*, *B. adolescentis*, *B. bifidum*, *B. gallicum*, *B. animalis*, *B. lactis*, *L. pentosum*, *L. acidophilus*, *P. chrysogenum*, *A. nidulans*, *A. clavatus*, *L. mesenteroides*, and *O. oeni*.

[0291] The nucleic acid sequence (e.g. the gene) encoding for the protein having phosphoketolase (PKL) activity may suitably be incorporated in the genome of the recombinant yeast cell.

[0292] The recombinant cell may comprise one or more (heterologous) genes coding for an enzyme having phosphoketolase activity.

Phosphotransacetylase

[0293] As indicated above, the recombinant yeast cell can advantageously comprise a, preferably heterologous, nucleic acid sequence encoding a protein comprising phosphoketolase (PKL) activity (EC 4.1.2.9 or EC 4.1.2.22) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

[0294] As used herein, a phosphotransacetylase catalyzes at least the conversion of acetyl phosphate to acetyl-CoA.

[0295] The recombinant cell may comprise one or more heterologous genes coding for a protein having phosphotransacetylase activity. Such a protein having phosphotransacetylase activity is herein also referred to as “phosphotransacetylase protein”, “phosphotransacetylase enzyme” or simply as “phosphotransacetylase”. phosphotransacetylase is further herein abbreviated as “PTA”.

[0296] Preferably the protein having phosphotransacetylase (PTA) activity comprises or consists of:

[0297] an amino acid sequence of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40; or

[0298] a functional homologue of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40; or

[0299] a functional homologue of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39 or SEQ ID NO: 40.

[0300] Suitable nucleic acid sequences coding for an enzyme having phosphotransacetylase may in be found in an organism selected from the group of *B. adolescentis*, *B. subtilis*, *C. cellulolyticum*, *C. phytofermentans*, *B. bifidum*, *B. animalis*, *L. mesenteroides*, *Lactobacillus plantarum*, *M. thermophila*, and *O. oeni*.

[0301] The nucleic acid sequence (e.g. the gene) encoding for the protein having phosphotransacetylase (PTA) activity may suitably be incorporated in the genome of the recombinant yeast cell.

Acetate Kinase

[0302] As indicated above, the recombinant yeast cell can comprise a, preferably heterologous, nucleic acid sequence encoding a protein comprising phosphoketolase (PKL) activity (EC 4.1.2.9 or EC 4.1.2.22) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8) and/or a, preferably heterologous, nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

[0303] As used herein, an acetate kinase catalyzes at least the conversion of acetate to acetyl phosphate.

[0304] The recombinant cell may comprise one or more, preferably heterologous, genes coding for a protein having acetate kinase activity (EC 2.7.2.12). Such a protein having acetate kinase activity is herein also referred to as “acetate

kinase protein”, “acetate kinase enzyme” or simply as “acetate kinase”. Acetate kinase is further herein abbreviated as “ACK”.

[0305] Preferably the protein having acetate kinase (ACK) activity comprises or consists of:

[0306] an amino acid sequence of SEQ ID NO: 41 or SEQ ID NO: 42; or

[0307] a functional homologue of SEQ ID NO: 41 or SEQ ID NO: 42, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 41 or SEQ ID NO: 42; or

[0308] a functional homologue of SEQ ID NO: 41 or SEQ ID NO: 42, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 41 or SEQ ID NO: 42, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 41 or SEQ ID NO: 42.

[0309] The nucleic acid sequence (e.g. the gene) encoding for the protein having acetate kinase (ACK) activity may suitably be incorporated in the genome of the recombinant yeast cell.

Acetylating Acetaldehyde Dehydrogenase

[0310] As indicated above, the recombinant yeast cell can advantageously comprise and functionally express a, preferably heterologous, nucleic acid sequence encoding a protein comprising NAD⁺ dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10).

[0311] If an acetylating acetaldehyde dehydrogenase is present, more preferably, the recombinant yeast cell functionally expresses:

[0312] a, preferably heterologous, nucleic acid sequence encoding a protein comprising NAD⁺ dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10); and

[0313] a, suitably endogenous or heterologous, nucleic acid sequence encoding a protein having NAD⁺-dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC 1.1.1.2); and

[0314] a, suitably endogenous or heterologous, nucleic acid sequence encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1).

[0315] Acetylating acetaldehyde dehydrogenase is an enzyme that catalyzes the conversion of acetyl-Coenzyme A to acetaldehyde (EC 1.2.1.10). This conversion can be represented by the equilibrium reaction formula:



[0316] A protein having acetylating acetaldehyde dehydrogenase activity is herein also referred to as “acetylating acetaldehyde dehydrogenase protein”, “acetylating acetaldehyde dehydrogenase enzyme” or simply “acetylating acetaldehyde dehydrogenase”. Preferences for a acetylating acetaldehyde dehydrogenase and the nucleic sequences

encoding for such are as described in WO2011/010923 and WO2019/063507, incorporated herein by reference.

[0317] The nucleic acid sequence encoding a protein having NAD⁺-dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10) is preferably a heterologous nucleic acid sequence. The encoded NAD⁺-dependent acetylating acetaldehyde dehydrogenase may therefore preferably be a heterologous NAD⁺-dependent acetylating acetaldehyde dehydrogenase.

[0318] It is possible for the protein having acetylating acetaldehyde dehydrogenase activity to be monofunctional or bifunctional.

[0319] The nucleic acid sequence encoding the NAD⁺ dependent acetylating acetaldehyde dehydrogenase may in principle originate from any organism comprising a nucleic acid sequence encoding said dehydrogenase. Known acetylating acetaldehyde dehydrogenases that can catalyse the NADH-dependent reduction of acetyl-Coenzyme A to acetaldehyde may in general be divided in three types of NAD⁺ dependent acetylating acetaldehyde dehydrogenase functional homologues:

[0320] 1) Bifunctional proteins that catalyse the reversible conversion of acetyl-CoA to acetaldehyde, and the subsequent reversible conversion of acetaldehyde to ethanol. These type of proteins advantageously have both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity. An example of this type of proteins is the AdhE protein in *E. coli* (Gen Bank No: NP_415757). AdhE appears to be the evolutionary product of a gene fusion. The NH₂-terminal region of the AdhE protein is highly homologous to aldehyde:NAD⁺ oxidoreductases, whereas the COOH-terminal region is homologous to a family of Fe²⁺ dependent ethanol:NAD⁺ oxidoreductases (see Membrillo-Hernandez et al., “*Evolution of the adhE Gene Product of Escherichia coli from a Functional Reductase to a Dehydrogenase*”, (2000) J. Biol. Chem. 275: pages 33869-33875, herein incorporated by reference). The *E. coli* AdhE is subject to metal-catalyzed oxidation and therefore oxygen-sensitive (see Tamarit et al. “*Identification of the Major Oxidatively Damaged Proteins in Escherichia coli Cells Exposed to Oxidative Stress*” (1998) J. Biol. Chem. 273: pages 3027-3032, herein incorporated by reference).

[0321] 2) Proteins that catalyse the reversible conversion of acetyl-Coenzyme A to acetaldehyde in strictly or facultative anaerobic micro-organisms but do not possess alcohol dehydrogenase activity. An example of this type of proteins has been reported in *Clostridium kluyveri* (see Smith et al.” *Purification, Properties, and Kinetic Mechanism of Coenzyme A-Linked Aldehyde Dehydrogenase from Clostridium kluyveri*” (1980) Arch. Biochem. Biophys. Vol. 203: pages 663-675, incorporated herein by reference). An acetylating acetaldehyde dehydrogenase has been annotated in the genome of *Clostridium kluyveri* DSM 555 (GenBank No: EDK33116). A homologous protein AcdH is identified in the genome of *Lactobacillus plantarum* (GenBank No: NP_784141). Another example of this type of proteins is the said gene product in *Clostridium beijerinckii* NRRL B593 (see Toth et al. “*The ald Gene, Encoding a Coenzyme A-Acylating Aldehyde Dehydrogenase, Distinguishes Clostridium beijerinckii and Two Other Solvent-Producing Clostridia from Clostridium*

acetobutylicum", (1999), Appl. Environ. Microbiol. Vol. 65: pages 4973-4980, GenBank No: AAD31841, incorporated herein by reference).

[0322] 3) Proteins that are part of a bifunctional aldolase-dehydrogenase complex involved in 4-hydroxy-2-ketovaleate catabolism. Such bifunctional enzymes catalyze the final two steps of the meta-cleavage pathway for catechol, an intermediate in many bacterial species in the degradation of phenols, toluates, naphthalene, biphenyls and other aromatic compounds (Powlowski and Shingler "Genetics and biochemistry of phenol degradation by *Pseudomonas* sp. CF600" (1994) Biodegradation Vol. 5, pages 219-236, herein incorporated by reference). 4-Hydroxy-2-ketovaleate is first converted by 4-hydroxy-2-ketovaleate aldolase to pyruvate and acetaldehyde, subsequently acetaldehyde is converted by acetylating acetaldehyde dehydrogenase to acetyl-CoA. An example of this type of acetylating acetaldehyde dehydrogenase is the DmpF protein in *Pseudomonas* sp CF600 (GenBank No: CAA43226) (Shingler et al., "Nucleotide Sequence and Functional Analysis of the Complete Phenol/3,4-Dimethylphenol Catabolic Pathway of *Pseudomonas* sp. Strain CF600", (1992), J. Bacteriol., Vol. 174, pages 711-724, incorporated herein by reference). The *E. coli* MphF protein (Ferrandez et al., "Genetic Characterization and Expression in Heterologous Hosts of the 3-(3-Hydroxyphenyl) Propionate Catabolic Pathway of *Escherichia coli* K-12" (1997) J. Bacteriol. 179: pages 2573-2581, GenBank No: NP_414885, incorporated herein by reference) is homologous to the DmpF protein in *Pseudomonas* sp. CF600.

[0323] In a preferred embodiment, the protein having acetylating acetaldehyde dehydrogenase activity is bifunctional and comprises both NAD⁺ dependent acetylating acetaldehyde dehydrogenase (EC 1.2.1.10) activity and NAD⁺ dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC 1.1.1.2).

[0324] A suitable nucleic acid sequence may in particular be found in an organism selected from the group of *Escherichia*, in particular *E. coli*; *Mycobacterium*, in particular *Mycobacterium marinum*, *Mycobacterium ulcerans*, *Mycobacterium tuberculosis*; *Carboxydotherrmus*, in particular *Carboxydotherrmus hydrogenoformans*; *Entamoeba*, in particular *Entamoeba histolytica*; *Shigella*, in particular *Shigella sonnei*; *Burkholderia*, in particular *Burkholderia pseudo mallei*, *Klebsiella*, in particular *Klebsiella pneumoniae*; *Azotobacter*, in particular *Azotobacter vinelandii*; *Azarcus* sp; *Cupriavidus*, in particular *Cupriavidus taiwanensis*; *Pseudomonas*, in particular *Pseudomonas* sp. CF600; *Pelotomaculum*, in particular *Pelotomaculum thermopropionicum*. Preferably, the nucleic acid sequence encoding the NAD⁺ dependent acetylating acetaldehyde dehydrogenase originates from *Escherichia*, more preferably from *E. coli*.

[0325] Particularly suitable is an mphF gene from *E. coli*, or a functional homologue thereof. This gene is described in Ferrandez et al., "Genetic Characterization and Expression in Heterologous Hosts of the 3-(3-Hydroxyphenyl) Propionate Catabolic Pathway of *Escherichia coli* K-12" (1997) J. Bacteriol. 179: pages 2573-2581. Good results have been obtained with *S. cerevisiae*, wherein an mphF gene from *E. coli* has been incorporated. In a further advantageous embodiment the nucleic acid sequence encoding an (acety-

lating) acetaldehyde dehydrogenase is from *Pseudomonas*, in particular dmpF, e.g. from *Pseudomonas* sp. CF600.

[0326] Further, an acetylating acetaldehyde dehydrogenase (or nucleic acid sequence encoding such activity) may for instance be selected from the group of *Escherichia coli* adhE, *Entamoeba histolytica* adh2, *Staphylococcus aureus* adhE, *Piromyces* sp. E2 adhE, *Clostridium kluyveri* EDK33116, *Lactobacillus plantarum* acdH, *Escherichia coli* eutE, *Listeria innocua* acdH, and *Pseudomonas putida* YP 001268189.

[0327] Preferably the protein having NAD⁺-dependent acetylating acetaldehyde dehydrogenase activity comprises or consists of:

[0328] an amino acid sequence of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48; or

[0329] a functional homologue of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48 having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48; or

[0330] a functional homologue of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48 having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47 or SEQ ID NO: 48.

[0331] Most preferably the acetylating acetaldehyde dehydrogenase protein is a bifunctional protein having both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity.

[0332] The nucleic acid sequence (e.g. the gene) encoding for the protein having acetylating acetaldehyde dehydrogenase activity may suitably be incorporated in the genome of the recombinant yeast cell.

[0333] Examples of suitable enzymes are further illustrated below in tables 6(a) to 6(e) for BLAST of the listed enzymes, giving suitable alternative alcohol/acetaldehyde dehydrogenases.

TABLE 6(a)

BLAST Query - adhE from <i>Escherichia coli</i>		
Description	Identity (%)	Accession number
bifunctional acetaldehyde-CoA/alcohol dehydrogenase [<i>Escherichia coli</i> O157:H7 str. Sakai]	100	NP_309768.1

TABLE 6(a)-continued

BLAST Query - adHE from <i>Escherichia coli</i>		
Description	Identity (%)	Accession number
bifunctional acetaldehyde-CoA/alcohol dehydrogenase [<i>Escherichia coli</i> UTI89]	99	YP_540449.1

TABLE 6(a)-continued

BLAST Query - adHE from <i>Escherichia coli</i>		
Description	Identity (%)	Accession number
bifunctional acetaldehyde-CoA/alcohol dehydrogenase [<i>Enterobacter</i> sp. 638]	95	YP_001177024.1

TABLE 6(b)

BLAST Query - acdH from <i>Lactobacillus plantarum</i>		
Description	Identity (%)	Accession number
acetaldehyde dehydrogenase [<i>Lactobacillus plantarum</i> WCFS1]	100	YP_004888365.1
acetaldehyde dehydrogenase [<i>Lactobacillus pentosus</i> IG1]	95	CCC16763.1
aldehyde-alcohol dehydrogenase [<i>Enterococcus cecorum</i>]	58	WP_016251441.1
aldehyde-alcohol dehydrogenase 2 [<i>Enterococcus faecalis</i>]	57	WP_016623694.1
bifunctional acetaldehyde-CoA/alcohol dehydrogenase [<i>Lactobacillus zeae</i>]	55	WP_010493695.1
alcohol dehydrogenase [<i>Bacillus thuringiensis</i>]	54	WP_003280110.1
bifunctional acetaldehyde-CoA/alcohol dehydrogenase, partial [<i>Listeria monocytogenes</i>]	53	WP_009931954.1

TABLE 6(c)

BLAST Query - eutE from <i>Escherichia coli</i>		
Description	Identity (%)	Accession number
aldehyde oxidoreductase, ethanolamine utilization protein [<i>Escherichia coli</i> str. K-12 substr. MG1655]	100	NP_416950.1
ethanolamine utilization; acetaldehyde dehydrogenase [<i>Escherichia coli</i> 0157:H7 str. EDL933]	99	NP_289007.1
aldehyde dehydrogenase [<i>Escherichia albertii</i>]	99	WP_001075674.1

TABLE 6(d)

BLAST Query - Lin1129 from <i>Listeria innocua</i>		
Description	Identity (%)	Accession number
aldehyde dehydrogenase [<i>Listeria innocua</i>] > emb/CAC96360.1 lin1129 [<i>Listeria innocua</i> Clip11262]	100	NP_470466.1
ethanolamine utilization protein EutE [<i>Listeria innocua</i>]	99	WP_003761764.1
aldehyde dehydrogenase [<i>Listeria monocytogenes</i>]	95	AGR09081.1
hypothetical protein [<i>Enterococcus malodoratus</i>]	64	WP_010739890.1
aldehyde dehydrogenase [<i>Yersinia aldovae</i>]	59	WP_004699364.1
aldehyde dehydrogenase EutE [<i>Klebsiella pneumoniae</i>]	58	WP_004205473.1

TABLE 6(e)

BLAST Query - adhE from <i>Staphylococcus aureus</i>		
Description	Identity (%)	Accession number
bifunctional acetaldehyde-CoA/alcohol dehydrogenase [<i>Staphylococcus aureus</i> subsp. aureus Mu50]	100	NP_370672.1
aldehyde dehydrogenase family protein [<i>Staphylococcus aureus</i> CA-347]	99	YP_008127042.1
bifunctional acetaldehyde-CoA/alcohol dehydrogenase [<i>Staphylococcus epidermidis</i>]	85	WP_002495347.1

TABLE 6(e)-continued

BLAST Query - adhE from <i>Staphylococcus aureus</i>		
Description	Identity (%)	Accession number
aldehyde-alcohol dehydrogenase 2 [<i>Enterococcus faecalis</i>]	75	WP_016623694.1

Acetyl-Coenzyme A Synthetase

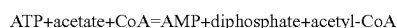
[0334] If the recombinant yeast cell functionally expresses a protein having acetylating acetaldehyde dehydrogenase activity, preferably the recombinant yeast cell is further functionally expressing:

[0335] a nucleic acid sequence encoding a protein having NAD⁺-dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC 1.1.1.2); and/or

[0336] a nucleic acid sequence encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1).

[0337] A protein having acetyl-Coenzyme A synthetase activity can herein also be referred to as “acetyl-Coenzyme A synthetase protein”, “acetyl-Coenzyme A synthetase enzyme” or simply “acetyl-Coenzyme A synthetase” or even “acetyl CoA synthetase”. The protein is further abbreviated herein as “ACS”.

[0338] The acetyl-Coenzyme A synthetase, also known as acetate-CoA ligase or acetyl-activating enzyme, catalyses the formation of acetyl-CoA from acetate, coenzyme A (CoA) and ATP as shown below:



[0339] It is understood that the recombinant yeast cell may naturally comprise an endogenous gene encoding an acetyl-Coenzyme A synthetase protein. In the alternative, or in addition thereto, the recombinant yeast cell may comprise a heterologous nucleic acid sequence encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1).

[0340] For example, the recombinant yeast cell according to the invention may comprise an acetyl-Coenzyme A synthetase, which may be present in the wild-type cell, as is for instance the case with *S. cerevisiae* which contains two acetyl-Coenzyme A synthetase isoenzymes encoded by the ACS1 (amino acid sequence illustrated as SEQ ID NO: 49) and ACS2 (amino acid sequence illustrated as SEQ ID NO: 50) genes (van den Berg et al (1996) J. Biol. Chem. 271:pages 28953-28959, incorporated herein by reference), or a host cell may be provided with one or more heterologous gene(s) encoding this activity, e.g. the ACS1 and/or

ACS2 gene of *S. cerevisiae* or a functional homologue thereof may be incorporated into a cell lacking acetyl-Coenzyme A synthetase isoenzyme activity.

[0341] Preferably the protein having NAD⁺-dependent acetyl-Coenzyme A synthetase activity comprises or consists of:

[0342] an amino acid sequence of SEQ ID NO: 49 or SEQ ID NO: 50; or

[0343] a functional homologue of SEQ ID NO: 49 or SEQ ID NO: 50 having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 49 or SEQ ID NO: 50; or

[0344] a functional homologue of SEQ ID NO: 49 or SEQ ID NO: 50 having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 49 or SEQ ID NO: 50, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 49 or SEQ ID NO: 50.

[0345] Preferably the recombinant yeast cell is a recombinant yeast cell wherein the, endogenous or heterologous, acetyl-Coenzyme A synthetase protein, is overexpressed, most preferably by using a suitable promoter as described for example in WO2011/010923, incorporated herein by reference. Any heterologous nucleic acid sequence (e.g. the gene) encoding for the protein having acetyl-Coenzyme A synthetase activity may suitably be incorporated in the genome of the recombinant yeast cell.

[0346] Examples of suitable proteins having acetyl-Coenzyme A synthetase activity are listed in table 7. At the top of table 7 the ACS2 used in the examples and that is BLASTED is mentioned.

TABLE 7

BLAST Query - ACS2 from <i>Saccharomyces cerevisiae</i>		
Description	Identity (%)	Accession number
acetate -- CoA ligase ACS2 [<i>Saccharomyces cerevisiae</i> S288c]	100	NP_013254.1
acetyl CoA synthetase [<i>Saccharomyces cerevisiae</i> YJM789]	99	EDN59693.1
acetate -- CoA ligase [<i>Kluyveromyces lactis</i> NRRL Y-1140]	85	XP_453827.1
acetate -- CoA ligase [<i>Candida glabrata</i> CBS 138]	83	XP_445089.1
acetate -- CoA ligase [<i>Scheffersomyces stipitis</i> CBS 6054]	68	XP_001385819.1
acetyl-coenzyme A synthetase FacA [<i>Aspergillus fumigatus</i> A1163]	63	EDP50475.1

TABLE 7-continued

BLAST Query - ACS2 from <i>Saccharomyces cerevisiae</i>		
Description	Identity (%)	Accession number
acetate -- CoA ligase <i>facA-Penicillium chrysogenum</i> [<i>Penicillium chrysogenum</i> Wisconsin 54-1255]	62	XP_002564696.1

Alcohol Dehydrogenase

[0347] If the recombinant yeast cell functionally expresses a protein having acetylating acetaldehyde dehydrogenase activity, preferably the recombinant yeast cell is further functionally expressing:

[0348] a nucleic acid sequence encoding a protein having NAD⁺-dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC1.1.1.2); and/or

[0349] a nucleic acid sequence encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1).

[0350] A protein having alcohol dehydrogenase activity is herein also referred to as “alcohol dehydrogenase protein”, “alcohol dehydrogenase enzyme” or simply “alcohol dehydrogenase”. The protein is further abbreviated herein as “ADH”.

[0351] The alcohol dehydrogenase enzyme catalyses the conversion of acetaldehyde into ethanol.

[0352] It is understood that the recombinant yeast cell may naturally comprise an endogenous nucleic acid sequence encoding an alcohol dehydrogenase protein. In the alternative, or in addition thereto, the recombinant yeast cell may comprise a heterologous nucleic acid sequence encoding a protein having alcohol dehydrogenase activity

[0353] For example, the recombinant yeast cell may naturally comprise a gene encoding alcohol dehydrogenase, as is the case with *S. cerevisiae* (Amino acid sequences of the native *S. cerevisiae* alcohol dehydrogenases ADH1, ADH2, ADH3, ADH4 and ADH5 are illustrated respectively as SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 and SEQ ID NO: 55), see Lutstorf and Megnet, “Multiple Forms of Alcohol Dehydrogenase in *Saccharomyces Cerevisiae*”, (1968), Arch. Biochem. Biophys., vol. 126, pages 933-944, incorporated herein by reference, or Ciriacy, “Genetics of Alcohol Dehydrogenase in *Saccharomyces cerevisiae* I. Isolation and genetic analysis of *adh* mutants”, (1975), Mutat. Res. 29, pages 315-326, incorporated herein by reference).

[0354] Preferably, however, the recombinant yeast cell comprises alcohol dehydrogenase activity within a, suitably heterologous, bifunctional enzyme having both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity as described herein above. That is, most preferably the alcohol dehydrogenase protein is a bifunctional protein having both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity. When the recombinant yeast cell comprises a heterologous nucleic acid sequence encoding a bifunctional protein having both acetylating acetaldehyde dehydrogenase activity as well as alcohol dehydrogenase activity, any native nucleic acid sequences encoding for any native protein encoding alcohol dehydrogenase activity may or may not be disrupted and/or deleted.

[0355] The recombinant yeast cell may therefore advantageously be a recombinant yeast cell functionally expressing:

[0356] one or more heterologous nucleic acid sequence(s) encoding a bifunctional protein having NAD⁺-dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10); and NAD⁺-dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC1.1.1.2); and

[0357] one or more, native or heterologous, nucleic acid sequence(s) encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1),

wherein optionally one or more native nucleic acid sequence(s) encoding a protein having NAD⁺-dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC1.1.1.2) are disrupted or deleted.

[0358] Alternatively the recombinant yeast cell may advantageously be a recombinant yeast cell functionally expressing:

[0359] one or more, native or heterologous, nucleic acid sequence(s) encoding a monofunctional protein having NAD⁺-dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10); and

[0360] one or more, native or heterologous, nucleic acid sequence(s) encoding a protein having acetyl-Coenzyme A synthetase activity (EC 6.2.1.1); and

[0361] one or more, native or heterologous, nucleic acid sequences(s) encoding a protein having NAD⁺-dependent alcohol dehydrogenase activity (EC 1.1.1.1 or EC1.1.1.2).

[0362] Preferences for the bifunctional protein are provided above and are as listed for the acetylating acetaldehyde dehydrogenase protein. If the protein is not bifunctional, the NAD⁺-dependent alcohol dehydrogenase protein is preferably a protein having NAD⁺-dependent alcohol dehydrogenase activity that comprises or consists of:

[0363] an amino acid sequence of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55; or

[0364] a functional homologue of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55 having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55; or

[0365] a functional homologue of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55 having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no

more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55.

[0366] Any heterologous nucleic acid sequence (e.g. the gene) encoding for the protein having NAD⁺-dependent alcohol dehydrogenase activity may suitably be incorporated in the genome of the recombinant yeast cell.

Deletion or Disruption of Glycerol 3-Phosphate Phosphohydrolase and/or Glycerol 3-Phosphate Dehydrogenase

[0367] The recombinant yeast cell further may or may not comprise a deletion or disruption of one or more endogenous nucleotide sequence encoding a glycerol 3-phosphate phosphohydrolase gene and/or encoding a glycerol 3-phosphate dehydrogenase gene.

[0368] Preferably enzymatic activity needed for the NADH-dependent glycerol synthesis in the yeast cell is reduced or deleted. The reduction or deletion of the enzymatic activity of glycerol 3-phosphate phosphohydrolase and/or glycerol 3-phosphate dehydrogenase can be achieved by modifying one or more genes encoding a NAD-dependent glycerol 3-phosphate dehydrogenase (GPD) and/or one or more genes encoding a glycerol phosphate phosphatase (GPP), such that the enzyme is expressed considerably less than in the wild-type or such that the gene encodes a polypeptide with reduced activity. Such modifications can be carried out using commonly known biotechnological techniques, and may in particular include one or more knock-out mutations or site-directed mutagenesis of promoter regions or coding regions of the structural genes encoding GPD and/or GPP. Alternatively, yeast strains that are defective in glycerol production may be obtained by random mutagenesis followed by selection of strains with reduced or absent activity of GPD and/or GPP. *S. cerevisiae* GPD1, GPD2, GPP1 and GPP2 genes are shown in WO2011010923, and are disclosed in SEQ ID NO: 24-27 of that application.

[0369] Preferably the recombinant yeast is a recombinant yeast that further comprises a deletion or disruption of a glycerol-3-phosphate dehydrogenase (GPD) gene. The one or more of the glycerol phosphate phosphatase (GPP) genes may or may not be deleted or disrupted.

[0370] More preferably the recombinant yeast is a recombinant yeast that comprises a deletion or disruption of a glycerol-3-phosphate dehydrogenase 1 (GPD1) gene. The glycerol-3-phosphate dehydrogenase 2 (GPD2) gene may or may not be deleted or disrupted.

[0371] Most preferably the recombinant yeast is a recombinant yeast that comprises a deletion or disruption of a glycerol-3-phosphate dehydrogenase 1 (GPD1) gene, whilst the glycerol-3-phosphate dehydrogenase 2 (GPD2) gene remains active and/or intact. Preferably therefore, only one of the *S. cerevisiae* GPD1, GPD2, GPP1 and GPP2 genes is disrupted and deleted, whereas most preferably only GPD1 is chosen from the group consisting of GPD1, GPD2, GPP1 and GPP2 genes to be disrupted or deleted.

[0372] Without wishing to be bound to any kind of theory it is believed that a recombinant yeast according to the invention wherein the GPD1 gene, but not the GPD2 gene, is deleted or disrupted, can be advantageous when applied in a fermentation process where the glucose at the start of or

during the fermentation, is preferably equal to or more than 80 g/L, more preferably equal to or more than 90 g/L, even more preferably equal to or more than 100 g/L, still more preferably equal to or more than 110 g/L, yet even more preferably equal to or more than 120 g/L, equal to or more than 130 g/L, equal to or more than 140 g/L, equal to or more than 150 g/L, equal to or more than 160 g/L, equal to or more than 170 g/L, or equal to or more than 180 g/L.

[0373] Preferably at least one gene encoding a GPD and/or at least one gene encoding a GPP is entirely deleted, or at least a part of the gene is deleted that encodes a part of the enzyme that is essential for its activity. Good results can be achieved with a *S. cerevisiae* cell, wherein the open reading frames of the GPD1 gene and/or of the GPD2 gene have been inactivated. Inactivation of a structural gene (target gene) can be accomplished by a person skilled in the art by synthetically synthesizing or otherwise constructing a DNA fragment consisting of a selectable marker gene flanked by DNA sequences that are identical to sequences that flank the region of the host cell's genome that is to be deleted. Suitably, good results can be obtained with the inactivation of the GPD1 and GPD2 genes in *Saccharomyces cerevisiae* by integration of the marker genes kanMX and hphMX4. Subsequently this DNA fragment is transformed into a host cell. Transformed cells that express the dominant marker gene are checked for correct replacement of the region that was designed to be deleted, for example by a diagnostic polymerase chain reaction or Southern hybridization.

[0374] Thus, in the recombinant yeast cells of the invention, glycerol 3-phosphate phosphohydrolase activity in the cell and/or glycerol 3-phosphate dehydrogenase activity in the cell can be advantageously reduced.

Glucoamylase

[0375] Preferably, the recombinant yeast cell further functionally expresses a nucleic acid sequence encoding for a glucoamylase (EC 3.2.1.20 or 3.2.1.3).

[0376] A protein having glucoamylase activity is herein also referred to as "glucoamylase enzyme", "glucoamylase protein" or simply "glucoamylase". Glucoamylase has herein been abbreviated as "GA".

[0377] Glucoamylase, also referred to as amyloglucosidase, alpha-glucosidase, glucan 1,4-alpha glucosidase, maltase glucoamylase, and maltase-glucoamylase, catalyses at least the hydrolysis of terminal 1,4-linked alpha-D-glucose residues from non-reducing ends of amylose chains to release free D-glucose. A glucoamylase may be further defined by its amino acid sequence. Likewise a glucoamylase may be further defined by a nucleotide sequence encoding the glucoamylase. As explained in detail above under definitions, a certain glucoamylase that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the glucoamylase.

[0378] Preferably the protein having glucoamylase activity comprises or consists of:

[0379] an amino acid sequence of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58; or

[0380] a functional homologue of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at

least 99% sequence identity with the amino acid sequence of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58; or

[0381] a functional homologue of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58 having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 56, SEQ ID NO: 57 or SEQ ID NO: 58.

[0382] The polypeptide of SEQ ID NO: 56 encodes a “mature glucoamylase”, referring to the enzyme in its final form after translation and any post-translational modifications, such as N-terminal processing, C-terminal truncation, glycosylation, phosphorylation, etc.

[0383] In an embodiment the nucleotide sequence encodes a polypeptide having an amino acid sequence of SEQ ID NO: 57 or a variant thereof having an amino acid sequence identity of at least 50%, preferably at least 60%, 70%, 75%, 80%, 85%, 90%, 95, 98%, or 99% with the amino acid sequence of SEQ ID NO: 57. Amino acids 1-17 of the SEQ ID NO: 57 may encode for a native signal sequence.

[0384] In another embodiment the nucleotide sequence allowing the expression of a glucoamylase encodes a polypeptide having an amino acid sequence of SEQ ID NO: 58 or a variant thereof having an amino acid sequence identity of at least 50%, preferably at least 60%, 70%, 75%, 80%, 85%, 90%, 95, 98%, or 99% with the amino acid sequence of SEQ ID NO: 58. Amino acids 1-19 of the SEQ ID NO: 58 may encode for a signal sequence.

[0385] A signal sequence (also referred to as signal peptide, targeting signal, localization signal, localization sequence, transit peptide, leader sequence or leader peptide) can be present at the N-terminus of a polypeptide (here, the glucoamylase) where it signals that the polypeptide is to be excreted, for example outside the cell and into the media.

[0386] The nucleic acid sequence (e.g. the gene) encoding for the protein having glucoamylase activity may suitably be incorporated in the genome of the recombinant yeast cell.

Glycerol Re-Uptake

[0387] The recombinant yeast cell may or may not further comprise one or more additional nucleic acid sequences that are part of a glycerol re-uptake pathway. That is, the recombinant yeast cell may or may not further comprise:

[0388] one or more heterologous nucleic acid sequences encoding for a glycerol dehydrogenase; and/or

[0389] one or more homologous or heterologous nucleic acid sequences encoding for a dihydroxyacetone kinase; and/or

[0390] one or more heterologous nucleic acid sequences encoding for a glycerol transporter.

[0391] Thus, in one preferred embodiment the recombinant yeast cell is a recombinant yeast cell functionally expressing:

[0392] one or more heterologous nucleic acid sequences encoding for a ribulose-1,5-phosphate carboxylase/

oxygenase (EC4.1.1.39; Rubisco), and optionally one or more nucleic acid sequences encoding for molecular chaperones for Rubisco;

[0393] one or more heterologous nucleic acid sequences encoding for phosphoribulokinase (EC2.7.1.19; PRK);

[0394] one or more nucleic acid sequences encoding for a transketolase (EC 2.2.1.1), wherein the transketolase is under control of a promoter (the “TKL promoter”) which has a TKL expression ratio *anaerobic/aerobic* of 2 or more;

[0395] one or more heterologous nucleic acid sequences encoding for a glycerol dehydrogenase;

[0396] one or more homologous or heterologous nucleic acid sequences encoding for a dihydroxyacetone kinase; and

[0397] optionally one or more heterologous nucleic acid sequences encoding for a glycerol transporter.

[0398] Without wishing to be bound by any kind of theory it is believed that a recombinant yeast cell that further comprises a combination of glycerol dehydrogenase, dihydroxyacetone kinase and optionally a glycerol transporter has an improved overall performance in the form of higher ethanol yields.

[0399] In an alternative preferred embodiment the recombinant yeast cell is a recombinant yeast cell that does not functionally express:

[0400] one or more heterologous nucleic acid sequences encoding for a glycerol dehydrogenase; and/or

[0401] one or more heterologous nucleic acid sequences encoding for a dihydroxyacetone kinase; and/or

[0402] one or more heterologous nucleic acid sequences encoding for a glycerol transporter.

[0403] Without wishing to be bound by any kind of theory it is believed that in the absence of one or more of these features of such a glycerol re-uptake pathway, a recombinant yeast cell is obtained that has a very low accumulation of glucose and/or other sugars and has an improved robustness when applied in a medium comprising a high amount of sugars. The application of a recombinant yeast cell that does not comprise one or more of a, heterologous and/or homologous, glycerol dehydrogenase; heterologous and/or homologous dihydroxyacetone kinase and/or heterologous and/or homologous glycerol transporter can therefore be advantageous when applied in a fermentation process where the glucose at the start of or during the fermentation, is preferably equal to or more than 80 g/L, more preferably equal to or more than 90 g/L, even more preferably equal to or more than 100 g/L, still more preferably equal to or more than 110 g/L, yet even more preferably equal to or more than 120 g/L, equal to or more than 130 g/L, equal to or more than 140 g/L, equal to or more than 150 g/L, equal to or more than 160 g/L, equal to or more than 170 g/L, or equal to or more than 180 g/L.

[0404] Most preferably, the recombinant yeast is therefore a recombinant yeast that is functionally expressing:

[0405] one or more heterologous nucleic acid sequences encoding for a ribulose-1,5-phosphate carboxylase/oxygenase (EC4.1.1.39; Rubisco), and optionally one or more nucleic acid sequences encoding for molecular chaperones for Rubisco;

[0406] one or more heterologous nucleic acid sequences encoding for phosphoribulokinase (EC2.7.1.19; PRK);

[0407] one or more nucleic acid sequences encoding for a transketolase (EC 2.2.1.1), wherein the transketolase

is under control of a promoter (the “TKL promoter”) which has a TKL expression ratio *anaerobic/aerobic* of 2 or more; wherein the recombinant yeast cell does not functionally express

- [0408] one or more heterologous nucleic acid sequences encoding for a glycerol dehydrogenase; and/or
- [0409] one or more heterologous nucleic acid sequences encoding for a dihydroxyacetone kinase; and/or
- [0410] one or more heterologous nucleic acid sequences encoding for a glycerol transporter.

Glycerol Dehydrogenase

[0411] As indicated above, the recombinant yeast cell may or may not functionally express

- [0412] a nucleic acid sequence encoding for a protein having glycerol dehydrogenase activity (E.C. 1.1.1.6);
- [0413] a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity (E.C. 2.7.1.28 or E.C. 2.7.1.29); and
- [0414] optionally a nucleic acid sequence encoding a protein having glycerol transporter activity.

[0415] Thus the recombinant yeast cell may or may not functionally express one or more, preferably heterologous, nucleic acid sequences encoding for a glycerol dehydrogenase.

[0416] If a glycerol dehydrogenase is present, the recombinant yeast cell may comprise a NAD⁺ linked glycerol dehydrogenase (EC 1.1.1.6) and/or a NADP⁺ linked glycerol dehydrogenase (EC 1.1.1.72). That is, the recombinant yeast cell may or may not comprise a nucleic acid sequence encoding a protein having NAD⁺ dependent glycerol dehydrogenase activity (EC 1.1.1.6) and/or a nucleic acid sequence encoding a protein having NADP⁺ dependent glycerol dehydrogenase activity (EC 1.1.1.72).

[0417] In one embodiment the protein having glycerol dehydrogenase activity is preferably a protein having NAD⁺ dependent glycerol dehydrogenase activity (EC 1.1.1.6) and preferably the recombinant yeast cell functionally expresses a nucleic acid sequence encoding a protein having NAD⁺ dependent glycerol dehydrogenase activity (EC 1.1.1.6). Such protein may be from bacterial origin or for instance from fungal origin. An example is *gldA* from *E. coli*.

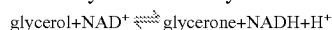
[0418] In an alternative or additional embodiment, a NADP⁺ dependent glycerol dehydrogenase can be present (EC 1.1.1.72).

[0419] If a glycerol dehydrogenase is present, a NAD⁺ linked glycerol dehydrogenase is preferred.

[0420] A protein having glycerol dehydrogenase activity is herein also referred to as “glycerol dehydrogenase protein”, “glycerol dehydrogenase enzyme” or simply as “glycerol dehydrogenase”. In analogy thereto a protein having NAD⁺ dependent glycerol dehydrogenase activity is herein also referred to as “NAD⁺ dependent glycerol dehydrogenase protein”, “NAD⁺ dependent glycerol dehydrogenase enzyme” or simply as “NAD⁺ dependent glycerol dehydrogenase”. The glycerol dehydrogenase is abbreviated as GLD.

[0421] Preferences for a glycerol dehydrogenase and the nucleic sequences encoding for such are as described in WO2015028582, incorporated herein by reference.

[0422] NAD⁺ dependent glycerol dehydrogenase (EC 1.1.1.6) is an enzyme that catalyzes the chemical reaction:



[0423] Thus, the two substrates of this enzyme are glycerol and NAD⁺, whereas its three products are glycerone, NADH, and H⁺. Glyceron and dihydroxyacetone are herein synonyms.

[0424] The glycerol dehydrogenase enzyme belongs to the family of oxidoreductases, specifically those acting on the CH—OH group of donor with NAD⁺ or NADP⁺ as acceptor. The systematic name of this enzyme class is glycerol:NAD⁺ 2-oxidoreductase. Other names in common use include glycerin dehydrogenase, and NAD⁺-linked glycerol dehydrogenase. This enzyme participates in glycerolipid metabolism. A glycerol dehydrogenase protein may be further defined by its amino acid sequence. Likewise a glycerol dehydrogenase protein may be further defined by a nucleotide sequence encoding the glycerol dehydrogenase protein. As explained in detail above under definitions, a certain glycerol dehydrogenase protein that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the glycerol dehydrogenase protein.

[0425] The nucleic acid sequence encoding the protein having glycerol dehydrogenase activity can be a heterologous nucleic acid sequence. The protein having glycerol dehydrogenase activity can be a heterologous protein having NAD⁺ dependent glycerol dehydrogenase activity.

[0426] If the recombinant yeast cell comprises one or more heterologous nucleic acid sequences encoding for a glycerol dehydrogenase, the recombinant yeast cell preferably further comprises suitable co-factors to enhance the activity of the glycerol dehydrogenase. For example, the recombinant yeast cell may comprise zinc, zinc ions or zinc salts and/or one or more pathways to include such in the cell.

[0427] Suitable examples of heterologous proteins having glycerol dehydrogenase activity include the glycerol dehydrogenase proteins of respectively *Klebsiella pneumoniae*, *Enterococcus aerogenes*, *Yersinia aldovae*, and *Escherichia coli*. Their amino acid sequences of such proteins have been illustrated respectively by SEQ ID NO: 59, SEQ ID NO: 60, SEQ ID NO: 61 and SEQ ID NO: 62.

[0428] The recombinant yeast cell therefore may or may not include one or more, suitably heterologous, glycerol dehydrogenase proteins having an amino acid sequence of SEQ ID NO: 59, SEQ ID NO: 60, SEQ ID NO: 61 and/or SEQ ID NO: 62; and/or functional homologues thereof comprising an amino acid sequence having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 59, SEQ ID NO: 60, SEQ ID NO: 61 and/or SEQ ID NO: 62; and/or functional homologues thereof comprising an amino acid sequence having one or more mutations, substitutions, insertions and/or deletions as compared to the amino acid sequence of SEQ ID NO: 59, SEQ ID NO: 60, SEQ ID NO: 61 and/or SEQ ID NO: 62, wherein more preferably the amino acid sequence of such functional homologues has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or

deletions as compared to the amino acid sequence of SEQ ID NO: 59, SEQ ID NO: 60, SEQ ID NO: 61 and/or SEQ ID NO: 62.

[0429] A preferred glycerol dehydrogenase protein is the glycerol dehydrogenase protein encoded by the *gldA* gene from *E. coli*. SEQ ID NO: 62 shows the amino acid sequence of this preferred NAD⁺ dependent glycerol dehydrogenase protein, encoded by the *gldA* gene from *E. coli*. The nucleic acid sequence of the *gldA* gene of *E. coli* is illustrated by SEQ ID NO: 63.

[0430] If the recombinant yeast cell comprises one or more heterologous nucleic acid sequences encoding for a glycerol dehydrogenase, the recombinant yeast cell therefore most preferably comprises a heterologous nucleotide sequence encoding a protein having NAD⁺ dependent glycerol dehydrogenase activity (E.C. 1.1.1.6) derived from *E. coli*, optionally codon-optimized for the host cell, as exemplified by the nucleic acid sequence shown in SEQ ID NO:63.

[0431] Preferable the nucleic acid sequence encoding the protein having glycerol dehydrogenase activity thus comprises or consists of:

[0432] a nucleic acid sequence of SEQ ID NO:63; or

[0433] a functional homologue of SEQ ID NO:63, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least

95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO:63; or
[0434] a functional homologue of SEQ ID NO:63, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO:63, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO:63.

[0435] If the recombinant yeast cell comprises one or more heterologous nucleic acid sequences encoding for a glycerol dehydrogenase, the recombinant yeast cell therefore most preferably comprises one or more nucleotide sequence encoding a glycerol dehydrogenase (E.C. 1.1.1.6) derived from *E. coli*, optionally codon-optimized for the host cell. Such heterologous nucleic acid sequence (e.g. the gene) encoding for the glycerol dehydrogenase protein may suitably be incorporated in the genome of the recombinant yeast cell, for example as described in the examples of WO2015/028583, herein incorporated by reference.

[0436] Further examples of suitable glycerol dehydrogenases are listed in Table 8(a) to 8(d). At the top of each table the *gldA* that is BLASTED is mentioned.

TABLE 8(a)

BLAST Query - <i>gldA</i> from <i>Escherichia coli</i>		
Description	Identity (%)	Accession number
glycerol dehydrogenase, NAD [<i>Escherichia coli</i> str. K-12 substr. MG1655]	100	NP_418380.4
glycerol dehydrogenase [<i>Escherichia coli</i> 0127:H6 str. E2348/69]	99	YP_002331714.1
glycerol dehydrogenase [<i>Citrobacter youngae</i>]	94	WP_006686227.1
glycerol dehydrogenase [<i>Citrobacter freundii</i>]	92	WP_003840533.1

TABLE 8(b)

BLAST Query - <i>gldA</i> from <i>Klebsiella pneumoniae</i>		
Description	Identity (%)	Accession number
glycerol dehydrogenase [<i>Klebsiella pneumoniae</i> 342]	100	YP_002236495.1
glycerol dehydrogenase [<i>Citrobacter freundii</i>]	93	WP_003024745.1
Glycerol dehydrogenase (EC 1.1.1.6) [<i>Enterobacter aerogenes</i> EA1509E]	92	YP_004590977.1
glycerol dehydrogenase [<i>Escherichia coli</i>]	91	WP_016241524.1
glycerol dehydrogenase [<i>Enterococcus aerogenes</i>]	87	See examples herein strains with CAS15
glycerol dehydrogenase [<i>Yersinia aldovae</i>]	74	WP_004701845.1
glycerol dehydrogenase [<i>Enterobacteriaceae bacterium</i> LSJC7]	61	WP_017375113.1
glycerol dehydrogenase [<i>Citrobacter youngae</i>]	60	WP_006686227.1

TABLE 8(c)

BLAST Query - <i>gldA</i> from <i>Enterococcus aerogenes</i>		
Description	Identity (%)	Accession number
glycerol dehydrogenase [<i>Enterobacter aerogenes</i> KCTC 2190]	100	YP_004591726.1

TABLE 8(c)-continued

BLAST Query - gldA from <i>Enterococcus aerogenes</i>		
Description	Identity (%)	Accession number
Glycerol dehydrogenase (EC 1.1.1.6) [<i>Enterobacter aerogenes</i> EA1509E]	99	YP_007390021.1
glycerol dehydrogenase [<i>Klebsiella pneumoniae</i>]	92	WP_004203683.1
glycerol dehydrogenase [<i>Escherichia coli</i>]	88	WP_001322519.1
		See examples herein strains with CAS13
glycerol dehydrogenase [<i>Enterobacter cloacae</i> subsp. <i>cloacae</i> ATCC 13047]	87	YP_003615506.1

TABLE 8(d)

BLAST Query - gldA from <i>Yersinia aldovae</i>		
Description	Identity (%)	Accession number
glycerol dehydrogenase [<i>Yersinia aldovae</i>]	100	WP_004701845.1
glycerol dehydrogenase [<i>Yersinia intermedia</i>]	95	WP_005189747.1
glycerol dehydrogenase [<i>Serratia liquefaciens</i> ATCC 27592]	81	YP_008232202.1
glycerol dehydrogenase [<i>Escherichia coli</i>]	76	WP_016241524.1
		See examples herein strains with CAS13.
hypothetical protein EAE_03845 [<i>Enterobacter aerogenes</i> KCTC 2190]	75	YP_004590977.1
glycerol dehydrogenase [<i>Aeromonas hydrophila</i>]	65	WP_017410769.1

Dihydroxyacetone Kinase

[0437] As indicated above, the recombinant yeast cell may or may not functionally express

[0438] a nucleic acid sequence encoding for a protein having glycerol dehydrogenase activity (E.C. 1.1.1.6);

[0439] a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity (E.C. 2.7.1.28 or E.C. 2.7.1.29); and

[0440] optionally a nucleic acid sequence encoding a protein having glycerol transporter activity.

[0441] That is, the recombinant yeast cell may or may not functionally express one or more, homologous or heterologous, nucleic acid sequences encoding for dihydroxyacetone kinase (E.C. 2.7.1.28 or E.C. 2.7.1.29),

[0442] A protein having dihydroxyacetone kinase activity is herein also referred to as “dihydroxyacetone kinase protein”, “dihydroxyacetone kinase enzyme” or simply as “dihydroxyacetone kinase”. The dihydroxyacetone kinase is abbreviated herein as DAK.

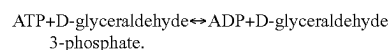
[0443] Preferences for a dihydroxyacetone kinase and the nucleic sequences encoding for such are as described in WO2015028582, incorporated herein by reference.

[0444] The protein having dihydroxy kinase activity may suitably belong to the enzyme categories of E.C. 2.7.1.28 and/or E.C. 2.7.1.29. The recombinant yeast cell thus suitably functionally expresses a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity (E.C. 2.7.1.28 and/or E.C. 2.7.1.29).

[0445] A dihydroxyacetone kinase is preferably herein understood as an enzyme that catalyzes the chemical reaction (EC 2.7.1.29):



and/or the chemical reaction (EC 2.7.1.28):



[0446] Other names in common use for a dihydroxyacetone kinase include glycerone kinase, ATP:glycerone phosphotransferase and (phosphorylating) acetol kinase. It is further understood that glycerone and dihydroxyacetone are the same molecule. A dihydroxyacetone kinase protein may be further defined by its amino acid sequence. Likewise a dihydroxyacetone kinase protein may be further defined by a nucleotide sequence encoding the dihydroxyacetone kinase protein. As explained in detail above under definitions, a certain dihydroxyacetone kinase protein that is defined by a nucleotide sequence encoding the enzyme, includes (unless otherwise limited) the nucleotide sequence hybridising to such nucleotide sequence encoding the dihydroxyacetone kinase protein.

[0447] If present, the recombinant yeast cell preferably functionally expresses a nucleic acid sequence encoding a native protein having dihydroxyacetone kinase activity. More preferably, the nucleic acid sequence encoding the protein having dihydroxyacetone kinase activity is a native nucleic acid sequence.

[0448] Yeast comprises two native isozymes of dihydroxyacetone kinase (DAK1 and DAK2). These native dihydroxyacetone kinase enzymes are preferred according to the invention. Preferably the host cell is a *Saccharomyces cerevisiae* cell and preferably the above native dihydroxyacetone kinase enzymes are the native dihydroxyacetone kinase enzymes of a *Saccharomyces cerevisiae* yeast cell. The amino acid sequences of the native dihydroxyacetone kinase proteins of *Saccharomyces cerevisiae*, DAK1 and DAK2, have been illustrated respectively by SEQ ID NO: 64

and SEQ ID NO: 65. The nucleic acid sequences coding for these native dihydroxyacetone kinase proteins DAK1 and DAK2 have been illustrated respectively by SEQ ID NO: 69 and SEQ ID NO: 70.

[0449] It is also possible for the recombinant yeast cell to functionally express a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity, where the nucleic acid sequence is a heterologous nucleic acid sequence, respectively wherein the protein is a heterologous protein. In an embodiment the recombinant yeast cell comprises a heterologous gene encoding a dihydroxyacetone kinase. Suitable heterologous genes include the genes encoding dihydroxyacetone kinases from *Saccharomyces kudriavzevii*, *Zygosaccharomyces bailii*, *Kluyveromyces lactis*, *Candida glabrata*, *Yarrowia lipolytica*, *Klebsiella pneumoniae*, *Enterobacter aerogenes*, *Escherichia coli*, *Yarrowia lipolytica*, *Schizosaccharomyces pombe*, *Botryotinia fuckeliana*, and *Exophiala dermatitidis*. Preferred heterologous proteins having dihydroxyacetone kinase activity include those derived from respectively *Klebsiella pneumoniae*, *Yarrowia lipolytica* and *Schizosaccharomyces pombe*, as illustrated respectively by SEQ ID NO: 66, SEQ ID NO: 67 and SEQ ID NO: 68.

[0450] The recombinant yeast cell may or may not comprise a genetic modification that causes overexpression of a dihydroxyacetone kinase, for example by overexpression of a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity. The nucleotide sequence encoding the dihydroxyacetone kinase may be native or heterologous to the cell. Nucleic acid sequences that may be used for overexpression of dihydroxyacetone kinase in the cells of the invention are for example the dihydroxyacetone kinase genes from *S. cerevisiae* (DAK1) and (DAK2) as e.g. described by Molin et al., “Dihydroxy-acetone kinases in *Saccharomyces cerevisiae* are involved in detoxification of dihydroxyacetone” (2003), J. Biol. Chem., vol. 278: pages 1415-1423, incorporated herein by reference. In a preferred embodiment a codon-optimised (see above) nucleotide sequence encoding the dihydroxyacetone kinase is overexpressed, such as e.g. a codon optimised nucleotide sequence encoding the dihydroxyacetone kinase of SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 66, SEQ ID NO: 67 or SEQ ID NO: 68.

[0451] As indicated above, the native nucleic acid sequences encoding dihydroxyacetone kinase proteins in *Saccharomyces cerevisiae*, DAK1 and DAK2, have been illustrated respectively by SEQ ID NO: 69 and SEQ ID NO: 70.

[0452] Preferably the recombinant yeast cell does comprise a genetic modification that increases the specific activity of any dihydroxyacetone kinase in the cell. For example, the recombinant yeast cell may comprise one or more native and/or heterologous nucleic acid sequence encoding one or more native and/or heterologous dihydroxyacetone kinase protein(s), such as DAK1 and/or DAK2, that is/are overexpressed. A native dihydroxyacetone kinase, such as DAK1 and/or DAK2, may for example be overexpressed via one or more genetic modifications resulting in more copies of the gene encoding for the dihydroxy acetone kinase than present in the non-genetically modified cell, and/or a non-native promoter may be applied.

[0453] Preferably the recombinant yeast cell is a recombinant yeast cell, wherein the expression of the nucleic acid sequence encoding the protein having dihydroxyacetone

kinase activity is under control of a promoter. The promoter can for example be a promoter that is native to another gene in the host cell.

[0454] For overexpression of the nucleotide sequence encoding the dihydroxyacetone kinase, the nucleotide sequence (to be overexpressed) can be placed in an expression construct wherein it is operably linked to suitable expression regulatory regions/sequences to ensure overexpression of the dihydroxyacetone kinase enzyme upon transformation of the expression construct into the host cell of the invention (see above). Suitable promoters for (over)expression of the nucleotide sequence coding for the enzyme having dihydroxyacetone kinase activity include promoters that are preferably insensitive to catabolite (glucose) repression, that are active under anaerobic conditions and/or that preferably do not require xylose or arabinose for induction. Examples of such promoters are given above. A dihydroxyacetone kinase that is overexpressed, is preferably overexpressed by at least a factor 1.1, 1.2, 1.5, 2, 5, 10 or 20 as compared to a strain which is genetically identical except for the genetic modification causing the overexpression. Preferably, the dihydroxyacetone kinase is overexpressed under anaerobic conditions by at least a factor 1.1, 1.2, 1.5, 2, 5, 10 or 20 as compared to a strain which is genetically identical except for the genetic modification causing the overexpression. It is to be understood that these levels of overexpression may apply to the steady state level of the enzyme's activity (specific activity in the cell), the steady state level of the enzyme's protein as well as to the steady state level of the transcript coding for the enzyme in the cell. Overexpression of the nucleotide sequence in the host cell produces a specific dihydroxyacetone kinase activity of at least 0.002, 0.005, 0.01, 0.02 or 0.05 U min⁻¹ (mg protein)⁻¹, determined in cell extracts of the transformed host cells at 30° C. as described e.g. in the Examples of WO2013/081456.

[0455] A most preferred dihydroxyacetone kinase protein is the dihydroxyacetone kinase protein encoded by the *Dak1* gene from *Saccharomyces cerevisiae*. SEQ ID NO: 64 shows the amino acid sequence of a suitable dihydroxyacetone kinase protein, encoded by the *Dak1* gene from *Saccharomyces cerevisiae*. SEQ ID NO: 69 illustrates the nucleic acid sequence of the *Dak1* gene itself.

[0456] If the recombinant yeast cell comprises one or more overexpressed nucleic acid sequences encoding for a dihydroxyacetone kinase, the recombinant yeast cell therefore most preferably comprises one or more overexpressed nucleotide sequence encoding a dihydroxyacetone kinase derived from *Saccharomyces cerevisiae*, as exemplified by the nucleic acid sequence shown in SEQ ID NO: 69.

[0457] Preferably the protein having dihydroxy acetone kinase activity thus comprises or consists of:

[0458] an amino acid sequence of SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 66, SEQ ID NO: 67 or SEQ ID NO: 68; or

[0459] a functional homologue of SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 66, SEQ ID NO: 67 or SEQ ID NO: 68, having at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the amino acid sequence of SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 66, SEQ ID NO: 67 or SEQ ID NO: 68; or

[0460] a functional homologue of SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 66, SEQ ID NO: 67 or SEQ ID NO: 68, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 66, SEQ ID NO: 67 or SEQ ID NO: 68, more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 amino acid mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 64, SEQ ID NO: 65, SEQ ID NO: 66, SEQ ID NO: 67 or SEQ ID NO: 68.

The protein having an amino acid sequence of SEQ ID NO: 64 and functional homologues thereof are most preferred.

[0461] Preferable the nucleic acid sequence encoding the protein having dihydroxy acetone kinase activity comprises or consists of:

[0462] a nucleic acid sequence of SEQ ID NO: 69 or SEQ ID NO: 70; or

[0463] a functional homologue of SEQ ID NO: 69 or SEQ ID NO: 70, having at least 40%, at least 45%, at

least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90% at least 95%, at least 98% or at least 99% sequence identity with the nucleic acid sequence of SEQ ID NO: 69 or SEQ ID NO: 70; or

[0464] a functional homologue of SEQ ID NO: 69 or SEQ ID NO: 70, having one or more mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 69 or SEQ ID NO: 70; more preferably a functional homologue that has no more than 300, no more than 250, no more than 200, no more than 150, no more than 100, no more than 75, no more than 50, no more than 40, no more than 30, no more than 20, no more than 10 or no more than 5 nucleic acid mutations, substitutions, insertions and/or deletions when compared with the nucleic acid sequence of SEQ ID NO: 69 or SEQ ID NO: 70.

[0465] The nucleic acid sequence (e.g. the gene) encoding for the dihydroxy acetone kinase protein may suitably be incorporated in the genome of the recombinant yeast cell.

[0466] Examples of suitable dihydroxyacetone kinases are listed in Table 9(a) to 9(d). At the top of each table the DAK's used in the examples and that is BLASTED is mentioned.

TABLE 9(a)

BLAST Query - DAK1 from <i>Saccharomyces cerevisiae</i>		
Description	Identity (%)	Accession number
Dak1p [<i>Saccharomyces cerevisiae</i> S288c]	100	NP_013641.1
dihydroxyacetone kinase [<i>Saccharomyces cerevisiae</i> YJM789]	99	EDN64325.1
DAK1-like protein [<i>Saccharomyces kudriavzevii</i> IFO 1802]	95	EJT44075.1
ZYBAOS11-03576g1_1 [<i>Zygosaccharomyces bailii</i> CLIB 213]	77	CDF91470.1
hypothetical protein [<i>Kluyveromyces lactis</i> NRRL Y-1140]	70	XP_451751.1
hypothetical protein [<i>Candida glabrata</i> CBS 138]	63	XP_449263.1
Dak2p [<i>Saccharomyces cerevisiae</i> S288c]	44	NP_116602.1
DAK1 [<i>Yarrowia lipolytica</i>]	41	See examples herein strains with CAS23

TABLE 9(b)

BLAST Query - dhaK from <i>Klebsiella pneumoniae</i>		
Description	Identity (%)	Accession number
dihydroxyacetone kinase subunit DhaK [<i>Klebsiella pneumoniae</i> 342]	100	YP_002236493.1
dihydroxyacetone kinase subunit K [<i>Klebsiella pneumoniae</i>]	99	WP_004149886.1
dihydroxyacetone kinase subunit K [<i>Enterobacter aerogenes</i>]	96	WP_020077889.1
dihydroxyacetone kinase subunit DhaK [<i>Escherichia coli</i> IA139]	88	YP_002407536.1
dihydroxyacetone kinase, DhaK subunit [<i>Escherichia coli</i>]	87	WP_001398949.1

TABLE 9(c)

BLAST Query - DAK1 from <i>Yarrowia lipolytica</i>		
Description	Identity (%)	Accession number
YALIOF09273p [<i>Yarrowia lipolytica</i>]	100	XP_505199.1
dihydroxyacetone kinase [<i>Schizosaccharomyces pombe</i>]	46	AAC83220.1
dihydroxyacetone kinase Dak1 [<i>Schizosaccharomyces pombe</i> 972h-]	45	NP_593241.1
dihydroxyacetone kinase [<i>Saccharomyces cerevisiae</i> RM11-1a]	44	EDV12567.1
Dak2p [<i>Saccharomyces cerevisiae</i> JAY291]	44	EEU04233.1
BN860_19306g1_1 [<i>Zygosaccharomyces bailii</i> CLIB 213]	44	CDF87998.1
Dak1p [<i>Saccharomyces cerevisiae</i> CEN.PK113-7D]	42	EIW08612.1
		See examples herein strains with CAS21

TABLE 9(d)

BLAST Query - DAK1 from <i>Schizosaccharomyces pombe</i>		
Description	Identity (%)	Accession number
dihydroxyacetone kinase Dak1 [<i>Schizosaccharomyces pombe</i> 972h-]	100	NP 593241.1
putative dihydroxyacetone kinase protein [<i>Botryotinia fuckeliana</i> BcDW1]	48	EMR88164.1
Dihydroxyacetone kinase 1 [<i>Fusarium oxysporum</i> f. sp. <i>cubense</i> race 1]	48	ENH64704.1
Dak1p [<i>Saccharomyces cerevisiae</i> CEN.PK113-7D]	46	EIW08612.1
Dak2p [<i>Saccharomyces cerevisiae</i> JAY291]	44	EEU04233.1
dihydroxyacetone kinase [<i>Exophiala dermatitidis</i> NIH/UT8656]	42	EHY55064.1

Glycerol Transporter

[0467] The recombinant yeast cell can optionally, i.e. may or may not, comprise a nucleotide sequence encoding a glycerol transporter. Such a glycerol transporter can allow any glycerol that is externally available in the medium (e.g. from the backset in corn mash) or secreted after internal cellular synthesis to be transported into the cell and converted to ethanol.

[0468] If a glycerol transporter is present, the recombinant yeast preferably comprises one or more nucleic acid sequences encoding a heterologous glycerol transporter represented by amino acid sequence SEQ ID NO: 71, SEQ ID NO: 72 or a functional homologue thereof having an amino acid sequence identity of at least 50%, preferably at least 60%, 70%, 75%, 80%, 85%, 90%, 95%, 98%, or 99% with the amino acid sequence of SEQ ID NO: 71 and/or SEQ ID NO: 72.

[0469] In an embodiment the recombinant yeast can further comprise a deletion or disruption of one or more endogenous nucleotide sequences encoding a glycerol exporter (e.g FPS1).

Nitrate Reductase

[0470] The recombinant yeast cell may also advantageously comprise, respectively functionally express, a nucleic acid sequences encoding an enzyme having NADH-dependent nitrate reductase activity and/or a nucleic acid sequences encoding an enzyme having NADH-dependent nitrite reductase activity. Details for the expression of such

an alternative redox sink have been described in non-published US patent application U.S. 63/087,642 filed with the United States Patent Office on 5 Oct. 2020, the contents of which are herewith incorporated by reference.

[0471] Nitrate reductase (NR) catalyzes the reduction of nitrate (NO₃⁻) to nitrite (NO₂⁻). Nitrite reductase catalyzes the reduction of nitrite to ammonia (NH₃). Nitrate reductase and/or nitrite reductase can be part of a so-called nitrogen assimilation pathway in certain cells. Cells comprising nitrate reductase activity and/or nitrite reductase activity include certain plant cells and bacterial cells and a few yeast cells. As indicated by Linder, the ability to assimilate inorganic nitrogen sources other than ammonia is thought to be rare among budding yeasts. Among the few fungi that are naturally capable to assimilate nitrate or nitrite are *Blastobotrys adeninivorans* (family Trichomonascaceae) *Candida boidinii* (family Pichiaceae), *Cyberlindnera jadinii* (family Phaffomycetaceae), and *Ogataea polymorpha* (family Pichiaceae).

[0472] Preferably the recombinant yeast cell as described herein comprises at least one or more genes encoding a NADH-dependent nitrate reductase.

[0473] By a NADH-dependent nitrate reductase is herein understood a nitrate reductase that is exclusively depended on NADH as a co-factor or that is predominantly dependent on NADH as a cofactor. Preferably the NADH-dependent nitrate reductase has a ratio of catalytic efficiency for NADPH/NADP+ as a cofactor (k_{cat}/K_m)^{NADP+} to NADH/NAD+ as cofactor (k_{cat}/K_m)^{NAD+} i.e. a catalytic efficiency ratio (k_{cat}/K_m)^{NADP+}:(k_{cat}/K_m)^{NAD+}, of more than 1:1, more

$$\text{nitrate} + \text{NADH} + \text{H}^+ \rightarrow \text{nitrite} + \text{NAD}^+ + \text{H}_2\text{O}$$

sequence of one or more of such aforementioned NADH-dependent nitrate reductases, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned NADH-dependent nitrate reductases.

[0477] Preferred NADH-dependent nitrate reductases hence include: NADH-dependent nitrate reductases comprising a polypeptide having an amino acid sequence of SEQ ID NO:89 and/or SEQ ID NO:90, as described herein; and/or functional homologues of SEQ ID NO:89 and/or SEQ ID NO:90 comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of SEQ ID NO:89 and/or SEQ ID NO:90 respectively; and/or functional homologues of SEQ ID NO:89 and/or SEQ ID NO:90 comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of SEQ ID NO:89 and/or SEQ ID NO:90 respectively. Preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:89 and/or SEQ ID NO:90 respectively.

[0478] Preferably the recombinant yeast cell comprises an exogenous gene coding for an enzyme with NADH-dependent nitrate reductase activity. More preferably the recombinant yeast cell comprises an exogenous gene coding for an enzyme with NADH-dependent nitrate reductase activity selected from the group consisting of NADH-dependent nitrate reductases as obtained or derived from *Agrostemma githago*, *Amaranthus hybridus*, *Amaranthus tricolor*, *Ankistrodesmus braunii*, *Arabidopsis thaliana*, *Aspergillus niger*, *Aspergillus nidulans*, *Auxenochlorella pyrenoidosa*, *Bradyrhizobium* sp., *Bradyrhizobium* sp. 750, *Brassica juncea*, *Brassica*, *oleracea*, *Camellia sinensis*, *Candida boidinii*, *Candida utilis*, *Capsicum frutescens*, *Chenopodium album*, *Cyberlindnera jadinii*, *Brassica juncea*, *Brassica oleracea*, *Camellia sinensis*, *Capsicum frutescens*, *Chenopodium album*, *Chlamydomonas reinhardtii*, *Chlorella fusca*, *Chlorella* sp. *Chlorella* sp. Berlin, *Chlorella vulgaris*, *Conticri-bra weissflogii*, *Cucumis sativus*, *Cucurbita maxima*, *Cucurbita pepo*, *Cucurbita* sp., *Dunaliella tertiolecta*, *Emiliania huxleyi*, *Emericella nidulans*, *Fusarium oxysporum*, *Fusarium oxysporum* JCM 11502, *Glyceria maxima*, *Glycine max*, *Gossypium hirsutum*, *Gracilaria chilensis*, *Gracilaria tenuistipitata*, *Helianthus annuus*, *Hordeum vulgare*, *Lactuca sativa*, *Lemna minor*, *Lupinus albus*, *Mycobacterium tuberculosis*, *Nicotiana plumbaginifolia*, *Nicotiana tabacum*, *Ogataea angusta*, *Ogataea polymorpha*.

Oryza sativa, *Phaeocystis Antarctica*, *Phragmites australis*, *Physcomitrella patens*, *Pisum arvense*, *Polytrichum commune*, *Pyropia yezoensis*, *Raphanus sativus*, *Rhodobacter capsulatus*, *Rhodobacter capsulatus* E1F1, *Ricinus communis*, *Selaginella kraussiana*, *Sinapis alba*, *Skeletonema costatum*, *Skeletonema tropicum*, *Solanum lycopersicum*, *Spinacia oleracea*, *Suaeda maritima*, *Tetraselmis gracilis*, *Thalassia Testudinum*, *Thalassiosira Antarctica*, *Thalassiosira pseudonana*, *Triticum aestivum*, *Triticum turgidum* subsp durum, *Ulva* sp. and *Zea mays*, and functional homologues of such NADH-dependent nitrate reductases comprising an amino acid sequence with at least 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of such aforementioned NADH-dependent nitrate reductases; and functional homologues of such NADH-dependent nitrate reductases comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned NADH-dependent nitrate reductases, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned NADH-dependent nitrate reductases.

[0479] Suitably the recombinant yeast cell may comprise a nucleotide sequence coding for an amino acid sequence of any of SEQ ID NO:89 and/or SEQ ID NO:90 or an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of any of SEQ ID NO:89 and/or SEQ ID NO:90. Preferably the amino acid sequence has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:89 and/or SEQ ID NO:90 respectively.

[0480] The recombinant yeast cell may combine one or more genes encoding the above NADH-dependent nitrate reductase with one or more genes encoding an NADPH-dependent nitrite reductase. Preferably, however, the recombinant yeast cell combines one or more genes encoding the above NADH-dependent nitrate reductase with one or more genes encoding a NADH-dependent nitrite reductase.

[0481] Examples of suitable NADH-dependent nitrate reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020)), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:89, are listed in Table 10 below.

TABLE 10

Examples of suitable NADH-dependent nitrate reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020)), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 89, are listed in Table 10 below.			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 89 [%]
K3YG56	Cluster: Nitrate reductase	PACMAD clade	94.06
P16081	Cluster: Nitrate reductase [NADH] 1	cellular organisms	90.54
UPI0004DEBCF9	Cluster: nitrate reductase [NADH]-like	<i>Zea mays</i>	88.57
A0A6G1CIJ1	Cluster: Uncharacterized protein	<i>Oryza meyeriana</i> var. <i>granulata</i>	85.7
P27967	Cluster: Nitrate reductase [NADH]	Pooideae	85.37
A0A3B6T8X1	Cluster: Nitrate reductase	Triticinae	86.11
A0A4S8IHP6	Cluster: Nitrate reductase	Musaceae	78.55
A0A2T7D896	Cluster: Nitrate reductase	Panicum	78.22
A0A199UYK8	Cluster: Nitrate reductase	Ananas comosus	77.45
A0A4S8J849	Cluster: Nitrate reductase	Mesangiospermae	77.67
A0A2H3Y7X8	Cluster: Nitrate reductase	Arecaceae	76.68
A0A4S8JVI1	Cluster: Nitrate reductase	Musa balbisiana	75.58
A0A200QET5	Cluster: Nitrate reductase	Macleaya cordata	72.39
F6H0M5	Cluster: Nitrate reductase	Vitis	72.61
A0A4Y7J3A8	Cluster: Nitrate reductase	Papaver somniferum	71.95
A0A218XU44	Cluster: Nitrate reductase	Punica granatum	71.73
A0A2C9PIN3	Cluster: Nitrate reductase	Phellodendron amurense	72.06

TABLE 10-continued

Examples of suitable NADH-dependent nitrate reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 89, are listed in Table 10 below.			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 89 [%]
UPI000B8CEBA2	Cluster: nitrate reductase [NADH]-like	Carica papaya	71.51
A0A2R6RP67	Cluster: Nitrate reductase	<i>Actinidia chinensis</i> var. <i>chinensis</i>	70.63
A0A5J5BUB1	Cluster: Nitrate reductase	Nyssa sinensis	71.29
V4TCH9	Cluster: Nitrate reductase	Citrus	71.51
P36859	Cluster: Nitrate reductase [NADH]	Petunia hybrida	71.62
A0A0E0GDD5	Cluster: Nitrate reductase	Oryzinae	74.76
UPI001263D0A7	Cluster: nitrate reductase [NAD(P)H]-like	Pistacia vera	71.4
A0A061FFV7	Cluster: Nitrate reductase	Byttnerioideae	70.85
K3YPT2	Cluster: Nitrate reductase	Panicoideae	75.21
A0A1U7ZDA9	Cluster: Nitrate reductase	Nelumbo nucifera	72.17
AOAOPVXPX4	Cluster: Nitrate reductase	Oryza	74.17
A0A6A3CP56	Cluster: Nitrate reductase	Hibiscus syriacus	69.97
A0A067KLT8	Cluster: Nitrate reductase	Jatropha curcas	70.08
A0A214HOD3	Cluster: Nitrate reductase	Juglans regia	69.86
A0A5J9UYH6	Cluster: Uncharacterized protein (Fragment)	PACMAD clade	74.26
A0A5N6RHU6	Cluster: Nitrate reductase	fabids	70.63
A0A1U8KZC0	Cluster: Nitrate reductase	Malvaceae	70.74
UPI000C0393D8	Cluster: nitrate reductase [NADH]-like	Durio zibethinus	73.33
B9H7A1	Cluster: Nitrate reductase	Saliceae	70.41
P17569	Cluster: Nitrate reductase [NADH]	Cucurbitaceae	70.74
UPI000581598E	Cluster: nitrate reductase [NADH] 2-like	Sesamum indicum	70.41
UPI000F7C3BDF	Cluster: nitrate reductase [NADH] 2 isoform X1	Abrus precatorius	74.12
P08509	Cluster: Nitrate reductase [NADH] 2	Solanaceae	70.08
P39867	Cluster: Nitrate reductase [NADH], clone PBNBR1405	Brassicaceae	71.65
UPI000C1D548F	Cluster: nitrate reductase [NADH] 2-like	<i>Olea europaea</i> var. <i>sylvestris</i>	71.22
P39866	Cluster: Nitrate reductase [NADH] 2	Phaseoleae	73.88
P27968	Cluster: Nitrate reductase [NAD(P)H]	Triticeae	72.82
A8IJ84	Cluster: Nitrate reductase	Beta vulgaris	71.14
A0A2U1KIT7	Cluster: Nitrate reductase	Artemisia annua	69.64
A0A165XNP4	Cluster: Nitrate reductase	Daucus carota subsp. <i>sativus</i>	69.31

TABLE 10-continued

Examples of suitable NADH-dependent nitrate reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 89, are listed in Table 10 below.			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 89 [%]
B9RJB1	Cluster: Nitrate reductase	Mesangiospermae	74
P43101	Cluster: Nitrate reductase [NADH]	Asteraceae	69.86
11LW34	Cluster: Nitrate reductase	<i>Glycine</i> subgen. <i>Soja</i>	74.49
UPI000980E182	Cluster: nitrate reductase [NADH]-like	Ananas comosus	75.54
A0A4P1R4X1	Cluster: Nitrate reductase	Lupinus angustifolius	71.22
Q8H1T7	Cluster: Nitrate reductase	Tilia platyphyllos	69.42
11IER4	Cluster: Nitrate reductase	Brachypodium distachyon	73.47
A0A1U7Z2Y3	Cluster: Nitrate reductase	Nelumbo nucifera	72.45
A0A068VGB5	Cluster: Nitrate reductase	Coffea	68.87
A0A2T7F896	Cluster: Nitrate reductase	Panicum	73.27
A0A2Z7C1Z2	Cluster: Nitrate reductase	Doroceras hygrometricum	69.31
UPI00053C7726	Cluster: nitrate reductase [NADH] 2	Tarenaya hassleriana	69.64
A0A445BT68	Cluster: Nitrate reductase (NADH)	Arachis	72.55
P39869	Cluster: Nitrate reductase [NADH]	Lotus japonicus	73.76
UPI0010FB6302	Cluster: nitrate reductase [NADH] 2	Cajanus cajan	73.64
Q93XS1	Cluster: Nitrate reductase	Prunus	73.76
A0A022Q6Q6	Cluster: Nitrate reductase	Erythranthe guttata	69.2
A0A1U8NV83	Cluster: Nitrate reductase	Gossypium	73.04
P27783	Cluster: Nitrate reductase [NAD(P)H]	fabids	69.53
A0A2T8KWM5	Cluster: Nitrate reductase	Panicum hallii	72.93
A0A214GBD5	Cluster: Nitrate reductase	Juglans regia	69.42
A0A5N6PSR2	Cluster: Nitrate reductase	Mikania micrantha	69.2
A0A2P6PI42	Cluster: Nitrate reductase	Mesangiospermae	73.88
E7ELN7	Cluster: Nitrate reductase	IRL clade	70.41
A0A4U5QSR8	Cluster: Nitrate reductase	Populus	73.64
A0A2K2BFU1	Cluster: Nitrate reductase	Saliceae	74.12
A0A1Z5RNB1	Cluster: Nitrate reductase	Sorghum bicolor	76.67
A0A0M4P4G3	Cluster: Nitrate reductase	rosids	73.28
M4CVT6	Cluster: Nitrate reductase	Brassica campestris	70.47
A0A251SAE9	Cluster: Nitrate reductase	Helianthus annuus	72.55
A0A2G5DUU4	Cluster: Nitrate reductase	Aquilegia coerulea	71.24
P23312	Cluster: Nitrate reductase [NADH]	Chenopodioideae	72.79
P11832	Cluster: Nitrate reductase [NADH] 1	Brassicaceae	71.22

TABLE 10-continued

Examples of suitable NADH-dependent nitrate reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 89, are listed in Table 10 below.			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 89 [%]
P11035	Cluster: Nitrate reductase [NADH] 2	rosids	68.76
W1NJDO	Cluster: Nitrate reductase	Magnoliopsida	69.31
UPI000C048389	Cluster: nitrate reductase [NADH]-like	Durio zibethinus	71.36
A0A6A3C902	Cluster: Nitrate reductase	Hibiscus syriacus	73.16
P39870	Cluster: Inducible nitrate reductase [NADH] 2	Glycine subgen. Soja	73.52
UPI000F7C9CC8	Cluster: inducible nitrate reductase [NADH] 2	Abrus precatorius	72.79
A0A6A5MI86	Cluster: Nitrate reductase	Lupinus	70.31

Nitrite Reductase

[0482] As indicated above, nitrite reductase catalyzes the reduction of nitrite to ammonia (NH₃).

[0483] Preferably the recombinant yeast cell as described herein comprises at least one or more genes encoding a NADH-dependent nitrite reductase.

[0484] By a NADH-dependent nitrite reductase is herein understood a nitrite reductase that is exclusively depended on NADH as a co-factor or that is predominantly dependent on NADH as a cofactor. Preferably the NADH-dependent nitrite reductase has a ratio of catalytic efficiency for NADPH/NADP+ as a cofactor (k_{cat}/K_m)^{NADP+} to NADH/NAD+ as cofactor (k_{cat}/K_m)^{NAD+} i.e. a catalytic efficiency ratio (k_{cat}/K_m)^{NADP+}:(k_{cat}/K_m)^{NAD+}, of more than 1:1, more preferably of equal to or more than 2:1, still more preferably of equal to or more than 5:1, even more preferably of equal to or more than 10:1, yet even more preferably of equal to or more than 20:1, even still more preferably of equal to or more than 100:1, and most preferably equal to or more than 1000:1. There is no upper limit, but for practical reasons the NADH-dependent nitrite reductase may have a catalytic efficiency ratio (k_{cat}/K_m)^{NADP+}:(k_{cat}/K_m)^{NAD+} of equal to or less than 1.000.000.000:1 (i.e. 1.109). Most preferably the NADH-dependent nitrite reductase is exclusively depended on NADH/NAD+ as a co-factor. That is, most preferably the NADH-dependent nitrite reductase has an absolute requirement for NADH/NAD+ as a cofactor instead of NADPH/NADP+ as a cofactor.

[0485] Preferably the NADH-dependent nitrite reductase is a NADH-dependent nitrite reductase with enzyme classification EC 1.7.1.15 (i.e. with EC number EC 1.7.1.15). Suitably the NADH-dependent nitrite reductase, also referred to as NADH-dependent nitrite oxidoreductase, is an enzyme that catalyzes at least the following chemical reaction:

nitrite+3NADH+5H⁺→ammonia+3NAD⁺+2H₂O

The person skilled in the art will understand that the ammonia may also be present and/or referred to as so-called ammonium hydroxide NH₄OH

[0486] Suitably NADH-dependent nitrite reductases may include one or more NADH-dependent nitrite reductases as derived from *Aspergillus nidulans* (also called *Emericella nidulans*), *Arcobacter ellisii*, *Arcobacter pacificus* *Bacillus subtilis*, *Bacillus subtilis* JH642, *Cupriavidus taiwanensis*, *Escherichia coli*, *Ralstonia taiwanensis*, *Ralstonia syzygii*, *Ralstonia solanacearum*, *Rhodobacter capsulatus*, *Rhodobacter capsulatus*, *Paraburkholderia ribeironis*; and/or functional homologues of such NADH-dependent nitrite reductases comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of such aforementioned NADH-dependent nitrite reductases; and/or functional homologues of such NADH-dependent nitrite reductases comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned NADH-dependent nitrite reductases, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned NADH-dependent nitrite reductases.

[0487] *Escherichia coli* utilizes several distinct enzymes in its nitrite assimilation pathway. The nirD gene encodes a NADH-dependent nitrite reductase (NADH) small subunit, whilst the nirB gene encodes a NADH-dependent nitrite reductase (NADH) large subunit.

[0488] Preferred NADH-dependent nitrite reductases include the NADH-dependent nitrite reductases as derived from *Aspergillus nidulans* (also called *Emericella nidulans*), a nitrite reductase capable of utilizing both NADH and NADPH as electron donors, and/or *Escherichia coli*. At high nitrate and/or nitrite concentrations, the nitrite reductase encoded by the nirB gene of *Escherichia coli* is especially preferred.

[0489] Preferred NADH-dependent nitrite reductases hence include: NADH-dependent nitrite reductases comprising a polypeptide having an amino acid sequence of SEQ ID NO:91 (*E. coli* nitrite reductase small subunit encoded by nirD) and/or SEQ ID NO:92 (*E. coli* nitrite reductase large subunit encoded by nirB) and/or SEQ ID NO:93 (*Emericella nidulans* nitrate reductase encoded by niiA), as described herein; and/or functional homologues of SEQ ID NO:91 and/or SEQ ID NO:92 and/or SEQ ID NO:93 comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of SEQ ID NO:91 and/or SEQ ID NO:92 and/or SEQ ID NO:93 respectively; and/or functional homologues of SEQ ID NO:91 and/or SEQ ID NO:92 and/or SEQ ID NO:93 comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of SEQ ID NO:91 and/or SEQ ID NO:92 and/or SEQ ID NO:93 respectively. Preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:91 and/or SEQ ID NO:92 and/or SEQ ID NO:93 respectively.

[0490] Preferably the recombinant yeast cell comprises an exogenous gene coding for an enzyme with NADH-dependent nitrite reductase activity. More preferably the recombinant yeast cell comprises an exogenous gene coding for an enzyme with NADH-dependent nitrite reductase activity selected from the group consisting of NADH-dependent nitrite reductases as derived from *Aspergillus nidulans* (also called *Emericella nidulans*), *Arcobacter ellisii*, *Arcobacter pacificus*, *Bacillus subtilis*, *Bacillus subtilis* JH642, *Cupriavidus taiwanensis*, *Escherichia coli*, *Ralstonia taiwanensis*, *Ralstonia syzygii*, *Ralstonia solanacearum*, *Rhodobacter capsulatus*, *Rhodobacter capsulatus*, *Paraburkholderia ribeironis*; and/or functional homologues of such NADH-dependent nitrite reductases comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of such aforementioned NADH-dependent nitrite reductases; and/or functional homologues of such NADH-dependent nitrite reductases comprising an amino acid sequence having one or several substitutions, insertions

and/or deletions as compared to the amino acid sequence of one or more of such aforementioned NADH-dependent nitrite reductases, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned NADH-dependent nitrite reductases.

[0491] Suitably the recombinant yeast cell may comprise a nucleotide sequence coding for an amino acid sequence of any of SEQ ID NO:91 (*E. coli* nitrate reductase small subunit encoded by nirD) and/or SEQ ID NO:92 (*E. coli* nitrate reductase large subunit encoded by nirB) and/or SEQ ID NO:93 (*Emericella nidulans* nitrate reductase encoded by niiA), or an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of any of SEQ ID NO:91 and/or SEQ ID NO:92 and/or SEQ ID NO:93. Preferably the amino acid sequence has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:91 and/or SEQ ID NO:92 and/or SEQ ID NO:93 respectively.

[0492] The recombinant yeast cell may combine one or more genes encoding one or more of the above NADH-dependent nitrite reductases with one or more genes encoding an NADPH-dependent nitrate reductase. Preferably, however, the recombinant yeast cell combines one or more genes encoding one or more of the above NADH-dependent nitrite reductases with one or more genes encoding a NADH-dependent nitrate reductase.

[0493] Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:91 (small subunit encoded by nirD), are listed in Table 11 below.

[0494] Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:92 (large subunit encoded by nirB), are listed in Table 12 below.

TABLE 11

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 91 (small subunit encoded by nirD).

UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 91 [%]
E2QFM2	Cluster: Nitrite reductase	<i>Gammaproteobacteria</i>	99.07
A0A078LHD3	Cluster: Nitrite reductase small subunit	<i>Enterobacteriaceae</i>	94.44
POA230	Cluster: Nitrite reductase (NADH) small subunit	Bacteria	93.52
A0A170JME2	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Enterobacteriaceae</i>	91.67

TABLE 11-continued

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 91 (small subunit encoded by nirD)).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 91 [%]
A0A0F6TX22	Cluster: Nitrite reductase	<i>Gammaproteobacteria</i>	91.67
A0A6DOEP17	Cluster: Nitrite reductase small subunit NirD	<i>Enterobacteriaceae</i>	87.04
A0A198H1N5	Cluster: Nitrite reductase [NAD(P)H] small subunit	Bacteria	86.11
A0A447MQ21	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Enterobacteriaceae</i>	84.26
UPI00077B73C7	Cluster: nitrite reductase small subunit NirD	<i>Enterobacter timonensis</i>	83.33
A0A094XNF5	Cluster: Nitrite reductase (NADH) small subunit	<i>Enterobacterales</i>	82.41
UPI00105F841D	Cluster: nitrite reductase small subunit NirD	<i>Enterobacter</i> sp. OV724	81.48
A0A1X0XB32	Cluster: Nitrite reductase small subunit	<i>Enterobacterales</i>	81.48
A0A1A9FFN3	Cluster: Nitrite reductase small subunit	<i>Lelliottia amnigena</i>	81.48
UPI0002D5E0A7	Cluster: nitrite reductase small subunit NirD	<i>Metakosakonia massiliensis</i>	80.56
UPI0004E24438	Cluster: nitrite reductase small subunit NirD	<i>Siccibacter colletis</i>	79.63
A0A514EKN7	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Enterobacteriaceae</i>	77.78
A0A2P8VHS5	Cluster: Nitrite reductase small subunit NirD	<i>Enterobacteriaceae</i>	80.56
A0A4V3BP91	Cluster: Nitrite reductase (NADH) small subunit	cellular organisms	80.56
A0A3R9F800	Cluster: Nitrite reductase small subunit NirD	<i>Enterobacteriaceae</i>	79.63
R9VMQ4	Cluster: Nitrite reductase small subunit	<i>Enterobacteriaceae</i>	78.7
A0A285KH74	Cluster: Nitrite reductase (NADH) small subunit	<i>Enterobacter</i> sp. CC120223-11	80.37
A0A658U6S1	Cluster: Assimilatory nitrite reductase (NAD(P)H) small subunit	<i>Enterobacter</i>	77.78
A0A085A429	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Enterobacteriaceae</i>	78.7
A0A1VOL823	Cluster: Nitrite reductase small subunit	<i>Kosakonia</i>	76.85
UPI001431C56A	Cluster: nitrite reductase small subunit NirD	<i>Cedecea colo</i>	75

TABLE 11-continued

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 91 (small subunit encoded by nirD)).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 91 [%]
A0A085G1A1	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Enterobacteriaceae</i>	75
A0A089PL75	Cluster: Nitrite reductase	<i>Pluralibacter gergoviae</i>	76.85
A0A3C0H311	Cluster: Nitrite reductase small subunit	<i>Enterobacteriaceae</i>	77.78
A0A3N2DXC8	Cluster: Assimilatory nitrite reductase (NAD(P)H) small subunit	unclassified <i>Enterobacter</i>	75.93
AOA0LOATR1	Cluster: Nitrite reductase	<i>Enterobacteriaceae</i>	78.5
K8AGI3	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Cronobacter condimenti</i> 1330	75
A0A116EAUO	Cluster: Nitrite reductase (NADH) small subunit	<i>Enterobacter</i> sp. kpr-6	75
A0A0D5X0Q1	Cluster: Nitrite reductase	<i>Enterobacteriaceae</i>	75
C9Y259	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Enterobacteriaceae</i>	74.07
A0A4R0GWC0	Cluster: Nitrite reductase small subunit NirD	<i>Kosakonia quasisacchari</i>	75
UPI00057BF926	Cluster: nitrite reductase small subunit NirD	<i>Enterobacter</i> sp. Bisph1	73.15
A0A495AHMO	Cluster: Nitrite reductase small subunit NirD	<i>Enterobacter</i> sp. R1(2018)	72.22
A0A447QKU3	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Yersiniaceae</i>	75.47
12B4C9	Cluster: Nitrite reductase (NAD(P)H)	<i>Shimwellia blattae</i> (strain ATCC 29907 / DSM 4481 / JCM 1650 / NBRC 105725 / CDC 9005-74)	73.83
UPI0012B7CAE2	Cluster: nitrite reductase small subunit NirD	<i>Erwinia</i> sp. CPCC 100877	72.22
UPI00073DB139	Cluster: nitrite reductase small subunit NirD	[<i>Erwinia</i>] <i>teleogrylli</i>	73.83
UPI000237D150	Cluster: nitrite reductase small subunit NirD	<i>Enterobacter mori</i>	80.56
A0A4P8YE33	Cluster: Nitrite reductase small subunit NirD	<i>Enterobacteriaceae</i>	72.22
UPI0005DC1E97	Cluster: nitrite reductase small subunit NirD	<i>Yersinia rohdei</i>	71.03
AOA0U1HQI4	Cluster: Nitrite reductase small subunit	<i>Yersinia</i>	71.03
A0A0KOHM15	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Enterobacterales</i>	72.9

TABLE 11-continued

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 91 (small subunit encoded by nirD)).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 91 [%]
H8NP95	Cluster: Nitrite reductase small subunit	<i>Yersiniaceae</i>	69.16
A0A2S4QQ85	Cluster: Nitrite reductase small subunit	<i>Enterobacteriaceae</i>	69.44
A0A0A7RXX3	Cluster: Assimilatory nitrite reductase (NAD(P)H) small subunit	<i>Frischella perrara</i>	71.3
D4DYL5	Cluster: Nitrite reductase [NAD(P)H], small subunit	<i>Serratia odorifera</i> DSM 4582	73.58
A0A370QTS9	Cluster: Assimilatory nitrite reductase (NAD(P)H) small subunit	<i>Enterobacillus tribolii</i>	71.7
A0A1B9JUBO	Cluster: Nitrite reductase small subunit	<i>Gilliamella apicola</i>	69.16
A0A085U6Z8	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Yersinia ruckeri</i>	72.22
UPI00156B8D43	Cluster: UPI00156B8D43 related cluster	unknown	70.75
A0AOT9KRE5	Cluster: Nitrite reductase small subunit	<i>Yersinia</i>	70.37
A0A066TFS8	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Gilliamella</i>	69.16
A0A1B9M9X4	Cluster: Nitrite reductase small subunit	<i>Gilliamella</i>	68.22
A0A0E1NM95	Cluster: Nitrite reductase (NAD(P)H)	<i>Yersiniaceae</i>	68.52
A0A080KOG0	Cluster: Ferredoxin subunits of nitrite reductase and ring-hydroxylating dioxygenase	<i>Snodgrassella alvi</i> SCGC AB-598-002	69.16
A0A0Q4N870	Cluster: Nitrite reductase small subunit	<i>Enterobacterales</i>	67.29
UPI0004A391D4	Cluster: nitrite reductase small subunit NirD	<i>Tatumella saanichensis</i>	69.81
A0A1B9ZVE6	Cluster: Nitrite reductase small subunit	<i>Gilliamella apicola</i>	67.29
A0AOT9JI92	Cluster: Nitrite reductase small subunit	<i>Yersinia</i>	68.52
A0A2N5EJM2	Cluster: Nitrite reductase small subunit NirD	<i>Chimaeribacter</i>	67.59
A0A2V4E983	Cluster: Nitrite reductase small subunit	<i>Gilliamella</i>	65.74
A0A419N5U0	Cluster: Nitrite reductase small subunit NirD	<i>Yersiniaceae</i>	67.29

TABLE 11-continued

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 91 (small subunit encoded by nirD).

UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 91 [%]
A0A0F7H970	Cluster: Nitrite reductase	<i>Serratia</i>	70.75
A0A318NZT5	Cluster: Nitrite reductase small subunit NirD	<i>Yersiniaceae</i>	69.81
A0A250B1W9	Cluster: Nitrite reductase small subunit	<i>Gibbsiella quercinecans</i>	69.81
A0A6G91849	Cluster: Nitrite reductase small subunit NirD	<i>Orbus</i> sp. IPMB12	65.74
A0A419M923	Cluster: Nitrite reductase small subunit NirD	<i>Rahnella inusitata</i>	67.92
A0A389M9P6	Cluster: Nitrite reductase (NADH) small subunit	<i>Enterobacterales</i>	67.29
A0AON8GUB4	Cluster: Nitrite reductase (NADH) small subunit	<i>Vibrionales</i>	67.96
A0A443IEH9	Cluster: Nitrite reductase	<i>Pantoea</i> sp. LMG 27579	68.87
A0A1X1MLP8	Cluster: Nitrite reductase	<i>Vibrio</i> sp. qd031	69.9
A0A542BGK2	Cluster: Assimilatory nitrite reductase (NAD(P)H) small subunit	<i>Serratia fonticola</i>	68.87
A0A2U3B6U3	Cluster: Nitrite reductase small subunit	<i>Vibrio</i> sp. E4404	67.92
A0A240CDV1	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Gammaproteobacteria</i>	67.92
A0A2S918J3	Cluster: Nitrite reductase small subunit	Bacteria	67.29
A0A495RE76	Cluster: Assimilatory nitrite reductase (NAD(P)H) small subunit	<i>Orbus hercynius</i>	67.59
A0A178KL05	Cluster: Nitrite reductase small subunit	<i>Photobacterium</i>	66.04
A0A198GJ99	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Cosenzaea myxofaciens</i> ATCC 19692	63.55
A0A2NOWOU3	Cluster: Nitrite reductase small subunit	<i>Enterobacterales</i>	66.36
A0A1B9JKC9	Cluster: Nitrite reductase small subunit	<i>Gilliamella apicola</i>	66.67
UPI0011BEE383	Cluster: nitrite reductase small subunit NirD	<i>Pantoea</i> sp. CCBC3-3-1	65.09
A0A085G554	Cluster: Nitrite reductase [NAD(P)H] small subunit	<i>Ewingella americana</i>	67.29
A0A1B1NU77	Cluster: Nitrite reductase (NADH)	<i>Vibrio</i>	67.96
A0A2N7G6C9	Cluster: Nitrite reductase small subunit	<i>Vibrio</i> sp. 10N.261.55.A7	67.96

TABLE 11-continued

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 91 (small subunit encoded by nirD).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 91 [%]
A0A349FFE1	Cluster: Nitrite reductase small subunit	<i>Vibrio</i> sp.	69.52
A0A1B9JS56	Cluster: Nitrite reductase small subunit	<i>Gilliamella apicola</i>	64.49
A0A366XH21	Cluster: Nitrite reductase small subunit	<i>Vibrionales</i>	66.67
UPI00062A46AC	Cluster: nitrite reductase small subunit NirD	<i>Tatumella morbirosei</i>	67.59
UPI00046A4834	Cluster: nitrite reductase small subunit NirD	<i>Pantoea</i> sp. IMH	68.87

TABLE 12

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 92 (large subunit encoded by nirB).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 92 [%]
W9B2S1	Cluster: NirB protein	Bacteria	93.27
A0A486K457	Cluster: Nitrite reductase [NAD(P)H]	Enterobacterales	93.15
A0A1B7KIV8	Cluster: Nitrite reductase [NAD(P)H] large subunit	Enterobacterales	85.41
A0A087L3N6	Cluster: Nitrite reductase	Serratia	84.6
A0A654DEI2	Cluster: Nitrite reductase, large subunit, nucleotide-and Fe/S-cluster binding	Enterobacterales	84.36
UPI00156BAB07	Cluster: UPI00156BAB07 related cluster	unknown	81.3
A0A2I0FWV2	Cluster: Nitrite reductase (NAD(P)H)	Enterobacterales	81.52
I2B4D0	Cluster: Nitrite reductase (NAD(P)H)	bacterium CwR94 <i>Shimwellia blattae</i> (strain ATCC 29907 / DSM 4481 / JCM 1650 / NBRC 105725/ CDC 9005-74)	85.24
A0A370QTZ7	Cluster: Assimilatory nitrite reductase (NAD(P)H) large subunit	<i>Enterobacillus tribolii</i>	83.39
UPI000DEFBD5D	Cluster: nitrite reductase large subunit	<i>Edaphovirga crenea</i>	83.53
A0A380ASL5	Cluster: Nitric oxide reductase FIRd-NAD(+) reductase	cellular organisms	84.36
A0A443IEI0	Cluster: Nitrite reductase	unclassified <i>Pantoea</i>	81.42
A0A221T4L0	Cluster: Nitrite reductase [NAD(P)H] large subunit	Enterobacterales	80.45
A0A2N5EJN0	Cluster: Nitrite reductase (NAD(P)H)	Enterobacterales	82.68

TABLE 12-continued

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 92 (large subunit encoded by nirB).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 92 [%]
A0A377N6S4	Cluster: Benzene 1,2-dioxygenase system ferredoxin -- NAD(+) reductase subunit	Enterobacterales	82.82
A0A4P8SFP8	Cluster: Nitrite reductase (NAD(P)H)	Nissabacter sp. SGAir0207	82.11
A0A0L7TEE8	Cluster: Nitrite reductase	Erwinia iniecta	79.62
A0A1W6B3D3	Cluster: Nitrite reductase large subunit	Erwiniaceae	79.88
A0A506V9E3	Cluster: Nitrite reductase large subunit	Mixta	79.41
UPI000468AD1D	Cluster: nitrite reductase large subunit	Erwiniaceae	81.02
A0A0N0Z921	Cluster: Nitrite reductase [NAD(P)H] large subunit	Moellerella wisconsensis	79.34
B4EXR2	Cluster: Nitrite reductase [NAD(P)H] large subunit	Enterobacterales	78.34
A0A198GKA3	Cluster: Nitrite reductase [NAD(P)H] large subunit	Cosenzaea myxofaciens ATCC 19692	78.11
A0A0F3LTP0	Cluster: Nitrite reductase	Enterobacterales	78.46
E1SGS9	Cluster: Nitrite reductase (NAD(P)H) subunit	Erwiniaceae	77.78
A0A1X1EZA9	Cluster: Nitrite reductase large subunit	Pantoea cypripedii	78.25
A0A3A9I4K8	Cluster: Nitrite reductase large subunit	Erwiniaceae	77.9
A0A2S9I8J4	Cluster: Nitrite reductase (NAD(P)H)	Erwiniaceae	79.48
A0A0R4FQI6	Cluster: Nitrite reductase, large subunit, nucleotide-binding	Xenorhabdus nematophila	76.42
UPI001265FD68	Cluster: nitrite reductase large subunit	Erwinia endophytica	77.49
UPI00146D663B	Cluster: nitrite reductase large subunit	Rahnella sp. SAP-1	78.32
A0A3S4CWL7	Cluster: Rubredoxin-NAD(+) reductase	Enterobacterales	78.65
A0A1Y2S8S4	Cluster: NADH oxidase	Xenorhabdus vietnamensis	76.78
A0A0A3YW59	Cluster: Nitrite reductase	Erwinia typographi	79.31
A0A506PS78	Cluster: Nitrite reductase large subunit	Pantoea	76.12
D8MY05	Cluster: Nitrite reductase [NAD(P)H] large subunit	Erwinia	79.12
E6WGY1	Cluster: Nitrite reductase (NAD(P)H), large subunit	unclassified Pantoea	75.62
A0A3R8ZPT1	Cluster: Nitrite reductase large subunit	unclassified Erwinia	75.3
A0A5M7NEQ9	Cluster: Nitrite reductase large subunit	Enterobacterales	75.3
A0A2V2BEC9	Cluster: Assimilatory nitrite reductase (NAD(P)H) large subunit	Pantoea	76.36
A0A1X0W567	Cluster: Nitrite reductase large subunit	Rouxiella	77.88
A0A6G9I851	Cluster: Nitrite reductase large subunit	Orbus sp. IPMB12	77.66

TABLE 12-continued

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 92 (large subunit encoded by nirB)).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 92 [%]
UPI0011BF5221	Cluster: nitrite reductase large subunit	Pantoea sp. CCBC3-3-1	76.66
A0A556RIT3	Cluster: Nitrite reductase large subunit	Gilliamella	74.91
A0A1R4J5I6	Cluster: Nitrite reductase [NAD(P)H] large subunit	Vibrio	72.82
A0A1B9MVL8	Cluster: Nitrite reductase large subunit	Gilliamella	75.03
UPI0013596BEF	Cluster: nitrite reductase large subunit	unclassified Psychromonas	72.59
A0A1B9L8J9	Cluster: Nitrite reductase large subunit	Gilliamella apicola	74.79
A0A095TED5	Cluster: Nitrite reductase	Tatumella	74.67
A0A2S7VXZ8	Cluster: Nitrite reductase large subunit	Photobacterium	73.6
A0A495REZ8	Cluster: Assimilatory nitrite reductase (NAD(P)H) large subunit	Orbus hercynius	75.15
A0A2T3QB92	Cluster: Nitric oxide reductase FIRd-NAD(+) reductase	Photobacterium damsela	75.18
A0A1X0WF55	Cluster: Nitrite reductase large subunit	Gammaproteobacteria	76.51
A0A2I8TVB9	Cluster: Nitrite reductase large subunit	Vibrio	75.33
A0A2T3NPM2	Cluster: Nitrite reductase large subunit	Photobacterium sanctipauli	74.85
R9PFS3	Cluster: Nitrite reductase	Agarivorans	73.18
A0A0B7J736	Cluster: Nitrite reductase, large subunit, NAD(P)H-binding	Photobacterium	73.6
U4KF90	Cluster: Nitrite reductase [NAD(P)H] large subunit	Vibrio nigripulchritudo	73.78
A0A1Y6IN13	Cluster: Nitrite reductase [NAD(P)H]	Vibrio mangrovi	73.54
A0A1L9L1Z8	Cluster: Nitrite reductase [NAD(P)H]	Vibrionaceae	73.78
A0A0A7RZE6	Cluster: Assimilatory nitrite reductase (NAD(P)H) large subunit	Frischella perrara	73.24
A0A1Q9GSK0	Cluster: Nitrite reductase large subunit	Photobacterium	74.97
UPI000200DAF9	Cluster: nitrite reductase large subunit	Vibrio fumiissii	74.97
A0A378PW16	Cluster: Nitric oxide reductase FIRd-NAD(+) reductase	Vibrionaceae	72.82
A0A1B9JKB8	Cluster: Nitrite reductase large subunit	Gilliamella apicola	72.65
D4ZBU5	Cluster: Nitrite reductase [NAD(P)H], large subunit	Shewanella violacea (strain JCM 10179 / CIP 106290 / LMG 19151 / DSS12)	72.4
UPI00117F52C3	Cluster: nitrite reductase large subunit	Vibrio fumiissii	74.97
UPI0001B93571	Cluster: nitrite reductase large subunit	Vibrio fumiissii	74.97
A0A1B1NUU8	Cluster: Nitrite reductase (NADH)	Vibrionaceae	72.94
A0A178KL92	Cluster: Nitrite reductase large subunit	Photobacterium	73.39

TABLE 12-continued

Examples of suitable NADH-dependent nitrite reductases, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 92 (large subunit encoded by nirB)).			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 92 [%]
A0A1T4RZM3	Cluster: Nitrite reductase (NADH) large subunit	Vibrio cincinnatiensis	74.37
UPI0012ADB8F8	Cluster: nitrite reductase large subunit	Vibrio furnissii	74.97
A0A418YIX2	Cluster: Nitrite reductase large subunit	Motilimonas sp. PLHSC7-2	73
Q6LS87	Cluster: Putative nitrite reductase (NAD(P)H), large subunit	Photobacterium	73.67
A0A4R3I7X5	Cluster: Nitrite reductase (NADH) large subunit	Reinekea marinisedimentorum	73.72
UPI0003F87F52	Cluster: nitrite reductase large subunit	Psychromonas	72.02
UPI00037B4F3D	Cluster: nitrite reductase large subunit	Uliginosibacterium gangwonense	70.2
A0A2N7DBN5	Cluster: Nitrite reductase large subunit	Vibrio sp. 10N.286.49.B3	73.06
A0A484GE91	Cluster: Nitrite reductase large subunit	Candidatus Schmidhempelia bombi str. Bimp unclassified Vibrio	72.16
A0A2M8H0G2	Cluster: Nitrite reductase (NAD(P)H)	Vibrio	73.84
U3CNX4	Cluster: Nitrite reductase large subunit	Vibrio	73.78
UPI0010A63CB5	Cluster: nitrite reductase large subunit	Vibrio sp. H11	73.12
A0A090IJG5	Cluster: Nitrite reductase (NAD(P)H) large subunit	Moritella	71.92
A0A2N0WYN5	Cluster: Nitrite reductase (NAD(P)H)	Psychromonas	72.87
UPI000B356675	Cluster: nitrite reductase large subunit	Thaumasiovibrio subtropicus	73.12
A0A1R4LRM2	Cluster: Nitrite reductase [NAD(P)H]	Vibrio	69.41
UPI00082A845F	Cluster: nitrite reductase large subunit	Erwinia sp. ErVv1	72.27
B8KAR6	Cluster: Nitrite reductase (NAD(P)H), large subunit	Vibrio	73.3
A0A1G8E1S6	Cluster: Assimilatory nitrite reductase (NAD(P)H) large subunit	Vibrio xiamenensis	72.59
A0A0C5WZ76	Cluster: Nitrite reductase (NAD(P)H), large subunit	Photobacterium	72.94
A0A2G4AWG1	Cluster: Nitrite reductase [NAD(P)H]	Vibrionaceae	73.18

Nitrate/Nitrite Transporter

[0495] Preferably, the recombinant yeast cell further comprises one or more genetic modifications that result in an increased transport of oxidized nitrogen source, such as nitrate or nitrite, into the yeast cell. More preferably the recombinant yeast cell further comprising one or more genes encoding a nitrate and/or nitrite transporter.

[0496] Suitable transporters may include the sulphite transporters Ssu1 and Ssu2 (as described by Cabrera et al in their article titled “Molecular Components of Nitrate and Nitrite Efflux in Yeast”, published February 2014 Volume 13

Number 2 Eukaryotic Cell p. 267-278, herein incorporated by reference); and the nitrate/nitrite transporter YNT1 derived from *Pichia angusta* (also referred to as *Hansenula polymorpha*) and/or a functional homologues of one or more of such nitrate/nitrite transporters comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with one or more of the aforementioned nitrate/nitrite transporters; and/or functional homologues of one or more of such nitrate/nitrite transporters comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of

such aforementioned nitrate/nitrite transporters, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned nitrate/nitrite transporter YNT1.

[0497] Preferably the recombinant yeast cell comprises a nucleic acid sequence encoding the nitrate/nitrite transporter YNT1 derived from *Pichia angusta* and/or a functional homologues of such nitrate/nitrite transporter YNT1 comprising an amino acid sequence with at least 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with nitrate/nitrite transporter YNT1; and/or functional homologues of such nitrate/nitrite transporter YNT1 comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of one or more of such aforementioned nitrate/nitrite transporter YNT1, wherein preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to such aforementioned nitrate/nitrite transporter YNT1.

[0498] Preferred nitrate/nitrite transporter hence include: nitrate/nitrite transporters comprising a polypeptide having an amino acid sequence of SEQ ID NO:94, as described

herein; and/or functional homologues of SEQ ID NO:94 comprising an amino acid sequence with at least 40, 50, 60, 65, 70, 75, 80, 85, 90, 95, 98 or at least 99% amino acid sequence identity with SEQ ID NO:94; and/or functional homologues of SEQ ID NO:94 comprising an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of SEQ ID NO:94. Preferably the amino acid sequence of any of the above functional homologues has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:94.

[0499] Suitably the recombinant yeast cell may comprise a nucleotide sequence coding for an amino acid sequence of SEQ ID NO:94 or an amino acid sequence having one or several substitutions, insertions and/or deletions as compared to the amino acid sequence of any of SEQ ID NO:94. Preferably the amino acid sequence has no more than 300, 250, 200, 150, 100, 75, 50, 40, 30, 20, 10 or 5 amino acid substitutions, insertions and/or deletions as compared to SEQ ID NO:94 respectively.

[0500] Examples of suitable nitrite/nitrate transporters, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO:94 are listed in Table 13 below.

TABLE 13

Examples of suitable nitrite/nitrate transporters, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 94.			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 94 [%]
W1QLT0	Cluster: Nitrate/nitrite transporter	Ogataea	95.66
W6MFQ5	Cluster: Nitrate/nitrite transporter	Kuraishia capsulata CBS 1993	56.89
A0A1E4TX12	Cluster: Nitrate/nitrite transporter	Pachysolen tannophilus NRRL Y-2460	56.55
A0A1E3NV96	Cluster: Nitrate/nitrite transporter	Wickerhamomyces anomalus (strain ATCC 58044 / CBS 1984 / NCYC 433 / NRRL Y-366-8)	56.52
K0KNB9	Cluster: Nitrate/nitrite transporter	Wickerhamomyces ciferrii (strain F-60-10 / ATCC 14091 / CBS 111 / JCM 3599 / NBRC 0793 / NRRL Y-1031)	56.46
A0A0H5C0H7	Cluster: Nitrate/nitrite transporter	Cyberlindnera jadinii (strain ATCC 18201 / CBS 1600 / CCRC 20928 / JCM 3617 / NBRC 0987 / NRRL Y-1542)	54.44
A0A5P8N917	Cluster: Nitrate/nitrite transporter	Cyberlindnera americana	54.58
A0A061BB65	Cluster: CYFAOS24e00188g1_1	Cyberlindnera fabianii	52.72
A0A060T893	Cluster: Nitrate/nitrite transporter	Blastobotrys adeninivorans	47.18
B3WFS7	Cluster: Nitrate/nitrite transporter	Blastobotrys adeninivorans	47.79
A0A261XZN8	Cluster: Nitrate/nitrite transporter	Bifiguratus adelaidae	47.49

TABLE 13-continued

Examples of suitable nitrite/nitrate transporters, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 94.			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 94 [%]
A0A0E9N9H8	Cluster: Nitrate/nitrite transporter	Saitoella complicata (strain BCRC 22490 / CBS 7301 / JCM 7358 / NBRC 10748 / NRRL Y-17804)	46.17
A0A060T786	Cluster: Nitrate/nitrite transporter	Blastobotrys adeninivorans	43.91
A0A0A1T938	Cluster: Nitrate/nitrite transporter	Torrubiella hemipterigena	42.57
A0A3D8R1D7	Cluster: Nitrate/nitrite transporter	Coleophoma	42.66
A0A370TMI7	Cluster: Nitrate/nitrite transporter	Venustampulla echinocandica	41.72
A0A3N4HFN0	Cluster: Nitrate/nitrite transporter	Ascobolus immersus RN42	42.16
A0A167RLM4	Cluster: Nitrate/nitrite transporter	Cordyceps fumosorosea ARSEF 2679	40.04
A0A179I4H5	Cluster: Nitrate/nitrite transporter	Akanthomyces lecanii	40.94
A0A0G2E479	Cluster: Nitrate/nitrite transporter	Phaeomoniella chlamydospora	42.33
A0A6A6CV94	Cluster: Nitrate/nitrite transporter	Zasmidium cellare ATCC 36951	42.91
S3CRP8	Cluster: Nitrate/nitrite transporter	Glarea lozoyensis	40.4
A0A1B8DXT4	Cluster: Nitrate/nitrite transporter	unclassified Pseudogymnoascus	42.74
A0A094B0S5	Cluster: Nitrate/nitrite transporter	Pseudogymnoascus	42.94
A0A6A6WTZ3	Cluster: Nitrate/nitrite transporter	Melanomma pulvis- pyrius CBS 109.77	42.51
G3JS45	Cluster: Nitrate/nitrite transporter	Cordyceps militaris	40.65
A0A0E9NCH7	Cluster: Nitrate/nitrite transporter	Saitoella complicata (strain BCRC 22490 / CBS 7301 / JCM 7358 / NBRC 10748 / NRRL Y-17804)	44.27
A0A0C3CTX3	Cluster: Nitrate/nitrite transporter	Oidiodendron maius Zn	43.15
A0A6G1J7V7	Cluster: Nitrate transporter-like protein	Lentithecium fluviatile CBS 122367	42.63
A0A4U0XSZ7	Cluster: Nitrate/nitrite transporter	Cryomyces minteri	41.8
UPI000CE1C7E3	Cluster: nitrate transporter-like	Quercus suber	41.33
A0A6A5XQJ5	Cluster: Nitrate/nitrite transporter	Aaosphaeria arxii CBS 175.79	42.24
A0A1W5CWS1	Cluster: Nitrate/nitrite transporter	Lasallia pustulata	43.11
A0A6A6R7R9	Cluster: Nitrate/nitrite transporter	Mytiliniaceae	43.99
U1GKS4	Cluster: Nitrate/nitrite transporter	Endocarpon pusillum (strain Z07020 / HMAS- L-300199)	42.43
D5GLI8	Cluster: Nitrate/nitrite transporter	Tuber melanosporum (strain Mel28)	41.75
A0A5J5EIC4	Cluster: Nitrate/nitrite transporter	Sphaerospora brunnea	41.7
UPI00144AC6B7	Cluster: nitrate transporter	Lindgomycetes ingoldianus	42.6
A0A6A6RHP6	Cluster: Nitrate/nitrite transporter	Massarina eburnea CBS 473.64	42.6
A0A6A6JM23	Cluster: Nitrate/nitrite transporter	Westerdykella ornata	42.63

TABLE 13-continued

Examples of suitable nitrite/nitrate transporters, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 94.			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 94 [%]
A0A317SYP8	Cluster: Nitrate/nitrite transporter	Tuber	41.34
A0A232LMW8	Cluster: Nitrate/nitrite transporter	Elaphomyces granulatus	43.09
A0A1L7WDP7	Cluster: Nitrate/nitrite transporter	Phialocephala subalpina	42.04
A0A6A5UCT1	Cluster: Nitrate/nitrite transporter	Byssothecium circinans	43.38
A0A6A6BJV5	Cluster: Nitrate/nitrite transporter	Aplosporella prunicola CBS 121167	44.08
A0A6G1LEQ8	Cluster: Nitrate transporter	Teratosphaeria nubilosa	41.68
S3DBU1	Cluster: Nitrate/nitrite transporter	Glarea lozoyensis	43.51
A0A4U0TQA4	Cluster: Nitrate/nitrite transporter	Hortaea thailandica	40.68
A0A4U0X4U1	Cluster: Nitrate/nitrite transporter	Friedmanniomyces	40.28
A0A507R2R1	Cluster: Nitrate/nitrite transporter	Monascus purpureus	41.84
A0A2V1DKP2	Cluster: Nitrate/nitrite transporter	Periconia macrospinoso	41.46
A0A1L9RJV0	Cluster: Nitrate/nitrite transporter	Aspergillus wentii DTO 134E9	42.54
A0A167P4D7	Cluster: Nitrate/nitrite transporter	Sporothrix insectorum RCEF 264	41.12
UPI000CE20498	Cluster: nitrate transporter-like	Quercus suber	42.71
A0A4Z1P351	Cluster: Nitrate/nitrite transporter	Venturia	41.63
A0A6G1K4M7	Cluster: Nitrate transporter-like protein	Pleomassaria siparia CBS 279.74	42.16
M7U767	Cluster: Nitrate/nitrite transporter	Sclerotiniaceae	42.65
A0A0D2AUR8	Cluster: Nitrate/nitrite transporter	Verruconis gallopava	41.41
A0A6G1H5R5	Cluster: Nitrate transporter	Aulographum hederar CBS 113979	41.73
A0A1Y2TEF0	Cluster: Nitrate/nitrite transporter	unclassified Hypoxylon	41.77
A0A177BV64	Cluster: Nitrate/nitrite transporter	Paraphaeosphaeria sporulosa	41.75
A0A6A5WR45	Cluster: Nitrate/nitrite transporter	Amniculicola lignicola CBS 123094	42.63
A0A6G1KNX3	Cluster: Nitrate transporter-like protein	Pleomassaria siparia CBS 279.74	41.41
W9WHT3	Cluster: Nitrate/nitrite transporter	Herpotrichiellaceae	42.14
A0A2T2P8J8	Cluster: Nitrate/nitrite transporter	Corynespora cassiicola Philippines	43.06
A0A6A6T246	Cluster: Nitrate/nitrite transporter	Lophiostoma macrostomum CBS 122681	41.41
A0A2P8ABJ3	Cluster: Nitrate/nitrite transporter	Elsinoe australis	42.25
A0A1Y1ZNY9	Cluster: Nitrate/nitrite transporter	Clohesyomyces aquaticus	40.77
W2RSI4	Cluster: Nitrate/nitrite transporter	Cyphellophora europaea CBS 101466	41.24
A0A2V1E4S3	Cluster: Nitrate/nitrite transporter	Periconia macrospinoso	41.41
A0A6A5V026	Cluster: Nitrate/nitrite transporter	Bimuria novae-zelandiae CBS 107.79	41.15
A0A2K1QG31	Cluster: Nitrate/nitrite transporter	Elsinoaceae	42.25

TABLE 13-continued

Examples of suitable nitrite/nitrate transporters, their UniProt Database Accession number (as can be found on the Uniprot website (www.uniprot.org/ as per 4 Oct. 2020), their description, the organism from which they may be derived, and their amino acid sequence identity with SEQ ID NO: 94.			
UNIPROT DATABASE ACCESSION NUMBER	DESCRIPTION	ORGANISM	IDENTITY TO SEQ ID 94 [%]
A0A094A2I4	Cluster: Nitrate/nitrite transporter	<i>Pseudogymnoascus</i> sp. VKM F-4281 (FW-2241)	43.54
A0A6A6XX03	Cluster: Nitrate/nitrite transporter	<i>Melanomma pulvis-pyrius</i> CBS 109.77	42.07
A0A2J5HKF9	Cluster: Nitrate/nitrite transporter	<i>Aspergillus</i>	40.56
A0A6A6IDB6	Cluster: Nitrate/nitrite transporter	<i>Trematosphaeria pertusa</i>	42.45
A0A6A5ST79	Cluster: Nitrate/nitrite transporter	<i>Clathrospora elyanae</i>	41.85
A0A3N4JV10	Cluster: Nitrate/nitrite transporter	<i>Choiromyces venosus</i> 120613-1	42.06
A0A6A5XFW9	Cluster: Nitrate/nitrite transporter	<i>Aaosphaeria arxii</i> CBS 175.79	41.43
A0A553IBL5	Cluster: Nitrate/nitrite transporter	<i>Xylaria flabelliformis</i>	40.24
R7Z451	Cluster: Nitrate/nitrite transporter	<i>Coniosporium apollinis</i> (strain CBS 100218)	40.61
A0A139I046	Cluster: Nitrate/nitrite transporter	<i>Pseudocercospora musae</i>	40.24
A0A2T2NBP3	Cluster: Nitrate/nitrite transporter	<i>Corynespora cassicola</i> Philippines	41.84
A0A177E3H1	Cluster: Nitrate/nitrite transporter	<i>Alternaria</i>	42.21
B0XPC3	Cluster: Nitrate/nitrite transporter	<i>Neosartorya fumigata</i>	42.89
A0A150V198	Cluster: Nitrate/nitrite transporter	<i>Acidomyces richmondensis</i> BFW	42.08
A0A3N4KQT8	Cluster: Nitrate/nitrite transporter	<i>Morchella conica</i> CCBAS932	40.44
A0A1Y2DSQ4	Cluster: Nitrate/nitrite transporter	<i>Pseudomassariella vexata</i>	41.05
A0A2V1BSJ8	Cluster: Nitrate/nitrite transporter	<i>Helotiales incertae sedis</i>	42.04
A0A1V1TKZ5	Cluster: Nitrate/nitrite transporter	Fungi	40.24
K1WJ63	Cluster: Nitrate/nitrite transporter	<i>Marssonina brunnea</i> f. sp. multigermtubi (strain MB_m1)	42.06
A0A6A6E950	Cluster: Nitrate/nitrite transporter	<i>Zopfia rhizophila</i> CBS 207.26	41.92
A0A2K1R263	Cluster: Nitrate/nitrite transporter	<i>Sphaceloma murrayae</i>	41.83
A0A2S7Q2X6	Cluster: Nitrate/nitrite transporter	unclassified <i>Rutstroemia</i>	42.32

Co-Factors

[0501] Preferably the recombinant yeast cell further comprises suitable co-factors to enhance the activity of the above mentioned NADH-dependent nitrate reductase and/or NADH-dependent nitrite reductase. Preferred cofactors include flavin adenine dinucleotide (FAD), heme prosthetic groups, and/or molybdenum cofactor (MoCo). Preferably the recombinant yeast cell may therefore further comprise one or more genes encoding enzymes for the synthesis of one or more of flavin adenine dinucleotide (FAD), heme prosthetic groups, and/or molybdenum cofactor (MoCo). For example, the recombinant yeast cell may comprise one or more genes encoding for an enzyme having FAD synthase activity. Preferred co-factors are as exemplified in non-published US patent application U.S. 63/087,642 filed with

the United States Patent Office on 5 Oct. 2020, the contents of which are herewith incorporated by reference.

Recombinant Expression

[0502] The recombinant yeast cell is a recombinant cell. That is to say, a recombinant yeast cell comprises, or is transformed with or is genetically modified with a nucleotide sequence that does not naturally occur in the cell in question. Techniques for the recombinant expression of enzymes in a cell, as well as for the additional genetic modifications of a recombinant yeast cell are well known to those skilled in the art. Typically such techniques involve transformation of a cell with nucleic acid construct comprising the relevant sequence. Such methods are, for example, known from standard handbooks, such as Sam-

brook and Russel (2001) "*Molecular Cloning: A Laboratory Manual*", (3rd edition), published by Cold Spring Harbor Laboratory Press, or F. Ausubel et al., eds., "*Current protocols in molecular biology*", Green Publishing and Wiley Interscience, New York (1987). Methods for transformation and genetic modification of fungal host cells are known from e.g. EP-A-0635574, WO98/46772, WO 99/60102, WO00/37671, WO90/14423, EP-A-0481008, EP-A-0635574 and U.S. Pat. No. 6,265,186.

Fermentation Process

[0503] The invention further provides a process for the production of ethanol, comprising converting a carbon source, preferably a carbohydrate or another organic carbon source, using a recombinant yeast cell as described in this specification, thereby forming ethanol.

[0504] The feed for this fermentation process suitably comprises one or more fermentable carbon sources. The fermentable carbon source preferably comprises or is consisting of one or more fermentable carbohydrates. More preferably, the fermentable carbon source comprises one or more mono-saccharides, disaccharides and/or polysaccharides. For example, the fermentable carbon source may comprise one or more carbohydrates selected from the group consisting of glucose, fructose, sucrose, maltose, xylose, arabinose, galactose, mannose and trehalose. The fermentable carbon source, preferably comprising or consisting of one or more carbohydrates, may suitably be obtained from starch, cellulose, hemicellulose lignocellulose, and/or pectin. Suitably the fermentable carbon source may be in the form of a, preferably aqueous, slurry, suspension, or a liquid.

[0505] The concentration of fermentable carbohydrate, such as for example glucose, during fermentation is preferably equal to or more than 80 g/L. That is, the initial concentration of glucose at the start of the fermentation, is preferably equal to or more than 80 g/L, more preferably equal to or more than 90 g/L, even more preferably equal to or more than 100 g/L, still more preferably equal to or more than 110 g/L, yet even more preferably equal to or more than 120 g/L, equal to or more than 130 g/L, equal to or more than 140 g/L, equal to or more than 150 g/L, equal to or more than 160 g/L, equal to or more than 170 g/L, or equal to or more than 180 g/L. The start of the fermentation may be the moment when the fermentable fermentable carbohydrate is brought into contact with the recombinant cell of the invention.

[0506] The fermentable carbon source may be prepared by contacting starch, lignocellulose, and/or pectin with an enzyme composition, wherein one or more mono-saccharides, disaccharides and/or polysaccharides are produced, and wherein the produced mono-saccharides, disaccharides and/or polysaccharides are subsequently fermented to give a fermentation product.

[0507] Before enzymatic treatment, the lignocellulosic material may be pretreated. The pretreatment may comprise exposing the lignocellulosic material to an acid, a base, a solvent, heat, a peroxide, ozone, mechanical shredding, grinding, milling or rapid depressurization, or a combination of any two or more thereof. This chemical pretreatment is often combined with heat-pretreatment, e.g. between 150-220° C. for 1 to 30 minutes. Subsequently the pretreated material can be subjected to enzymatic hydrolysis to release sugars that may be fermented according to the invention. This may be executed with conventional methods, e.g.

contacting with cellulases, for instance cellobiohydrolase(s), endoglucanase(s), beta-glucosidase(s) and optionally other enzymes. The conversion with the cellulases may be executed at ambient temperatures or at higher temperatures, at a reaction time to release sufficient amounts of sugar(s). The result of the enzymatic hydrolysis is hydrolysis product comprising C5/C6 sugars, herein designated as the sugar composition.

[0508] In one embodiment the fermentable carbohydrate is, or is comprised by a biomass hydrolysate, such as a corn stover or corn fiber hydrolysate. Such biomass hydrolysate may in its turn comprise, or be derived from corn stover and/or corn fiber.

[0509] By a "hydrolysate" is herein understood a polysaccharide-comprising material (such as corn stover, corn starch, corn fiber, or lignocellulosic material, which polysaccharides have been depolymerized through the addition of water to form mono and oligosaccharide sugars. Hydrolysates may be produced by enzymatic or acid hydrolysis of the polysaccharide-containing material.

[0510] A biomass hydrolysate may be a lignocellulosic biomass hydrolysate. Lignocellulose herein includes hemicellulose and hemicellulose parts of biomass. Also lignocellulose includes lignocellulosic fractions of biomass. Suitable lignocellulosic materials may be found in the following list: orchard primings, chaparral, mill waste, urban wood waste, municipal waste, logging waste, forest thinnings, short-rotation woody crops, industrial waste, wheat straw, oat straw, rice straw, barley straw, rye straw, flax straw, soy hulls, rice hulls, rice straw, corn gluten feed, oat hulls, sugar cane, corn stover, corn stalks, corn cobs, corn husks, switch grass, miscanthus, sweet sorghum, canola stems, soybean stems, prairie grass, gamagrass, foxtail; sugar beet pulp, citrus fruit pulp, seed hulls, cellulosic animal wastes, lawn clippings, cotton, seaweed, algae (including macroalgae and microalgae), trees, softwood, hardwood, poplar, pine, shrubs, grasses, wheat, wheat straw, sugar cane bagasse, corn, corn husks, corn hobs, corn kernel, fiber from kernels, products and by-products from wet or dry milling of grains, municipal solid waste, waste paper, yard waste, herbaceous material, agricultural residues, forestry residues, municipal solid waste, waste paper, pulp, paper mill residues, branches, bushes, canes, corn, corn husks, an energy crop, forest, a fruit, a flower, a grain, a grass, a herbaceous crop, a leaf, bark, a needle, a log, a root, a sapling, a shrub, switch grass, a tree, a vegetable, fruit peel, a vine, sugar beet pulp, wheat midlings, oat hulls, hard or soft wood, organic waste material generated from an agricultural process, forestry wood waste, or a combination of any two or more thereof. Algae, such as macroalgae and microalgae have the advantage that they may comprise considerable amounts of sugar alcohols such as sorbitol and/or mannitol. Lignocellulose, which may be considered as a potential renewable feedstock, generally comprises the polysaccharides cellulose (glucans) and hemicelluloses (xylans, heteroxylans and xyloglucans). In addition, some hemicellulose may be present as glucomannans, for example in wood-derived feedstocks. The enzymatic hydrolysis of these polysaccharides to soluble sugars, including both monomers and multimers, for example glucose, cellobiose, xylose, arabinose, galactose, fructose, mannose, rhamnose, ribose, galacturonic acid, glucuronic acid and other hexoses and pentoses occurs under the action of different enzymes acting in concert. In addition, pectins and other pectic substances such as arabinans may make up

considerably proportion of the dry mass of typically cell walls from non-woody plant tissues (about a quarter to half of dry mass may be pectins). Lignocellulosic material may be pretreated. The pretreatment may comprise exposing the lignocellulosic material to an acid, a base, a solvent, heat, a peroxide, ozone, mechanical shredding, grinding, milling or rapid depressurization, or a combination of any two or more thereof. This chemical pretreatment is often combined with heat-pretreatment, e.g. between 150-220° C. for 1 to 30 minutes.

[0511] The process for the production of ethanol may comprise an aerobic propagation step and an anaerobic fermentation step. More preferably the process according to the invention is a process comprising an aerobic propagation step wherein the population of the recombinant yeast cell is increased; and an anaerobic fermentation step wherein the carbon source is converted to ethanol by using the recombinant yeast cell population.

[0512] By propagation is herein understood a process of recombinant yeast cell growth that leads to increase of an initial recombinant yeast cell population. Main purpose of propagation is to increase the population of the recombinant yeast cell using the recombinant yeast cell's natural reproduction capabilities as living organisms. That is, propagation is directed to the production of biomass and is not directed to the production of ethanol. The conditions of propagation may include adequate carbon source, aeration, temperature and nutrient additions. Propagation is an aerobic process, thus the propagation tank must be properly aerated to maintain a certain level of dissolved oxygen. Adequate aeration is commonly achieved by air inductors installed on the piping going into the propagation tank that pull air into the propagation mix as the tank fills and during recirculation. The capacity for the propagation mix to retain dissolved oxygen is a function of the amount of air added and the consistency of the mix, which is why water is often added at a ratio of between 50:50 to 90:10 mash to water. "Thick" propagation mixes (80:20 mash-to-water ratio and higher) often require the addition of compressed air to make up for the lowered capacity for retaining dissolved oxygen. The amount of dissolved oxygen in the propagation mix is also a function of bubble size, so some ethanol plants add air through spargers that produce smaller bubbles compared to air inductors. Along with lower glucose, adequate aeration is important to promote aerobic respiration during propagation, making the environment during propagation different from the anaerobic environment during fermentation.

[0513] By an anaerobic fermentation process is herein understood a fermentation step run under anaerobic conditions.

[0514] The anaerobic fermentation is preferably run at a temperature that is optimal for the cell. Thus, for most recombinant yeast cells, the fermentation process is performed at a temperature which is less than about 50° C., less than about 42° C., or less than about 38° C. For recombinant yeast cell or filamentous fungal host cells, the fermentation process is preferably performed at a temperature which is lower than about 35, about 33, about 30 or about 28° C. and at a temperature which is higher than about 20, about 22, or about 25° C.

[0515] The ethanol yield, based on xylose and/or glucose, in the process according to the invention is preferably at least about 50, about 60, about 70, about 80, about 90, about

95 or about 98%. The ethanol yield is herein defined as a percentage of the theoretical maximum yield.

[0516] The process according to the invention, and the propagation step and/or fermentation step suitably comprised therein can be carried out in batch, fed-batch or continuous mode. A separate hydrolysis and fermentation (SHF) process or a simultaneous saccharification and fermentation (SSF) process may also be applied.

[0517] The recombinant yeast and process according to the invention advantageously allow for a more robust process. Advantageously the process, or any anaerobic fermentation during the process can be carried out in the presence of high concentrations of carbon source. The process, respectively any anaerobic fermentation step therein, is therefore preferably carried out in the presence of a glucose concentration of 25 g/L or more, 30 g/L or more, 35 g/L or more, 40 g/L or more, 45 g/L or more, 50 g/L or more, 55 g/L or more, 60 g/L or more, 65 g/L or more, 70 g/L or more, 75 g/L or more, 80 g/L or more, 85 g/L or more, 90 g/L or more, 95 g/L or more, 100 g/L or more, 110 g/L or more, 120 g/L or more or may for example be in the range of 25 g/L-250 g/L, 30 g/L-200 g/L, 40 g/L-200 g/L, 50 g/L-200 g/L, 60 g/L-200 g/L, 70 g/L-200 g/L, 80 g/L-200 g/L, or 90 g/L-200 g/L.

[0518] For the recovery of the fermentation product existing technologies are used. For different fermentation products different recovery processes are appropriate. Existing methods of recovering ethanol from aqueous mixtures commonly use fractionation and adsorption techniques. For example, a beer still can be used to process a fermented product, which contains ethanol in an aqueous mixture, to produce an enriched ethanol-containing mixture that is then subjected to fractionation (e.g., fractional distillation or other like techniques). Next, the fractions containing the highest concentrations of ethanol can be passed through an adsorber to remove most, if not all, of the remaining water from the ethanol. In an embodiment in addition to the recovery of fermentation product, the yeast may be recycled.

[0519] All patent and literature references cited in the present specification are hereby incorporated by reference in their entirety.

[0520] The following examples are offered for illustrative purposes only, and are not intended to limit the scope of the present invention in any way

EXAMPLES

General Molecular Biology Techniques

[0521] Unless indicated otherwise, the methods used are standard biochemical techniques. Examples of suitable general methodology textbooks include Sambrook et al., *Molecular Cloning, a Laboratory Manual* (1989) and Ausubel et al., *Current Protocols in Molecular Biology* (1995), John Wiley & Sons, Inc.

Hplc Analysis

[0522] HPLC analysis is typically conducted as described in "*Determination of sugars, byproducts and degradation products in liquid fraction in process sample*"; Laboratory Analytical Procedure (LAP, Issue date: Dec. 8, 2006; by A. Sluiter, B. Hames, R. Ruiz, C. Scarlata, J. Sluiter, and D. Templeton; Technical Report (NREL/TP-51042623); January 2008; National Renewable Energy Laboratory.

[0523] After fermentation, samples for HPLC analysis were separated from yeast biomass and insoluble components (corn mash) by passing the clear supernatant after centrifugation through a 0.2 μ m pore size filter

Starter Strains

[0524] Strains were prepared using Ethanol Red® as starting strain. Ethanol Red® is a commercial *Saccharomyces cerevisiae* strain, available from Lesaffre.

[0525] A strain construction approach that can be followed is described in WO2013/144257A1 and WO2015/028582, incorporated herein by reference.

[0526] Expression cassettes from various genes of interest can be recombined in vivo into a pathway at a specific locus upon transformation of this yeast (U.S. Pat. No. 9,738,890 B2). The promoter, ORF and terminator sequences are assembled into expression cassettes with Golden Gate technology, as described by Engler et al (2011) and ligated into BsaI-digested backbone vectors that decorated the expression cassettes with the connectors for the in vivo recombination step. The expression cassettes including connectors are amplified by PCR. In addition, a 5'- and a 3'-DNA fragment of the up- and downstream part of the integration locus was amplified using PCR and decorated by a connector sequence. Upon transformation of yeast cells with these DNA fragments, in vivo recombination and integration into the genome takes place at the desired location. CRISPR-Cas9 technology is used to make a unique double stranded break at the integration locus to target the pathway to this specific locus (DiCarlo et al., 2013, Nucleic Acids Res 41:4336-4343) and WO16110512 and US2019309268. The gRNA was expressed from a multi-copy yeast shuttling vector that contains a natMX marker which confers resistance to the yeast cells against the antibiotic substance nourseothricin (NTC). The backbone of this plasmid is based on pRS305 (Sikorski and Hieter, Genetics 1989, vol. 122, pp. 19-27), including a functional 2 micron ORI sequence. The *Streptococcus pyogenes* CRISPR-associated protein 9 (Cas9) was expressed from a pRS414 plasmid (Sikorski and Hieter, 1989) with kanMX marker which confers resistance to the yeast cells against the antibiotic substance geneticin (G418). The guide RNA and protospacer sequences were designed with a gRNA designer tool (<https://www.atum.bio/eCommerce/cas9/input>).

Example 1: Construction of "Rubisco" Strain (Intermediate Strain IX1)

[0527] In the current example the starter strain was transformed with the cbbM gene encoding the single subunit of ribulose-1,5-biphosphate-carboxylase (RuBisCO) from *Thiobacillus denitrificans* and the genes encoding chaperonins GroEL and GroES from *E. coli* to aid in the proper folding of the RuBisCO protein in the cytosol of *S. cerevisiae*.

[0528] An expression cassette with *S. cerevisiae* TDH3 promoter, cbbM gene and *S. cerevisiae* CYC1 terminators was obtained from plasmid pBTWW002, the expression cassettes with *S. cerevisiae* TEF1 promoter, groEL and *S. cerevisiae* ACT1 terminator was obtained from plasmid pUD232, and the expression cassettes with *S. cerevisiae* TPI1 promoter, groES gene and *S. cerevisiae* PGI1 termi-

nator was obtained from plasmid pUD233, as described by Guadalupe-Medina et al., Biotechnol, Biofuels, 2013, vol. 6, p. 125) and US2019309268.

[0529] These three cassettes were integrated in the INT1 locus between NTR1 (YOR071c) and GYP1 (YOR070c) located on chromosome XV of *S. cerevisiae* with CRISPR-Cas9 using the gRNA expression cassette (as described by Di Carlo et al., Nucleic Acids Res. 2013; pp. 1-8). The cassette was ordered as synthetic DNA cassette (gBLOCK) at Integrated DNA Technologies (Leuven, Belgium) as INT1 gBLOCK and homologous recombination with PCR fragment (5'-INT1) generated with primers BoZ-783 and DBC-18463 genomic DNA of strain CEN.PK113-7D as template and a PCR fragment (3'-INT1) generated with primers DBC-18464 and BoZ-788 using genomic DNA of strain CEN.PK113-7D as template as described in WO2018/114762.

[0530] Diagnostic PCR was performed to confirm the correct assembly and integration at the INT1 locus of the GroES-cbbM-GroEL cassettes. Cas9 and gRNA plasmids were removed with twice a o/n liquid culture with YEPhD, dilution plated on YEPhD plate afterwards for single colonies. Streaks of cells made of the single colonies of the YEPhD plate and re-streaked on YEPhD and YEPhD+G418 (400 mg/ml) and YEPhD+NTC (200 mg/ml). Plasmid free colonies selected which have no growth on the antibiotic plates.

[0531] The above example resulted in intermediate strain IX1 which contained two copies of the GroES, cbbM, and GroEL (see Table 14 for the detailed genotype).

Example 2: Construction of Reference "PRK-Rubisco" Strain (Reference Strain RX2)

[0532] The intermediate strain IX1, as obtained in example 1, was transformed with phosphoribulokinase (prk) from *S. oleacea* (as described by Guadalupe-Medina et al., Biotechnol, Biofuels, 2013, vol. 6, p. 125) and ribulose-1, 5-biphosphate-carboxylase (RuBisCO) from *Thiobacillus denitrificans* encoded by the cbbM gene. The prk gene was expressed by the *S. cerevisiae* ANB1 promoter as described in U.S. Ser. No. 10/689,670 (illustrated by SEQ ID NO: 26 (cbbM), SEQ ID NO: 28 (prk) and SEQ ID NO: 23 (ANB1)) and the *S. cerevisiae* EBO1 terminator (Sc_EBO1t) (illustrated in SEQ ID NO: 79).

[0533] The cbbM gene was cloned between the *S. cerevisiae* ENO2 promoter (illustrated by SEQ ID NO: 80) and *S. cerevisiae* YOX1 terminator (illustrated by SEQ ID NO: 81).

[0534] Cassettes were assembled into a pathway "prk-cbbM" and integrated in the INT14.02 locus, a non-coding region between ORF YNL179c and RPS3 (YNL178W) on chromosome XIV from *S. cerevisiae* using CRISPR-Cas9 with the INT14.02 protospacer (illustrated by SEQ ID NO: 82) and the following two sequences for homologous integration: Sc_INT14.02_FLANK5 (illustrated by SEQ ID NO: 83) and Sc_INT14.02_FLANK3 (illustrated by SEQ ID NO: 84).

[0535] Diagnostic PCR was performed to confirm the correct assembly and integration at the INT14.02 locus of the prk and-cbbM cassettes, and the gRNA and Cas9 plasmids were removed as described above. This resulted in reference strain RX2 which contained two copies of prk, groEL and groES and 4 copies of cbbM (see Table 14 for detailed genotypes).

Example 3: Construction of New Strains NX3 and NX4

[0536] New strains NX3 and NX4 were constructed by transforming the reference strain RX2 obtained in example 2 as follows:

[0537] A DNA fragment was compiled using Golden Gate Cloning (as described for example by Engler et al., “*Generation of Families of Construct Variants Using Golden Gate Shuffling*”, (2011), published in chapter 11 of Chaofu Lu et al. (eds.), *cDNA Libraries: Methods and Applications, Methods in Molecular Biology*, vol. 729, pages 167-180, incorporated herein by reference) and comprised the *S. cerevisiae* ANB1 promoter (illustrated by SEQ ID NO: 23), *Pichia pastoris* TKL1 gene (illustrated by SEQ ID NO: 18) and the *S. cerevisiae* TDH1 terminator. The DNA fragment was named “fragmentA” (illustrated by SEQ ID NO: 73). Fragment A was integrated in the INT95 locus between SOD1 (YJR104C) and ADO1 (YJR105W) located on chromosome X of *S. cerevisiae* reference strain RX2 using CRISPR-Cas9 and INT95 protospacer (illustrated by SEQ ID NO: 85) and the following two sequences for homologous integration: Sc_INT95B_FLANK5 (illustrated by SEQ ID NO: 86) and Sc_INT95B_FLANK3 (illustrated by SEQ ID NO: 87).

[0538] Diagnostic PCR was performed to confirm the correct assembly and integration at the INT95 locus of the promoted *Pichia* TKL1 expression cassette. Plasmid free colonies were selected and this resulted in new strains NX3 and NX4 which contains two copies of the promoted *Pichia* TKL1 expression cassette (see Table 14 for detailed genotypes).

Example 4: Construction of New Strains NX5 and NX6

[0539] New strains NX5 and NX6 were constructed by transforming reference strain RX2 obtained in example 2 with four expression cassettes:

[0540] The first the cassette contains the DNA fragment named “fragmentA”, comprising the *S. cerevisiae* ANB1 promoter, *Pichia pastoris* TKL1 orf and the *S. cerevisiae* TDH1 terminator as referred to in example 3. (illustrated by SEQ ID NO: 73)

[0541] The second cassette contains a DNA fragment named “fragmentB”, comprising the *S. cerevisiae* MYO4 promoter (Sc_MYO4.pro), *S. cerevisiae* DAK1 orf (Sc_DAK1.orf) and *S. cerevisiae* GPM1 terminator (Sc_GPM1.ter). The nucleic acid sequence of the DNA fragment is illustrated in SEQ ID NO: 74.

[0542] The third cassette contains a DNA fragment named “fragmentC”, comprising the *S. cerevisiae* HHH2 promoter (Sc_HHH2.pro), *E. coli* gldA orf (Ec_gldA.orf) and *S. cerevisiae* EFM1 terminator (Sc_EFM1.ter). The nucleic acid sequence of the DNA fragment is illustrated in SEQ ID NO: 75.

[0543] The fourth cassette contains a DNA fragment named “fragmentD”, comprising the *S. cerevisiae* ANB1 promoter (Sc_ANB1.pro_0001), *Zygosaccharomyces rouxii* orf encoding glycerol transporter GLYT (ZYRO0E01210) (Zrou_T5.orf) and *S. cerevisiae* terminator (Sc_TEF1.ter_0001). The nucleic acid sequence of the DNA fragment is illustrated in SEQ ID NO: 76.

[0544] These four cassettes were integrated in the INT95 locus between SOD1 (YJR104C) and ADO1 (YJR105W)

located on chromosome X of *S. cerevisiae* reference strain RX2 using CRISPR-Cas9 using the following sequences for homologous integration as described above with Sc_INT95B_FLANK5 (illustrated by SEQ ID NO: 86) and Sc_INT95B_FLANK3 (illustrated by SEQ ID NO: 87).

[0545] Diagnostic PCR was performed to confirm the correct assembly and integration at the INT95 locus of the four expression cassettes. Plasmid free colonies were selected which resulted in new strains NX5 and NX6 (see Table 14 for detailed genotypes).

Example 5: Construction of New Strains NX7 and NX8

[0546] New strains NX7 and NX8 were constructed by transforming reference strain RX2 obtained in example 2 with the cassette containing the DNA fragment comprising *S. cerevisiae* ANB1 promoter, *Pichia pastoris* TKL1 orf and the *S. cerevisiae* TDH1 terminator, i.e. “fragmentA” (illustrated by SEQ ID NO: 73) as referred to in example 3, in the INT95 locus using CRISPR-Cas9 using Sc_INT95B_FLANK5.acc (SEQ ID NO: 86) and Sc_INT95B_FLANK3. acc (SEQ ID NO: 87) for homologous integration as also illustrated in example 3.

[0547] The GPD2 was deleted using CRISPR-Cas9 and GPD2 protospacer (illustrated by SEQ ID NO: 88) and for a repair DNA fragment (illustrated by SEQ ID NO: 78) for homologous recombination and deleting GPD2. The INT95 integration and GPD2 deletion was checked by diagnostic PCR. Plasmid free colonies were selected which resulted in new strains NX7 and NX8 (see Table 14 for detailed genotypes).

Example 6: Construction of New Strains NX9 and NX10

[0548] New strains NX9 and NX10 were constructed by transforming reference strain RX2 with three expression cassettes:

[0549] The first the cassette contains the DNA fragment named “fragmentA”, comprising the *S. cerevisiae* ANB1 promoter, *Pichia pastoris* TKL1 orf and the *S. cerevisiae* TDH1 terminator as referred to in example 3. (illustrated by SEQ ID NO: 73)

[0550] The second cassette contains a DNA fragment named “fragmentE” comprising *S. cerevisiae* PFY1promoter (Sc_PFY1.pro), *E. coli* gldA orf (Ec_gldA.orf) and *S. cerevisiae* EFM1 terminator (Sc_EFM1.ter). The nucleic acid sequence of the DNA fragment is illustrated in SEQ ID NO: 77.

[0551] The third cassette contains the DNA fragment named “fragmentD”, comprising the *S. cerevisiae* ANB1 promoter (Sc_ANB1.pro_0001), *Zygosaccharomyces rouxii* orf encoding glycerol transporter GLYT (ZYRO0E01210) (Zrou_T5.orf) and *S. cerevisiae* terminator (Sc_TEF1.ter_0001). The nucleic acid sequence of the DNA fragment is illustrated in SEQ ID NO: 76.

[0552] These three cassettes were integrated in the INT95 locus using CRISPR-Cas9 using INT95B_FLANK5 (SEQ ID NO: 86) and INT95B_FLANK3 (SEQ ID NO: 87) for homologous integration. Diagnostic PCR was performed to confirm the correct assembly and integration at the INT95 locus of the three expression cassettes. Plasmid free colonies were selected which resulted in strain NX9 and NX10 (see Table 14 for detailed genotypes).

TABLE 14

<i>S. cerevisiae</i> strains used in this study	
Strain name	Genotype
Parental strain	Wildtype
intermediate strain IX1	INT1::GroES-cbbM-GroEL
reference strain RX2	INT1::GroES-cbbM-GroFL
	INT70::PRK-cbbM
NX3	INT1::GroES-cbbM-GroEL
	INT70::PRK-cbbM
	INT95::pp_TKL1
NX4	INT1::GroES-cbbM-GroEL
	INT70::PRK-cbbM
	INT95::pp_TKL1
NX5	INT1::GroES-cbbM-GroEL
	INT70::PRK-cbbM
	INT95::ppTKL1-DAK1-gldA-T5
NX6	INT1::GroES-cbbM-GroEL
	INT70::PRK-cbbM
	INT95::ppTKL1-DAK1-gldA-T5
NX7	INT1::GroES-cbbM-GroEL
	INT70::PRK-cbbM
	INT95::ppTKL1-DAK1-gldA-T5 medium ΔGPD2
NX8	INT1::GroES-cbbM-GroEL
	INT70::PRK-cbbM
	INT95::ppTKL1-DAK1-gldA-T5 medium ΔGPD2
NX9	INT1::GroES-cbbM-GroEL
	INT70::PRK-cbbM
	INT95::ppTKL1-gldA-T5
NX10	INT1::GroES-cbbM-GroEL
	INT70::PRK-cbbM
	INT95::ppTKL1-gldA-T5

Example 7: Fermentations

[0553] Precultures of the above new “NX3-NX10” strains were made as follows: Glycerol stocks (−80° C.) were thawed at room temperature and used to inoculate 0.2 L mineral medium [as described by Luttk, M L H. et al (2000) “*The Saccharomyces cerevisiae ICL2 Gene Encodes a Mitochondrial 2-Methylisocitrate Lyase Involved in Propionyl-Coenzyme A Metabolism*”. J. Bacteriol. 182:7007-13] supplemented with 2% (w/v) glucose, at pH 6.0 (adjusted with 2M H₂SO₄/4N KOH), in an unbaffled 0.5 L shake-flask. The precultures were incubated for 18 hours at 32° C. and shaken at 200 RPM. After estimating of the yeast cell dry weight (CDW) through OD600 measurement (using an existing CDW vs OD600 calibration line), a quantity of preculture corresponding to the required 0.5 gCDW/L inoculum concentration for the propagation was centrifuged (3 min, 5300×g), washed once with one sample volume sterile demineralized water, centrifuged once more, and resuspended in propagation medium.

[0554] Propagation of the above “NX3-NX10” strains was carried out as follows: A propagation step was performed in 500 mL shake flasks using 100 mL of filtered and diluted corn mash (70% v/v Corn mash: 30% v/v water) supplemented with 1.25 g/L urea and the antibiotics: neomycin and penicillin G with a final concentration of 50 µg/mL and 100 µg/mL respectively. After all additions, the pH was adjusted to 5.0 using 2M H₂SO₄/4N KOH. Glucoamylase (Achieve® T, Novozymes, was dosed at the start of the propagation at a concentration of 0.1 mL/L. All strains were propagated for 6 h at 32° C. and shaken at 200 RPM.

[0555] Main fermentations of the above “NX3-NX10” strains was carried out as follows: A main fermentation step was performed using 200 ml medium in 500 ml Schott bottles equipped with pressure recording/releasing caps (Ankom Technology, Macedon NY, USA), while shaking at 140 rpm and 32° C. pH was not controlled during fermentation. Fermentations were executed with corn mash having increased dry solids content of 36% w/w DS. Subsequently, the corn mash was supplemented with 1.0 g/L urea, and the antibiotics: neomycin and penicillin G with a final concentration of 50 µg/mL and 100 µg/mL respectively; antifoam (Basildon, approximately 0.5 mL/L). After all additions, the pH was adjusted to 5.0 using 2M H₂SO₄/4N KOH. Glucoamylase (Achieve® T, Novozymes) was dosed at the start of the fermentation at a concentration of 0.24 mL/L. The required yeast pitch from propagation to fermentation was 1.5% on fermentation volume. All strains were tested under a condition of high solids, ie. 36% w/w DS).

[0556] Sampling of the fermentation was carried out as follows: Samples were taken from the main fermentations only. Samples for HPLC analysis were taken at 18, 24, 42, 48, and 66 hours. Table 15 summarizes the ethanol production (g/l) at each point in time. Table 16 summarizes the remaining glucose concentration (g/l) at each point in time.

[0557] Conclusions were as follows: The remaining glucose concentration was an indicator for the robustness of the yeast strain. Due to the presence of glucoamylase, glucose was continuously produced. Without wishing to be limited by any kind of theory it is believed that less robust strains will become more inhibited towards the end of the fermentation and as a result a higher concentration of unconverted glucose will be identified in the sample. A more robust strain will become less inhibited towards the end of the fermentation and as a result a lower concentration of unconverted glucose will be identified in the sample.

[0558] As illustrated by Table 16, the yeasts according to the invention, NX3, NX4, NX5, NX6, NX7, NX8, NX9 and NX10, resulted in lower concentrations of unconverted glucose after 48 hours and 66 hours of fermentation than the reference strain.

TABLE 15

Ethanol concentration (g/l) in the fermentation medium									
Time (h)/Strain	RX2	NX3	NX4	NX5	NX6	NX7	NX8	NX9	NX10
0 hours	0.395	0.37	0.363	0.363	0.378	0.372	0.364	0.364	0.342
18 hours	61.80	61.80	61.20	61.80	58.80	61.80	63.00	56.60	59.00
24 hours	86.80	86.00	79.00	82.00	80.20	177.2	56.00	81.00	82.60
42 hours	134.20	137.40	137.60	135.30	132.80	137.20	137.60	133.20	135.70
48 hours	133.30	143.70	145.30	140.20	141.40	141.30	143.00	139.10	138.60
66 hours	141.20	150.10	143.40	143.70	149.60	144.90	143.70	142.80	144.10

TABLE 15-continued

Ethanol concentration (g/l) in the fermentation medium									
Time (h)/Strain	RX2	NX3	NX4	NX5	NX6	NX7	NX8	NX9	NX10
Percentage (%) increase in ethanol yield (g/l) vis-à-vis RX2 at 48 hours.	0.0	7.8	9.0	5.2	6.1	6.0	7.3	4.4	4.0
Percentage (%) increase in ethanol yield (g/l) vis-à-vis RX2 at 66 hours.	0.0	6.3	1.6	1.8	5.9	2.6	1.8	1.1	2.1

TABLE 16

Remaining glucose (grams/liter) concentrations in the fermentation medium									
Time/Strain	RX2	NX3	NX4	NX5	NX6	NX7	NX8	NX9	NX10
0 hours	40.82	40.80	40.81	40.88	40.86	40.90	40.92	40.89	40.94
18 hours	59.80	51.80	50.80	61.80	60.60	46.20	45.60	64.60	58.00
24 hours	50.00	37.80	37.80	52.60	48.60	70.2	24.50	52.00	44.00
42 hours	15.00	6.44	5.22	13.50	17.90	5.72	5.18	15.50	9.79
48 hours	13.80	6.33	4.55	9.42	10.71	5.42	4.56	10.13	7.10
66 hours	18.80	13.16	10.01	14.30	14.50	11.76	11.25	14.60	12.73
Percentage (%) decrease in remaining glucose (g/l) vis-à-vis RX2 at 48 hours.	0.0	-54.1	-67.0	-31.7	-22.4	-60.7	-67.0	-26.6	-48.6
Percentage (%) decrease in remaining glucose (g/l) vis-à-vis RX2 at 66 hours.	0.0	-30.0	-46.8	-23.9	-22.9	-37.4	-40.2	-22.3	-32.3

Example 8. Construction of Phosphoketolase Pathway-Expressing Reference Strain FGG1-pPATH1 (i.e. Reference Strain RX11)

[0559] WO2018/172328 describes the construction of several phosphoketolase pathway-expressing *Saccharomyces cerevisiae* strains, including FGG1-pPATH1 strain. Strain FGG1-pPATH1 had a relevant genotype comprising PKL, PTA and AADH. A summary of the relevant strains for the below examples is provided in below Table 17. The strains can be constructed in a manner as described in WO2018/172328, herewith incorporated by reference.

[0560] As explained in WO2018/172328, in an industrial environment strains such as the FGG1-pPATH1 strain can be affected in its osmotolerance and its stress response to the external environment.

TABLE 17

Phosphoketolase pathway-expressing <i>Saccharomyces cerevisiae</i> strains			
Strain:	FG	Reference strain RX11 (FGG1-pPATH1 of WO2018/172328)	New strain NX12 (including promoted <i>Pichia</i> TKL1)
Relevant genotype:	Wild type	PKL, PTA, AADH	PKL, PTA, AADH INT95::pp_TKL1

Example 9: Construction of New Strain NX12 (Prophetic, According to the Invention)

[0561] New strain NX12 can be constructed by transforming the reference strain RX11 (FGG1-pPATH1 as described in WO2018/172328) as follows:

[0562] A DNA fragment is compiled comprising the *S. cerevisiae* ANB1 promoter (illustrated by SEQ ID NO: 23),

Pichia pastoris TKL1 gene (illustrated by SEQ ID NO: 18) and the *S. cerevisiae* TDH1 terminator. The DNA fragment is named “fragmentA” (illustrated by SEQ ID NO: 73). The DNA fragmentA is assembled using Golden Gate Cloning (as described for example by Engler et al., “*Generation of Families of Construct Variants Using Golden Gate Shuffling*”, (2011), published in chapter 11 of Chaofu Lu et al. (eds.), *cDNA Libraries: Methods and Applications, Methods in Molecular Biology*, vol. 729, pages 167-180, incorporated herein by reference). This expression cassette can be integrated in the INT95 locus between SOD1 (YJR104C) and ADO1 (YJR105W) located on chromosome X of *S. cerevisiae* reference strain RX11 using CRISPR-Cas9 and INT95 protospacer (illustrated by SEQ ID NO: 85) and two sequences for homologous integration: Sc_INT95B_FLANK5 (illustrated by SEQ ID NO: 86) and Sc_INT95B_FLANK3 (illustrated by SEQ ID NO: 87).

[0563] Diagnostic PCR can be performed to confirm the correct assembly and integration at the INT95 locus of the promoted TKL1 expression cassette. Plasmid free colonies are then selected and this results in new strain NX12 which contains two copies of the promoted TKL1 expression cassette (see Table 17 for detailed genotypes).

Example 10: Fermentations (Prophetic)

[0564] Precultures of the above new “NX12” strain can be made as follows: Glycerol stocks (−80° C.) are thawed at room temperature and used to inoculate 0.2 L mineral medium [as described by Luttkik, M L H. et al (2000) “The *Saccharomyces cerevisiae* ICL2 Gene Encodes a Mitochondrial 2-Methylisocitrate Lyase Involved in Propionyl-Coenzyme A Metabolism”. *J. Bacteriol.* 182:7007-supplemented with 2% (w/v) glucose, at pH 6.0 (adjusted with 2M H2SO4/4N KOH), in an unbaffled 0.5 L shake-flask. The precultures are incubated for 18 hours at 32° C. and shaken at 200 RPM. After estimating of the yeast cell dry weight (CDW) through

OD600 measurement (using an existing CDW vs OD600 calibration line), a quantity of preculture corresponding to the required 0.5 gCDW/L inoculum concentration for the propagation is centrifuged (3 min, 5300×g), ished once with one sample volume sterile demineralized water, centrifuged once more, and resuspended in propagation medium.

[0565] Propagation of the above “NX12” strain can be carried out as follows: A propagation step is performed in 500 mL shake flasks using 100 mL of filtered and diluted corn mash (70% v/v Corn mash: 30% v/v water) supplemented with 1.25 g/L urea and the antibiotics: neomycin and penicillin G with a final concentration of 50 µg/mL and 100 µg/mL respectively. After all additions, the pH is adjusted to 5.0 using 2M H₂SO₄/4N KOH. Glucoamylase (Achieve® T, Novozymes, is dosed at the start of the propagation at a concentration of 0.1 mL/L. All strains are propagated for 6 hours at 32° C. and shaken at 200 RPM.

tified in the sample. A more robust strain such as NX12 will become less inhibited towards the end of the fermentation and as a result a lower concentration of unconverted glucose will be identified in the sample.

Example 11. Construction of Acetylating
Acetaldehyde Dehydrogenase-Expressing Reference
Strain IMZ132 (i.e. Reference Strain RX13)

[0569] WO2011/010923 describes an acetylating acetaldehyde dehydrogenase-expressing strain IMZ132, further referred to herein as reference strain RX13. The strain IMZ132 can be constructed in a manner as described in WO2011/010923, herewith incorporated by reference. In addition, the strain IMZ132 was deposited at Centraalbureau voor Schimmelcultures on Jul. 16, 2009 under deposit number CBS125049.

TABLE 18

acetylating acetaldehyde dehydrogenase -expressing <i>Saccharomyces cerevisiae</i> strains		
Strain:	Reference strain RX13 (IMZ132 of WO2011/010923)	New strain NX14 (including promoted <i>Pichia</i> TKL1)
Relevant genotype:	MATa ura3 leu2 gpd1(-1, 1133)::loxP gpd2(-2, 1281)::hphMX4 YEplac181(LEU2) pUDE43(URA3 pTHD3::mhpF (<i>E. coli</i>)):CYC1t)	MATa ura3 leu2 gpd1(-1, 1133)::loxP gpd2(-2, 1281)::hphMX4 YEplac181(LEU2) PUDE43(URA3 pTHD3::mhpF (<i>E. coli</i>)):CYC1t) INT95::pp_TKL1

[0566] Main fermentations of the above “NX12” strain can be carried out as follows: A main fermentation step is performed using 200 ml medium in 500 ml Schott bottles equipped with pressure recording/releasing caps (Ankom Technology, Macedon NY, USA), while shaking at 140 rpm and 32° C. pH is not controlled during fermentation. Fermentations are executed with corn mash having increased dry solids content of 36% w/w DS. Subsequently, the corn mash is supplemented with 1.0 g/L urea, and the antibiotics: neomycin and penicillin G with a final concentration of 50 µg/mL and 100 µg/mL respectively; antifoam (Basildon, approximately 0.5 mL/L). After all additions, the pH is adjusted to 5.0 using 2M H₂SO₄/4N KOH. Glucoamylase (Achieve® T, Novozymes) is dosed at the start of the fermentation at a concentration of 0.24 mL/L. The required yeast pitch from propagation to fermentation is 1.5% on fermentation volume. All strains are tested under a condition of high solids, ie. 36% w/w DS).

[0567] Sampling of the fermentation can be carried out as follows: Samples are taken from the main fermentations only. Samples for HPLC analysis are taken at 18, 24, 42, 48, and 66 hours. Ethanol production (g/l) at each point in time and remaining glucose concentration (g/l) at each point in time can be analyzed.

[0568] Conclusions can be as follows: The remaining glucose concentration is an indicator for the robustness of the yeast strain. Due to the presence of glucoamylase, glucose is continuously produced. Without wishing to be limited by any kind of theory it is believed that less robust strains such as reference strain RX11 will become more inhibited towards the end of the fermentation and as a result a higher concentration of unconverted glucose will be iden-

Example 12: Construction of New Strain NX14
(Prophetic, According to the Invention)

[0570] New strain NX14 can be constructed by transforming the reference strain RX13 (IMZ132 as described in WO2011/010923) as follows:

[0571] A DNA fragment is compiled comprising the *S. cerevisiae* ANB1 promoter (illustrated by SEQ ID NO: 23), *Pichia pastoris* TKL1 gene (illustrated by SEQ ID NO: 18) and the *S. cerevisiae* TDH1 terminator. The DNA fragment is named “fragmentA” (illustrated by SEQ ID NO: 73). The DNA fragmentA is assembled using Golden Gate Cloning (as described for example by Engler et al., “*Generation of Families of Construct Variants Using Golden Gate Shuffling*”, (2011), published in chapter 11 of Chaofu Lu et al. (eds.), *cDNA Libraries: Methods and Applications, Methods in Molecular Biology*, vol. 729, pages 167-180, incorporated herein by reference). This expression cassette can be integrated in the INT95 locus between SOD1 (YJR104C) and ADO1 (YJR105W) located on chromosome X of *S. cerevisiae* reference strain RX13 using CRISPR-Cas9 and INT95 protospacer (illustrated by SEQ ID NO: 85) and two sequences for homologous integration: Sc_INT95B_FLANK5 (illustrated by SEQ ID NO: 86) and Sc_INT95B_FLANK3 (illustrated by SEQ ID NO: 87).

[0572] Diagnostic PCR can be performed to confirm the correct assembly and integration at the INT95 locus of the promoted TKL1 expression cassette. Plasmid free colonies are then selected and this results in new strain NX14 which contains two copies of the promoted TKL1 expression cassette (see Table 18 for detailed genotypes).

Example 13: Fermentations (Prophetic)

[0573] Precultures of the above new “NX14” strain can be made as follows: Glycerol stocks (-80° C.) are thawed at

room temperature and used to inoculate 0.2 L mineral medium (as described by Luttkik, M L H. et al (2000) “*The Saccharomyces cerevisiae ICL2 Gene Encodes a Mitochondrial 2-Methylisocitrate Lyase Involved in Propionyl-Coenzyme A Metabolism*”. J. Bacteriol. 182:7007-13) supplemented with 2% (w/v) glucose, at pH 6.0 (adjusted with 2M H₂SO₄/4N KOH), in an unbaffled 0.5 L shake-flask. The precultures are incubated for 18 hours at 32° C. and shaken at 200 RPM. After estimating of the yeast cell dry weight (CDW) through OD600 measurement (using an existing CDW vs OD600 calibration line), a quantity of preculture corresponding to the required 0.5 gCDW/L inoculum concentration for the propagation is centrifuged (3 min, 5300× g), ished once with one sample volume sterile demineralized water, centrifuged once more, and resuspended in propagation medium.

[0574] Propagation of the above “NX14 strain can be carried out as follows: A propagation step is performed in 500 mL shake flasks using 100 mL of filtered and diluted corn mash (70% v/v Corn mash: 30% v/v water) supplemented with 1.25 g/L urea and the antibiotics: neomycin and penicillin G with a final concentration of 50 µg/mL and 100 µg/mL respectively. After all additions, the pH is adjusted to 5.0 using 2M H₂SO₄/4N KOH. Glucoamylase (Achieve® T, Novozymes, is dosed at the start of the propagation at a concentration of 0.1 mL/L. All strains are propagated for 6 hours at 32° C. and shaken at 200 RPM.

[0575] Main fermentations of the above “NX14 strain can be carried out as follows: A main fermentation step is performed using 200 ml medium in 500 ml Schott bottles equipped with pressure recording/releasing caps (Ankom Technology, Macedon NY, USA), while shaking at 140 rpm and applying a temperature of 32° C. pH is not controlled during fermentation. Fermentations are executed with corn mash having increased dry solids content of 36% w/w DS. Subsequently, the corn mash is supplemented with 1.0 g/L urea, and the antibiotics: neomycin and penicillin G with a final concentration of 50 µg/mL and 100 µg/mL respectively; antifoam (Basildon, approximately 0.5 mL/L). After all additions, the pH is adjusted to 5.0 using 2M H₂SO₄/4N KOH. Glucoamylase (Achieve® T, Novozymes) is dosed at the start of the fermentation at a concentration of 0.24 mL/L. The required yeast pitch from propagation to fermentation is 1.5% on fermentation volume. All strains are tested under a condition of high solids, ie. 36% w/w DS).

[0576] Sampling of the fermentation can be carried out as follows: Samples are taken from the main fermentations only. Samples for HPLC analysis are taken at 18, 24, 42, 48, and 66 hours. Ethanol production (g/l) at each point in time and remaining glucose concentration (g/l) at each point in time can be analyzed.

[0577] Conclusions can be as follows: The remaining glucose concentration is an indicator for the robustness of the yeast strain. Due to the presence of glucoamylase, glucose is continuously produced. Without wishing to be limited by any kind of theory it is believed that less robust strains such as reference strain RX13 will become more inhibited towards the end of the fermentation and as a result a higher concentration of unconverted glucose will be identified in the sample. A more robust strain such as NX14 will become less inhibited towards the end of the fermentation and as a result a lower concentration of unconverted glucose will be identified in the sample.

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SEQUENCE LISTING

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FEATURE Location/Qualifiers

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source 1..680

 mol_type = protein

 organism = *Saccharomyces cerevisiae*

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FEATURE Location/Qualifiers

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source 1..2059

 mol_type = genomic DNA

 organism = *Saccharomyces cerevisiae*

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REGION           1..707
                  note = Transketolase 1 (TKL1) from Komagataella phaffii 7435
source           1..707
                  mol_type = protein
                  organism = Komagataella phaffii

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```

SEQUENCE: 3
MARIPKAVSY NDDIHLVLV TFRCYVLDLV EQYGGGHPGS AMGMVAIGIA LWKYQMKYAP 60
NDPDYFNDRD FVLSNGHVCL FQYLFQHLTG LKEMTVKQLQ SYHSSDYHSL TPGHPPIENP 120
AVEVTTGPLG QGISNAVGMA IGSKNLAATY NRPFPVVDN TIYAIVGDAC LQEGPALESI 180
SLAGHLALDN LIVIYDNNQV CCDGSVDVNN TEDISAKPRA QNWNVIDIVD GSRDVATIVK 240
AIDWAKAETE RPTLINVRTE IGQDSAFGNH HAAHGSALGE EGIRELKTKY GFNPAQKFWF 300
PKEVYDFFAE KPAEGDELVK NWKKLVDSYV KEYPREGQEF LSRVRGELPK NWRTYIPQDK 360
PTEPTATRTS AREIVRALGK NLPQVIAGSG DLSVSILLNW DGVKYFFNPK LQTFCGLGGD 420
YSGRYIEFGI REHSMCAIAN GLAAYNKGTF LPITSTFYMF YLYAAPALRM AALQELKAH 480
IATHDSIGAG EDGPTHQPIA LSSLFRAMPN FYMRPADAT EVAALFEVAV ELEHSTLLSL 540
SRHEVDQYPG KTSAQGAKRG GYVVEDCEGK PDVQLIGTGS ELEFAIKTAR LLRQQKGWKV 600
RVLSFPCQRL FDEQSITVRR SVLRRGEVPT VVVEAYVAYG WERYATAGYT MNTFGKSLPV 660
EDVYKYFGYT PEKIGERVVQ YVNSIKASPK ILYEFHDLKG KPKHDKL 707

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SEQ ID NO: 4      moltype = AA length = 707
FEATURE
REGION           1..707
                  note = Transketolase 2 (TKL2) from Komagataella phaffii 7435
source           1..707
                  mol_type = protein
                  organism = Komagataella phaffii

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```

SEQUENCE: 4
MARIPKAVST QDDIHELVLV TFRCYVLDLV EQYGGGHPGS AMGMVAIGIA LWKYQMKYAP 60
NDPDYFNDRD FVLSNGHVCL FQYLFQHLTG LKEMTVKQLQ SYHSSDYHSL TPGHPPIENP 120
AVEVTTGPLG QGISNAVGMA IGSKNLAATY NRPFPVVDN TIYAIVGDAC LQEGPALESI 180
SLAGHLALDN LIVIYDNNQV CCDGSVDVNN TEDISAKPRA QNWNVIEVEN GSRDVATLVK 240
AIEWAKAENE RPTLINVRTE IGQDSAFGNH HAAHGSALGE EGIRELKAKY GPDVARKFWF 300
PQEVYDFFAE KPAEGDQLVA NWKKLLDEYV KNPQGEEL KARIRGELPK NWKSFIPQDK 360
PTEPTATRTS AREIVRSLGQ NLPQVIAGSG DLSVSILLNW GGVKYYFFNPK LQTFCGLGGD 420
YSGRYIEFGI REHSMCAIAN GLAAYNKGTF LPITSTFYMF YLYAAPALRM AALQELKAH 480
IATHDSIGAG EDGPTHQPIA LSSLFRAMPN FYIIRPADAT EVAALFEVAV ELEHSTLFSL 540
SRHEVEQYPG KTSAEGAKRG GYVVEDCEGK PDVQLIGAGS ELEFAVKTAR LLRQQKGWKV 600
RVLSFPCQRL FDQQLSLAYR SVLRRGEVPT VVVEAYVAYG WERYATAGYT MNTFGKSLPV 660
EDVYKYFGYT PEKIGEKVAA YVNSIKASPK ILYEFTDLKG KPKHDKL 707

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SEQ ID NO: 5      moltype = AA length = 676
FEATURE
REGION           1..676
                  note = Transketolase from Candida tenuis (strain ATCC 10573)
source           1..676
                  mol_type = protein
                  organism = Candida tenuis

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SEQUENCE: 5
MSTDQLAVNT IRLAVDTVS KANSGHGPAP LGLAPAAHV FQNMKFNPDK TKWINRDRFV 60
LSNGHACALL YTMLFLYGYD YTVEDLKSFR QLHSKTPGHP ESTDCPGVEV TTGPLGQGIC 120
NAVGLALAQK QFAATYNKPN YTLSDNFTYC FLGDGCLMEG VSSEASSLAG HLQLGNLIAF 180
WDDNQISIDG STEVAFTEDV IARYKAYAWH ILEVEDGNSD LAGIAKAIEE AKKVTDKPTL 240
IRIKTTIGYG SLAQGTGVH GAPLKPDDVK QLSKFGFDP EEFFAVPKEV TESIYQKHVES 300
NLKVQKEWEA TFAAYKKEYP ELGAEVQRRL DGKLPEGWEK ALPSFTPADK SVATRKLSEG 360
VINALSDILP EFIGGSADLT PSNLTKATGS VDFQPPSTGL GDYSGKYIRY GVREHGMGAI 420
MNGIAAFGAN YKNYGGTFLN FVSYAAGAVR LSALSGHPIT WVATHDSIGL GEDGPTHQPI 480
ETLAHFRALP NCSVWRPADG NEVSAAYKSA VSSTSTPHIL ALSRQNLPLQ EGSSIEKASK 540
GGYVLKEASK PDVILVASGS EVEIAVNAAK LLEAEGKKVS VVSIPDLTF DKQSDYRRLS 600
VLPDGA PVVS IEVMSFPGWS KYSHEQFGLS RFGASGKAED VYKYLEFTPE GVAERASKTI 660
TFYKGKEVLS PLTRAF 676

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SEQ ID NO: 6      moltype = AA length = 678
FEATURE
REGION           1..678
                  note = Transketolase from Debaryomyces
source           1..678

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mol_type = protein
organism = Debaryomyces sp.

SEQUENCE: 6
MSQDLDDLAI STIRLLSVDS VDKAQSGHPG APLGLAPAAH AVWQKMKFNP KDTNWINRDR 60
FILSNHGACA LLYSLLLVLYG YDLTVEDLKQ FRQLNSKTPG HPEALDVPGC EVTTGPLGQG 120
ISNAVGLAMA QKQFGATYNK PDHKISDSYT YAFGLDGCML EGVASETCSL AGHLQLNNLI 180
AFWDDNHISI DGNNTNVAFT DVMLRPFKAYG WHLIEIAEGD TDLDIAIAKAI DEAKKVEDKP 240
TLIRLTTTIG FGSQAQGTGH VHGAAPLKADD IKQLKESMGF NPEETFVIPK EVTDYLLKKHV 300
EENQKVQEDW KKLFPESYKQK YPELGAELQR RLDGKLPENW SDALPTYSPS DSAVATRKLS 360
ETVLTKILPV IPEIIGGSAD LTGSNLTRAA GSVDFQPPAT GLGDYSGRYI RYGVREHGMG 420
AIMNGIAAFG ANYKNYGGTF LNFVSYGAGA VRLSALSHPH VIWVATHDSI GLGEDGPTHQ 480
PIETLAHLRA IPNLSVWRPA DGNETSAAYL QAIESTSTPS IIALTRQNLP QLEGSSIEKA 540
AKGGYTLVKQ DKPDVIVVSS GSEVVISADA AKKLESEGFK AGVVSMPDFL TFDKQPLDYR 600
LSVLPDGVPI MSVEVMSSFG WSKYSHEQFG LNRFGASGKG PEIYKFFFT PEGVADRAAK 660
TIQFYK GKPL LSPLNRAF 678

SEQ ID NO: 7      moltype = AA length = 685
FEATURE          Location/Qualifiers
REGION          1..685
                note = Transketolase from Kuraishia capsulata CBS 1993
source          1..685
                mol_type = protein
                organism = Kuraishia capsulata

SEQUENCE: 7
MSALDIKSIN TIRVLAVDTV FKANSCHPGA PMGLAPAAHV LFQQMKFNPK NPDWINRDRF 60
VLSNGHAVAL LYTLHLHYGY SLTLDLKS FRLNSKTPGH PEADIPGVEV TTGPLGQGI 120
NAVGLAIAQA QFAATYNKPD FKISDNYTYA IFDGCMMEG VASEACSLAG HLQLGNLIAF 180
YDDNHISIDG DTNSAFTEDV TKRLESYGWE VLWVHNGNED LAAIAALEQ AKKSSKPTCI 240
RLTTTIGYGS LLQGSHSVHG SPLKADDIKQ LKEKWGFDP KTFDVPQEVY DSYAKTAAAG 300
AAAEAEWNL FSKYQAAYPE LGAEELARRLK GELPAGWEKN LPTYTPADPA VASRKLSETV 360
LTKIQAALPE LVGGSADLTG SNLTRWPSAV DFQPPSTGLG EWSGRYFRFG VREHGMGAIM 420
NGISAFGANY KAFGGTFLNF VSYAAGAVRL SALSCHPIIW VATHDSIGLG EDGPTHQPIE 480
TLAHFRAIPN LAVWRPADGN ETSAAYYQAL TSTHTPSIIA LSRQNLPLE GSSIEKASKG 540
AYTLVAADKP DVIIAATGSE VYIAVDSVKV LAAQGIKASV VSVPDFHTFD RQTEEYQLSV 600
FPDGVPILSV EVMSTAVWSK YSHEQFGINR FGISGAADI YEYFGFTPEG IAEATKTVTE 660
FYKGKELLSP LHKAFYSKY EPHKY 685

SEQ ID NO: 8      moltype = AA length = 682
FEATURE          Location/Qualifiers
REGION          1..682
                note = Transketolase from Candida arabinofementans NRRL
                  YB-2248
source          1..682
                mol_type = protein
                organism = Candida arabinofementans

SEQUENCE: 8
MSSELELKAV NTIRLLSVDA VSKANSCHPGA APMGMAPAAH ALWTQMKFNP KNPEWINRDR 60
FVLSNGHACA LLYSLLHLFG YPLTLDLKS FRLGSKCPG HPEYETPGIE VTTGPLGQGI 120
CNAIGLMAQA AQFAATFNKP GFTLSDNYTY CFFDGCMMG VASEAMSLA GHLQLGNLIA 180
FYDDNHISID GDTNVAFTED VAARLESYGW DMLYVKDADT DLNAISDAIA QAKKTSKPTC 240
IKITTTIGYG SLLQGSHSVH GAPLKADDVK QLKKKWGFDP EKTPEVPQEV YDLYAKVGEA 300
GAAAEAEWNT LLEKYTVEYP ELGAELKRL AGELPETFIK DLPVYSVDDD AVASRKLSEI 360
VLSKIEKNLP ELVGSADLT GSNLTRWGAA VDFQPPSTGL GEYSGRYFRF VREHGMGAI 420
MNGFSAFGAN YKVGGTFLN FVSYAAGALR LSALSCHPVI WIATHDSIGL GEDGPTHQPI 480
ETLAHFRAIP NLMVWRPADG NETSAAYLKA IESKHTPSVI ALSRQNLPL AGSSIEKAAK 540
GGYVVYEASG DAQLILVSTG SEVAISIDGA KKLAEAGIPT TVVSMDFHT FDKQSLEYRK 600
SILPDGIPIM SVEVMSTAVW AKYSHVQFGI NRFASGKAP EIYKYFEFT EGVASRGKKT 660
VEFFKGSKLL SPLNTPFDSV TM 682

SEQ ID NO: 9      moltype = AA length = 677
FEATURE          Location/Qualifiers
REGION          1..677
                note = Transketolase from Metschnikowia pulcherrima
source          1..677
                mol_type = protein
                organism = Metschnikowia pulcherrima

SEQUENCE: 9
MSDIDQLAIS TIRLLAVDAV AKANSCHPGA PLGLAPAAHA VKKEMKFNPK NPDWVNRDRF 60
VLSNGHACAL LYAMLHLHYG DMSLDDLKQF RQLNSKTPGH PEKFEIPGAE VTTGPLGQGI 120
SNAVGLAIAQ KQFAATFNKD DFAISDSYTY AFLDGCCLME GVASEASSLA GHLQLNNLIA 180
FWDDNKISID GSTEVAFTED VLKRYEAYGW DTLTIEKGD TLEGVQAIAK TAKALKKPTL 240
IRLTIIIGYG SLQQGTGHVH GAPLKPDIDK QLKEKFGFDP TKSFVVPQEV YDYYGTLVKK 300
NQELESEWKN TVESYIQKFP EEGAVLARRL KGELPEDWAK CLPTYTADDK PLATRKLSM 360
ALTKILDVVP ELIGGSADLT GSNLTRAPDM VDFQPPQTGL GNYAGRYIRY GVREHGMGAI 420
MNGIAGFGAG FRNYGGTFLN FVSYAAGAVR LSALSHLPVI WVATHDSIGL GEDGPTHQPI 480
ETLAHFRAIP NISVWRPADG NEVSAAKSA IESTSTPHIL ALTRQNLPL AGSSVEKAST 540

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GGYTVYQTTD KPAVLIIVASG SEVAISIDAA KKLEGEGIKA NVVSLVDFHT FDKQPLDYRL 600
SVLPDGVPIIM SVEVMSSFGW SKYSHEQFGL NRRFGASGKAE DLYKFFDFTP EGVADRAAKT 660
VQFYKKGDLL SPLNRAF 677

SEQ ID NO: 10 moltype = AA length = 677
FEATURE Location/Qualifiers
REGION 1..677
note = Transketolase from *Meyerozyma*
source 1..677
mol_type = protein
organism = *Meyerozyma* sp.

SEQUENCE: 10
MTDIDQLAIA TIRLLSVDQV DAAQSGHPGA PLGLAPAAHA VFKKMKFNPKN NTSWINRDRF 60
ILSNHGACAL LYSMLCLYGY ALNVDDLKRF RQVGSKTPGH PERLDVPGIE VTTGPLGQGI 120
SNAVGLALAQ RQFAATYINKP DFPISDSYVY AFLGDGCLME GVASEASSVA GHLQLGNLIA 180
FWDDNKITID GDTAVAFTEG VEARYRAYGW HTIHIKEGDS DLEGIKAIE EAKKVTDKPT 240
LIRLTTIIGF GSLLQOGTHSV HGAALKADDI KQLKTKFGFN PDEKFVIPPE VTADYAKHVA 300
ENQRIEGEWN KLVADYKQKY PTEGAELARR LEGKLPDQWK DALPTYTPKD KTQATRALSQ 360
TVLEKLYPVI PELIGGSADL TGSNLTRATE AVDFQPPSSK LGDYSGRYIR YGVREHGMGA 420
IMNGIAAFGA NYKNFGGTFL NRVSYAAGAV RLSALSHHPV VVWATHDSIG LGEDGPTHQP 480
VETLAHFRAL PNMSVWRPAD GNEVSAAYIA ALETDSHPSI IALSRQGCPC LEGSSIEKAL 540
KGGYTVYEVE KPDVIVSTG SEVNIAIEGA KQLNNDGIKA NVVSMPDFFT FDQQLSYQL 600
SVLPDGVPIIM SFEVMSTFGW SKYSHEHFGM NRRFGMSGKGP EIYKALGFTP EGVADKAAKT 660
VQFYKNKPVV SPLNKAI 677

SEQ ID NO: 11 moltype = AA length = 675
FEATURE Location/Qualifiers
REGION 1..675
note = Transketolase from *Diutina rugosa*
source 1..675
mol_type = protein
organism = *Diutina rugosa*

SEQUENCE: 11
MSIEEKFIINT IRLLAVIDEVS KANSNGHPGAP LGLAPTAHVI FKNMKFNPKE PNWPNRDRFV 60
LSNGHACALL YAMLCLYGYK TTVDLQKQFR QVNSKTPGHP EADIDGVEVT TGPLGQGISN 120
AVGLAIAQKQ FAARENTDKY TVSDNYTYCI LGDGCMLMEGV SSEASSLAGH LQLDNLIAPW 180
DDNHITIDGD TNVAFTEENV ERYKAYGWNV LHVANGDDDI DAIQKAIDEA KQTKGKPTLI 240
KVTTVIGYGS LLQGTGHVHG APLKADDDAQ LKKKFGFDPE QSFVIPAEVE EFAAKHVTKN 300
LKVYNWINDNL VKNYVAEEKQ GAELQRRIDG KLPADWTKAL PTYTSSDKAL ATRKLSEMLV 360
DNITAEPLPEL IGGSADLTGS NLTKAKGTID FQPPSTGLGD YDGVYIRYGV REHGMGAIMN 420
GIHAFGAGYK AFGGTFLNFV SYASGAVRLS ALSEFPVIWV ATHDSIGLGE DGPTHQPIET 480
LAHFRATPNL MVWRPADGNE VSAAYQAAIE STHTPSIAL TRQNLPLEH SSIEENALKGG 540
YVYVPVENPD LIIVSSSGSV LSIDTAKKM AGEYKVRVY SMAEFLTFDK QPEDYKLSVL 600
PDGVPIMSLE VMSTFGWSKY SHEQFGLNTF GASGKGPVY KKFPGFTVDGV SDKAAKVIEF 660
YQKQLVSLP KRAFN 675

SEQ ID NO: 12 moltype = AA length = 682
FEATURE Location/Qualifiers
REGION 1..682
note = Transketolase from *Pachysolen tannophilus* NRRL Y-2460
source 1..682
mol_type = protein
organism = *Pachysolen tannophilus*

SEQUENCE: 12
MSDLDQLAIN TIRLLAVDAV AKANSNGHPGA PMGLAPAAHV LFQQMRFNPK DPAWLNRDRF 60
VLSNGHACAL LYSLLHLFGY DLGIEDLKQF RQLHSKTPGH PERGIPGVEV TTGPLGQGVG 120
NAVGAIAAQK HVAAKFNKPQ FSLIDGYTYC FPGDGCMMEG VASEAASLAG HLQLGNLICF 180
YDDNHISIDG DTKFAFTEDV GKRLEAYGWH VITVKDANTD LAGIAAAIEE AKKVTDKPSC 240
IRLVTTIGFG SLQAGTGSVH GSPLKPDII QVKKKFGFDP EKTFFYVPKEV YELYAKTSA 300
GAAAEQEWQ LFAKYQQQYP ELGEELARRL TDKLPQGWK ALPRYTPQDK ALATRKLSEM 360
VLTSSIDIVP ELIGGSADLT PSNLTRVNSA VDFQPPSLKI GDYSGRYIRF GVREHGMGAI 420
MNGIAAFGSN LRPFGGTFLN FVSYAAGALR LAALSHHPVI WVATHDSIGL GEDGPTHQPI 480
ETLAHFRAPL NLMVWRPADG NEVSAAYYKS LSEKTPSVI ALSRQNLPLQ EGSSIEKAAK 540
GGYTLVSTDK PDVILVASGS EVSISVEAAK VLANKGIKAN VVSIPDFFTE DRQPRDYQLS 600
VLPNGVPSLS VEVMTAGWS KYSHEQFGIN TFGASGKAAD IYKSFEPTPE GIADRAEKTY 660
NPFYKGHELLS PLNKAFESIH YF 682

SEQ ID NO: 13 moltype = AA length = 692
FEATURE Location/Qualifiers
REGION 1..692
note = Transketolase from *Rasamsonia emersonii* CBS 393.64
source 1..692
mol_type = protein
organism = *Rasamsonia emersonii*

SEQUENCE: 13
MGYTELDQLA INTIRVLATN NSGVTQVDAT SKANAGHPGA PMGLAPVAHV LFNKFMNPNP 60

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KNPDWVNRDR FVLSNGHGCM LQYALLHLFG YNVSIEDLKN FRQIGSITPG HPEAKETPGV 120
 EVTTGPLGQG FSNVGLAIA QAHTAAVFNK PGYELFNNTY YTIQDGCAM EGVACEAASM 180
 AGHLKLGNI AIYDDNHISI DGDTKVAFTE DVLKRFEAYG WHTQWVKDGD HDLQAIEDAI 240
 RKAQEVKDKP SIKLTTTIG YGSKLQGTAG VHGNPLKADD AQSVMKRKFGF NPDQSFVVPQ 300
 EYVDLYGKRA AEGAAKEQEW NALLEKYATE YKAEYDDLRL RLSGKLPEGW EKNLPTYKPT 360
 DAAVASRKLK ETVLEKIHG IPELLSGSAD LTGSNNTRWK NAVDFQPPEY GIGDWSGRYL 420
 RYGVREHMG AVMMGLAAYG TVIPAGGTFL NFVSYAAGAV RLSALSRVR IWATHDSIG 480
 LGEDGPTHQP IETLVHFRAT PNLVWVRPAD GNETSAAAYS ALTAKNTPSI LALTRQNLPQ 540
 LQNSTIEAAL KGGYVILEAP DAAITIVSTG SEVSLSIDAA NYLKEKHGIV ARVVS LPCWE 600
 VFDAQDKAYR LQVIPDGPV LSVEAASSTIG WERYSHEQFG LNRFGASGPY KELYKRFELT 660
 PEGISKRAIA TIEYKKGQKV RSPINRAFEQ VV 692

SEQ ID NO: 14 moltype = AA length = 679
 FEATURE Location/Qualifiers
 REGION 1..679
 note = Transketolase from Kluyveromyces marxianus (strain
 DMKU3-1042 /BCC 29191 / NBRC 104275)
 source 1..679
 mol_type = protein
 organism = Kluyveromyces marxianus

SEQUENCE: 14
 MSQYSDIDRL AINTIRLLSV DQVASANS GH PGAPLGLAPA AHVWVKQMR L NPKNPEWINR 60
 DRFVLSNGHA CALLYSLHL FGYDFSIEDL KQFRHLGSKT PGHPEFELPG VEVTTGPLGQ 120
 GISNAVGLAI AQANFAATYN KPDPFELSDSF TYVFLGDGCL QEGVSSEACS LAGHLKLKLN 180
 IAFYDDNKIT IDGNTNVSFD EDVGKRYEAY GWEVLSVENG NDDLDSISKA LEQAKRSDKP 240
 TLIKLVTTIG FGS LQAGTHA VHGA PLKADD IKQLKTKPGF NPESFVVPQ EYVDLYNKST 300
 IEPGIEANNK WNALFEAYSA KPPELGAEIK RRLAGELPTG WENKLPTYTP KDSAVASRKL 360
 SELLLQNIIE DIPELIGGSA DLTPSNLRT KEVVD FQAPS SGLGDTGRY IRYGVREHGM 420
 GAIMNGISAF GANYKPYGGT FLNFVSYAAG ALRLSALSGH PVIWVATHDS IGLGEDGPTH 480
 QPIETLAHFR ALPNLQVWRP ADGNEVSAAY KVALSKHTP AVIALSRQNL PQLEGSSIEK 540
 AAKGGYVLQD VEQPDIAIVS TGSEVGIAVE AAKVLAEKNI KARI VSLPDF HTFDQQPKY 600
 QLSVLPDGPV ILSVEVLSTS GWAKYSHQQF GLNRF GASGK GPAVYEKPDF TPQGIASRAE 660
 KTVEFYKKGQ VYSPLNTAF 679

SEQ ID NO: 15 moltype = AA length = 676
 FEATURE Location/Qualifiers
 REGION 1..676
 note = Transketolase from Ogataea parapolyomorpha DL-1
 source 1..676
 mol_type = protein
 organism = Ogataea parapolyomorpha

SEQUENCE: 15
 MSQDIQTKAI NTIRVLAADI VAKANS GH PGAPMMAAAH AVYTQMKFNP KNPHWINRDR 60
 FVLSNGHACA LLYTLLHLFG YDLTMDDLKS FRQINSKTPG HPEVGIPGVE VTTGPLGQGI 120
 SNAVGLAIAQ ANFAATYNKP GYTLSDNYTY CFFGDGCMME GVASEAMSLA GHLQLGNLIA 180
 FYDDNKISID GSTEVATED VCGRLEKYGW DIFEVPDADT DVA AISEAIA KAKKTNKPSC 240
 IRIRTTIGY SLNAGSHSVH GAPLKKDDII QLKEKWGFDP EKS FVVPQEV YDYYHKISEA 300
 GAAA EAEWNK LFEAYQKEFP KEGAE LARRL RGELPEGWQN VLPTYQPGDA AVASRKLSEV 360
 CLGKLQAVLP ELVGGADLT PSNLTRWSGA IDFPPESTGL GDYSGKYLRF GVREHGMGAI 420
 INGISAYGAN YKSPGATFLN FVSYASGALR LSALSHHPVI WVATHDSIGL GEDGPTHQPI 480
 ETLAHFRAP NLMVWRPADG NETSAAIYKA ISSTKTPSVL ALSRQNL PQL PGSSIEKALK 540
 GGYTVHEVDN AKLILVATGS EVSLSIDAAK LLTEKGIPTA VVSI PDFFTF DQQPADYKLS 600
 VLPDGPVILS VEVMTSGWA KYAH AQFGLN RFGASGKTAD VYKFFDFTPE GVAKRGEQTY 660
 EFFKGKNLIS PLHTPF 676

SEQ ID NO: 16 moltype = AA length = 681
 FEATURE Location/Qualifiers
 REGION 1..681
 note = Transketolase, synthetic, based on Komagataella
 phaffii
 source 1..681
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 16
 MSDLLAINTI RLLAVDTEQ YGGGHPGAPM GLAPAAHTLF QMRFNPRNP AWINRDRFVL 60
 SNGHACALLY TMLFLYGYDY TIDDLQSYHS SDYHSLTPGH PEALPGEV VTTGPLGQGI 120
 NAVGLAIAQA QLAATYNKP YELFSNYTYA FLGDGCLQEG VAQEAISLAG HLG DGLKLI 180
 FWDNQSISID GDTNVSFTED VPAKFRAQGW EVLSVKDGN DLEGIAALA KAKTNNKPTL 240
 IRLPTIIGY SLQQGTGHVH GSPLKPD DIK QLK KKF GFD P EQNFVVPREV TESYAKHVAD 300
 NQQVEVEWNK LLTAYTKYEP ELGQELHRR L DGKLPENWQK ALPTYTVDDK PVASRKLSEI 360
 VLTSIEKELP ELVGSADLT GSNLTRWPA VDFQPKSTGL GDFSGRYFRF GVREHGMGAI 420
 INGISAYGAN FKAYGGTFLN FVSYASGAVR LSALSGHPII WVATHDSIGL GEDGPTHQPI 480
 ETLAHLRALP NLMVWRPADG NETSAAYLRA IESKHTPSII ALTRQNLPQL EHS SIEKAAK 540
 GGYTVYPVEN PDIILVASGS EVSIAIDGAK KLGTEGVKAS VVSI PDFFTF DSQARSYQLS 600
 VLPDGPVIMS VEVMTSGWA KYSHEQFGIN RFGISGPGE IYKFFEF TAE GVADRASKV 660
 QFYKGKELLS PLNKAFESVH A 681

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SEQ ID NO: 17      moltype = AA  length = 679
FEATURE            Location/Qualifiers
REGION             1..679
                   note = Transketolase 1 (TKL1) from Komagataella phaffii
source             1..679
                   mol_type = protein
                   organism = Komagataella phaffii

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SEQUENCE: 17
MSDLLAINTI RLLAVDTVAK ANSGHPGAPM GLAPAAHTLF QMRFNPRNP AWINRDRFVL 60
SNGHACALLY TMLFLYGYDY TIDDLKSFQ LNSKTPGHPE AELPGVEVTT GPLGQGIANA 120
VGLAIAQAQL AATYNKPNYE LFSNYTYAFL GDGCLQEGVA QEASLAGHL GDGLKLIAPW 180
DDNQISIDGD TNVSFTEDVP AKFRAQGW EV LSVKDGNDL EGIAAALAKA KTTNKPTLIR 240
LPTIIGYGS LQQGTHGVHGS PLKPDDIKQL KKKFGFDPEQ NFVVPREVTE SYAKHVADNQ 300
QVEVEWNKLL TAYTKEYPEL GQELHRRLDG KLPENWQKAL PTYTVDDKPV ASRKLSEIVL 360
TSIEKELPEL VGG SADLTGS NLTRWPDVD FQPKSTGLGD FSGRYFRGV REHGMGAIIN 420
GISAYGANFK AYGGTFLNFV SYASGAVRLS ALSGHPIIWV ATHDSIGLGE DGPTHQPIET 480
LAHLRALPNL MVWRPADGNE TSAAYLRAIE SKHTPSIIAL TRQNLPLEH SSIEKAAKGG 540
YTVYYPVENPD IILVASGSEV SIAIDGAKKL GTEGVKASVV SIPDFTFDS QARSYQLSVL 600
PDGVPIMSV E VMSTFGWSKY SHEQFGINRF GISGPGPEIY KPFEFTAEGV ADRASKVVQF 660
YKGKELLSPL NKAPESVHA 679

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SEQ ID NO: 18      moltype = DNA  length = 2056
FEATURE            Location/Qualifiers
misc_feature        1..2056
                   note = Transketolase 1 (TKL1)
source              1..2056
                   mol_type = genomic DNA
                   organism = Komagataella phaffii

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SEQUENCE: 18
ggctctgaat gtctgatttg ttggccatca acaccatcag attgttggt gttgacactg 60
ttgccaaggc taactctggt caccaggtg ctccaatggg ttgggtcca gctgtcaca 120
ctttgttcaa gcaaatgaga ttcaacccaa gaaacccagc ttgatcaac agagacagat 180
ttgtttgtgc caatggtcac gcttgtgctt tgctatacac catgttgttc ttgtacggtt 240
acgactacac catgtgatgc ttgaaatcct tccgtcaatt gaactctaag actccaggtc 300
atccagaagc tgaattacca ggtgttgaag tcaactaccg tccattgggt caaggtattg 360
ctaacgctgt cggtttggtt attgctcaag ctcaattggc tgctacctac aacaagccta 420
actacgaatt attctcaac tacacttacg ctttcttagg tgacggttgt ttgcaagaag 480
gtgtgtgtca agaagccatc tccttggtg gtcacttggt tgatggtttg aaattaattg 540
ctttctggga cgacaacca atctctattg acggtgacac taacgtctcc ttcaactgaag 600
atgttccagc taagttcaga gcccaagggt ggggaagtct gtcggtcaag gacggttaacg 660
acgacttgga aggtatcgct gccgcttgg ctaaggccaa gactaccaac aagccaactt 720
tgatcagatt gccatccatc atcgggttac gttctttgca acaaggtaac cacggtgttc 780
acggttctcc attgaagcca gatgatata agcaattgaa gaagaagttt ggtttcgacc 840
cagaacaaaa ctttgttgtt ccaagagaag ttaccgaatc ttacgccaa catgttgctg 900
acaaccaaca agtcgaagtt gaatggaaca aattgttgac tgcttacacc aaggaatacc 960
cagaattggg ccaagaattg cacagaagat tggacggtta gttgccagaa aactggcaaa 1020
aggccttgcc aacctacacc gtcgatgaca agccagttgc ctctcgtaaa ttatctgaaa 1080
tcgtcttaac ttccatttgc aaggaattgc cagaattagt cgggtgttct gctgacttga 1140
cgggttccaa cttgactaga tggccagatg ctgtcgactt ccaaccaaag tctaccggtt 1200
taggtgactt ctccggctgt tatctcagat ttggtgttag agaacacggt atgggtgcca 1260
ttattaacgg tatctctgct tacggtgcta acttcaaggc ttacggtggt actttcttga 1320
actttgttct ctacgcttct ggtgtgtgta gattgtccgc tttgtccggt catccaatca 1380
tctgggttgc tactcaacga tccatcggtt tgggtgaaga cgggtccact caccaaccaa 1440
tcgaaacttt ggtctactta agagctttgc caaacttgat ggtctggcgt ccagctgatg 1500
gtaatgaac ctctgctgcc tacttgagag ccattgaatc caagcacact ccatctatca 1560
tcgctttgac cagacaaaac ttgcctcaat tggaaacact tctatcgaa aaagctgcca 1620
agggttggtta caccgtgaaa ccagttgata tattttggtt gcttctggtt 1680
ctgaagtttc cattgccatt gatggtgcca agaagttggg tacccaaggt gtcagggtt 1740
ctgttgtctc catccagat ttcttcacct tcgacttcca agctagatct taccaatgt 1800
ctgttttgcc agatggtgtc ccaatcatgt ccgtcgaagt tatgtctact ttcggttggt 1860
ctaagtactc ccacgaacaa ttcggtatta acagattcgg aatttctggt ccagggtccag 1920
aaatctacaa attcttgaa ttaccgctg aaggtgtcgc tgacagagct tctaagggtg 1980
tccaattcta caagggttaag gaactattat ctccattaaa caaggcttcc gaatccgtcc 2040
acgcgtaaa gagacc 2056

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SEQ ID NO: 19      moltype = AA  length = 679
FEATURE            Location/Qualifiers
REGION             1..679
                   note = Transketolase from Kluyveromyces lactis
source             1..679
                   mol_type = protein
                   organism = Kluyveromyces lactis

```

```

SEQUENCE: 19
MSQYTDIDRL AVSTIRLLAV DQVSAANS GH PGAPLGLAPA AHVIWKQMR L NPKNPEWINR 60
DRFVLSNGHA CALLYSLHL FGYDMSIEDL KHFRHLGSKT PGHPEFELPG VEVTTGPLGQ 120

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-continued

GISNAVGMAI	AQANFAATYN	KPDYELSDSY	TYVFLGDGCL	QEGVSSEASS	LAGHLRLKLN	180
IAFYDDNTIT	IDGNINVSFD	EDVSKRYEAY	GWEVLHVENG	NDDLDVAISKA	LEQAKLSDRP	240
TLIKLETTQI	FGSLNAGSHS	VHGAPLKADD	VKQLKVKFPG	NPEESFVSQ	EVYDLYNPK	300
IEPIGEANKQ	WDALDALTQV	QFVELGAEVK	RRLAGEFPFG	WESKLPPTYTP	EDSVASRKL	360
SEIUVDNISF	TLPELNLGGS	DLTPSNLTRY	KGAVDFQPPY	TGLGDVSYGR	IRYGVREHGM	420
GAIMNGISAF	GANNRYPGYT	FLNFVSYRSG	AVRSLASLGH	PIWVWATHDS	IGLGEDGPTH	480
QPIETLAHFR	ATPNLQVWRP	ADNGEVTAAY	KVALTNKHTP	AIATLSRQNL	PQLGGSSVEK	540
AVKGGYILQD	VQDPDLAIVS	TGSEVGIAVE	AAKVLAEKNI	KARIVSLPDF	HSFGQQSKEY	600
QLSVFPDGPV	LISVEVLATS	GWSKYAHQSF	GLDRFGASGK	GPAVYEKFEP	TPQGIATRAE	660
KTVFVYKGKO	VTSPINTAF					678

```

SEQ ID NO: 20      multype = DNA   length = 2056
FEATURE            Location/Qualifiers
misc_feature       1..2056
                   note = Transketolase from Kluyveromyces lactis
source             1..2056
                   mol_type = genomic DNA
                   organism = Kluyveromyces lactis

```

SEQUENCE: 20	Organism - <i>Kluyveromyces fragilis</i>						
ggctctcgaat	gtctcaatat	actgatatac	atcgtttgge	tgctctctat	atcagattgt	60	
tggtctgtgca	ccaagttcat	gctgctatct	ctggtcaccc	tgggtcccce	ttgggtttga	120	
ctccagctgc	tcacgtctct	tggaacaaca	tgagattaaa	cccaagaac	ccagaatgga	180	
tcaacagaga	cagattctgt	ttgtccaacg	gtcacgctgt	tgctctatta	tactctttgt	240	
tgcattttgt	cggttacgac	atgtctatcg	aagattttgaa	gcacttcaga	catcttgggt	300	
ccaagacccc	aggttcacca	gaatttgaat	tgcagggtgt	tgaaagttacc	acttggtccat	360	
ttgggtcaagg	tatctccaat	gctgttggtt	tggttatgct	ccaagctaac	ttcgccgcta	420	
cttacaacaa	gccagactac	gaattgtctg	attctttacac	ctacgctctc	ttgggtgacg	480	
gtgttttgca	agaaggtgtt	tcctctgaag	ctcttctctt	agctggtcat	ttgagatgta	540	
agaacttgat	tgctttctac	gacgacaacc	aaatcaccat	tgatggtaac	atcaacgctt	600	
ctttcgatga	agatgtttcc	aagagatacg	aagctcacgg	ttgggaagtc	ttgcacgttg	660	
aaaacggtaa	tgtgatgttg	gatgtctatc	aacaggtctt	ggaaacagcc	aagtttgtctg	720	
acagacacct	cttgatcaaa	ttgaccacca	ccatcggttt	cggttctctt	aacgctggtt	780	
ctcaactctg	tcacggtgct	ccattaaagg	ccgatgacgt	taagcaattg	aaggtcaaat	840	
tcggtttcaa	cccagaagaa	tctttcgttg	ttctccaaag	agtctacgac	ttgtacaaca	900	
agtcctactat	tgaacacaggt	atcgaagata	acaagcaatg	ggatgccttg	ttggacgctt	960	
acgttggtca	attccagaaa	ttgggtgctg	aagtc aaag	aagat ttggc	ggtgaatttc	1020	
cagaagggtg	ggaatctaa	ttgccaaact	acactccaga	agattctgcc	gtcgcttcca	1080	
gaaagttgtc	cgaaattgtc	tttgacaacg	tttccgacac	tttgccagaa	ttgtcgaagt	1140	
gttctgctga	cttgactcca	tccaacttaa	ccagatccaa	gggtgtgctc	gattttccag	1200	
cccacaatca	tggttttggt	gactactccg	gtcggtacat	gcagatcggt	gttagagaac	1260	
acggttatgg	tgtatcatg	aacggtatct	ctgccttcgg	tgctaactac	agaccatcag	1320	
gtggtacttt	cttgaacttc	gtttcttacg	cttctggctg	tgttagatta	tcgcgtttgt	1380	
ttgtgcatcc	agttatctgg	gtcgccactc	acgactctat	tggtctagg	gaagacggct	1440	
caactccaca	accaattcgaa	accttagccc	acttcagagc	tatcccgaat	ttacaagttt	1500	
ggcgtccagc	cgacggtaac	gaagtctctg	ctgtctacaa	ggtttgcttg	accaaacggt	1560	
acactccagc	tatcatgtct	ttatctagac	aaaacttgcc	tcaattgcaa	gggtctctccg	1620	
tcgaaaaagg	tgtcaaaagt	gggtacattt	tgcaagacgt	cgaccaaccca	gacttggcta	1680	
tcgtttccac	cggttctgaa	gttggttatt	ctgtcgaaag	tgcccaaggt	ttggccgaaa	1740	
agaacattaa	gggtagaatt	gtttctttac	cagacttcca	ctccatttgg	caacaatacca	1800	
aggaatacca	attgtctgtt	ttcccagacg	gtgtcccaat	tttgtcgtt	gaagttttgt	1860	
ctacctccgg	ttggtccaaa	tacgctcacc	aatctctcgg	tttgatcgct	tcgggtttgt	1920	
ccggtaaagg	tccagctgtc	tacgaaaagt	tcgaattcac	cccacaaggt	attgctacca	1980	
gagctgaaaa	cagacctgaa	ttctacaagg	gtaagcaagt	catttctcca	ttgaacactg	2040	
ctttttaaag	gagacc					2056	

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SEQ ID NO: 21          moltype =   length =
SEQUENCE: 21
000
```

```
SEQ ID NO: 22      moltype = DNA  length = 12
FEATURE            Location/Qualifiers
misc_feature       1..12
                   note = Promoter motif
source             1..12
                   mol_type = other DNA
                   organism = synthetic construct
```

SEQUENCE: 22
aaaaattggtt ga 12

```
SEQ ID NO: 23      moltype = DNA  length = 622
FEATURE            Location/Qualifiers
misc_feature       1..622
                   note = Promoter ANB1 from Saccharomyces cerevisiae
source             1..622
                   mol_type = genomic DNA
                   organism = Saccharomyces cerevisiae
```

-continued

SEQUENCE: 23

```

gggtctcggtg cagatcggtg aaaacatctt gatcttgctc ccgggaattt tagattcagg 60
taggaaattg attacatcaa tactgttacc ctgaatcata ttccagcatg tcgtctcaca 120
cggaatatata attcatttct tgggtttcca aaaaaatttt catttttttt cacttttttg 180
tttcgtctctc cttttttttt tttttttttt attttttttc ctgtgttcac cttttttttt 240
ttcagttgac atctttctgc attcttttct gtgttttttt tttttttttt cgtttttcca 300
ttgttcgttc gttgcctgtt ttttcgccct attgttctcg agcctaaaaa tttttctctt 360
tcctgcttcc ctttctctgt tcaaaagtcc ctattccatt gttctctttg gtaaacctcat 420
tggttgctgga actcagatat attcaggtca atttactgta cttcaattga cttttttctt 480
gaaatttcaa cttgcctttt caactgttc ttctttttta atcttattct acactttagt 540
tcccttacct tgttctaat tattgtctag caaaaagaaa acatacacct atttcattca 600
cacactaaaa caatggggaga cc 622

```

SEQ ID NO: 24

moltype = DNA length = 825

FEATURE

Location/Qualifiers

misc_feature

1..825

note = Promoter DAN1 from *Saccharomyces cerevisiae*

source

1..825

mol_type = genomic DNA

organism = *Saccharomyces cerevisiae*

SEQUENCE: 24

```

agaatgcaag ggcaaacagg cctgaggtta ctttaaaata taatgtacct tgaaaactat 60
gtatacattg catttattaa gcagtagggg attttggttg caagaaaaata aagctcaaaa 120
tatctttgga gtttgacaat agtatagaaa aaatagctca gcaaccttaa taaaaaagg 180
gtcattgggc aaagcgtttt aaaaagattt aagtggtagt gtttaaagt aaagagattg 240
atatagttaa aaattgttga gctcaattca cgtcggatcc ggcatcgtt tttctcaat 300
cctcacgtgc tttctcgtt tgagtcaaaa agttcatatg atgctatctc ccgcttatct 360
tattagtcga aaatggggag aatttcctat tttatctgtc gtttagcaca tacggccagg 420
aaaaatcata agatttcgcc gaacgacggg gtcaattcgt cctttttgta cacatcgtt 480
aatttatgag gaaaaattga tgaacgtatc ctccgtagac gtcctctgta aaagtttcat 540
gtttcctgcg cgttctcttg ataggcaata aaacaataca acgcgtgctt tgaaaaatgc 600
caagatctat acgaggcctc taacaaaaa cgttcaggga acagagaagt ctaaaaatgc 660
aaaaggggtc ctgggtactc attgaataga aatgattgaa aatactcgtt ataaaaatgc 720
acgactaaat gatactattt ttatgtcgac acggtactat ttcttctttt gcagataaaa 780
gtgtagcaga taaaagtgtg gcatactaaa tatatacccc aagta 825

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SEQ ID NO: 25

moltype = AA length = 459

FEATURE

Location/Qualifiers

REGION

1..459

note = Ribulose-1,5-biphosphate carboxylase oxygenase (RuBisCO)

source

1..459

mol_type = protein

organism = *Thiobacillus denitrificans*

SEQUENCE: 25

```

MDQSARYADL SLKEEDLIK GRHILVAYKM KPKSGYGYLE AAHFFAESS TGTNVEVSTT 60
DDFTKGVDAL VYYIDEASED MRIAYPLELF DRNVTDGRFM LVSFLTLAIG NNQMGDIEH 120
AKMIDFYVPE RCIQMFQDGA TDISNLWRIL GRPVVNGGYI AGTIIKPKLG LRPEPPAKAA 180
YQFWLGGDFI KNDEPQGNQV FCPLKKVLPL VYDAMKRAQD DTGQAKLFMS NITADDDHYEM 240
CARADYALEV FGPADAKLAF LVDGYVGGPG MVTARRQYP GQYLHYHRAG HGAVTSPSAK 300
RGYTAFVLAK MSRLQGASGI HVGTMGYGKM EGEEDDKIIA YMIERDECQG PVYFQKWYGM 360
KPTTPIISGG MNALRLPGFF ENLGHGNVIN TAGGGSYGHI DSPAAGAISL RQSYECWKQG 420
ADPIEFAKEH KEFARAFESF PKDADKLFPG WREKLGVHS 459

```

SEQ ID NO: 26

moltype = DNA length = 1380

FEATURE

Location/Qualifiers

misc_feature

1..1380

note = synthetic; based on cbbM gene from *Thiobacillus denitrificans*, codon optimized for *S. cerevisiae*

source

1..1380

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 26

```

atggatcaat ctgcaagata tgctgacttg tctttaaagg aagaagattt gattaaaggt 60
ggtagacata ttttgggtgc ttacaaaatg aaacaaaat ctggttatgg ttatttggaa 120
gctgctgctc attttctgct tgaatcttct acaggtacaa atgttgaagt ttctactaca 180
gatgatttta caaaagggtg tgatgcttta gtttactaca tcgatgaagc ttcagaagat 240
atgagaattg cttatccatt ggaattattc gacagaaatg ttactgacgg aagattcatg 300
ttagtttctt ttttgacttt ggctattggt aacaatcaag gaatgggaga tatagaacat 360
gcaaaaatga tagattttta cgttccagaa agatgtattc aaatgtttga tgggtccagct 420
acagatatatt ctaatttttg gagaattttg ggtagaccag tagttaatgg tggttatatt 480
gctggtacta ttattaagcc aaaattgggt ttaagaccag aaccatttgc taaagctgct 540
tatcaatttt ggttgggtgg agattttatc aagaatgacg aaccacaagg taatcaagtt 600
ttttgtccat tgaaaaaagt tttgccattg gtttacgatg ctatgaaaag agcacaagat 660
gatactggtc aagcaaaaat gttttctatg aatattactg cagacgatca ttatgaaatg 720
tgtgcaagag ctgattatgc tttggaagtt ttcggtccag atgcagataa attggctttt 780

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ttggtagatg	gttacgttgg	aggtccagga	atggttacta	ctgctagaag	gcaatatacct	840
ggccaatatt	tgcaattatca	tagagcaggt	cacggtgctg	ttactttctcc	atctgctaaa	900
agaggttata	ctgcttttgt	tttggctaaa	atgtctagat	tgcaaggcgc	ttcaggtatt	960
catgttggtg	ctatgggtta	tggaaaaatg	gaaggagaag	gcgacgataa	gattattgct	1020
tatatgatag	aaagggacga	atgtcaaggt	ccagtttatt	ttcaaaaatg	gtacgggatg	1080
aaaccaacta	ctccaattat	ctccggagga	atgaatgctt	tgagattgcc	tggttttttc	1140
gaaaatttgg	gtcatggtta	cggtattaat	actgcagggtg	gtggttctta	cggtcatatt	1200
gattctcctg	ctgctgggtg	tattttcttg	agacaatctt	acgaatgttg	gaaacaaggt	1260
gcagatccaa	ttgaatttgc	taaggaacat	aaggaatttg	caagagcttt	tgaatctttt	1320
ccaaaagatg	ctgataagtt	atttccagga	tggagagaaa	aattgggagt	tcattcttaa	1380

SEQ ID NO: 27 moltype = AA length = 352
 FEATURE Location/Qualifiers
 REGION 1..352
 note = *S. oleracea* phosphoribulokinase (prk)
 source 1..352
 mol_type = protein
 organism = *Spinacia oleracea*

SEQUENCE: 27
 MSQQQTIVIG LAADSGCGKS TFMRLTSVF GGAAEPPKGG NPDSNTLISD TTVICLDDF 60
 HSLDRNGRKV EKVLTALDPKA NDFDLMYEQV KALKEGKAVD KPIYNHVSGL LDPPELIQPP 120
 KILVIEGLHP MYDARVRELL DFSIYLDISN EVKFAWKIQR DMKERGHSLE SIKASIESRK 180
 PDFDAYIDPQ KQHADVVIEW LPTELIPDDD EGKVLVRVMI QKEGVKFFNP VYLFDEGSTI 240
 SWIPCGRKLT CSYPGIKFSY GPDTFYGNFV TVVEMDGMFD RLDELIYVES HLSNLSTKPY 300
 GEVTQQMLKH QNPFSGSNGT GFPTIIGLK IRDLFEQLVA SRSTATATAA KA 352

SEQ ID NO: 28 moltype = DNA length = 1059
 FEATURE Location/Qualifiers
 misc_feature 1..1059
 note = synthetic; based on prk gene from *S. oleracea* -
 codon optimizedfor *S. cerevisiae*
 source 1..1059
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 28
 atgtcacaaac aacaaacaat tgtgattggt ttagcagcag attcaggttg tggtaaagagt 60
 acattcatga ggaggttaac aagtgttttc ggtggcgcg cgagccacc aaagggtggt 120
 aaccagatt caaacacatt gattagtgc actactactg ttatctgttt ggatgatttt 180
 cattcccttg atagaaatgg caggaaagtg gaaaaagtta ctgctttaga cccaaaagct 240
 aatgattttg atcttatgta tgaacaagtt aaggctttga aagaaggtta agctgttgat 300
 aaacctattt ataatcatgt ttctggtttg ttggaccctc ctgagcttat tcaacctcct 360
 aagatcttgg tcattgaagg gttacacccc atgtatgacg cacgtgtgag ggaattgcta 420
 gacttcagca tctacttgga cattagcaat gaagttaaat ttgcttgga aattcagaga 480
 gacatgaaag aaagaggaca cagtcttgaa agcatcaaa cagattatga atccagaaag 540
 ccagattttg atgcttacat tgaccacaaa aagcagcatg ctgatgtagt gattgaagta 600
 ttgccaactg aactcatttc tgatgatgat gaaggcaaa tggtgagagt aaggatgatt 660
 cagaaagaag gagtcaagtt tttcaacca gtttacttgt ttgatgaagg atctaccatt 720
 tcatggatcc catgtgttag aaaattaaca tgttcttacc ctggtatcaa attttcctat 780
 ggccagagca cctctatgag caacgaggtg acagtagtag agatggatgg gatgtttgac 840
 agattagacg aactaatcta cgtcgaaaag catttgagca atctatcaac caagttttat 900
 ggtgaagtca ctcaacaaat gttgaagcac caaaatttcc caggaaagca caatggaact 960
 ggtttcttcc aaaccataat tggattgaag atcagagact tgttcagaga gctcgttgct 1020
 agcaggtcta cagcaactgc aaacagtgct aaagcctag 1059

SEQ ID NO: 29 moltype = AA length = 548
 FEATURE Location/Qualifiers
 REGION 1..548
 note = Translated protein sequence, based on GroEL from *E. coli*
 source 1..548
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 29
 MAAKDVKFGN DARVKMLRGV NVLADAVKVT LGPKGRNVVL DKSPGAPTIT KGVSVVAREI 60
 ELEDKFENMG AQMVKEVASK ANDAAGDGTT TATVLAQAI TEGLKAVAAG MNPMDLKRGI 120
 DKAVTAABEE LKALSVPCSD SKAIAQVGTI SANSDETVGK LIAEAMDKVG KEGVITVEDG 180
 TGLQDELDDV EGMQFDRGYL SPYFINKPET GAVELESPFI LLADKKISNI REMLPVLEAV 240
 AKAGKPLLI AEDVEGEALA TLVVNTMRGI VKVAVKAPG FGDRRKAMLQ DIATLTGGTV 300
 ISEEIGMELE KATLEDLQGA KRVVINKDIT TIIDGVGEEA AIQGRVAQIR QQIEEATSDY 360
 DREKLQERVA KLAGGVAIVK VGAATEVEMK EKKARVEDAL HATRAAVEEG VVAGGGVALI 420
 RVASKLADLR GQNEQNVGI KVALRAMEAP LRQIVLNCGE EPSVVANTVK GGDGNYGYNA 480
 ATEEYGNMID MGILDPTKVT RSALQYAASV AGLMITTECM VTDLPKNDAA DLGAAGMGMG 540
 MGGMGGM 548

SEQ ID NO: 30 moltype = DNA length = 1647
 FEATURE Location/Qualifiers

-continued

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misc_feature      1..1647
                  note = Synthetic, based on GroEL from E. coli - codon
                  optimized
source            1..1647
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 30
atggctgcta aggcaggttaa gttcggtaac gacgctagag ttaagatgtt gagaggtgtt 60
aacgtttttg ctgacgctgt taaggttact ttgggtccaa agggtagaaa cgttgttttg 120
gacaagtctt tcggtgctcc aactatcact aaggacgggt tttctgttc tagagaaatc 180
gaattggaag acaagttcga aaacatgggt gctcaaatgg ttaaggaagt tgcttctaag 240
gctaacgacg ctgctggtga cggctactact actgctactg ttttggtca agctatcatc 300
actgaagggt tgaaggctgt tgctgctggt atgaaccaa tggacttgaa gagaggtatc 360
gacaaggctg ttactgctgc tgttgagaa tgtaaggctt tgtctgttc atgttctgac 420
tctaaggcta tcgctcaagt tggctactatc tctgctaact ctgacgaaac tgttggttaag 480
ttgatcgctg aagctatgga caaggttggt aagggaagggt ttatcactgt tgaagacgggt 540
actggtttgc aagacgaatt ggaagttggt gaaggtatgc aattcgacag aggttacttg 600
tctccatact tcatcaacaa gccagaaact ggtgctgttg aattggaatc tccattcatc 660
ttgttgctg acaagaagat ctctaactc agagaaatgt tgccagtttt ggaagctggt 720
gctaaggctg gtaaggcatt gttgatcatc gctgaagacg ttgaagggtg agctttggct 780
actttggttg ttaacactat gagaggtatc gttgaagggtg ctgctgttaa ggctccaggt 840
ttcggtgaca gaagaaaggc tatgttgcaa gacatcgcta cttgactgg tggtaactgtt 900
atctctgaag aaatcggtat ggaattggaa aaggctactt tggaagactt gggtaagct 960
aagagagttg ttatcaacaa ggaactact actatcatcg acggtgttgg tgaagaagct 1020
gctatccaag gtatagttgc tcaaatcaga caacaaatcg aagaagctac ttctgactac 1080
gacagagaaa agttgcaaga aagagttgct aagttggctg gtggtgttgc tgttatcaag 1140
gttggtgctg ctactgaagt tgaatgaag gaaaagaagg ctatagttga agacgctttg 1200
cacgctacta gagctgctgt tgaagaagggt gttgttctg gtggtgtgtg tgctttgatc 1260
agagttgctt ctaagttggc tgacttgaga ggtcaaaacg aagacaaaaa cgttggtatc 1320
aaggttgctt tgagagctat ggaagctcca ttgagacaaa tcgttttgaa ctgttggtgaa 1380
gaacctatcg ttgttgctaa cactgttaag ggtggtgacg gtaactacgg ttacaacgct 1440
gctactgaag aatacggtaa catgatcgac atgggtatct tggaccacaa taaggttact 1500
agatctgctt tgcaatacgc tgcttctgtt gctggttga tgatcactac tgaatgtatg 1560
gttactgact tgcaaaagaa cgacgctgct gacttgggtg ctgctggtgg tatgggtggt 1620
atgggtggtg tgggtggtat gatgttaa 1647

SEQ ID NO: 31      moltype = AA length = 97
FEATURE            Location/Qualifiers
REGION            1..97
                  note = Translated protein sequence, based on GroES of E.coli
source            1..97
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 31
MNIRPLHDRV IVKRKEVETK SAGGIVLTGS AAAKSTRGEV LAVGNRILE NGEVKPLDVK 60
VGDIVFNDG YGVKSEKIDN EEVLIMSED ILAIVEA 97

SEQ ID NO: 32      moltype = DNA length = 294
FEATURE            Location/Qualifiers
misc_feature      1..294
                  note = Synthetic, based on GroES from E. coli - codon
                  optimized
source            1..294
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 32
atgaacatca gaccattgca cgacagagtt atcgtaaga gaaaggaagt tgaactaag 60
tctgctgggt gtatcgtttt gactggttct gctgctgcta agtctactag aggtgaagtt 120
ttggctgttg gtaacggtg aatcttggaa aacggtgaag ttaagccatt ggacgttaag 180
gttggtgaca tcgttatctt caacgacggt tacggtgtta agtctgaaaa gatcgacaac 240
gaagaagttt tgatcatgtc tgaatctgac atcttggcta tcgttgaagc ttaa 294

SEQ ID NO: 33      moltype = AA length = 825
FEATURE            Location/Qualifiers
REGION            1..825
                  note = B. animalis xylulose-5P/fructose-6P
                  phosphoketolase(Bani_XFP.orf) amino acid sequence
source            1..825
                  mol_type = protein
                  organism = Bifidobacterium animalis

SEQUENCE: 33
MTNPVIGTPW QKLDPRVSEE AIEGMDKYWR VANYMSIGQI YLRSNPLMKE PFTRDDVKHR 60
LVGHWGTTTP LNFLLAHINR LIADHQQNTV FIMGPGHGGP AGTAQSYIDG TYTEYYPNIT 120
KDEAGLQKFF RQFSYPPGIP SHFAPETPGS IHGEGELGYA LSHAYGAIMD NPSLFVPCII 180
GDGEAETGPL ATGWQSNKLV NPTDGIIVLP ILHLNGYKIA NPTILARISD EELHDFFRGM 240
GYHPYEFVAG PDNEDHLSIH RRFAPLFFETI FDEICDIKAA AQTDDMTRPF YPMLIFRTPK 300

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GWTCPKFIDG KKTEGWSRAH QVPLASARDT EAHFEVLKGW MESYKPEELF NADGSIKEDV 360
TAFMPKGELR IGANPNANGG RIREDLKLPE LDQYEITGVK EYGHGWGQVE APRSLGAYCR 420
DIIKNPDSF RVFGPDETAS NRLNATYEVK KKQWDNGYLS ALVDENMAVT GQVVEQLSEH 480
QCEGFLEAYL LTGRHGIWSS YESFVHVIDS MLNQHAKWLE ATVREIPWRK PISSVNLVLS 540
SHVWRQDHNG FSHQDPGVTS VLLNKTFNND HVTNIYFATD ANMLLAIAEK CFKSTNKINA 600
IFAGKQPAAT WITLDEVRAE LEAGAAEWKW ASNAKSNDDEV QVVLAAAGDV PTQEIMAASD 660
ALNKMGIKFK VVNVVDLKL QSSKENDEAM SDEDFADLFT ADKPVLFAH SYAQDVRGLI 720
YDRPNHNDFT VVGKEQGST TTPFDMVRVN DMDRYALQAK ALELIDADKY ADKINELNEF 780
RKTAQFAVD NGYDIPEFTD WVYPDVKVDE TSMLSATAAT AGDNE 825

SEQ ID NO: 34      moltype = AA length = 825
FEATURE           Location/Qualifiers
REGION            1..825
                  note = B. adolescentis xylulose-5P/fructose-6P
                  phosphoketolase(Bado_XFP.orf) amino acid sequence
source            1..825
                  mol_type = protein
                  organism = Bifidobacterium adolescentis

SEQUENCE: 34
MTSPVIGTPW KKLNAVSEEE AIEGVDKYWR AANYLSIGQI YLRSNPLMKE PFTREDVKHR 60
LVGHWGTTTPG LNFLIGHINR LIADHQQNTV IIMGPGHGGP AGTAQSYLDG TYTEYFPNIT 120
KDEAGLQKFF RQFSYPGGIP SHYAPETPGS IHEGGELGYA LSHAYGAVMN NPSLFVPAIV 180
GDGEAETGPL ATGWQSNKLI NPTDGTGIVLP ILHLNGYKIA NPTILSRISD EELHEFFHGM 240
GYEPYEFVAG FDNEDHLSIH RRFAPLFTETV FDEICDIKAA AQDDMTTRPF YPMIIFRTPK 300
GWTCPKFIDG KKTEGWSRAH QVPLASARDT EAHFEVLKMW LESYKPEELF DENGAVKPEV 360
TAFMPTGELR IGENPNANGG RIREELKLPE LEDYEVKEVA EYGHGWGQLE ATRRLGVYTR 420
DIIKNPDSF RIFGPDETAS NRLQAAYDVT NKQWDAGYLS AQVDEHMAVT GQVTEQLSEH 480
QMEGFLEAYL LTGRHGIWSS YESFVHVIDS MLNQHAKWLE ATVREIPWRK PISSMNLVLS 540
SHVWRQDHNG FSHQDPGVTS VLLNKCFNND HVIGIYFPVD SNMLLVAEK CYKSTNKINA 600
IAGKQPAAT WITLDEARAE LEKGAEEWKW ASNVKSNDDEA QIVLAATGDV PTQEIMAASD 660
KLDAMGIKFK VVNVVDLKL QSAKENNEAL SDEEFAELFT EDKPVLFAH SYARDVRGLI 720
YDRPNHNDFT VHGKEQGST TTPYDMVRVN NIDRYELQAE ALRMIDADKY ADKINELEAF 780
RQEAQFAVD NGYDHPDYTD WVYSGVNTNK QGAISATAAT AGDNE 825

SEQ ID NO: 35      moltype = AA length = 825
FEATURE           Location/Qualifiers
REGION            1..825
                  note = B. lactis xylulose-5P/fructose-6P phosphoketolase
                  (Blac_XFP.orf) amino acid sequence
                  note = subspecies lactis
source            1..825
                  mol_type = protein
                  organism = Bifidobacterium animalis

SEQUENCE: 35
MTNPVIGTPW QKLDKPVSEE AIEGMDKYWR VANYMSIGQI YLRSNPLMKE PFTRDDVKHR 60
LVGHWGTTTPG LNFLLAHINR LIADHQQNTV FIMGPGHGGP AGTAQSYIDG TYTEYFPNIT 120
KDEAGLQKFF RQFSYPGGIP SHFAPETPGS IHEGGELGYA LSHAYGAIMD NPSLFVPCII 180
GDGEAETGPL ATGWQSNKLV NPTDGTGIVLP ILHLNGYKIA NPTILARISD EELHDFFRGM 240
GYEPYEFVAG FDNEDHLSIH RRFAPLFTETV FDEICDIKAA AQDDMTTRPF YPMIIFRTPK 300
GWTCPKFIDG KKTEGWSRAH QVPLASARDT EAHFEVLKGW MESYKPEELF NADGSIKEDV 360
TAFMPKGELR IGANPNANGG RIREDLKLPE LDQYEITGVK EYGHGWGQVE APRSLGAYCR 420
DIIKNPDSF RVFGPDETAS NRLNATYEVK KKQWDNGYLS ALVDENMAVT GQVVEQLSEH 480
QCEGFLEAYL LTGRHGIWSS YESFVHVIDS MLNQHAKWLE ATVREIPWRK PISSVNLVLS 540
SHVWRQDHNG FSHQDPGVTS VLLNKTFNND HVTNIYFATD ANMLLAIAEK CFKSTNKINA 600
IFAGKQPAAT WITLDEARAE LEAGAAEWKW ASNAKSNDDEV QVVLAAAGDV PTQEIMAASD 660
ALNKMGIKFK VVNVVDLKL QSSKENDEAM SDEDFADLFT ADKPVLFAH SYAQDVRGLI 720
YDRPNHNDFT VVGKEQGST TTPFDMVRVN DMDRYALQAK ALELIDADKY ADKINELNEF 780
RKTAQFAVD NGYDIPEFTD WVYPDVKVDE TSMLSATAAT AGDNE 825

SEQ ID NO: 36      moltype = AA length = 813
FEATURE           Location/Qualifiers
REGION            1..813
                  note = L. mesenteroides xylulose-5P/fructose-6P
                  phosphoketolase(Lmes_XFP.orf) amino acid sequence
source            1..813
                  mol_type = protein
                  organism = Leuconostoc mesenteroides

SEQUENCE: 36
MADFDSKEYL ELVDKWWRAT NYLSAGMIFL KSNPLFSVTN TPIKAEDVKV KSIHGWGTIS 60
GQTFLYAHAN RLINLYGLNM FYVGGPGHGG QVMVTNAYLD GAYTEDYPEI TQDIEGMSHL 120
FKRFSFPGGI GSHMTAQTPG SLHEGGELGY SLSHAFGAVL DNPQVAFVAV VGDGEAETGP 180
SMASWHSIKF LNAKNDGAVL PVLDLNGFKI SNPTIFSRMS DEEITKFFEG LGYSPRFIEN 240
DDIHDIYATH QLAANILDQA IEDIQAIQND ARENGKYQDG EIPAWPVIIA RLPKGWGGPT 300
HDASNNPIEN SFRAHQVPLP LEQHDLATLP EFEDWMNSYK PEELFNADGS LKDELKAIAP 360
KGDKRMSAMP ITNEGADRSD LKLPNWREFA NDINDNTRGK EFADSKRMMD MATLSNYLGA 420
VSQNLNPTFRF PFGPDETMSN RLWGLFNVTP RQWMEIEKEP QDQLLSTPTR IIDSQLEHQ 480

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AEGWLEGYTL TGRVGIFASY ESFLRVVDTM VTQHPKWLRLH ASEQAWRNDY PSLNLIATST 540
 AFQQDHNGYT HQDPGMLTHL AEKKSNFIRE YLPADGNSLL AVQERAFSER HKVNLLIASK 600
 QPRQOWFTVE EAEVLANEGL KIIDWASTAP SGDVDITPAS AGTEPTIETL AALWLINQAF 660
 PDVKFRYVNV VELLRLQKKS EPNMNDEREL SAEFPNKYFQ ADTPVIFGFH AYENLIESFF 720
 FERKFTGDVY VHGYREDGDI TTTYDMRVYS HLDRFHQAKE AAEILSANGK IDQAAADTFI 780
 AKMDDTLAKH FQVTRNEGRD IEEFTDWTWS PLK 813

SEQ ID NO: 37 moltype = AA length = 323
 FEATURE Location/Qualifiers
 REGION 1..323
 note = B. subtilis phosphotransacetylase (Bs_PTA.orf) amino
 acidsequence
 source 1..323
 mol_type = protein
 organism = Bacillus subtilis

SEQUENCE: 37
 MADLFSTVQE KVAGKDVKIV FPEGLDERIL EAVSKLAGNK VLNPIVIGNE NEIQAKAKEL 60
 NLTLGGVKIY DPHTYEGMED LVQAFVERRK GKATEEQARK ALLDENYFGT MLVYKGLADG 120
 LVSGAAHSTA DTVRPAIQII KTKEGVKKTS GVFIMARGEE QYVPADCAIN IAPDSQDLAE 180
 IAIESANTAK MFDIEPRVAM LSFSTKGSAS SDETEKVADA VKIAKEKAPE LTLDGEGFQPD 240
 AAFVPSVAEK KAPDSEIKGD ANVVFVPSLE AGNIGYKIAQ RLGNFEEAVGP ILQGLNMPVN 300
 DLSRGCNAED VYNLALITAA QAL 323

SEQ ID NO: 38 moltype = AA length = 325
 FEATURE Location/Qualifiers
 REGION 1..325
 note = L. plantarum phosphotransacetylase (Lpla_PTA.orf)
 amino acidsequence
 source 1..325
 mol_type = protein
 organism = Lactobacillus plantarum

SEQUENCE: 38
 MDLFESLSQK ITGQDQITVF PEGTEPRIVG AAARLAADGL VKPIVLGATD KVQAVAKDLN 60
 ADLTGVQVLD PATYPAEDKQ AMLDSLVERR KGKNTPEQAA KMLEDENYFG TMLVVMGKAD 120
 GMVSGAVHPT GDTVPRALQI IKTKPGSHRI SGAFIMQKGE ERYVFADCAI NIDPDADTLA 180
 EIATQSAATA KVPDIDPKVA MLSFSTKGSAS KGDMVTKVQE ATAKAQAAAP ELAIDGEMQF 240
 DAAFVEKVLG QKAPGSKVAG HANVVFPEL QSGNIGYKIA QRFHGFEEAVG PVLQGLNKPV 300
 SDSLRCSEEE DVYKVAIITA AQGLA 325

SEQ ID NO: 39 moltype = AA length = 556
 FEATURE Location/Qualifiers
 REGION 1..556
 note = B. adoloscentis phosphotransacetylase (Bado_PTA.orf)
 amino acidsequence
 source 1..556
 mol_type = protein
 organism = Bifidobacterium adoloscentis

SEQUENCE: 39
 MSFTSVTIIS PEAANGRNIV ALGVTKTLLA AGKTGVFRPA VCRKDTFTDV LIEASNAGLS 60
 REQSGVGVCPK RARNDKEGSR ADIVAAYTQA VETARPDAMV IVGTDRSAVN DPAMFSFNAD 120
 VAADLQSPVL LAVCTIERTP EQVKSTVEAS TKVIEDAGSK VVGVFITGCD DTQPNPLKAC 180
 FVDYVPVPWT LPAVDNFNDD AISKADAEFA TNVDAVELTV ALESPPDAPT TPYAFQYGLL 240
 GKAKADKKTI VLPEGNEDRI IKAADYLLER DIVDLIIVGD ENAILARGQE LGLKSLGKAK 300
 FQAKDDETVL EPMVAKLCEL RAKKGMTEEQ ARKQLADDSY FGTMVLVVMGM ADGLVSGSVN 360
 STANTVRPAL QVIKTKPGTS LVSGAPLMCF KDHAADFAD AINLNPNAEQ LAEIAIQSAE 420
 TAKAFGLEPK VGMLSYSTLG SGKGPDVDLV EEATTIVKDK APDLAVVGSQ QFDAAWSPTV 480
 AATKAKGDPV AGHNVVVFVP DLCAGNIAYK AVQRSSGAAA VGPVLQGLNR PVNDLSRGAT 540
 VQDIINTIAL TAIEAQ 556

SEQ ID NO: 40 moltype = AA length = 333
 FEATURE Location/Qualifiers
 REGION 1..333
 note = M. thermophila phosphotransacetylase (Mthe_PTA.orf)
 amino acidsequence
 source 1..333
 mol_type = protein
 organism = Methosarcina thermophila

SEQUENCE: 40
 MVTFLEKISE RAKKLNKTIA LPETEDIRTL QAAAKILERG IADIVLVGNE ADIKALAGDL 60
 DLSKAKIVDP KTYEKKDEYI NAFYELRKHK GITLENAAEI MSDYVYFAVM MAKLGVEDGV 120
 VSGAAHSSSD TLRPAVQIVK TAKGAALASA FFIIISVPDCE YGSDGTFLFA DSGMVEMPSV 180
 EDVANIAVIS AKTFELLVQD VPKVAMLSYS TKGSAKSLT EATIASTKLA QELAPDIAID 240
 GELQVDAIIV PKVAASKAPG SPVAGKANVF IFPDLNCGNI AYKIAQRLAK AEAYGPITQG 300
 LAKPINDLSR GCSDEDIVGA VAITCVQAAA QDK 333

SEQ ID NO: 41 moltype = AA length = 409

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FEATURE Location/Qualifiers
 REGION 1..409
 note = B. adolescentis acetate kinase (Bado_ACK) amino acid
 sequence
 source 1..409
 mol_type = protein
 organism = Bifidobacterium adolescentis

SEQUENCE: 41
 MAKTVLVINS GSSSIKYQLV DLETGEGIAS GLVEKIGEPV DGHYKHEYNG EKHELEEEPIH 60
 DHEQGLKRVL GFFDFEGPKL ADAGIVAVGH RVVQGSIFP KPALVNDKTI GOVKDLAVLA 120
 PLHNGPEAKG AEVMSRLLPD VPQIFVFDSS FFFQLPKASS TYALNKEVAQ QYHIRRYGAH 180
 GTSHEFISSV VPSVIGKPAE GLKQIVLHIG NGASASAEIS GKPVETSMGL TPLEGLVMGG 240
 RTGDIIDPAVV FHLIRNAHMS VDELDTLFNK RSGMMGLTGF GDLREVHRLV EEGNEDAKLA 300
 LDIIYVHRIVG YIGNYTAQMG GVDVITFTAG VGENDDVVRK MVDCKLAPFG VKLDEEKNAT 360
 RSKEPRIIST PDSAVTICVI PTNEELAIAIR KSAATAEEGK DSYGNVFSK 409

SEQ ID NO: 42 moltype = AA length = 420
 FEATURE Location/Qualifiers
 REGION 1..420
 note = Emericella nidulans (A. nidulans) ACK (Ani_ACK) amino
 acidsequence
 source 1..420
 mol_type = protein
 organism = Emericella nidulans

SEQUENCE: 42
 MPRKSILSVN AGSSSVKITF YSYTKTPSVI ATAQVSGITA PPATFKYSVG SKQKKEELKE 60
 KISSGPDFAK LLLHRCFTDS DLKDVASADD LAYICHRVVH GGDPEPVI NEETYHQLED 120
 LEDLAPLHNF AALEIVRLCK KELPNVQSIT FFDSSFHKS LPPYVKTYPID QETARRNKL 180
 KYGFHGISYS FILRSVAEYL NKPVEKTS LI ALHIGSGASV CAIKDGKSID TSMGLTPLAG 240
 LPGATRSGLD DPSLVFHYTN EAGKLSAST KEMHISTAE ILNKKSGWKV LTGTTDFSQI 300
 AVEDPPSEQH KLAFDILVDR IVGYIGNYV KLDGQVEGIV FAGGIGKSA LLRKAVIEQT 360
 RCLGFAVDPE KNQHGPGDDE TVVDITASGR SDVKRVFICQ TDEQFEMAYN CTKTQGLDKQ 420

SEQ ID NO: 43 moltype = AA length = 891
 FEATURE Location/Qualifiers
 REGION 1..891
 note = E. coli bifunctional NAD+ dependent
 acetylatingacetaldehyd/alcohol dehydrogenase (E. coli
 adhE) amino acidsequence
 source 1..891
 mol_type = protein
 organism = Escherichia coli

SEQUENCE: 43
 MAVTNVAELN ALVERVKKAQ REYASFTQEQ VDKIFRAAAL AAADARIPLA KMAVAESGMG 60
 IVEDKVIKNH FASEYIYNAY KDEKTCGVLS EDDTFGTITI AEPIGIICGI VPTNTPTSTA 120
 IFKSLISLKT RNAIIFSPHP RAKDATNKAA DIVLQAAIAA GAPKDLIGWI DQPSVELSNA 180
 LMHHPDINLI LATGGPGMVK AAYSSGKPAI GVGAGNTPVV IDETADIKRA VASVLMSTKF 240
 DNGVICASEQ SVVVVDVSYD AVRERPATHG GYLQGGKEL AVQDVILKNG ALNAAIVGQP 300
 AYKIAELAGF SVPENTKILI GEVTVVDESE PFAHEKLSPT LAMYRAKDFE DAVEKAEKLV 360
 AMGGIGHTSC LYTDQDNQPA RVSYFGQKMK TARILINTPA SQGGIGDLYN FKLAPSLTLG 420
 CGSWGGSNIS ENVGPKHLIN KKTVAKRAEN MLWHKLPSKI YFRGSLPIA LDEVITDGHK 480
 RALIVTDRFL FNNGYADQIT SVLKAAGVET EVFPEVEADP TLSIVRKGA E LANSFKPDVI 540
 IALGGGSPMD AAKIMVWME HPETHFEELA LRFMDIRKRI YKFPKMGVKA KMIAVTTTSG 600
 TGSEVTPFAV VTDDATGQKY PLADYALTPD MAIVDANLVM DMPKSLCAF G LDAVTHAME 660
 AYVSVLASEF SDGQALQALK LLKEYLPASY HEGSKNPVAR ERVHSAATIA GIAFANAFLG 720
 VCHSMAHKL G SQPHIPHGLA NALLICNVIR YNANDNPTKQ TAFSQYDRPQ ARRRYAEIAD 780
 HLGLSAPGDR TAAKIEKLLA WLETLKAE LG IPKSIREAGV QEADFLANVD KLSFDAFDDQ 840
 CTGANPRYPL ISELKQILLD TYYGRDYVEG ETAAKKEAAP AKAEEKAKKS A 891

SEQ ID NO: 44 moltype = AA length = 467
 FEATURE Location/Qualifiers
 REGION 1..467
 note = E. coli ethanolamine utilizing protein (Ec_eutE)
 amino acidsequence
 source 1..467
 mol_type = protein
 organism = Escherichia coli

SEQUENCE: 44
 MNQDIEQVV KAVLLKMQSS DTPSAVHEM GVFAASLDDAV AAKVAQQGL KSVAMRQLAI 60
 AAIREAGEKH ARDLAELAVS ETGMGRVEDK FAKNVAQARG TPGVECLSPQ VLTGDNGLTL 120
 IENAPWGVVA SVTPSTNPAA TVINNAISLI AAGNSVIFAP HPAAKKVSQR AITLLNQAI 180
 AAGGPENLLV TVANPDIETA QRLFKFPGIG LLVVTGGEAV VEAARKHTNK RLIAAGAGNP 240
 PVVVDETADL ARAAQSVK G ASFNNIICA DEKVLIVVDS VADELMRLME GHAVKLTA 300
 QAQQLQPVLL KNIDERGKGT VSRDWVGRDA GKIAAAIGLK VPQETRLLFV ETAEHPFAV 360
 TELMMPVLV VRVANVADAI ALAVKLEGGC HHTAAMHSRN IENMNQMANA IDTSIFVKNG 420
 PCIAGLGLGG EGWTTMTITT PTGEGVTSAR TFRVLRRCVL VDAFRIV 467

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SEQ ID NO: 45 moltype = AA length = 455
 FEATURE Location/Qualifiers
 REGION 1..455
 note = L. plantarum acetaldehyde dehydrogenase (Lpla_acdH)
 source 1..455
 mol_type = protein
 organism = Lactobacillus plantarum

SEQUENCE: 45
 MLKEMEETTV SRSIDRLVLN ASLAANRLEV MDQSQVDQAV AAMARAHAHA RGMLAAMAVE 60
 ETGRGNRYRDK VAKNDFAAKN VYNYIKDDKT VGIINDDPVS GVMKVAEPVG IIAGVTPVTN 120
 PTSTVIFNAM LALKTRNPPI FGFHPPAQS CVETGRIIRD AAIASGAPKD WIQWIKTPSL 180
 EATNTLMNHP GVATIIATGG AGMVKTAYST GKPALGVGPG NVPCFIEQTA DIQQAVSDVV 240
 TSKSPDNGMI CASESNLIVA DQIYDQVKRE LSHNGVYFVG TENFKALEAT VMNLDKQAVD 300
 PKVAGQTPWQ IAQWAGFDVP SDTKVLAVEL PSIGGDQVLS REKLSPVLAV VHAKDTEAGF 360
 NLMKRSLALG GLGHTAALHT TDEAVMNKFA LEMTACRALI NVPSSQGAIG YKYDNVAPSL 420
 TLGCGTWGHN SISHNLEDWD LLNIKTVAKR LTKIR 455

SEQ ID NO: 46 moltype = AA length = 469
 FEATURE Location/Qualifiers
 REGION 1..469
 note = L. innocua acetaldehyde dehydrogenase (Linn_acdH)
 source 1..469
 mol_type = protein
 organism = Listeria innocua

SEQUENCE: 46
 MESLELEQLV KKVLEKLAE QKEVPTKTTT QGAKSGVFDV VDEAVQAAVI AQNCYKEKSL 60
 EERRNVVKAI REALYPEIET IATRAVAETG MGNVTDKILK NTLAIEKTPG VEDLYTEVAT 120
 GDNGMTLYEL SPYGVIGAVA PSTNPTETLI CNSIGMLAAG NAVFYSPHPG AKNISLWLE 180
 KLNTIVRDSG GIDNLIVTVA KPSIQAAQEM MNHPKVPLLV ITGGPGVVLQ AMQSGKKVIG 240
 AGAGNPPSIV DETANIEKAA ADIVDGASF DNILCIAEKS VVAVDSIADF LLFQMEKNGA 300
 LHVTNPSDIQ KLEKVAVDK GVTNKKLVGK SATEILKEAG IACDFTPLRI IVETEKSHPF 360
 ATVELLMPIV PVVRVPDFDE ALEVAIELEQ GLHHTATMHS QNISRLNKA RDMQTSIFVK 420
 NGPSFAGLGF RGEGSTTTTI ATPTGEGTTT ARHFARRRRR VLTDFGSIR 469

SEQ ID NO: 47 moltype = AA length = 869
 FEATURE Location/Qualifiers
 REGION 1..869
 note = S. aureus acetaldehyde/alcohol dehydrogenase
 (Saur_adhE)
 source 1..869
 mol_type = protein
 organism = Staphylococcus aureus

SEQUENCE: 47
 MLTIPEKENR GSKEQEVAIM IDALADKGGK ALEALSKKSQ EEIDHIVHQM SLAAVDQHMV 60
 LAKLAHEETG RGIYEDKAIK NLYASEYIWN SIKDNKTVGI IGEDKEKGLT YVAEPIGVIC 120
 GVTPTTNPTS TTIFKAMIAI KTGNIPIFAP HPSAQESSKR AAEVVLEAM KAGAPKDIIQ 180
 WIEVPSIEAT KQLMNHKGIA LVLATGGSGM VKSAYSTGKP ALGVGPGNVP SYIEKTAHIK 240
 RAVNDIIGSK TFDNGMICAS EQVVVIDKEI YKDVTFNEKA HQAYFVKDE LQRLNAINM 300
 EQKTGIKPD I VGKSAVEIAE LAGIPVPENT KLIIAEISGV GSDYPLSREK LSPVLALVKA 360
 QSTKQAFQIC EDTLHFGGLG HTAVIHTEDE TLQKDFGLRM KACRVLVNTP SAVGGIGDMY 420
 NELIPSLTLG CGSYGRNSIS HNVSATDLLN IKTIARRNN TQIFKVPAQI YFEENAIMSL 480
 TTMDKIEKVM IVCDPGMVEF GYTKTVENVL RQRETPQPIK IFSEVEPNPS TNTVYKGLEM 540
 MVDQPDPTII ALGGGSAMDA AKAMWMFFEH PETSFFGAKQ KFLDIGKRTY KIGMPENATF 600
 ICIPTTSGTG SEVTPFAVIT DSETNVKYPL ADFALTPDVA IIDPQFVMSV PKSVTADTGM 660
 DVLTHAMESY VSVMASDYTR GLSLQAIKLT FEYLKSSVEK GDKVSREKMH NASTLAGMAF 720
 ANAFGLIAHS IAHKIGGEYG IPHGRANAIL LPHIIRYNAK DPQKHALFPK YEFPRADTDY 780
 ADIAKFLGLK GNTTEALVES LAKAVYELGQ SVGIEMNLKS QGVSEELNE SIDRMAELAF 840
 EDQCTTANPK EALISEIKDI IQTSYDYKQ 869

SEQ ID NO: 48 moltype = AA length = 314
 FEATURE Location/Qualifiers
 REGION 1..314
 note = acetylating acetaldehyde dehydrogenase (Ec_mhpF)
 source 1..314
 mol_type = protein
 organism = Escherichia coli

SEQUENCE: 48
 MSKRKVAIIG SGNIGTDLMI KILRHGQHLE MAVMVGIDPQ SDGLARARRM GVATTHEGVI 60
 GLMNMPEFAD IDIVFDATSA GAHVKNDAAL REAKPDIRLI DLTPAAIGPY CVPWNLEAV 120
 DQLNVNMVTC GGQATIPMVA AVSRVARVHY AEIIASIASK SAGPGTRANI DEFTETTSRA 180
 IEWGGAAGKGI AIIVLNPAEP PLMMRDTVYV LSDEASQDDI EASINEMAEA VQAYVPGYRL 240
 KQRVQFEVIP QDKPVNLPV GQFSGLKTAV WLEVEGAHY LPAYAGNLDI MTSSALATAE 300
 KMAQSLARKA GEAA 314

SEQ ID NO: 49 moltype = AA length = 713

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FEATURE Location/Qualifiers
 REGION 1..713
 note = Acetyl-coenzyme A synthetase 1 (Sc_ACS1)
 source 1..713
 mol_type = protein
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 49

MSPSAVQSSK	LEEQSSSEIDK	LKAKMSQSAA	TAQQKKEHEY	EHLTSVKIVP	QRPISDRLOP	60
AIATHYSPLH	DGLQDYQRLH	KESIEDPAKF	FGSKATQFLN	WSKPFDKVFI	PDPKTGRPSF	120
QNNAWFLNGQ	LNACYNQVDR	HALKTPNKKA	IIFEGDEPGQ	GYSITYKELL	EEVCQVAQVL	180
TYSMGVRKGD	TVAVYMPMPV	EAIITLLAIS	RIGAIHSVVF	AGFSSNSLRD	RINDGDSKVV	240
ITTDESNRGG	KVIETKRIVD	DALRETPGVR	HVLVYRKTNN	PSVAFHAPRD	LDWATEKKKY	300
KTYYPCTPVD	SEDPLFLLYT	SGSTGAPKGV	QHSTAGYLLG	ALLTMRYTFD	THQEDVFFTA	360
GDIGWITGHT	YVYVGPLLYG	CATLVFEGTP	AYPNYSRYWD	IIDEHKVTQF	YVAPTALRL	420
KRAGDSYIEN	HSLKSLRCLG	SVGEPIAAEV	WEWYSEKIGK	NEIPIVDITYW	QTESGSHLVT	480
PLAGGVTPMK	PGSASFPPFG	IDAVVLDPNT	GEELNTSHAE	GVLA VKAAMP	SFARTIWKNH	540
DRYLDTYLNP	YPGYFTGDG	AAKDKDGYIW	ILGRVDDVVN	VSGHRLSTAE	IEAAIIEDPI	600
VAECAVVGFN	DDL TGQAVAA	FVVLKNKSSW	STATDDELQD	IKKHLVPTVR	KDIGPFAAPK	660
LIILVDDLPK	TRSGKIMRRI	LRKILAGESD	QLGDVSTLSN	PGIVRHLIDS	VKL	713

SEQ ID NO: 50 moltype = AA length = 683
 FEATURE Location/Qualifiers
 REGION 1..683
 note = Acetyl-coenzyme A synthetase 2 (Sc_ACS2)
 source 1..683
 mol_type = protein
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 50

MTIKEHKVVY	EAHNVKALKA	PQHFYNSQPG	KGYVTDMQHY	QEMYQQSINE	PEKFFDKMAK	60
EYLHWDAPYT	KVQSGSLNNG	DVAWFLNGKL	NASYNCVDRH	AFANPDKPAL	IYEADDESDN	120
KIITFGELLR	KVSQIAGVLK	SWGKKGDTV	AIYLPMIPEA	VIAMLA VARI	GAIHSVVFAG	180
FSAGSLKDRV	VDANSKVIT	CDEGKRGGKT	INTKKIVDEG	LNGVDLVSR	LVPQRTGTEG	240
IPMKAGRDY	WHEEAAQRT	YLPVPSDAE	DPLFLLYTSG	STGSPKGVVH	TTGGYLLGAA	300
LTRRVFDIH	PEDVLF TAGD	VGNITGHTYA	LYGPLTLGTA	SIIFESTPAY	PDYGRYWRII	360
QRHKATHFYV	APTALRLIKR	VGEAEIAKYD	TSSLRVLGSV	GEPISPDLWE	WYHEKVGKNK	420
CVICDTMWQT	ESGSHLIAPL	AGAVPTKPGS	ATVPFFGINA	CIIDPVTGVE	LEGNDVEGLV	480
AVKSPWPSMA	RSVWNHHDY	MDTYLKPYPG	HYFTGDGAGR	DHDGYWIRG	RVDDVVNVSG	540
HLRSTSEIEA	SISNHNVSSE	AAVVGIPDEL	TGQTVVAVVS	LKDGYLQNN	TEGDAEHITP	600
DNLRLRELILQ	VRGEIGPFAS	PKTIILVRDL	PRTRSGKIMR	RVLRKVASNE	AEQLGDLTTL	660
ANPEVVPPII	SAVENQFFSQ	KKK				683

SEQ ID NO: 51 moltype = AA length = 348
 FEATURE Location/Qualifiers
 REGION 1..348
 note = Alcohol dehydrogenase 1 (Sc_ADH1)
 source 1..348
 mol_type = protein
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 51

MSIPETQKGV	IFYESHGKLE	YKDIPVPKPK	ANELLINVKY	SGVCHTDLHA	WHGDWPLPVK	60
LPLVGGHEGA	GVVVGMGENV	KGWKIGDYAG	IKWLNGSCMA	CEYCELGNES	NCPHADLSGY	120
THDGSFQQYA	TADAVQAAHI	PQGTDLAQVA	PILCAGITVY	KALKSANLMA	GHWVAISGAA	180
GGLGSLAVQY	AKAMGYRVLG	IDGGEGKEEL	FRSIGGEVFI	DFTKEKDIVG	AVLKATDGGGA	240
HGVINVSUSE	AAIEASTRYV	RANGTTVLVG	MPAGAKCCSD	VFNQVVK SIS	IVGSYVGNRA	300
DTREALDFFA	RGLVKSPIKV	VGLSTLPEIY	EKMEKGQIVG	RYVVDTSK		348

SEQ ID NO: 52 moltype = AA length = 348
 FEATURE Location/Qualifiers
 REGION 1..348
 note = Alcohol dehydrogenase 2 (Sc_ADH2)
 source 1..348
 mol_type = protein
 organism = *Saccharomyces cerevisiae*

SEQUENCE: 52

MSIPETQKAI	IFYESNGKLE	HKDIPVPKPK	PNELLINVKY	SGVCHTDLHA	WHGDWPLPTK	60
LPLVGGHEGA	GVVVGMGENV	KGWKIGDYAG	IKWLNGSCMA	CEYCELGNES	NCPHADLSGY	120
THDGSFQEYA	TADAVQAAHI	PQGTDLAEVA	PILCAGITVY	KALKSANLRA	GHWAAISGAA	180
GGLGSLAVQY	AKAMGYRVLG	IDGGPGKEEL	FTSLGGEVFI	DFTKEKDIVS	AVVKATNGGA	240
HGIINVSUSE	AAIEASTRYC	RANGTVVLVG	LPAGAKCCSD	VFNHVVK SIS	IVGSYVGNRA	300
DTREALDFFA	RGLVKSPIKV	VGLSSLPEIY	EKMEKGQIAG	RYVVDTSK		348

SEQ ID NO: 53 moltype = AA length = 375
 FEATURE Location/Qualifiers
 REGION 1..375
 note = Alcohol dehydrogenase 3 (Sc_ADH3)
 source 1..375

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mol_type = protein
organism = Saccharomyces cerevisiae

SEQUENCE: 53
MLRTSTLFTR RVQPSLFERN ILRLQSTAAI PKTQKGVIFY ENKGLHYKD IPVPEPKPNE 60
ILINVKYSKV CHTDLHAWHG DWPLPVKLPL VGGHEGAGVV VKLGSNVKGV KVGDLAGIKW 120
LNGSCMTCEF CESGHESNCP DADLSGYTHD GSFQQFATAD AIQAAKIQQG TDLAEVAPIL 180
CAGVTYVKAL KEADLKAGDW VAISGAAGGL GSLAVQYATA MGYRVLGIDA GEEKEKLFKK 240
LGGEVFDIFT KTKNMVSDIQ EATKGGPHGV INVSVEAAI SLSTEYVRPC GTVVLVGLPA 300
NAYVKSSEVS HVVKSINIKG SYVGNRADTR EALDFFSRGL IKSPIKIVGL SELPKVYDLM 360
EKGKILGRIV VDTSK 375

SEQ ID NO: 54      moltype = AA length = 382
FEATURE           Location/Qualifiers
REGION            1..382
                  note = Alcohol dehydrogenase 4 (Sc_ADH4)
source            1..382
                  mol_type = protein
                  organism = Saccharomyces cerevisiae

SEQUENCE: 54
MSSVTGFYIP PISFFGEGAL EETADYIKNK DYKKALIVTD PGIAAIGLSG RVQKMLEERD 60
LNVAIYDKTQ PNPNIANVTA GLKVLKEQNS EIVVSIGGGS AHDNAKAIAL LATNGGEIGD 120
YEGVNQSKKA ALPLFAINTT AGTASEMTRF TIISNEEKKI KMAIIDNNVT PAVAVNDPST 180
MFGLPPALTA ATGLDALTHC IEAYVSTASN PITDACALKG IDLINESLVA AYKDGKDKKA 240
RTDMCYAEYL AGMAFNNASL GYVHALAHQL GGFYHLPHGV CNAVLLPHVQ EANMQCPKAK 300
KRLGELALHF GASQEDPEET IKALHVLNRT MNIPRNKLKL GVKTEDFEIL AEHAMHDACH 360
LTNPVQFTKE QVVAIIKKAY EY 382

SEQ ID NO: 55      moltype = AA length = 351
FEATURE           Location/Qualifiers
REGION            1..351
                  note = Alcohol dehydrogenase 5 (Sc_ADH5)
source            1..351
                  mol_type = protein
                  organism = Saccharomyces cerevisiae

SEQUENCE: 55
MPSQVIPEKQ KAIVFYETDG KLEYKDVTVP EPKPNEILVH VKYSGVCHSD LHAWHGDWPF 60
QLKFPLIGGH EGAGVVVKLG SNVKGWKVGD FAGIKWLNGT CMSCEYCEVG NESQCPYLDG 120
TGFTHDGTFQ EYATADAVQA AHIPPNVNLA EVAPILCAGI TVYKALKRAN VIPGQWVTIS 180
GACGGLGSLA IQYALAMGYR VIGIDGGNAK RKLFEQLGGE IFIDFTEEKD IVGAIKATN 240
GGSHGVINVS VSEAAIEAST RYCRPNGTVV LVGMPAHAYC NSDVFNQVVK SISIVGSCVG 300
NRADTREALD PFARGLIKSP IHLAGLSDVP EIFAKMEKGE IVGRYVVETS K 351

SEQ ID NO: 56      moltype = AA length = 561
FEATURE           Location/Qualifiers
REGION            1..561
                  note = P. strigosozonata glucoamylase
source            1..561
                  mol_type = protein
                  organism = Punctularia strigosozonata

SEQUENCE: 56
SSKSVSAYIS SESPIASHKL LDNIGPDGAK APGAPPGVVV ASPSTDNPNY YYSWIRDSSL 60
VFKTLIDDDV NGKNTSKSLR SLIDDFVTAS SVFQQTNPNS GNVSTGGLGE PKFYVNETAF 120
LDSWGRPQRD GPALRSTALI TYANYLLDND NTTWVKDTLW PIIELDVNYV SDFWNYTTFD 180
LWEEVASSSF FTTAVQHRAL RQASKLAKTL DKTDNIDSWN TQADNVLCFL QSYWNGSAII 240
ANTGGGRSGI DANTVLASIH TFDSSAGCDA TTFQPCSDKA LSNLKVYVDA FRSIYEINSG 300
IDPNAAVATG RYPEDVFYDG NPWYLATAAV AEQLYDALYV WNTTGSLEIT DISLPFFQQF 360
DSDVKTGTYS DDDTFDSLIS SIQSFADGFL EIHAKYTPDD GALSEEPSKT DGSQTSAADL 420
TWSYAAALTA FDARSDDAAV KWGAKGLQVP DGTCKTNEGG DDGLGVPTVF LVKDAETVEG 480
QSVYITGSIA TLKSWSPDDA LLMSPSDYPT WTLTVNLSAS ESVQYKYIKK DTAGTVIWES 540
DPNNSLLVPS GGSVTTDDTW R 561

SEQ ID NO: 57      moltype = AA length = 578
FEATURE           Location/Qualifiers
REGION            1..578
                  note = P. strigosozonata glucoamylase with native signal
                  sequence FGA09
source            1..578
                  mol_type = protein
                  organism = Punctularia strigosozonata

SEQUENCE: 57
MLSSLIVSGL LASGVCAASK SVSAYISSES PIAHSKLLDN IGPDGAKAPG APFGVVVASP 60
STDNPNYYYY WIRDSSLVFK TLIDDYVNGK NTSKSLRSLI DDFVTASSVF QQTPNPNSGNV 120
STGGLGEPKF YVNETAPLDS WGRPQRDGPAL LRSTALITYA NYLLDNDNTT WVKDTLWPII 180
ELDVNYVSDF WNYTTFDLWE EVASSSFFT AVQHRALRQA SKLAKTLDKT DNIDSWNTQA 240
DNVLCFLQSY WNGSAIIANT GGGSGIDAN TVLASIHTFD SSAGCDATTF QPCSDKALSN 300
LKVYVDAPRS IYEINSGIDP NAAVATGRYP EDVFDGNFW YLATAVAEQ LYDALVWNT 360

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TGSLEITDIS LPFFQQPDS D VKTGTYSDDD TFDLSLISSIQ SFADGFLEIH AKYTPDDGAL 420
 SEEFSTKTDGS QTSAADLTWS YAAALTAFDA RSRDAAVKWG AKGLQVPDGT CKTNEGGDDG 480
 LGVPVTFVLK DAETVEGQSV YITGSIATLK SWSPPDALLM SPSDYPTWTL TVNLSASESV 540
 QYKYIKKDTA GTVIWESDPN NSLLVPSGGS VTDDTWR 578

SEQ ID NO: 58 moltype = AA length = 580
 FEATURE Location/Qualifiers
 REGION 1..580
 note = P. strigosozonata glucoamylase with Sc alpha mating
 factor signalsequence
 source 1..580
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 58
 MRFPSIFTAV LFAASSALAS SKSVSAYISS ESPIAHSKLL DNIGPDGAKA PGAPPGVVVA 60
 SPSTDNPNY YSWIRDSSLV FKTLIDDYVN GKNTSKSLRS LIDDFVTASS VFQQTNPNSG 120
 NVSTGGLGEP KFYVNETAFL DSWGRPQRDG PALRSTALIT YANYLLDNDN TTWVKDTLWP 180
 IIELDVNVYS DFWNYTTFDL WEEVASSSFF TTAVQHRALR QASKLAKTLD KTDNIDSWNT 240
 QADNVLCFLQ SYWNQSAIIA NTGGGRSGID ANTVLASIHT FDSSAGCDAT TFQPCSDKAL 300
 SNLKVYVDAF RSIYEINSGI DPNAAVATGR YPEDVFYDGN PWYLATAAVA EQLYDALYVW 360
 NTTGSLEITD ISLPFFQQFD SDVKTGTYS DDTFDSLSS IQSFADGFLE IHAKYTPDDG 420
 ALSEEFSTKD GSQTSADLT WSYAAALTA DARSRDAVK WGAQGLQVPD GTCKTNEGGD 480
 DGLGVPVTF LKDAETVEGQ SVYITGSIAT LKSWSPDDAL LMSPSYPTW TLTVNLSASE 540
 SVQYKYIKD TAGTVIWESD PNNSLLVPSG GSVTTDDTWR 580

SEQ ID NO: 59 moltype = AA length = 365
 FEATURE Location/Qualifiers
 REGION 1..365
 note = K. pneumoniae glycerol dehydrogenase (Kpne_gldA)
 source 1..365
 mol_type = protein
 organism = Klebsiella pneumoniae

SEQUENCE: 59
 MLKVIQSPAK YLQGPDAVL FGQYAKNLAE SFFVIADDFV MKLAGEKVVN GLQSHDIRCH 60
 AERFNGEC SH AEINRLMAIL QKQGRGVVG IGGGKTLDTA KAIGYYQKLP VVVIPTIAST 120
 DAPTSALSVI YTEAGEFEEY LIYPKNPDMV VMDTAIIAKA PVRLLVSGMG DALSTWFEAK 180
 ACYDARATSM AGGQSTEAL SLARLCYDTL LAEGEKARLA AQAGVVTEAL ERIEANTYL 240
 SGIGFESSGL AAAHAHNGF TLEECHHLY HGEKVAFGTL AQLVLQNSPM DEIETVLGFC 300
 QRVGLPVT LA QMGVKEGIDA KIAAVAKATC AEGETIHNMP FAVTPESVHA AILTADLLGQ 360
 QWLAR 365

SEQ ID NO: 60 moltype = AA length = 367
 FEATURE Location/Qualifiers
 REGION 1..367
 note = E. aerogenes glycerol dehydrogenase (Eaer_gldA)
 source 1..367
 mol_type = protein
 organism = Enterococcus aerogenes

SEQUENCE: 60
 MDRIIQSPGK YIQGAGAIKR LGEYLKPLAE RWLIIGDKFV LGFAEEQLRT SLGGAGLVAE 60
 IAPFNGEC SQ NEINRLRDIA SSAQCHAVLG IGGGKTLDTA KALAHYMHLP VVVVPTIAST 120
 DAPCSALSVI YTDGEFESY LMLPHNPNMV VVDTOIVAAA PARLLAAGIG DALATWFEAR 180
 ACSRS GATTM AGGKCTQAAL ALAELCYNTL VEEGEKAMLA AEQHVVTPAL ERVIEANTYL 240
 SGVGFESSGL AAAHAHNGF TAIPDAHHFY HGEKVAFGTL TQLVLENAPV EEIETAAALC 300
 HSVGLPITLA QLDIKGDIPA KMRTVAEAAC AEGETIHNMP GGACADQVYA ALLVADQYQG 360
 RFLQEW E 367

SEQ ID NO: 61 moltype = AA length = 364
 FEATURE Location/Qualifiers
 REGION 1..364
 note = Y. aldovae glycerol dehydrogenase
 source 1..364
 mol_type = protein
 organism = Yersinia aldovae

SEQUENCE: 61
 MLKVIQSPSK YIQGANALQS IGEFAKLLAN NYFIIADDFV MKLTADTVGT SLQTCCLKSH 60
 FSRFNGEC SR QEIERLTVEL KKYGCNGVIG IGGGKTLDTA KALAHYQHLP VVVVPTIAST 120
 DAPTSALSVI YTEGEFAEY LIYPKNPDI V LMDTTIIAKA PVRLLVAGMG DALSTYFEAQ 180
 ACFDAKAISM AGGASTLAAI TLARLCYETL LAEGYKAKLA VEAGVVTEAV ERIEANTYL 240
 SGIGFESSGL AAAHAHNGF TVLEECHHLY HGEKVAFGTL TQLVLQNSSM EEIETVLSFC 300
 QQLGLPITLA EMGVTQDLEC KIRAVAQASC AEGETIHNMP PKVTADSVYA AIIVADRLGQ 360
 AFLN 364

SEQ ID NO: 62 moltype = AA length = 367
 FEATURE Location/Qualifiers
 REGION 1..367

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source          note = E. coli glycerol dehydrogenase (Ec_gldA)
                1..367
                mol_type = protein
                organism = Escherichia coli

SEQUENCE: 62
MDRIIQSPGK YIQGADVNR LGEYLKPLAE RVLVVGDKFV LGFAQSTVEK SFKDAGLVVE 60
IAPFGECSQ NEIDRLRGIA ETAQCGAILG IGGGKTLDTA KALAHFMGVP VAIAPTIAS 120
DAPCSALSVI YTDEGEFDRY LLLPNPNPMV IVDTKIVAGA PARLLAAGIG DALATWFEAR 180
ACSRSGATM AGGKCTQAA ALAELCYNTL LEEGEKAMLA AEQHVVPAL ERVIEANTYL 240
SGVGFESGGL AAHAVHNGL TAIPDAHYY HGEKVAFGTL TQLVLENAPV EEIETVAALS 300
HAVGLPITLA QLDIKEDVPA KMRIVAEAAE AEGETIHNP GGATPDQVYA ALLVADQYQG 360
RFLQEWEE                                     367

SEQ ID NO: 63      moltype = DNA length = 1111
FEATURE           Location/Qualifiers
misc_feature       1..1111
                  note = nucleotide sequence encoding E.coli gldA
                  codon-optimised
source            1..1111
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 63
atggacagaa tcatccaatc tccaggttaag tacatccaag gtgctgacgt tatcaacaga 60
ttgggtgaat acttgaagcc attggctgaa agatgggttg ttgttggtga caagttcgtt 120
ttgggtttcg ctcaatctac cgtgaaaag tctttcaagg acgctgggtt ggttggtgaa 180
atcgctccat tcggtggtga atgttctcaa aacgaaatcg acagattgag aggtatcgct 240
gaaacccgctc aatgtggtgc tatcttggtg atcggtggtg gtaagacctt ggacaccgct 300
aaggtcttgg ctcacttcat ggggtgtcca gttgctatcg ctccaacctat cgcttctacc 360
gacgtcccat gtctgcttt gtctgttata tacaccgacg aaggtgaatt cgacagatac 420
ttgttggtgc caaacaaccc aaacatgggt atcggtgaca ccaagatcgt tgctggtgct 480
ccagctagat gtttggcagc tggatccggt gacgctttgg ctacctggtt cgaagctaga 540
gcttggtcta gatctggtgc taccaccatg gctgggtgta agtgtaacca agctgcttgg 600
gctttggctg aattgtgtta caacaccttg ttggaagaag gtgaaaaggc tatgttggtc 660
gctgaacaac acgttggttac ccagctttg gaaagagtta tcgaagctaa cactactctg 720
tctggtgttg gtttcgaatc tgggtggttg gctgctgctc acgctgttca caacggtttg 780
accgctatcc cagacgctca ccactactac cacggtgaaa aggttgcttt cggtaccttg 840
accgaattgg ttttggaaaa cgctccagtt gaagaaatcg aaacggttgc tgctttgtct 900
cacgctgttg gtttgccaat cactctggct caattggaca tcaaggaaga cgttccagct 960
aagatgagaa tcgttgctga agctgcttgt gctgaaggtg aaaccatcca caacatgcca 1020
ggtggtgcta cccagagcca agtttacgct gctttgttgg ttgctgacca atacggtcaa 1080
agattcctac aagaatggga ataaggcgcg c                                     1111

SEQ ID NO: 64      moltype = AA length = 584
FEATURE           Location/Qualifiers
REGION           1..584
                  note = S. cerevisiae dihydroxyacetone kinase (S. cerevisiae
                  DAK1)
source            1..584
                  mol_type = protein
                  organism = Saccharomyces cerevisiae

SEQUENCE: 64
MSAKSFEVTD PVNSSLKGFA LANPSITLVP EEKILFRKTD SDKIALISGG GSGHEPTHAG 60
FIGKMLSGA VVGEIFASPS TKQILNAIRL VVENASGVLL IVKNYTGDLV HFGLSAERAR 120
ALGINCRVAV IGDDVAVGRE KGMVVGRRAL AGTVLVHKIV GAFABEYSSK YGLDGTAKVA 180
KIINDNLVTI GSSLDHCKVP GRKFESELNE QMELGMGIH NEPGVKVLDL IPSTEDLISK 240
YMLPKLLDPN DKDRAVFKFD EDDEVLLVN NLGGVSNFVI SSITSKTDF LKENYNITPV 300
QTIAGTLMTS FNGNGFSITL LNATKATKAL QSDFEEIKSV LDLLNAPTNA PGWPIADFKE 360
TSAPSVNDDL LHNEVTAHAV GTYDFDKFAE WMKSGAEQVI KSEPHITELD NQVGDGDCGY 420
TLVAGVKGIT ENLDKLSKDS LSQAVAQISD FIEGSMGGTS GGLYSILLSG FSHGLIQVCK 480
SKDEPVTKEI VAKSLGIALD TLYKYTKARK GSSTMIDALE PFVKEFTASK DFNKAVKAAE 540
EGAKSTATFE AKFGRASYVG DSSQVEDPGA VGLCEFLKGV QSAL                                     584

SEQ ID NO: 65      moltype = AA length = 591
FEATURE           Location/Qualifiers
REGION           1..591
                  note = S. cerevisiae dihydroxyacetone kinase (S. cerevisiae
                  DAK2)
source            1..591
                  mol_type = protein
                  organism = Saccharomyces cerevisiae

SEQUENCE: 65
MSHKQFKSDG NIVTPYLLGL ARSNPGLTVI KHDRVVFRTA SAPNSGNPPK VSLVSGGGSG 60
HEPTHAGFVG EGALDAIAAG AIFASPSTKQ IYSAIKAVES PKGTLIIIVKN YTGDIHIFGL 120
AAERAKAAGM KVELVAVGDD VSVGKKKGS L VGRRLGATV LVHKKIAGAAA SHGLELAEVA 180
EVAQSVVDNS VTIAASLDHC TVPGHKPEAI LGENEYEIGM GIHNESGTYK SSPLPSISEL 240
VSQMLPLLLD EDEDRSYVKF EPKEDVVLV NNMGMSNLE LGYAAEIVSE QLIDKYQIVP 300

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KRTITGAFIT ALNGPFGGIT LMNASKAGD ILKYFDYPTT ASGWNQMYHS AKDWEVLAKG 360
 QVPTAPSLKT LRNEKGSQVK ADYDTFAKIL LAGIAKINEV EPKVTWYDTI AGDGDGCGTTL 420
 VSGGEALEEA IKNHTLRLED AALGIEDIAY MVEDSMGGTS GGLYSIYLSA LAQGVDRDSDG 480
 KELTAETFKK ASNVALDALY KYTRARPGYR TLIDALQPFV EALKAGKGPR AAAQAAYDGA 540
 EKTRKMDALV GRASYVAKEE LRLKLDSEGL PDPGAVGLAA LLDGFVTAAG Y 591

SEQ ID NO: 66 moltype = AA length = 572
 FEATURE Location/Qualifiers
 REGION 1..572
 note = K. pneumoniae dihydroxyacetone kinase (Kpne_dhaK)
 source 1..572
 mol_type = protein
 organism = Klebsiella pneumoniae

SEQUENCE: 66
 MTTKQFQFDS DPLNSALAAT AEASGLAYLP KSKVIYIPLT NDKVTLISGG GAGHEPAQTG 60
 FVGPGLLDAA VSGQIFASPS TKQIIAGVNA VKSQRGSIII VMNYTGdVIH FGMAAEQLRS 120
 RYDYHAELVS IGDdisVNKK AGRRLAGTV LVHKIAGHLA RDGWDVGVLA EALRTTAANL 180
 ATVAASLEHC TVPGRKFETE LAADEMEIGM GIHNEPGVKT IKIGKVESLL DELVDKFEPs 240
 KQDFVPFNKG DEVVLLVNSL GGVSSLELHA IANIAQTKE KVLGVKTVRL IVGNFMAAFN 300
 GPGFSLTLN VTTTAKKGNF DVLGALDAPV STAANPSLQQ KDKPANGGVQ EEKETDSDKP 360
 AEPTGIKADG KLFKAMIESA VDDLKKEEPQ ITKYDTIAGD GDCGETLLAG GDGILDAIKN 420
 KKIDLDAAAG VADISHIVEN SMGGTSGGLY SIFPSGLVVG IKETKAKELS VDVFAKACET 480
 ALETLSKYTQ ARVGDRTLMD ALVPFVETLS KTKDFAKAVE AARKGADETS KLPANFGRAS 540
 YVNEEGLeni PDPGALGLAV IFEGLLKAWK KK 572

SEQ ID NO: 67 moltype = AA length = 580
 FEATURE Location/Qualifiers
 REGION 1..580
 note = Y. lipolytica dihydroxyacetone kinase (Ylip_DAK1)
 source 1..580
 mol_type = protein
 organism = Yarrowia lipolytica

SEQUENCE: 67
 MDKHFINDPE VLVLDGLKSL ADMNKTLTVH EEGKFIYFHD YNKKNVSVIS GGGAGHEPTH 60
 SSFVGKGMTL AAVSGSIFAS PSSKQIYTG I KQVESEAGTL VICKNYTGDI LHFMALEKQ 120
 RTAGKKAELI AVADDVSVGR KKSgKVGRRG LSGTVLVHKI AGAAAARGLP LEAVTTIAKA 180
 AIDNLVSIGA SLAHVHVPGH EPIAKEDEMK HDEMELGMGI HNEPGCKRIS PIPSIDDLIA 240
 QMLKQMLDQS DKDRAYVKIE GDDEVVLLMN NLGGLSMLEF SAISHKVKEA LAKEYKINPV 300
 RIFAGPFTTS LNGLGFGITL LRTTDRVKVE GEEYSLVDLI DQPVEAIGWP LCQPSDLKSK 360
 NKIGNVSIEE GQKDVKSPTV VDKEKVRQAI VNSMENLIKA EPKITKFTDM AGDGDGCGTTL 420
 KRGAEGVLKF VKSDKFSDDP IRIVRDIADV IEDNMDGTSG ALYAIFFPHGF AKGMKDTLEK 480
 SKDISSKTWA AGLKVALDTL FKYTPARPGD STMCDALVPF VETFKVTNDL NAAVEEARKG 540
 ADATADMQAK LGRAVYVGDD VKVPDAGALG VVAIVEGFTK 580

SEQ ID NO: 68 moltype = AA length = 580
 FEATURE Location/Qualifiers
 REGION 1..580
 note = S. pombe dihydroxyacetone kinase (Spom_DAK1)
 source 1..580
 mol_type = protein
 organism = Schizosaccharomyces pombe

SEQUENCE: 68
 MDKHFINDPE VLVLDGLKSL ADMNKTLTVH EEGKFIYFHD YNKKNVSVIS GGGAGHEPTH 60
 SSFVGKGMTL AAVSGSIFAS PSSKQIYTG I KQVESEAGTL VICKNYTGDI LHFMALEKQ 120
 RTAGKKAELI AVADDVSVGR KKSgKVGRRG LSGTVLVHKI AGAAAARGLP LEAVTTIAKA 180
 AIDNLVSIGA SLAHVHVPGH EPIAKEDEMK HDEMELGMGI HNEPGCKRIS PIPSIDDLIA 240
 QMLKQMLDQS DKDRAYVKIE GDDEVVLLMN NLGGLSMLEF SAISHKVKEA LAKEYKINPV 300
 RIFAGPFTTS LNGLGFGITL LRTTDRVKVE GEEYSLVDLI DQPVEAIGWP LCQPSDLKSK 360
 NKIGNVSIEE GQKDVKSPTV VDKEKVRQAI VNSMENLIKA EPKITKFTDM AGDGDGCGTTL 420
 KRGAEGVLKF VKSDKFSDDP IRIVRDIADV IEDNMDGTSG ALYAIFFPHGF AKGMKDTLEK 480
 SKDISSKTWA AGLKVALDTL FKYTPARPGD STMCDALVPF VETFKVTNDL NAAVEEARKG 540
 ADATADMQAK LGRAVYVGDD VKVPDAGALG VVAIVEGFTK 580

SEQ ID NO: 69 moltype = DNA length = 1755
 FEATURE Location/Qualifiers
 misc_feature 1..1755
 note = S. cerevisiae dihydroxyacetone kinase (S. cerevisiae
 DAK1)
 source 1..1755
 mol_type = genomic DNA
 organism = Saccharomyces cerevisiae

SEQUENCE: 69
 atgtccgcta aatcgtttga agtcacagat ccagtcgaatt caagtctcaa aggggtttgcc 60
 cttgctaacc cctccattac gctgggtccct gaagaaaaaa ttctcttcag aaagaccgat 120
 tccgacaaga tctgattaat ttctggtggt ggtagtggaac atgaacctac acacgcgggt 180
 ttcatttgta aggggtatgtt gagggtggcc gtggttgccg aaatttttgc atccccctta 240

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acaaaacaga ttttaaattgc aatccgttta gtcaatgaaa atgcgtctgg cgttttattg 300
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gccttgaggta ttaactgccg cgttgctgtc ataggtgatg atgttgagcagg tggcagagaa 420
aaggggtggta tgggtggtag aagagcattg gcaggtaccg ttttggttca taagattgta 480
ggtgccttgc cagaagaata ttctagtaag tatggcttag acggtacagc taaagtggct 540
aaaattatca acgacaattt ggtgaccatt ggatcttctt tagaccattg taaagtccct 600
ggcaggaaat tcgaaagtga attaaacgaa aaacaaatgg aattgggtat gggatttcat 660
aacgaacctg gtgtgaaagt tttagacctt attccttcta ccgaagactt gatctccaag 720
tatatgctac caaaactatt ggatccaaac gataaggata gagcttttgt aaagtttgat 780
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caaacaaattg ctggcacatt gatgacctcc ttcaatggta atgggttcag tatcacatta 960
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gaaggtgcta aatccactgc tacattcgag gccaaatttg gcagagcttc gtatgtcggc 1680
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caaaagcgct tgttaa 1755

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SEQ ID NO: 70      moltype = DNA length = 1776
FEATURE           Location/Qualifiers
misc_feature       1..1776
                   note = S. cerevisiae dihydroxyacetone kinase (S. cerevisiae
                   DAK2)
source             1..1776
                   mol_type = genomic DNA
                   organism = Saccharomyces cerevisiae

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SEQUENCE: 70
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tcagctccta attcagggaa cctctcctaaa gtttcattgg tttctggagg tggcagtggt 180
catgagccaa cgcattgccg ttttgttggg gaaggtgcct tagatgcgat tgcagcaggt 240
gccatttttg cttctccttc aactaaacag atctattctg ctattaaagc tgttgaatct 300
cctaagggta ctttgatcat tgtaaaaaat tacaccgggt atattatata ttttggcttc 360
gctgctgaaa gagctaaaag tgctggaatg aaagtcgaa tcggttgctgt aggagatgat 420
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gaagttgctc agtcagtagt tgacaatagt gtcacaattg cggcatctct tgatcactgc 600
acggttcctg gccacaaacc tgaagccatt ttgggcgaga atgagtatga aatcggtatg 660
ggtattcata acgagtctgg tacctataag tcttctccgc tgcctatgat tcttgagctc 720
gtttcccgag tgcctcctct tcttctcgat gaggatgaag accgttctta tgtgaagttt 780
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aagagaacga ttactggagc attcattact gcattgaatg gtcctggggt tggatcaact 960
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cgcagtggtg ggaatcaaat gtaccattct gccaaagatt gggaggtact tgccaaaggg 1080
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SEQ ID NO: 71      moltype = AA length = 291
FEATURE           Location/Qualifiers
REGION            1..291
                   note = D. rerio aquaporin 9 (Drer_T3)
source            1..291
                   mol_type = protein
                   organism = Danio rerio

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SEQUENCE: 71
MEYLENIRNL RGRVLRDI IREFLAELLG TFLVILFGCG SVAQTVLSRE AKGQLLTIHF 60

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GPTLVMLAV	YMAGGVSGGH	VNPAVSLAMV	VLRKLPLKKF	PVYVLAQFLG	AFPGSCAVYC	120
LYYDAFTEFA	NGELAVTGN	VTAGIFASYP	REGLSLLNGF	IDQVIGAGAL	VLCILAVVDK	180
KNIGAPKGME	PLLVLGLSILA	IGVSMALNCG	YPINPARDLG	PRLFTAIAAGW	GLTVFSAGNG	240
WWWVPVVGPM	VGGVVGAAIY	FLMIEMHHPE	NDKNLEDDNS	LKDKYELNTV	N	291

SEQ ID NO: 72 moltype = AA length = 592
 FEATURE Location/Qualifiers
 REGION 1..592
 note = Z. rouxii ZYRO0E01210p (Zrou_T5)
 source 1..592
 mol_type = protein
 organism = Zygosaccharomyces rouxii

SEQUENCE: 72

MGKRTQGFM	DYASVIQGA	VTACYEIGCF	FGSLFVLFFG	DAIGRKPLII	FGAIIIVIIGT	120
VISTAPFHHA	WGLQFVVGR	VITGVGTGFN	TSTIPVWQSE	MTKPNIRGAM	INLDGSVIAF	180
GTMIAWYLD	GFSFINSSVQ	WRFPVSVQII	FALVLLFGIV	RMPEPRWLM	AKKRPAAERY	240
VLACNLDP	NDDAILAEMT	SLHEAVNRSS	NQKSQMKSLF	SMGKQONFSR	ALIASSTQFF	300
QQFTGCNAAI	YYSVLFQTT	VQLDRLLAMI	LGGVFATVYT	LSTLPSFYL	EKVGRRKMMF	360
FGALGQGISF	IITFACLVNP	TKQNAKGA	GLYLFICFG	LAILELPWIY	PPEIASMRVR	420
AATNAMSTCT	NWVTNFAVM	FTPVFIQTSQ	WGCYLFFAVM	NFIYLPVIF	FYPETAGRSL	480
EEIDIIIFAKA	HVDGTLPMV	AHRLPKLSMT	EVEDYSQSLG	LHDDENEKEE	YDEKEAEANA	540
ALFQVETSSK	SPSSNRKDD	APIEHNEVQE	SNDNSSNSSN	VEAPIPVHNN	DP	592

SEQ ID NO: 73 moltype = DNA length = 3050
 FEATURE Location/Qualifiers
 misc_feature 1..3050
 note = DNA fragment comprising Sc_ANB1p, Pp_TKL1 gene and
 Sc_TDH1t
 source 1..3050
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 73

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caatactgtt	accctgaatc	atattcgacg	atgtcgtctc	acacggaaa	ataatcatt	180
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tttttttttt	tttttttttt	ttctgtgttt	cacctttttt	tttttcagtt	gacatctttc	300
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aattattgtc	tagcaaaaag	aaaacataca	cctatttcat	tcacacacta	aaacaatg	660
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aatgagattc	aacccaagaa	acccagcttg	gatcaacaga	gacagatttg	ttttgtccaa	840
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gggtgtccca	atcatgtccg	tcgaagttat	gtctactctc	gggtgggtcta	agtaactcca	2520
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cttcgaattc accgctgaag gtgtcgtga cagagcttct aaggttgctc aattctacaa 2640
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gatgggttaa agattttttc tttttgggaa ataagtaaac aatatattgc tgccttctc 3000
cttccaactg tctgtgcat atccgcctt getgtcctt tcgatatctt 3050

SEQ ID NO: 74      moltype = DNA length = 2717
FEATURE           Location/Qualifiers
misc_feature       1..2717
                   note = DNA fragment comprising Sc_MYO4p, Sc_DAK1 and
                   Sc_GPM1t
source            1..2717
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 74
atccgtcacac gtttgtgcgc aaccggaagt attgcaacag gtcgaagttc gtgcataaca 60
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attcttaaac tttcttttag gtacacaatg aaaacatcaa gattgagtta aaacccttaa 180
aataaaacta ggtctttaag aaactcactc tccgggttat ccatagaatg tttaccattc 240
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attccctgct ttcgagt 2717

SEQ ID NO: 75      moltype = DNA length = 1837
FEATURE           Location/Qualifiers
misc_feature       1..1837
                   note = DNA fragment comprising Sc_HHF2p, Ec_gldA and
                   Sc_EFM1t
source            1..1837
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 75
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agcaacgcga gagagcacia cacgctgtta tcacgcaaac tatgttttga caccgagcca 120

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acaactgttg tctgtttgca cggggaagt ggtactagac cgagagtttc gcatttgtat 300
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gttttgggtt tcgctcaatc caccgtcgaa aagcttttca aggatgctgg tttggttggt 780
gaaatcgctc cattcggtgg tgaatgttct caaaatgaaa ttgaccgttt gagaggattt 840
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cagttgggag caaggtggta cttctgtcgt cgtatttt 1837

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SEQ ID NO: 76      moltype = DNA length = 2789
FEATURE           Location/Qualifiers
misc_feature       1..2789
                   note = DNA fragment comprising Sc_ANB1p0001, ZYR00E01210,
                   Zrou_T5,Sc_TEF1t0001
source             1..2789
                   mol_type = other DNA
                   organism = synthetic construct

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SEQUENCE: 76
tcctcggatc ggcaggttg gagcaagggt gtaactctgt cgctgtattt gtgcagatcg 60
gtgaaaaacat cttgatcttg ctcccggaat ttttagattc aggtaggaaa ttgattacat 120
caatactggt accctgaatc atattcgacg atgtcgtctc acacggaaaat ataattcatt 180
tcttgggttt ccaaaaaaat ttcaattttt tttaactttt ttgtttcgtc ctcccttttt 240
tttttttttt ttattttttt ttctgtgttt cacccttttt tttttcagtt gacatctttc 300
tgcatctctt tctgtgtttt tttttttttt ttctgttttt ccattgttcg ttcgttgcct 360
gttttttcgc cctattgttc tcgagcctaa aaattttttc ctttctgct ttctcttctt 420
cgttcaaatg ttctatttcc attgttctct ttggtaaact cattgttgtc ggaactcaga 480
tatattcagg tcaatttact gtaactcaat tgaacttttt cttgaaattt caacttgcct 540
tttcaacttg ttcttctttt ttaactttat tctacacttt agttccctta ccttgttctc 600
aattattgtc tagcaaaaag aaaacataca cctatttcat tcacacacta aaacaatggg 660
taagagaact caaggtttca tggactacgt tttctccaga acttctactg ctgggttgaa 720
agggtccaga ttgagataca ctgctgctgc cgttgcgtgt atcggtttcg ctttattcgg 780
ttacgaccaa ggtttgatgt ctgggttgat taccggtgac caattcaaca aggaattccc 840
tccaacaaaa tctaaccggtg acaacgacag atacgcctcc gtcacccaag gtgctgtcac 900
tgcttggtac gaaattgggt gtttcttcgg ttctttgttc gtcattattc ttggtgatgc 960
catcggtaga aagccattaa tcattttcgg tgccatcatc gtcacattg gtactgttat 1020
ctccactgct ccattccacc acgcttgggg tttgggtcaa tttgttgtcg gttaggttat 1080
cactgggtgc ggtactgggt tcaacacttc caccatccca gtcctggcaat ctgaaatgac 1140
caagccaaac atcagaggtg ctatgattaa cttggatggt tccgtcattg ctttcggtac 1200
tatgatgtct tactgggttg atttcgggtt ctctttcacc aactctctcg ttcaatggag 1260
attccaggtt tccgttcaaa tcacttttgc tttggtctta ttgttcggtt ttgtcagaat 1320
gccagaatct ccaagatggt ttagtgccaa gaagcgtcca gctgaagcca gatacgtttt 1380
ggcttggttg aatgacttgc cagaaaacga cgtatgccatc ttggccgaaa tgacttcttt 1440
gcacgaagct gtcacacagt cttctaacca aaagtctcaa atgaaatctt tgttctccat 1500
gggtgaagcaa caaaacttct ccagagcttt gattgcttcc tctaccaaat tcttccaaca 1560
attcaccggt tgtaatgctg ccatctacta ctccactggt ttgttccaaa ccaccgttca 1620
attagataga ttgttggtta tgactttggg tgggtgtctt gctaccgttt acaccttgct 1680
cactttgcca tctttctact tggttgaaaa ggttgggtcg cgtgaagatg tcttcttcgg 1740
tgctttagggt caaggtattt ctttcattat tacctttgct tggtttggtc acccaaccaa 1800
gcaaaacgct aaggggtgct ctgtcgggtt atactgttcc atcatctggt tcgggtttggc 1860
tatcttgaaa ttgcccattg tctaccacc agaaattgct tccatgagag tcagagctgc 1920
taccacacgt atgtccacct gtaccaactg ggttaaccaac tttgccgttg tcatgttcac 1980
cccagtcctt atccaaactt ctcaatgggg ttgttacttg tcttctcgtg tcatgaactt 2040
catctacttg cctgttatct tctcttctca ccagaaaact gccggctcgt ctttgaagaa 2100
aatcgatatt atttctgcta aggtcctatg tgacgggtact ctaccatgga tgggtgtctc 2160
cagattacca aagttgtcca tgactgaagt cgaagattac tctcaatctt tgggtttgca 2220

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cgacgatgaa aacgaaaagg aagaatacga cgaaaaagaa gctgaagcta acgctgcctt 2280
gttccaagtt gaaaaccttt ccaagtctcc atcctccaac agaaaggatg acgacgctcc 2340
aattgaacac aacgaagttc aagaatctaa cgacaactcc tctaactctt ctaacgttga 2400
agctccaatt ccagttcacc acaacgaccc ataaaggaga ttgataagac tttctagt 2460
gcatactctt tatatttaaa tcttatctat tagttaattt tttgtaattt atccttata 2520
atagctctgt tatcttaaaa tatcatttca gtatctaaaa attccccctt ttttctcagt 2580
atatcttaac aggcgacagt ccaaatgttg atttatccca gtccgattca tcagggttgt 2640
gaagcatttt gtcaatggtc gaaatcacat cagtaatagt gcctcttact tgctcatag 2700
aatttcttct tcttaacgtc accgttttgt ctttctctca tttggaacag caggcgaggt 2760
gtattggtgc gagaagaggt gtttccgag 2789

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SEQ ID NO: 77      moltype = DNA length = 1576
FEATURE
misc_feature       1..1576
                    note = DNA fragment comprising Sc_PFY1p, Ec_gldA and
                    Sc_EFM1t
source             1..1576
                    mol_type = other DNA
                    organism = synthetic construct

```

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SEQUENCE: 77
acccgatccc gaagaaagtc caactccagt tcgattccct gctttcgagt gtgcaggaga 60
cgttactttg tttatatata ttagtatgta caatcgcaaa gaaatggagt gatgacatgt 120
tgtagtattt agtatgaggt tactgtgtgg gaggtttttac catgattttt ggcgagaaca 180
cgccatgaaa tgtctttgta cgaaactcat taccgcgatt aatatttttt ttctttttaa 240
agctcagttg accctttctc attccttctt taaaacaact gtgtgatcct tgagaaaaga 300
taaaattacat acacaacata aacccaacta cgatcgcaaa taatggacag aatcatccaa 360
tctccaggtg agtacatcca aggtgctgat gttatcaaca gattagggtg atacttgaag 420
ccattggctg aaagatgggt agtcgtcggt gacaaattcg ttttgggttt cgctcaatcc 480
accgtcgaaa agtctttcaa ggatgctggt ttggttgttg aaatcgctcc attcgggtgt 540
gaatgttctc aaaatgaaat tgaccgtttg agaggatttg ctgaaactgc tcaatgtggt 600
gcctctcttg gtattgttgg tggtaagact ttggacactg ccaaggcttt ggcccacttc 660
atgggtgttc cagttgccaat tgctccaacc attgcttcta ccgatgctcc atgttctgct 720
ttgtccgtta tctacaccga cgaaggtgaa ttgaccgtt acttggtgtt gccaaacaac 780
ccaaacatgg tcattgttga caccgaagtc gttgccggtg ctccagccag attattggct 840
gcccgtatcg gtgatgcttt ggctacctgg ttccgaagcca gagcttgttc cagatctggt 900
gctactacca tggccggtgg taaatgtact caagctgctt tagctttggc tgaattgtgt 960
tacaacactt ttgtggaaga aggcgaaaag gctatgttgg ctgctgaaca acacgttgtt 1020
actccagctt tggaaagagt cattgaagcc aacacctact tgctccggtt tggattcgaa 1080
tctgtgtggt tagctgcccgc tcatgcccgc cacaacggtt tgactgccat cccagatgct 1140
caccactact accacggtga aaagggttgc ttccgtactt tgactcaatt agtcttggaa 1200
aacgctccag tcgaagaaat cgaaaaccgtt gctgctctat cccacgctgt cggtttgctc 1260
atcactttgg ctcaattgga catcaaggaa gatgtcccag ctaagatgag aattgttgc 1320
gaagctgctt gtgctgaaag tgaaccattt cacaacatgc caggtggtgc caccocagac 1380
caagctacg ctgctttggt gggtgctgac caatacgtgc aaagattctt gcaagaatgg 1440
gagtaaaatt gatctgtagc ctaagtataa aattctacgt atgtatatat ttacatgcaa 1500
ttttttcttt ttccaattca tgctctcctt cggatcgccc agttgggagc aaggtggtac 1560
ttctgtcgtc gtatttt 1576

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SEQ ID NO: 78      moltype = DNA length = 500
FEATURE
misc_feature       1..500
                    note = Repair DNA fragment for homologous recombination and
                    deletingGPD2
source             1..500
                    mol_type = other DNA
                    organism = synthetic construct

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SEQUENCE: 78
atgcaagcgt gtatcttcta agattcagtc atcatcatta ccgagtttgt ttctcttcac 60
atgatgaaga aggtttgagt atgctcgaaa caataagacg acgatggctc tgccattgtt 120
atattacgct ttgtccgcca ggtgccgatg ggttgcgtgag gggaagagtg tttagcttac 180
ggacctattg ccattgttat tccgattaat ctattgttca gcagctcttc tctacctgt 240
cattctagta taagaatcta tcattaacgt taatttctat atatacataa ctaccattat 300
acacgctatt atcgtttaca tatcacatca ccgttaatga aagatacgac accctgtaca 360
ctaacacaat taaataatcg ccataacctt ttctgttacc tatagccctt aaagctgttt 420
cttcgagctt tttcactgca gtaattctcc acatgggccc agccactgag ataagagcgc 480
tatgttagtc actactgacg 500

```

```

SEQ ID NO: 79      moltype = DNA length = 322
FEATURE
misc_feature       1..322
                    note = EB01 terminator (Sc_EB01t)
source             1..322
                    mol_type = genomic DNA
                    organism = Saccharomyces cerevisiae

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SEQUENCE: 79
ggtctcgtaa aagcttttga ttaagccttc tagtccaaaa aacacgtttt ttgtcattt 60

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atttcatttt cttagaatag tttagtttat tcattttata gtcacgaatg ttttatgatt 120
ctatataggg ttgcaaacaa gcatttttca ttttatgtta aaacaatttc aggtttacct 180
tttattctgc ttgtgggtgac gcgtgtatcc gcccgctctt ttggtcacc atgtatttaa 240
ttgcataaat aattcttaaa agtggagcta gtctatttct atttacatac ctctcatttc 300
tcatttcctc ccttcggaga cc 322

```

```

SEQ ID NO: 80      moltype = DNA length = 621
FEATURE           Location/Qualifiers
misc_feature       1..621
                   note = S. cerevisiae ENO2 promoter
source            1..621
                   mol_type = genomic DNA
                   organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 80
ggctcgggtg cgtgtcgacg ctgcgggtat agaaagggtt ctttactcta tagtacctcc 60
tcgtctacga ctgtcttctt cccaaagatg aacgcggcgt tatgtcacta acgacgtgca 120
ccaaacttgc gaaagtggaa tcccgttcca aaactggcat ccactaattg atacatctac 180
acaccgcacg ccttttttct gaagcccaact ttcgtggact ttgccatag caaaattcat 240
gaagtgtgat accaagtcag catacacctc actagggtag tttctttggt tgtattgatc 300
atttgggtca tcgtgggtca ttaatttttt ttctccattg ctttctggct ttgatcttac 360
tatcatttgg atttttgcg aaggtttag aattgtatgt gacaagtggc accaagcata 420
tataaaaaaa aaaagcatta tcttcctacc agagtgtatt gttaaaaacg tatttatagc 480
aaacgcaatt gtaattaatt cttattttgt atcttttctt ccttgtctc aatcttttat 540
ttttatttta tttttctttt cttagtttct ttcataaac caagcaacta atactataac 600
atacaataat aatgggagac c 621

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SEQ ID NO: 81      moltype = DNA length = 422
FEATURE           Location/Qualifiers
misc_feature       1..422
                   note = S. cerevisiae YOX1 terminator (Sc_YOX1t)
source            1..422
                   mol_type = genomic DNA
                   organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 81
ggctctgtaa atctaattctc tagtcattat ttattcgcaa attcatttcc ctatacggca 60
ttcatacata tcattgttca cttcagtcct agcatatato ataaaatata caattgtttt 120
ctaattacct tacgtttttt aaaagacttc tataatacct cttttaactt tacatgtagt 180
caaaataaag tgcagttcca tcgatggtag tttctcacc cggttgagtg atgttaacga 240
tgtttacgtg ataaaactta attatattat atcttttttt gcttatatgt tatacataga 300
ataaaaagtt gattaaacac acattggctc gaaaacacgt gtagtacttt ctcttttgaa 360
gaataaaaaa agaaaataaa gataataaaa acgaaaatag cgtacaatta tcttcggaga 420
cc 422

```

```

SEQ ID NO: 82      moltype = DNA length = 20
FEATURE           Location/Qualifiers
misc_feature       1..20
                   note = INT14.02 protospacer
source            1..20
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 82
ggagagaaag gcccgggcgt 20

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```

SEQ ID NO: 83      moltype = DNA length = 178
FEATURE           Location/Qualifiers
misc_feature       1..178
                   note = Sc_INT14.02_FLANK5
source            1..178
                   mol_type = unassigned DNA
                   organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 83
tttgagcgga cgccgaaacg tttttttttt ttgcttttagt ccagatacta accttttata 60
attttctttt tagggaggaa gaccgtctta agctcttaga gggtctcgca taccgaagta 120
aaagccaaga ccgaagcaaa cacgcaggat aatttttcgg tttactctca ctgcagct 178

```

```

SEQ ID NO: 84      moltype = DNA length = 571
FEATURE           Location/Qualifiers
misc_feature       1..571
                   note = Sc_INT14.02_FLANK3
source            1..571
                   mol_type = unassigned DNA
                   organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 84
catcttgatt tatgtcttgt ttgaggagat ttcggataaa tattcttcga ggggaaaatg 60
tcgtaaaaat aaaaacaata aagaaacaaa acgattaaaa gaaaaactca accatgctca 120
aaagtctctc tctttttttt cacgtgctgc gctgatgtaa gcagcaggac atggcaggaa 180

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gaaaatcgga gaattgtggc cgcgacacct gacaaacaat cgtcaggcaa atgtaaaatt 240
gtacatgaaa ggctgcgtag tcaatgtctt agaaggccgg cccgcgatgat ccagaagcag 300
accataaaaa aatcaaaaat tcgcggtctg gcagcaggca taagatgcat agcgttatcc 360
taaatgacat cacgatgata aatctctccg cgcgatgatgc ttttgatttg cctaagggcc 420
tgccatcggt ctggctcaaa ctattgaggg tctacatacc ttgaaaatcc aagtaaaagg 480
atggatatcg ttatactaaa agcaacacag aaaaggtcca cgtcagttcc acacaataac 540
atttacgtag tgttcacgcc gaagcagtta c 571

```

```

SEQ ID NO: 85      moltype = DNA length = 20
FEATURE
misc_feature      1..20
                  note = INT95 protospacer
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

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```

SEQUENCE: 85
gaaggcctat ggaaatggcc 20

```

```

SEQ ID NO: 86      moltype = DNA length = 150
FEATURE
misc_feature      1..150
                  note = Sc_INT95B_FLANK5
source            1..150
                  mol_type = unassigned DNA
                  organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 86
atagttacag ctaaacattt gcccaaaacc atctctttat atatatactt gcgatggcga 60
gcccagcgga agggatgtcc gcttactaat tccgacacac ctactttcga cggaatttgc 120
aatctccac aatatctccc tctgttctc 150

```

```

SEQ ID NO: 87      moltype = DNA length = 150
FEATURE
misc_feature      1..150
                  note = Sc_INT95B_FLANK3
source            1..150
                  mol_type = unassigned DNA
                  organism = Saccharomyces cerevisiae

```

```

SEQUENCE: 87
cttccaactg tctctgtgat atccgccctt gctgctcctt tcgatatctt gatgccgata 60
atttcacat ggatgagtcg cagtgctcgtc gtacgtgatg agggcaggct tgaaaattat 120
tcttcgagcc caattctctt tacttatagg 150

```

```

SEQ ID NO: 88      moltype = DNA length = 20
FEATURE
misc_feature      1..20
                  note = GPD2 protospacer
source            1..20
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 88
gatcttttac actccatcaa 20

```

```

SEQ ID NO: 89      moltype = AA length = 910
FEATURE
REGION            1..910
                  note = Nitrate reductase from Zea mays (Zm_NR1)
source            1..910
                  mol_type = protein
                  organism = Zea mays

```

```

SEQUENCE: 89
MAAVEPRQFG RLEPGSSPVR VATNGAKAYP PPASHLPARR DSPVRGCSFP PLVSPPRRLD 60
DASDDEDEEQ EDWRELYGSH LQLEVEPAVQ DARDEGTADA WIERNPCLVR LTGKHPLNCE 120
PPLARLMHHG FITPAPLHYV RNHGAVPRGD WATWTVETG LVRRPARLTM EELARDFPAV 180
EIPVTLACAG NRRKEQNMVQ QTVGFNWGAA GVSTSVWRGA RLRDVLRCG IVPKGGGALN 240
VCFEGAEDLP GGGGSKYGTS VTREWALDPS RDIMLAYMQN GEPLLPDHGF PVRVPIPGCI 300
GGRMVKWLKR IIVTPAESDN YYHFKDNRVL PSHVDAELAN AEAWWYKPEY IINELNINSV 360
ITTPGHDEIL PINSITTORG YTMKYAYSAG GKKKVTRVEV TLDGGETWLW CHLDHPEKPN 420
KYGKXWCWCF WSVEVEVLDL LGAKEIAVRA WDQSLNTQPE KLIWNLMGMM NNCWFVKVKN 480
VCRPHKGEIG LVFEHPTQPG NQPGGWMARQ KHLETAEEAA PGLKRSTSTP FMNTTDVGKQ 540
FTMSEVRKHA SQESAWIAVH GHVYDCTKFL KDHPGGADSI LINAGTDCTE EPDAIHSDKA 600
KALLDTYRIG ELITGTGYS SDNSVHGGSV LSHLAPIREA VRAPALSNPR DKIHCLRVGK 660
KELSRDVRLF RFSLPSPDQV LGLPIGKHIF VCASIEGKLC MRAYTPTSMV DEIGHFDLLV 720
KVYFKNEHPK FPNGLMTQY LDSLPVGSYI DVKGPLGHVE YTGRGSFVIN GKQRHASRLA 780
MICGSGGITP MYQIIQAVLR DQPEDHTEMH LVYANRTEDD ILLRDELDWR AAEYPDRCLKV 840
WYVIDQVKRP EEGWKYSVGF VTEAVLREHV PEGGDDTLAL ACGPPPMIQF AISPNELEKMK 900
YDMANSFVVF 910

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SEQ ID NO: 90 moltype = AA length = 926
 FEATURE Location/Qualifiers
 REGION 1..926
 note = Nitrate reductase from *Spinacia oleracea* (Spo_NIA)
 source 1..926
 mol_type = protein
 organism = *Spinacia oleracea*

SEQUENCE: 90
 MAASVDRQYH PAPMSGVVRT PFSNNHRSDS PVRNGYTFSN PPSSNGVVKP GEKIKLVDNN 60
 SNSNNGSNMN NNRYDSDEE DDDENEMNVW NEMIKKGNSE LEPSSVDSRD EGTADQWIER 120
 NPSMIRLTGK HPPNSEPLT RLMHGHFLTP VPLHYVRNHG PVPNAKWEDW TVEVTGLVKR 180
 PIRFTMDQLV NDFQSRFPV TLVCAGNRRK EQNMTKQSIG FNWGSAAVST SVWRGVPLRD 240
 VLKRCGVMS LKALNVCFE GAEDLPGGGG SKYGTSVKRE FAMDPARDII LAYMQNGEKL 300
 SPDHGYPVVM IIPGFIGGRM VKWLKRIIVT TTESDNYHYH KDNRLVPSHV DAELANSEAW 360
 WYKQEIINE LNVNSVITSP CHEEILPINA WTTQRPYTMR GYAYSGGGRK VTRVEVTMDG 420
 GDTWDICELD HQERGSKYGK FWCWCFSLE VEVLDDLGA EIGVRAWDES LNTQPEKLIW 480
 NVMGMMNWCW FRVKTNVCKP HKGEIGIVFE HPTQPGNKS GWMARERHLE ISDSGPTLKR 540
 TASTPFMNTT SKMYSMSEVK KHNTADSAWI VVHGNVYNAT RFLKDHPPGS DSILINAGTD 600
 CTEEFDAIHS DAKRLLLEDF RIGELISTGY TSDSSSPGNS VHGGSVYSGL AGLAPITEAV 660
 PLRNVALNPR VKIPCKLIEK VSLSHDVRRF RFLPSEDQV LGLPVGKHIF LKANVDDKLC 720
 MRAYTPSSTI DVVGYFDLVV KVFVKDVHPR FPNGGVMSQH LDSLSLGSIV DVKGPLGHIE 780
 YLGKGNFTVH GKPKFAKLA MISGGTGITP IYQVMQAILK DPEDKTEMHV VYANRTEEDI 840
 LLREELDKWA DEFDRVKVW YVVEKAEEGW KYDTGFISEK ILRDHVPVAVG DDVLALTCGP 900
 PPMIQFAVQP NLDKMGFDIK EQLLIF 926

SEQ ID NO: 91 moltype = AA length = 108
 FEATURE Location/Qualifiers
 REGION 1..108
 note = nirD Nitrite reductase (small subunit) from *E. coli*
 (Ec_nirD)
 source 1..108
 mol_type = protein
 organism = *Escherichia coli*

SEQUENCE: 91
 MSQWKDICKI DDILPETGVC ALLGDEQVAI FRPYHSDQVF AISNIDPFFE SSVLSRGLIA 60
 EHQLGELWAS PLKQRFRLS DGLCMEDEQF SVKHYEARVK DGVVQLRG 108

SEQ ID NO: 92 moltype = AA length = 847
 FEATURE Location/Qualifiers
 REGION 1..847
 note = nirB Nitrite reductase (large subunit) from *E. Coli*
 (Ec_nirB)
 source 1..847
 mol_type = protein
 organism = *Escherichia coli*

SEQUENCE: 92
 MSKVRLLAIIG NGMVGHFRFIE DLLDKSDAAN FDIITVFCEEP RIAYDRVHLS SYFSHHTAEE 60
 LSLVREGFYE KHGIKVLVGE RAITINRQEK VIHSSAGRTV FYDKLIMATG SYPWIPPIKG 120
 SDTQDCFYVR TIEDLNAIES CARRSKRGAV VGGGLLGLEA AGALKNLGIE THVIEFAPML 180
 MAEQLDQMG EQLRRKIESM GVRVHTSKNT LEIVQEGVEA RKTMRFADGS ELEVDPIVFS 240
 TGIRPRDKLA TQCLDVA PR GGIVINDSCQ TSDPDIIAIG ECASWNNRVF GLVAPGYKMA 300
 QVAVDHILGS ENAFEGADLS AKLKLGLVDV GGIGDAHGRT PGARSYVYLD ESKEIYKRLLI 360
 VSEDNKTLLG AVLVGDTSDY GNLLQLVLNA IELPENPDSL ILPAHSGSGK PSIGVDKLPD 420
 SAQICSCFDV TKGDLIAAIN KGCHTVAALK AETKAGTGCG GCIPLVTOVL NAEKQOGIE 480
 VNNLCEHFA YSRQELPHLI RVEGIKTFFEE LLAKHGKGYG CEVCKPTVGS LLASCWNEYI 540
 LKPEHTPLQD SNDNFLANIQ KDGTVSVIPR SPGGEITPEG LMAVGRIARE FNLYTKITGS 600
 QRLAMFGAQK DDLPEIWRQL IEAGFETGHA YAKALRMAKT CVGSTWCYRG VGDVSVGLGVE 660
 LENRYKGIRT PHKMKFVGSV CTRECSEAQG KDVGIATEK GWNLYVCNG GMKPRHADLL 720
 AADIDRETLL KYLDRFMFMY IRTADKLTRT APWLENLEGG IDYLKAVIID DKLGLNAHLE 780
 EEMARLREAV LCEWTETVNT PSAQTRFKHF INSDKRDPNV QMVPEREQHR PATPYERIPV 840
 TLVEDNA 847

SEQ ID NO: 93 moltype = AA length = 1104
 FEATURE Location/Qualifiers
 REGION 1..1104
 note = niiA Nitrite reductase from *Emericella nidulans*
 (*Aspergillus nidulans*) (En_niiA)
 source 1..1104
 mol_type = protein
 organism = *Emericella nidulans*

SEQUENCE: 93
 MPLLDGPRNG ETVTASAHNG IPIIDGVDP TLRGDIDQDP NRRQKIVVVG LGMVAVAFIE 60
 KLVKLDSEER KYDIVVIGEE PHIAYNRVGL SSYFEHRKIE DLYLNPKEWY GSFKDRSFDY 120
 YLNTRVTDVF PQHKTVTST GDIVSYDILV LATGSDAVLP TSTPGHDAK IFVYRTISDL 180
 ERLMEFAANH KGQGTGVTVGG GLLGLEAKA MTDLEDFGSV KLIDRNKWL ARQLDGDAGS 240

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LVTKKIRDLG	LEVLHEKRVA	KIHTDDNNV	TGILFEDGQE	LDCCCICFAI	GIRPRDELGG	300
STGIQCAKRG	GFVIDESLRT	SVNDIYAIGE	CASWENQTFG	IIAPGIEMAD	VLSFNLTNPD	360
KEPKRFNRPD	LSTKLKLLGV	DVASFGDFFA	DRDGPFLPG	QRPSAESIGA	ADPNREEEPQ	420
VKALTYRDPF	GGVYKYLFT	MDGKYLGGM	MIGDTKDYYK	LNQMVKSQKP	LEVPPSEFIL	480
GAQSGGEENA	DDLDDSTQIC	SCHNVTKGDV	VESVKSCTCK	TIADVKSCTK	AGTGC GGCM	540
LVQSFNKTMT	LDMGQEVSN	LCVHIPYSRA	DLNVIAIRQ	LRTFDDVMKS	AGKCPDSLGC	600
EICKPAIASI	LSSLFNPHLM	DKYHELQET	NDRFLANIQR	NGTFSVVRV	PGGEITADKL	660
IAIGQVAKKY	NLYCKITGGQ	RIDMFGARKQ	DLLDIWTEL	DAGMESGHAY	AKSLRTVKSC	720
VGTTWCRFGV	GDSVGMAIRL	EQRYKSIRAP	HKFKGAVSGC	VRECAEAQNK	DFGLIATEKG	780
FNIFVGGNGG	AKPRHSELLA	KDVPPEEVIP	ILDYVIFYI	RTADKLQRTA	RWLESLPGGI	840
EYLDVVVLND	KLGIAAEMER	QMQLVDSYF	CEWTETVRNP	KRRKYFQQA	NTDETVENVE	900
IVKEREQVRP	TYWPKDGANE	DFKGHWSSSL	SWQPVIKADY	FSDGPPAISS	ANIKRGDTQL	960
AIPKVKGKYY	ATQMCPhKR	TFVLSDLGIG	DDNGKYWVS	CPYHKRNFEL	NGEQAGRCQN	1020
DEAMNIATFP	VEEREDGWYI	MKLPPVEELD	SVLGTEKWKV	KKGEAVDPFE	AYDKKYSGMK	1080
GKRAGAKGIE	GSKPTRSPSN	TIDW				1104

SEQ ID NO: 94 moltype = AA length = 508

FEATURE Location/Qualifiers

REGION 1..508

 note = YNT1 Nitrate/Nitrite transporter from *Pichia angusta*

 (Hansenulapolymorpha)

source 1..508

 mol_type = protein

 organism = *Ogataea angusta*

SEQUENCE: 94

MQLSTLWEPP	IVNPRNLKAT	SIPIPNLWNV	YGRNFFFGWF	GFFVCFLSWF	APPPLLHGML	60
KDRLRLTAVD	ISNNNICGLT	GTLLGRFILG	PLNDKYGPRI	TLTGLVLVAGA	IPATFVPLVT	120
NVAGLHAIRF	FISPLGSSFI	CCSQFCVFF	DNNIIGTANA	ISAGWGNAGG	GVAFFVMPAI	180
SNALENRGYS	LHHSWSYSFV	IGPFLILMIT	AILIFVFGSD	CPRGRWSLRG	DILGINMDNM	240
LVKSVSVTRH	FSKEGELTSV	FVEPVNAIDK	VVVEPNQDQE	ILEVADIING	DEIIEDPSLN	300
DVVKICLSRP	TMLVGLCYMC	SFGTELAVES	IISNLFQGM	TNWSTSKAGA	WGSMLGLLNV	360
VARPAGGIIS	DFLYQRFKTT	KAKKFWMIPT	GLMQGIFLIW	IGLVPELSIA	GLIVSVSFLC	420
LWFEMNGGAN	YACVPVVRH	HSGIVSGVTG	AMGNLGGILF	SLVFRYTISN	GVNNYFKAFW	480
IIGIVCTAVN	LVCVLIPIRE	ERPRKAEN				508

1. A recombinant yeast cell functionally expressing:
 - a nucleic acid sequence encoding a native protein having transketolase activity (EC 2.2.1.1); and
 - a nucleic acid sequence encoding a heterologous protein having transketolase activity (EC 2.2.1.1).
2. The recombinant yeast cell according to claim 1, wherein the heterologous protein having transketolase activity comprises or consists of an amino acid sequence having a percentage identity with the amino acid sequence of the native protein having transketolase activity in the range of equal to or more than 30% to equal to or less than 80%.
3. The recombinant yeast cell according to claim 1, wherein the heterologous protein having transketolase activity comprises or consists of:
 - an amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19; or
 - a functional homologue of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19, having at least 70% sequence identity with the amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19; or

- a functional homologue of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10, SEQ ID NO: 11, SEQ ID NO: 12, SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17 or SEQ ID NO: 19.
4. The recombinant yeast cell according to claim 1, wherein the native protein having transketolase activity comprises or consists of:
 - an amino acid sequence of SEQ ID NO: 1; or
 - a functional homologue of SEQ ID NO: 1, having sequence identity with the amino acid sequence of SEQ ID NO: 1; or
 - a functional homologue of SEQ ID NO: 1, having one or more mutations, substitutions, insertions and/or deletions when compared with the amino acid sequence of SEQ ID NO: 1.
5. The recombinant yeast cell according to claim 1, wherein the expression of the nucleic acid sequence encoding the heterologous protein having transketolase activity is under control of a promoter (the "TKL promoter") which TKL promoter has an anaerobic/aerobic expression ratio for the transketolase of 2 or more.
6. The recombinant yeast cell according to claim 1, wherein the expression of the nucleic acid sequence encoding the native protein having transketolase activity is under

control of a promoter (the “TKL promoter”) which TKL promoter has an anaerobic/aerobic expression ratio for the transketolase of 2 or more.

7. The recombinant yeast cell according to claim 5, wherein the TKL promoter is the promoter of ANB1 and/or DAN1.

8. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell comprises one or more genetic modifications to functionally express a protein that functions in a metabolic pathway forming a non-native redox sink.

9. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell functionally expresses:

a nucleic acid sequence encoding a protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity; and/or

a nucleic acid sequence encoding a protein having phosphoribulokinase (PRK) activity; and/or

a nucleic acid sequence encoding one or more molecular chaperones for the protein having ribulose-1,5-biphosphate carboxylase oxygenase (Rubisco) activity.

10. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell functionally expresses:

a nucleic acid sequence encoding a protein comprising phosphoketolase activity (EC 4.1.2.9 or EC 4.1.2.22, PKL); and/or

a nucleic acid sequence encoding a protein having phosphotransacetylase (PTA) activity (EC 2.3.1.8); and/or

a nucleic acid sequence encoding a protein having acetate kinase (ACK) activity (EC 2.7.2.12).

11. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell functionally expresses a nucleic acid sequence encoding a protein comprising NAD⁺ dependent acetylating acetaldehyde dehydrogenase activity (EC 1.2.1.10).

12. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell functionally expresses a nucleic acid sequence encoding an enzyme having NADH-

dependent nitrate reductase activity and/or a nucleic acid sequence encoding an enzyme having NADH-dependent nitrite reductase activity.

13. The recombinant yeast cell according to claim 12, wherein the recombinant yeast cell further functionally expresses a nucleic acid sequence encoding an enzyme having nitrate and/or nitrite transporter activity.

14. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell further comprises a deletion or disruption of a nucleic acid sequence encoding a protein having glycerol-3-phosphate dehydrogenase (GPD) activity and/or a nucleic acid sequence encoding a protein having glycerol phosphate phosphatase (GPP) activity.

15. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell further functionally expresses:

a nucleic acid sequence encoding for a protein having glycerol dehydrogenase activity (E.C. 1.1.1.6);

a nucleic acid sequence encoding a protein having dihydroxyacetone kinase activity (E.C. 2.7.1.28 or E.C. 2.7.1.29); and/or

a nucleic acid sequence encoding a protein having glycerol transporter activity.

16. The recombinant yeast cell according to claim 1, wherein the recombinant yeast cell further functionally expresses a nucleic acid sequence encoding a protein having glucoamylase activity (EC 3.2.1.20 or 3.2.1.3).

17. A process for the production of ethanol, comprising converting a carbon source using a recombinant yeast cell according to claim 1.

18. The process according to claim 17, wherein the process is at least partly carried out in a medium comprising glucose in a glucose concentration of 25 g/L or more.

19. The process according to claim 17, wherein the process is at least partly carried out in the presence of a saccharolytic enzyme.

20. The recombinant yeast cell according to claim 6, wherein the TKL promoter is the promoter of ANB1 and/or DAN1

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